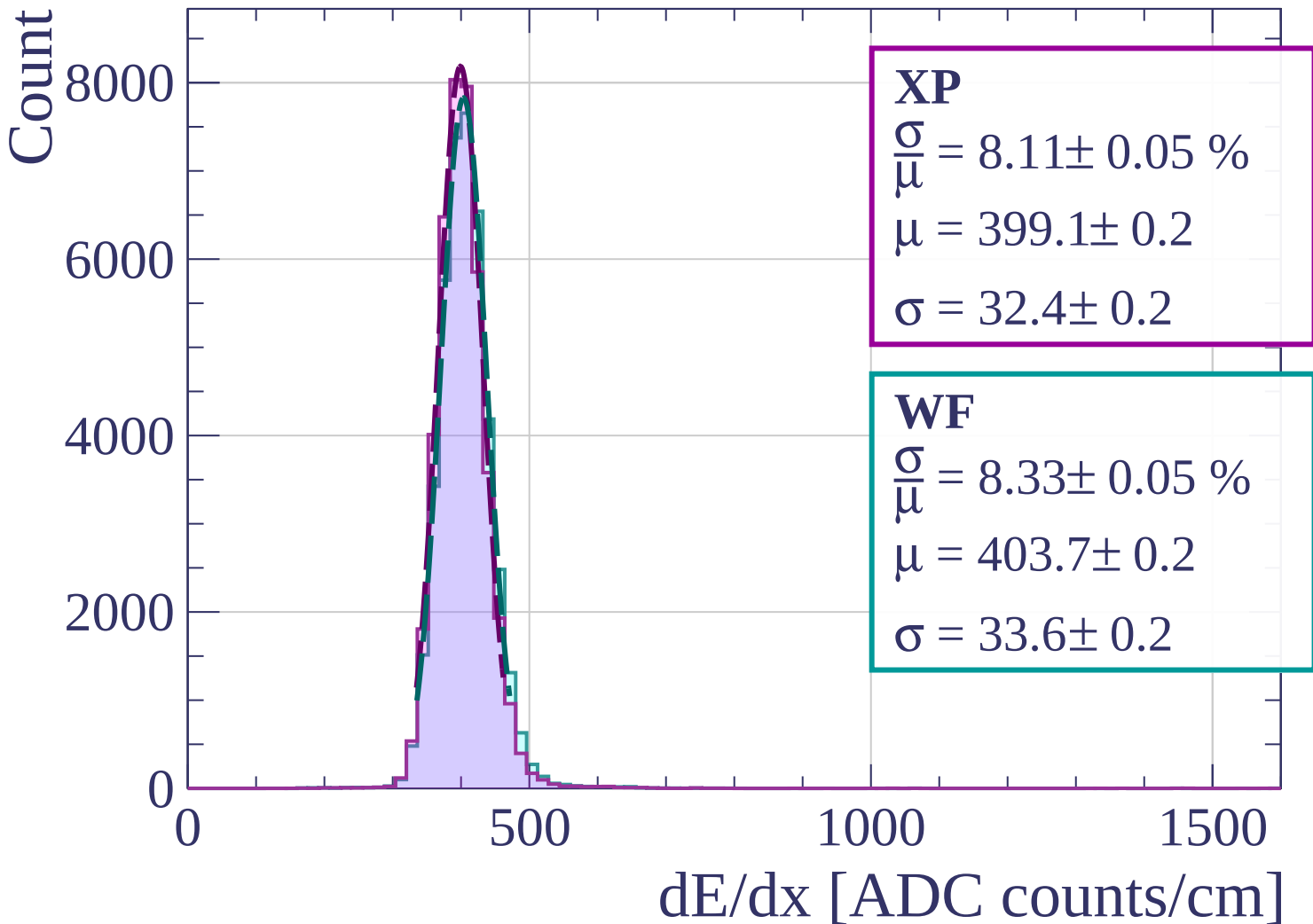
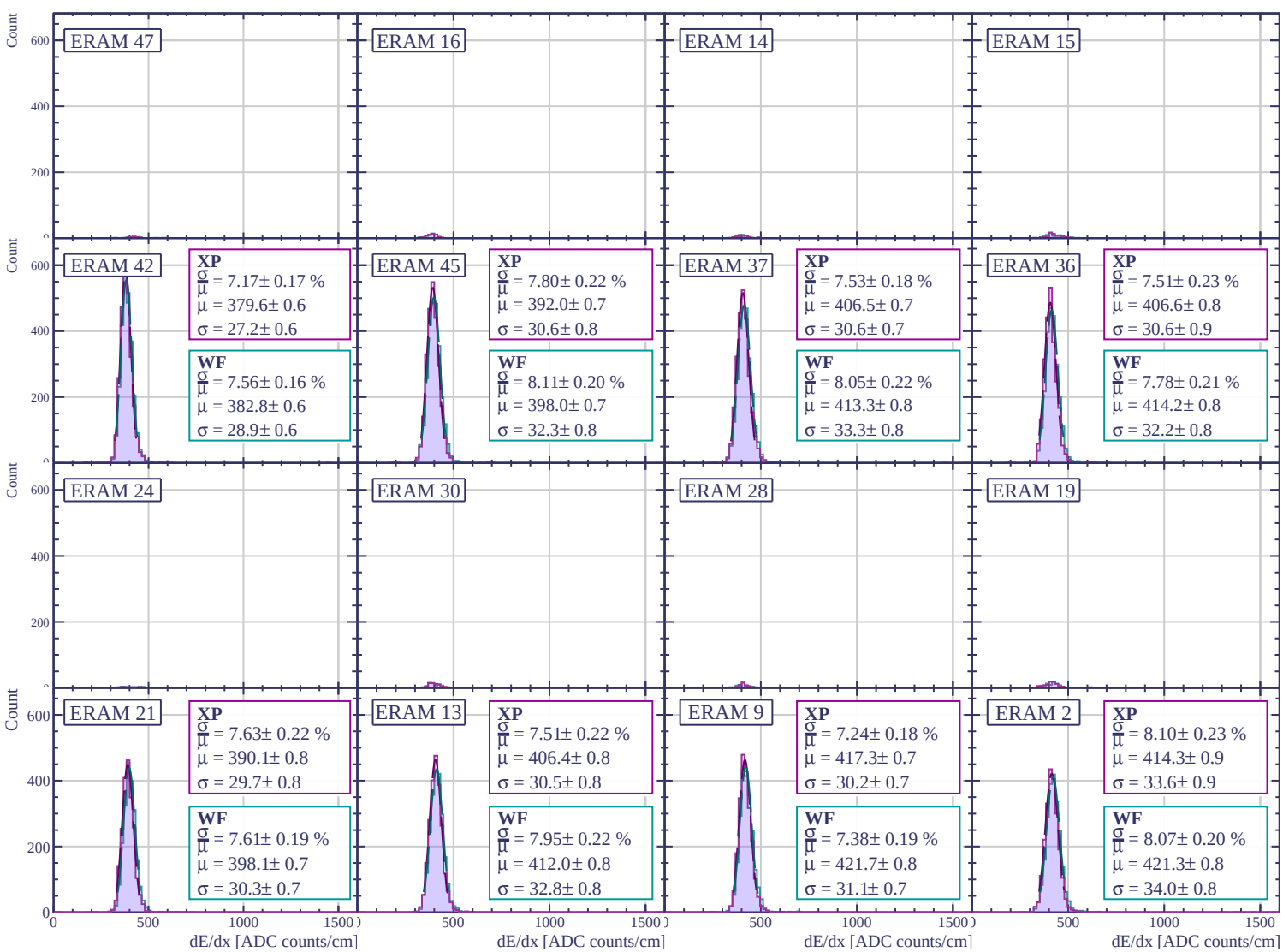
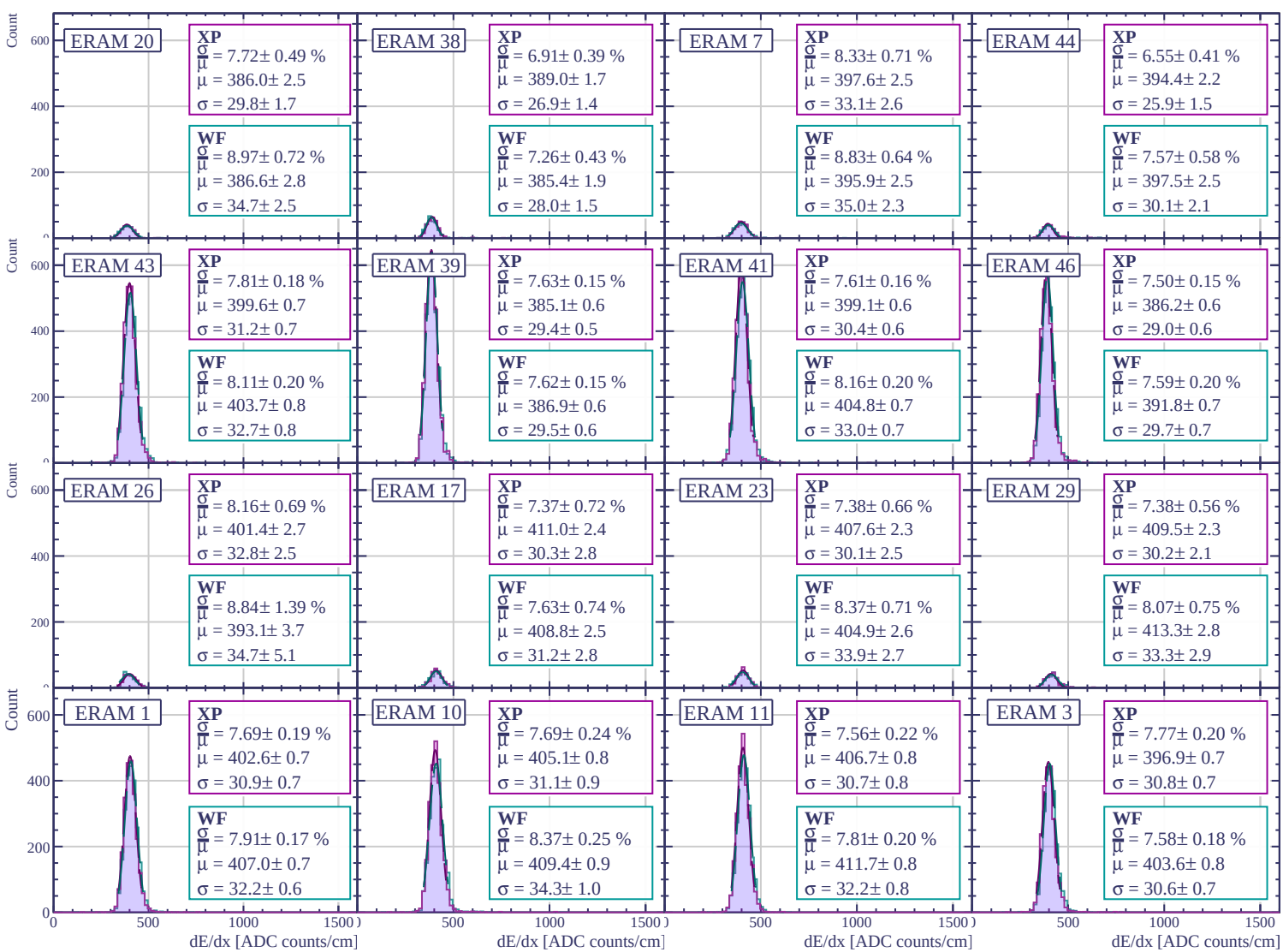
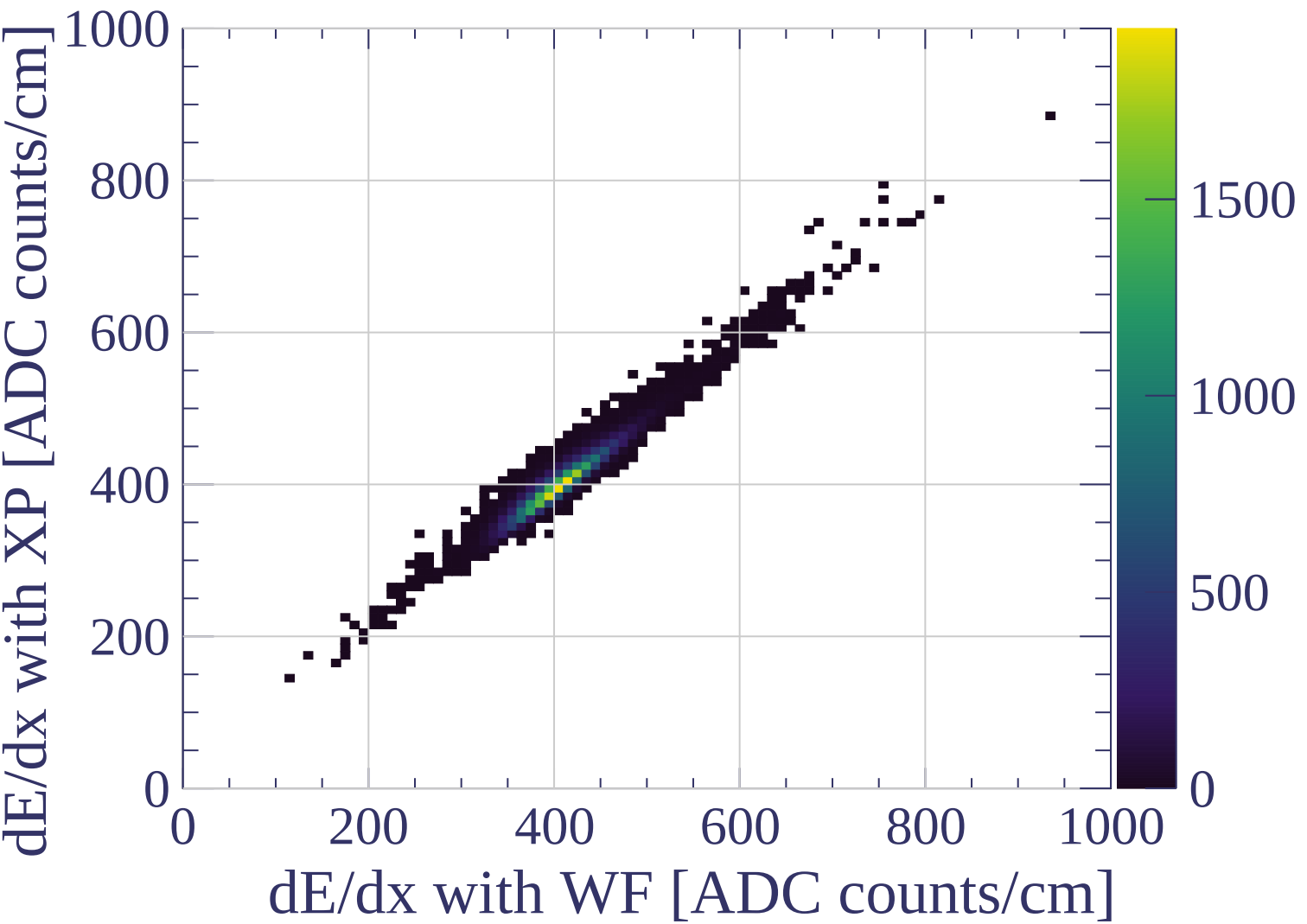
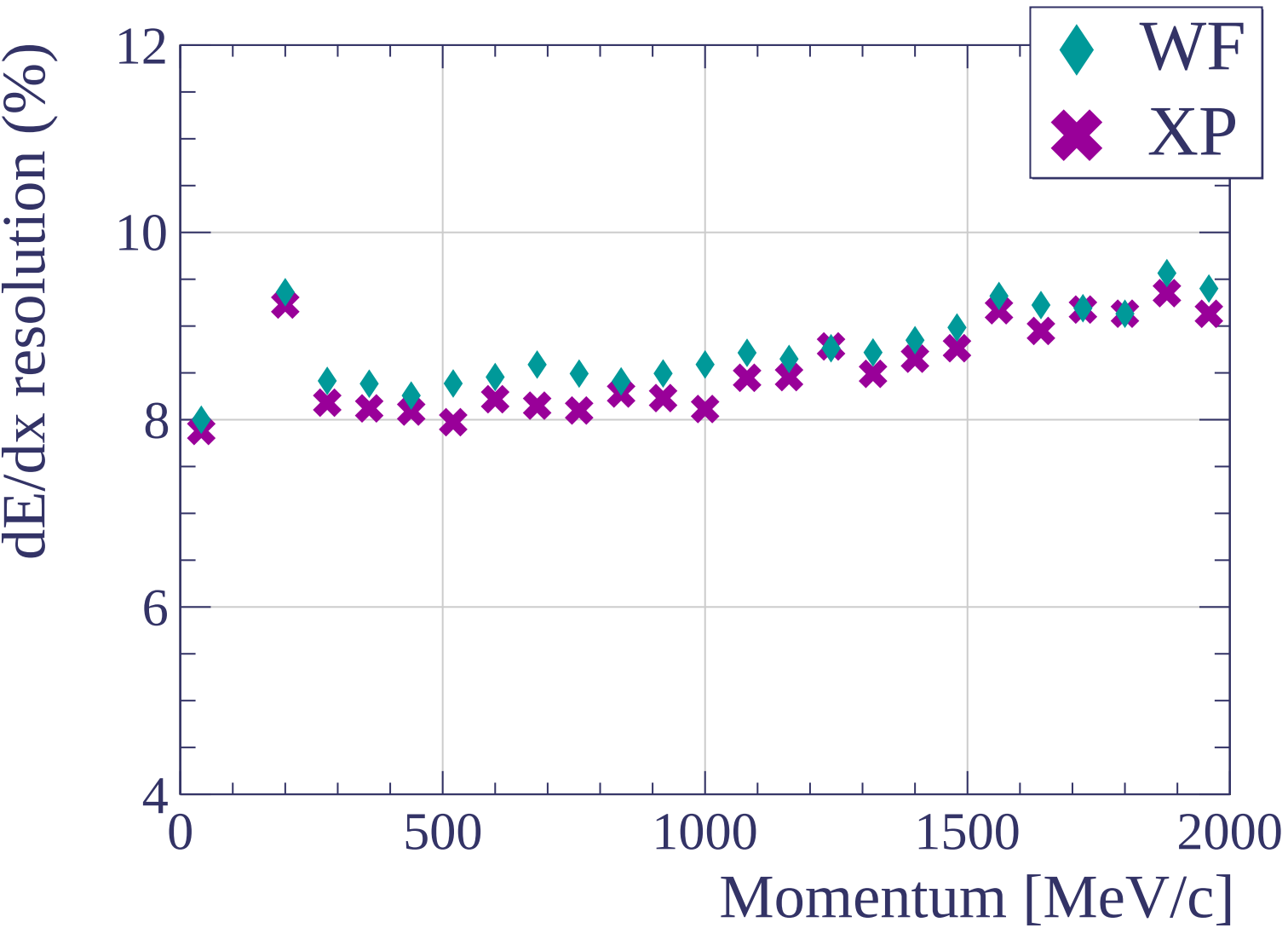


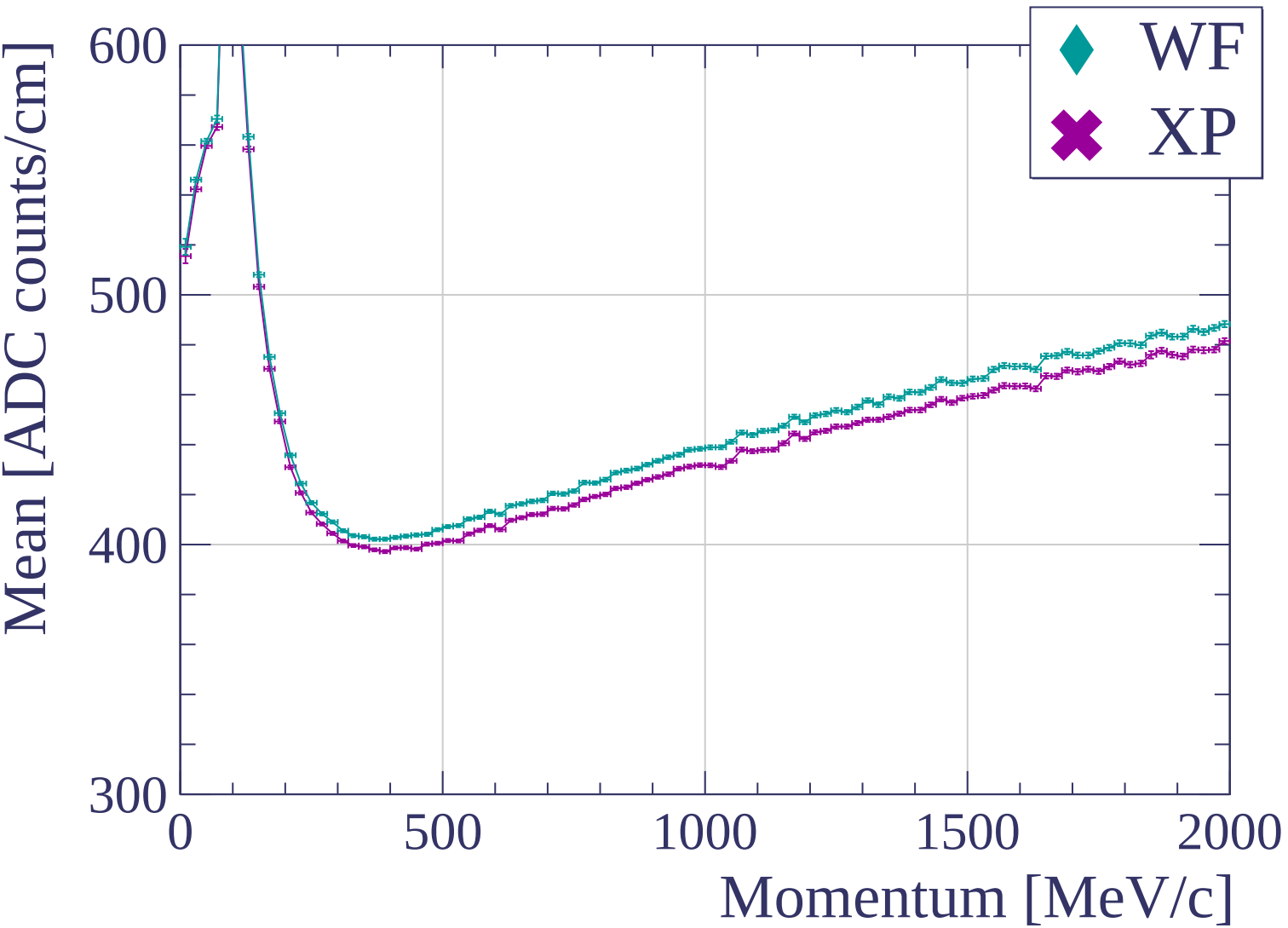


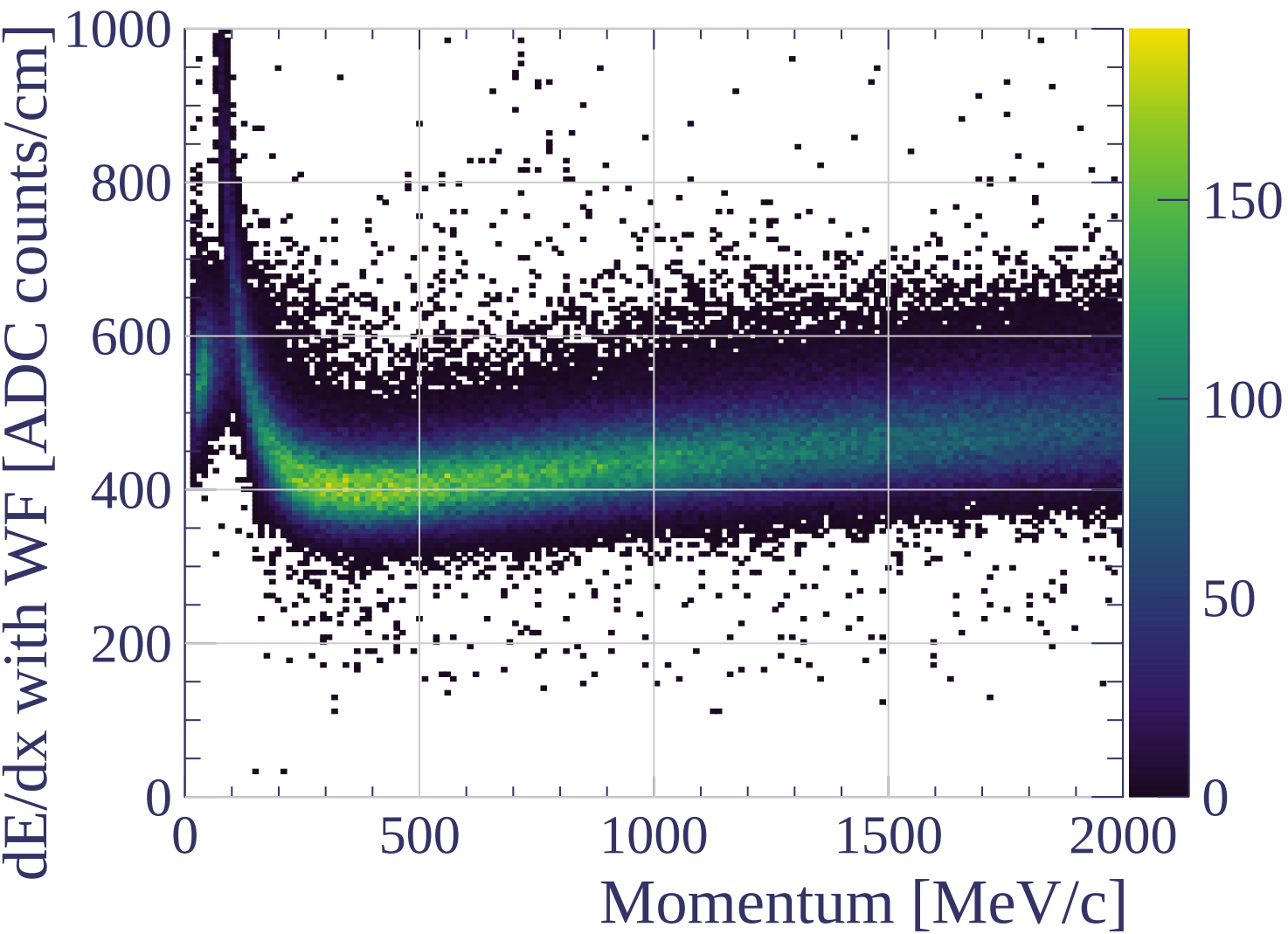


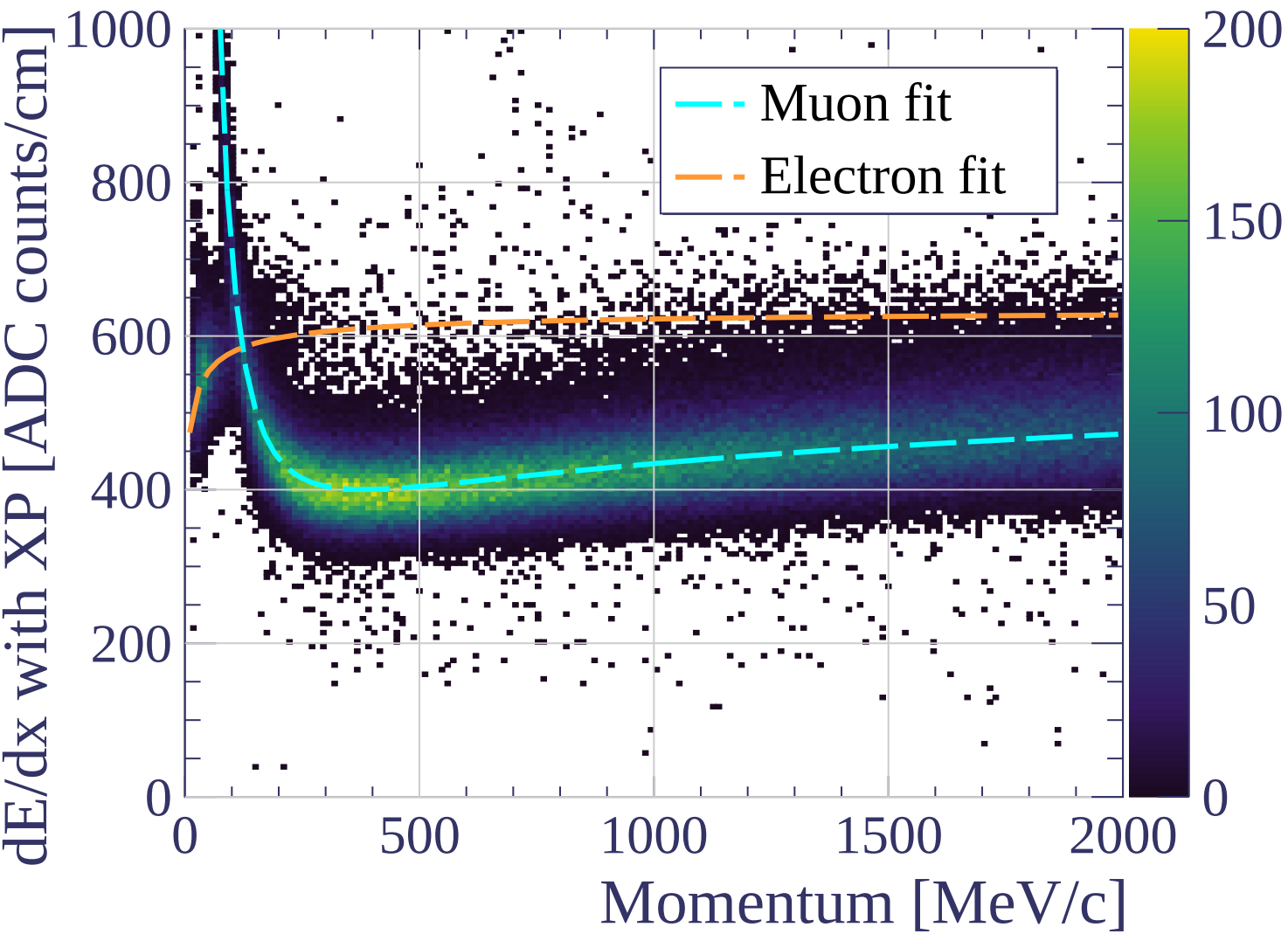




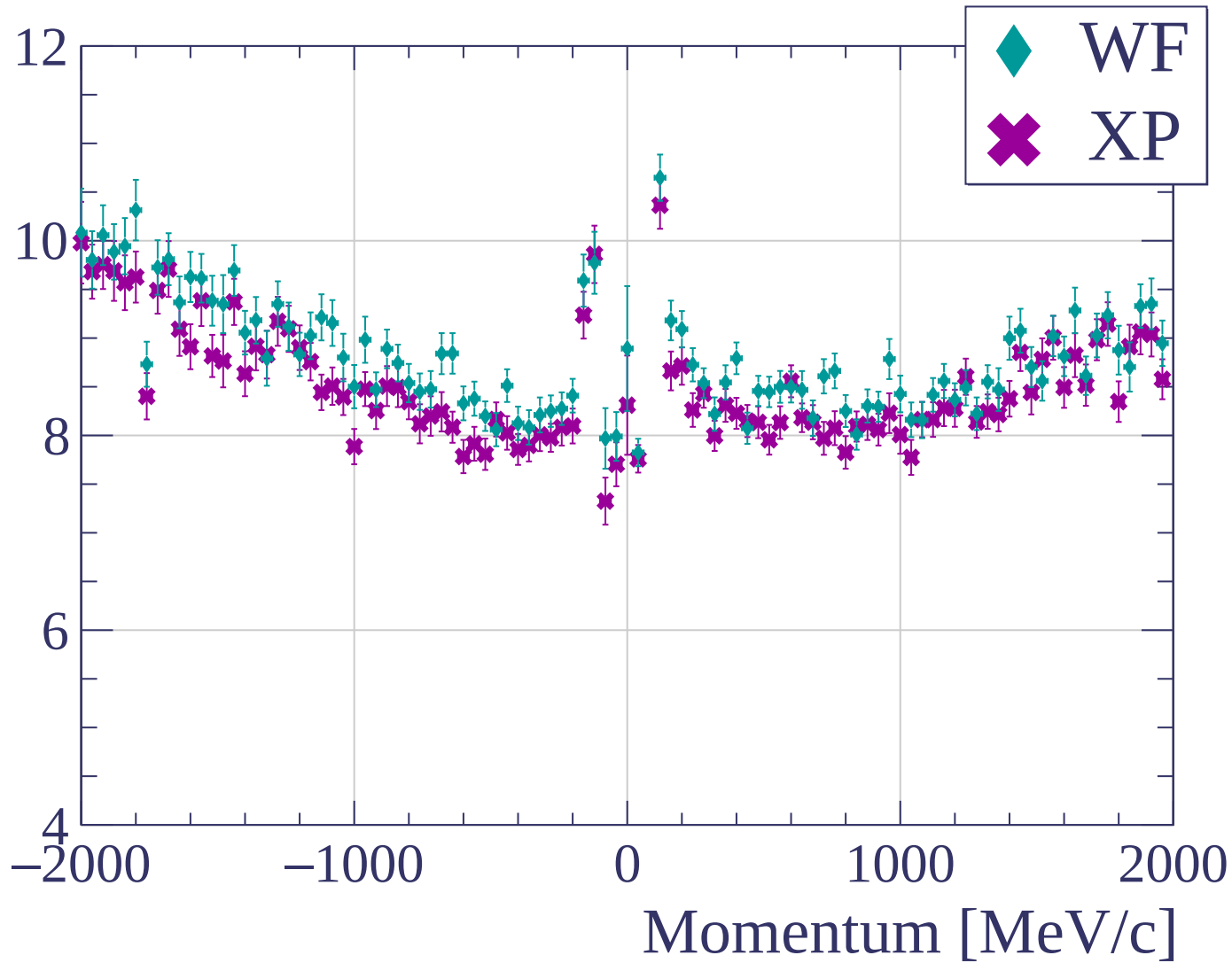


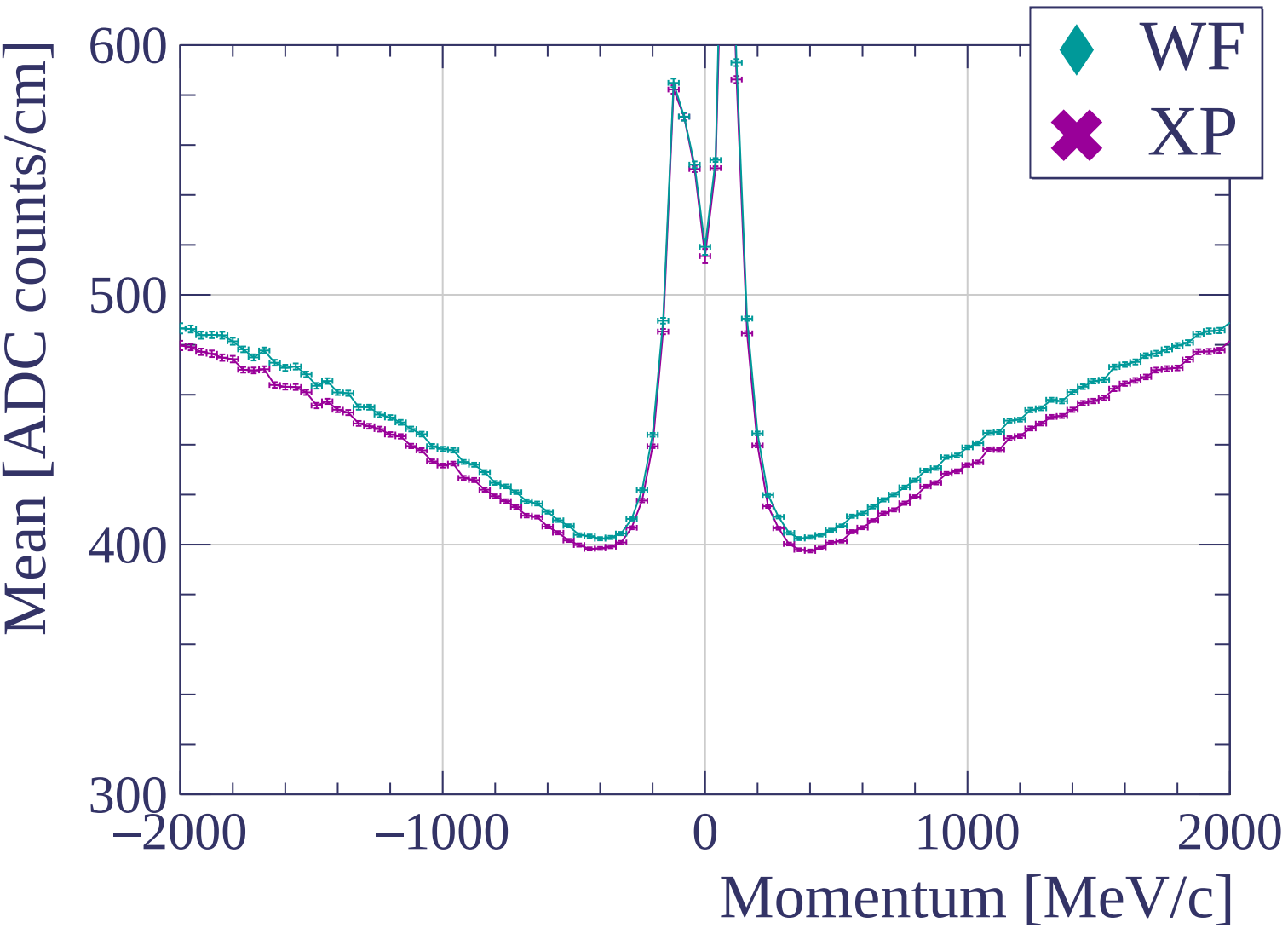


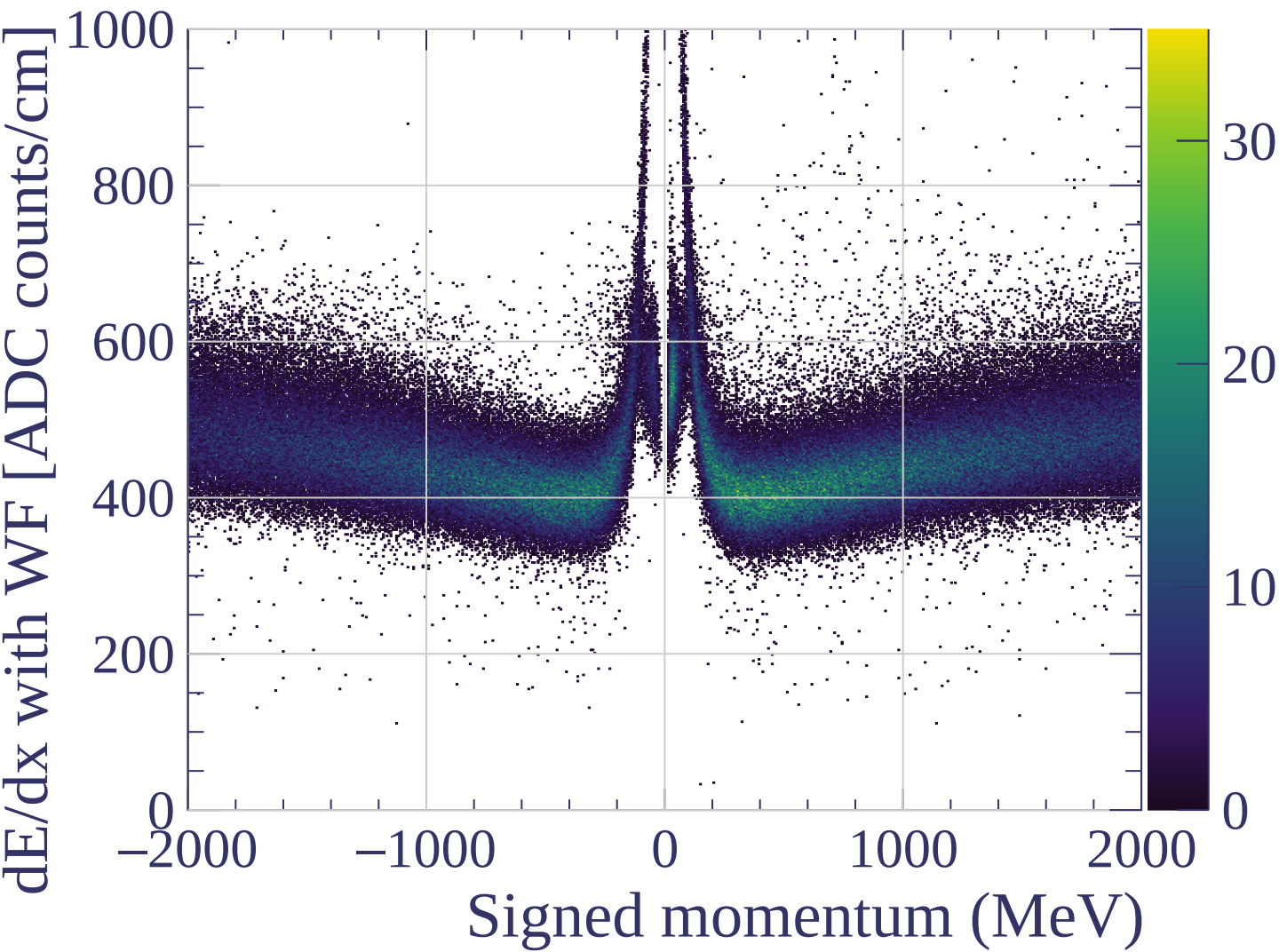


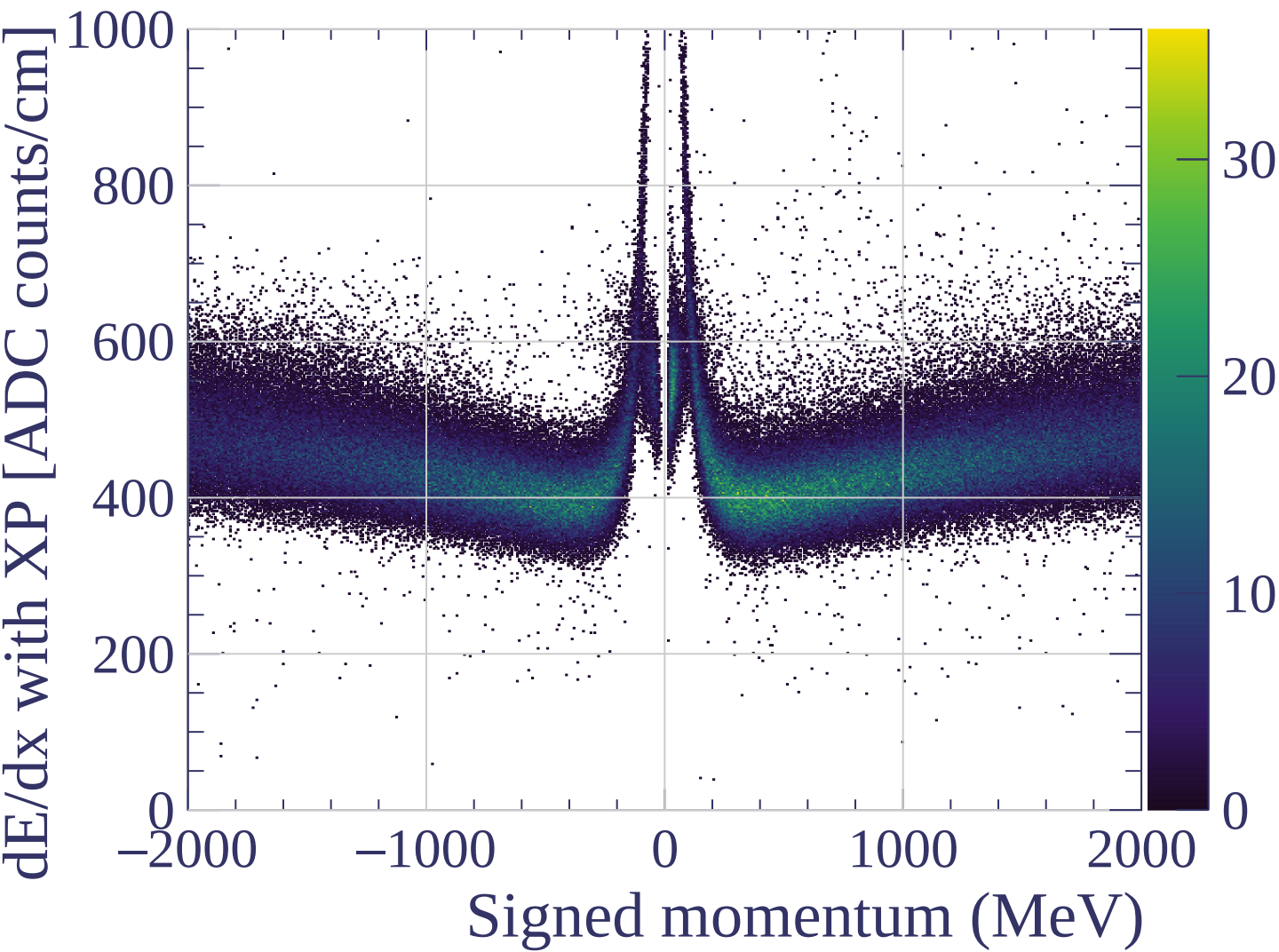


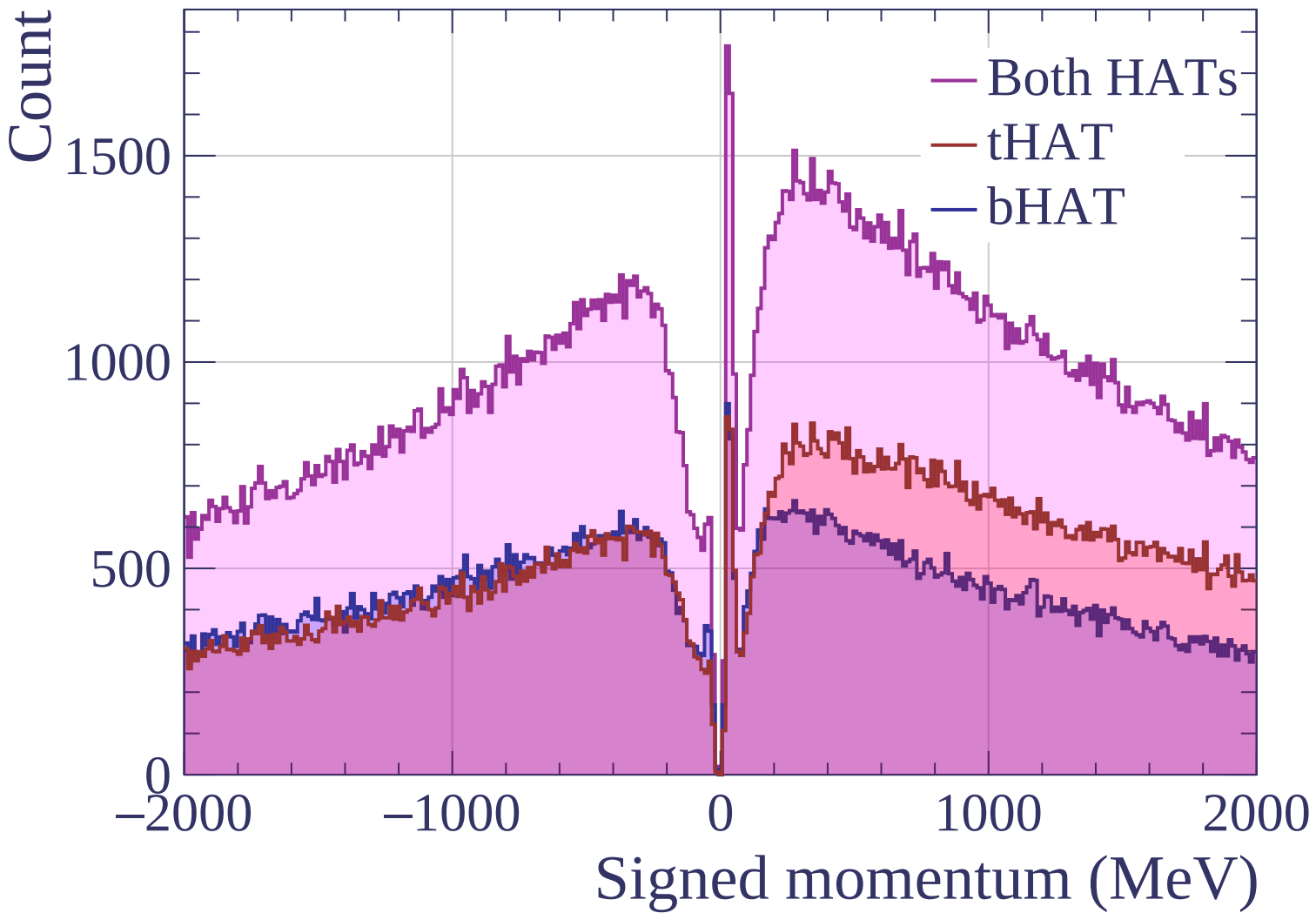
dE/dx resolution (%)

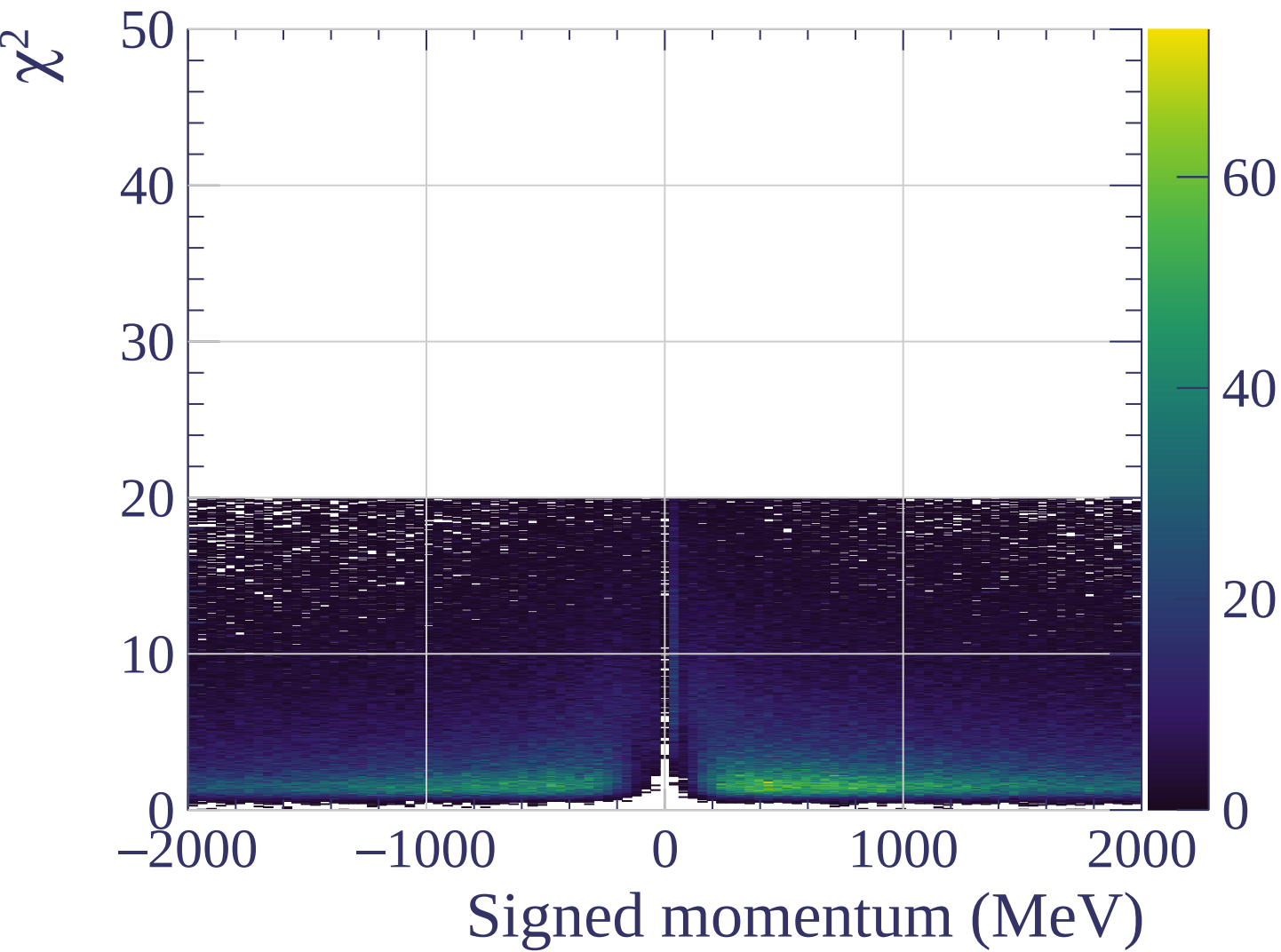


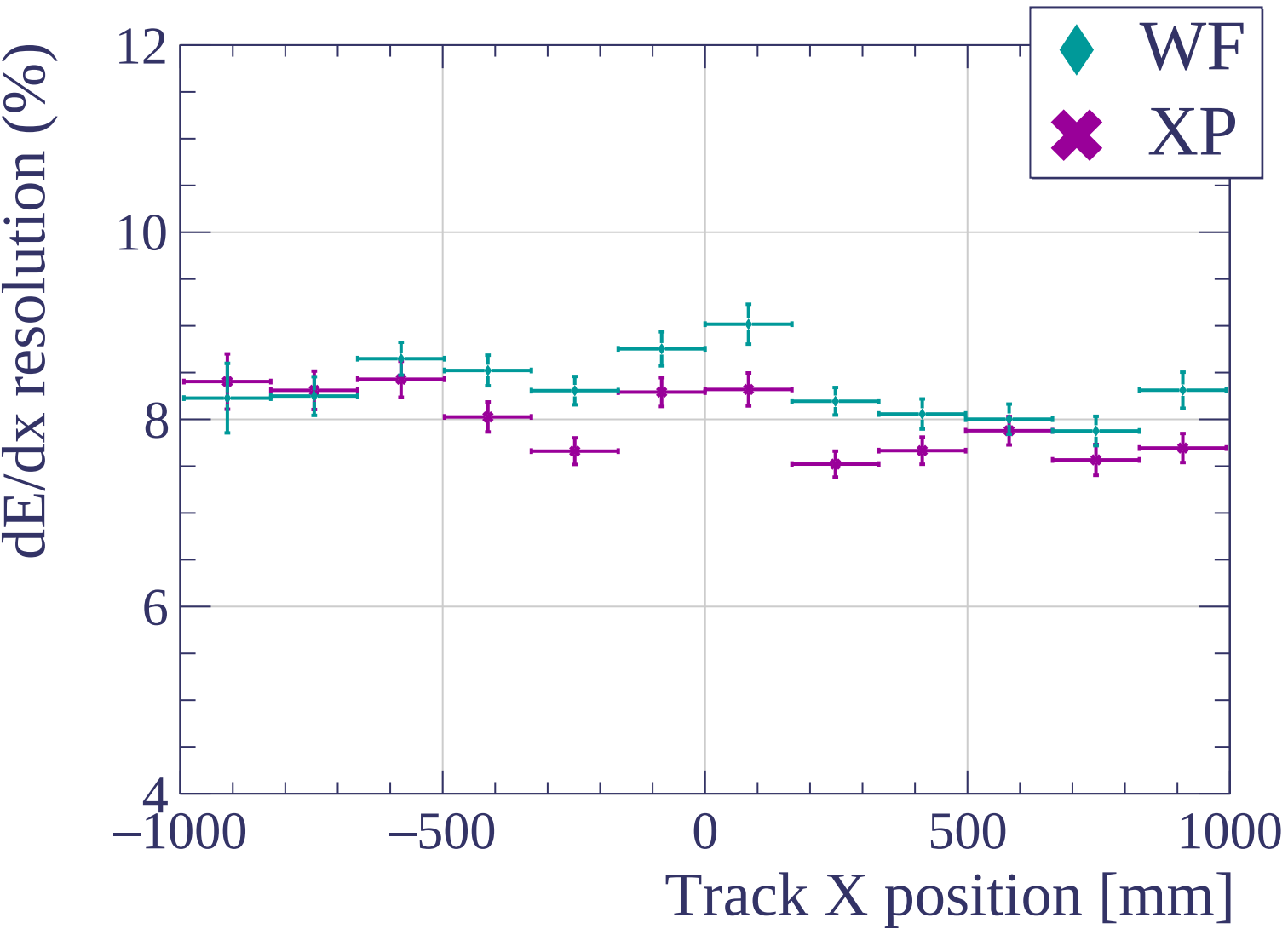


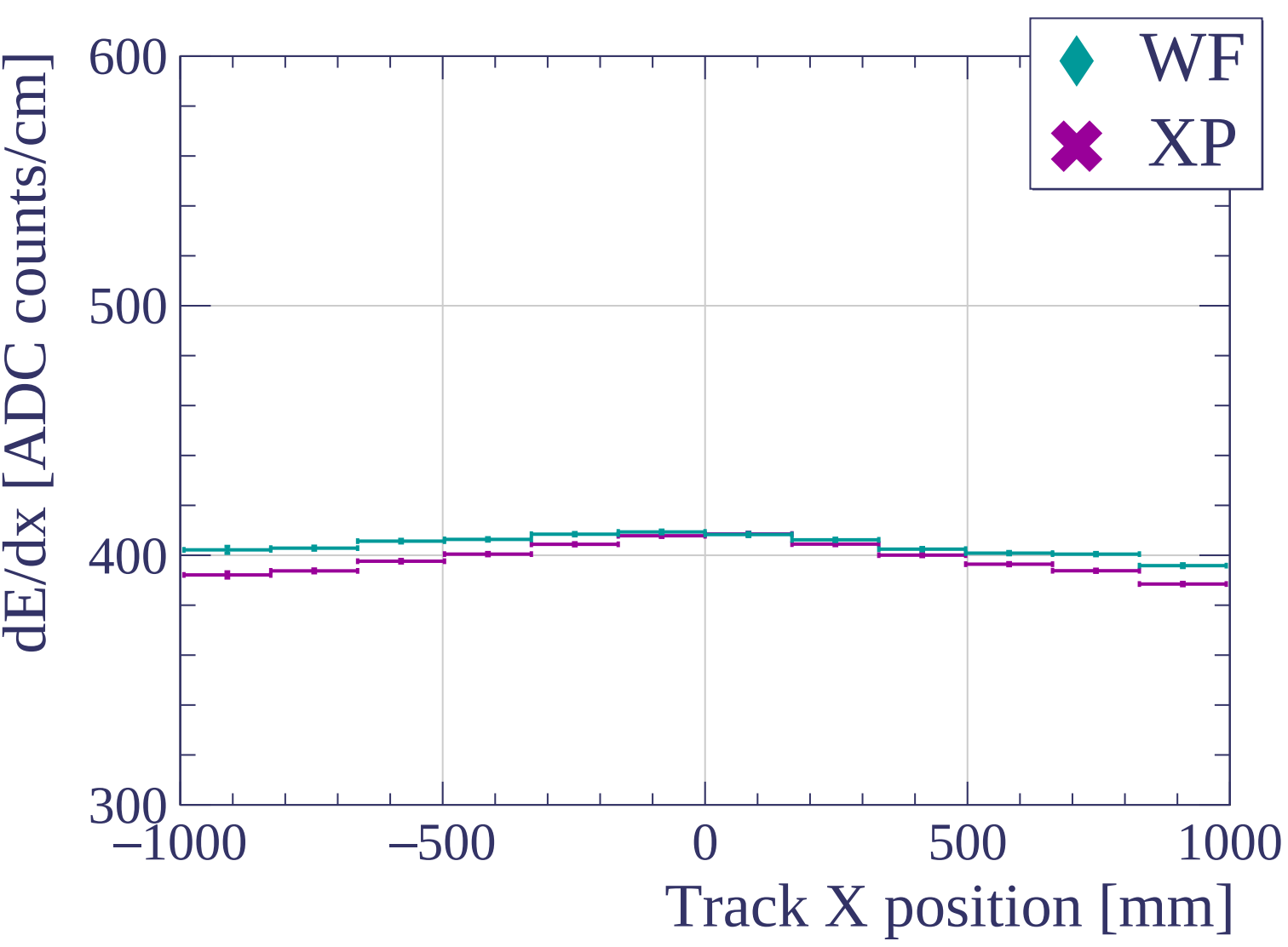


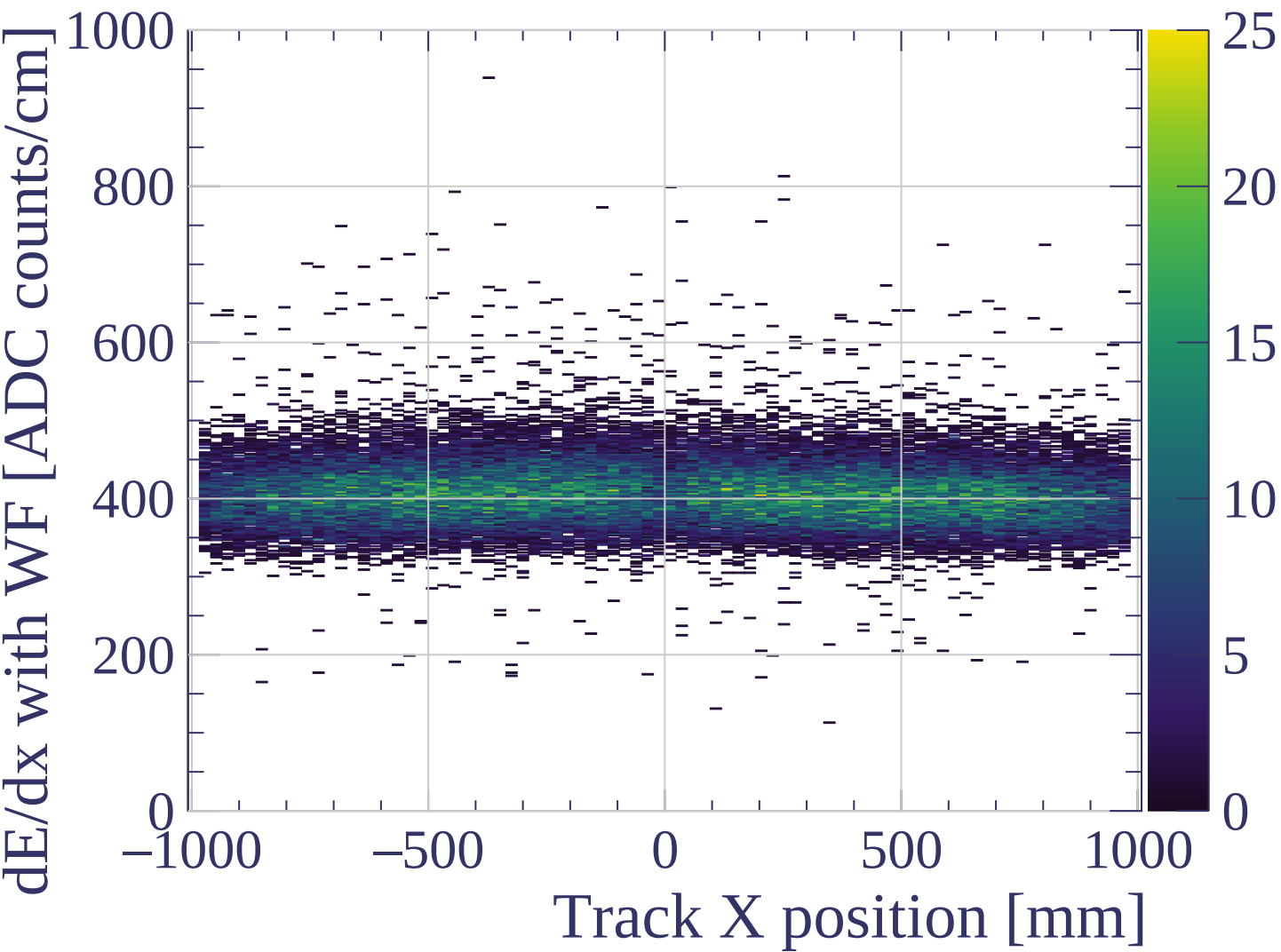


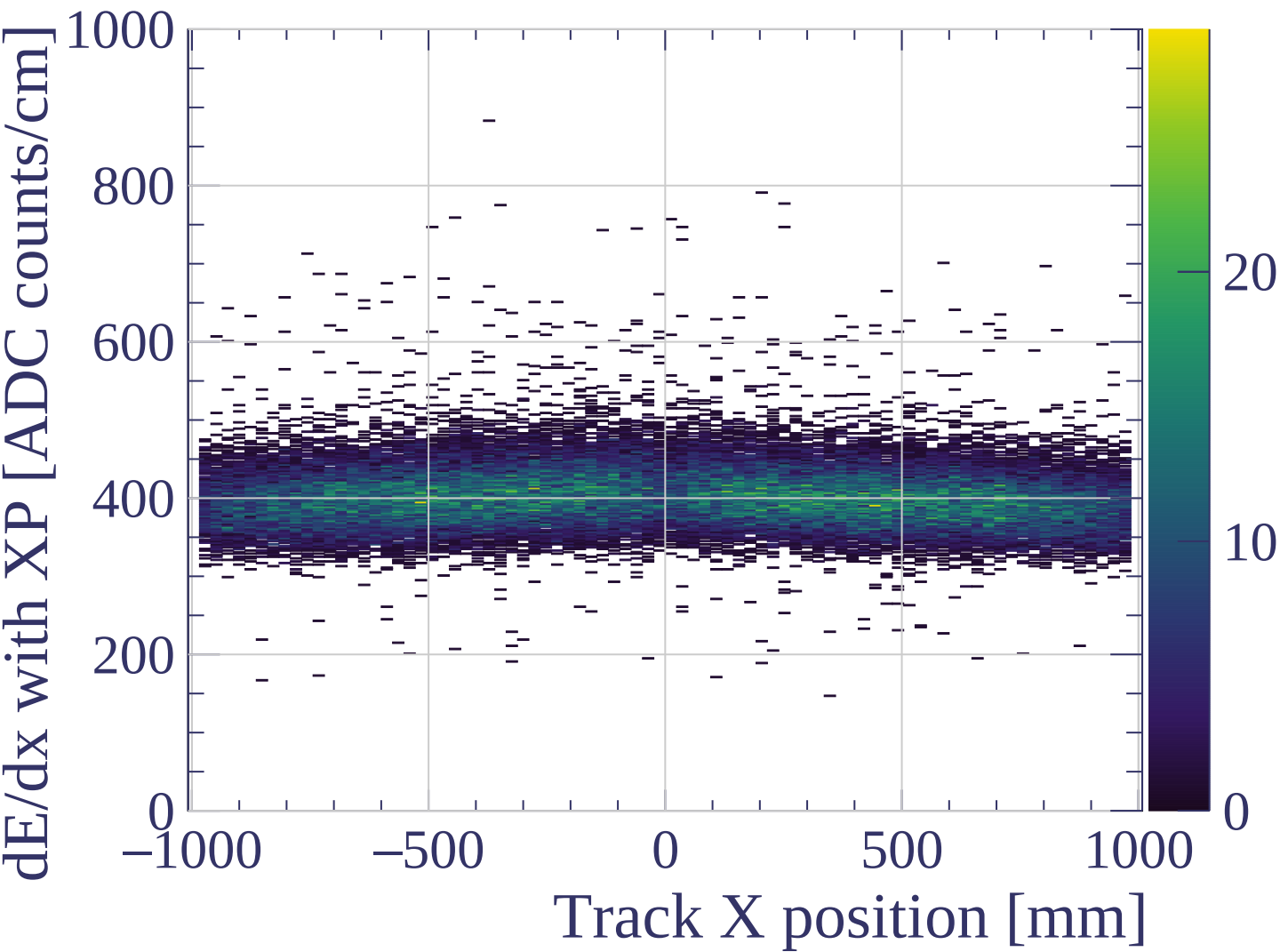


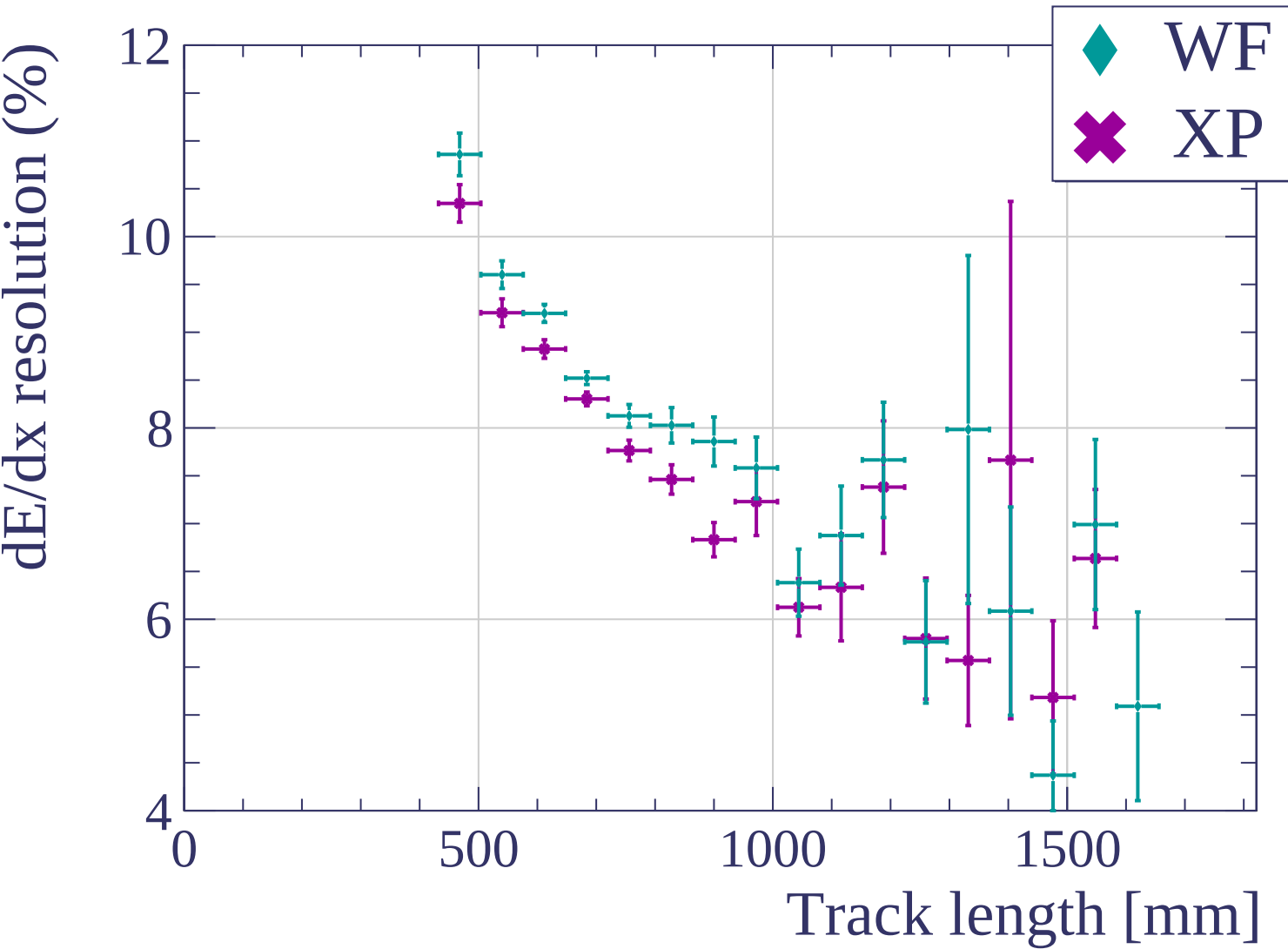


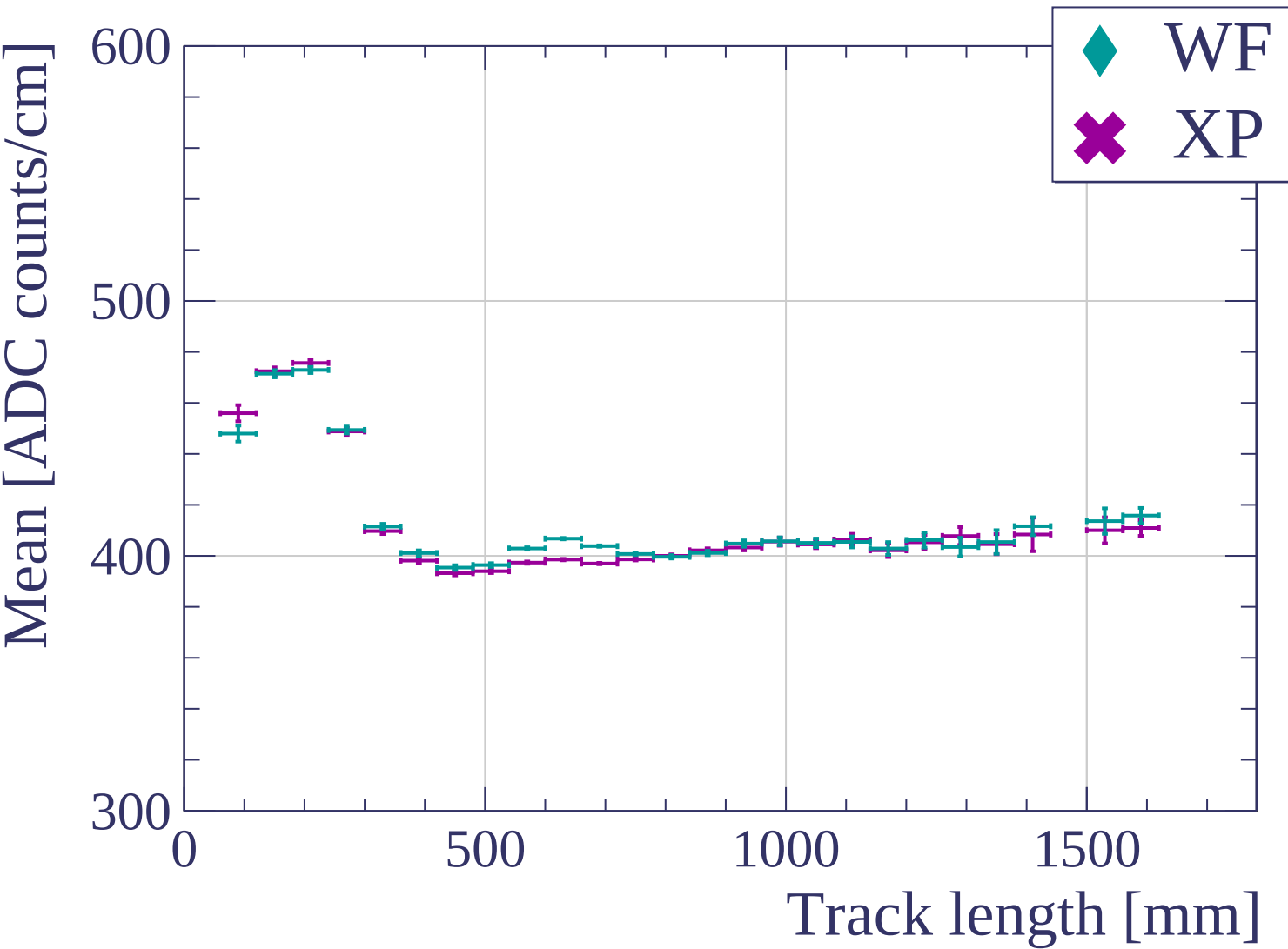


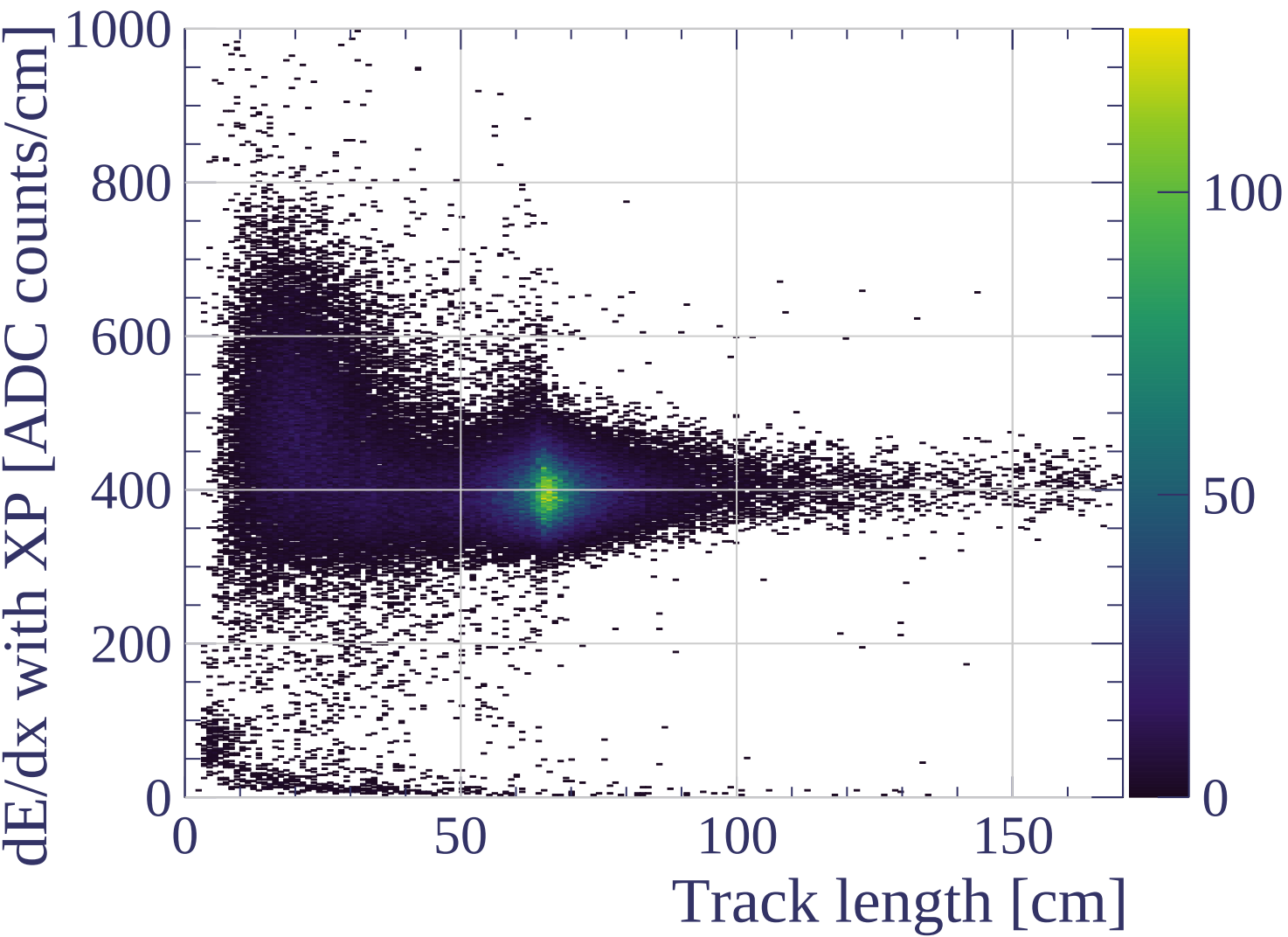


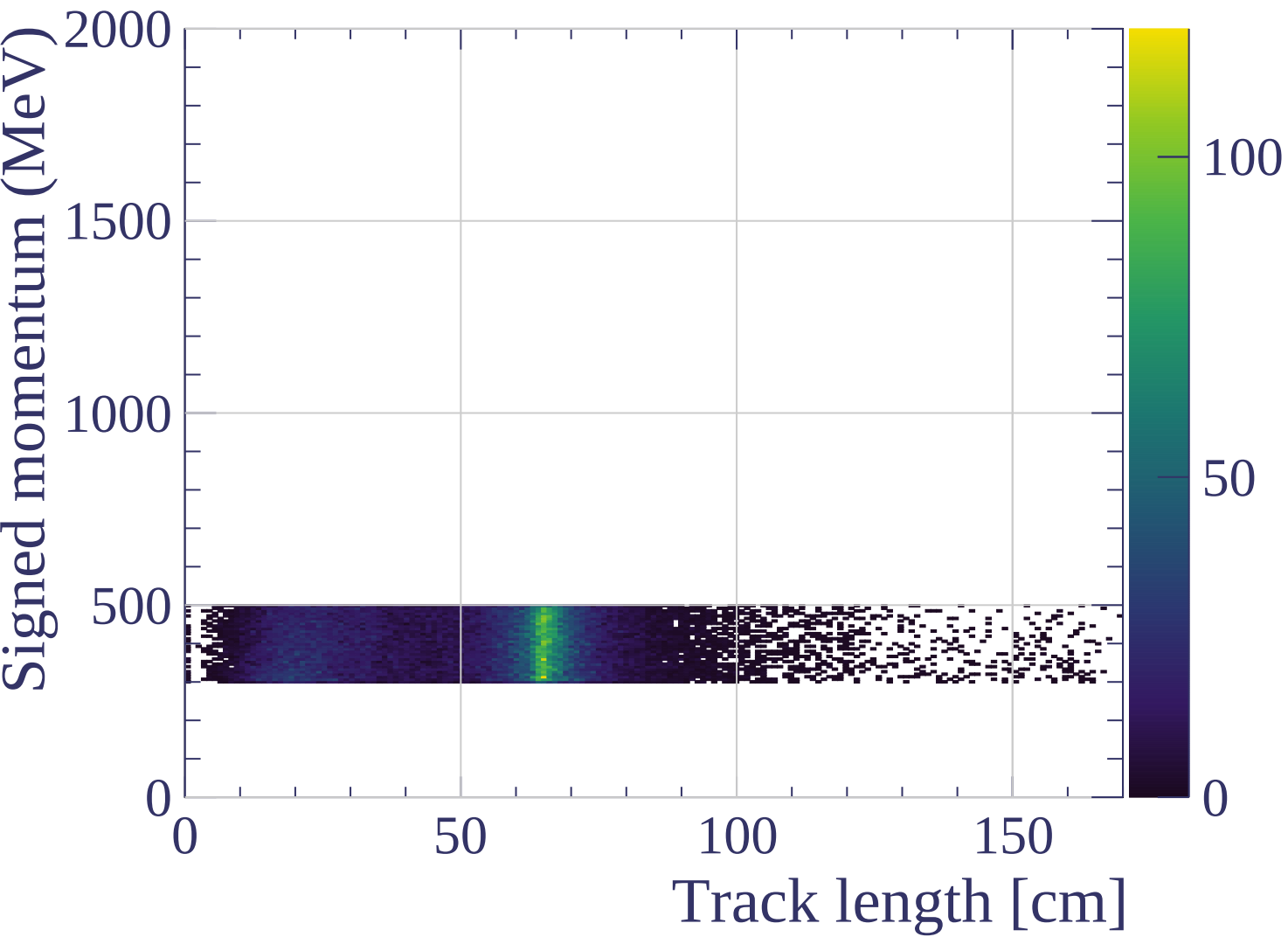


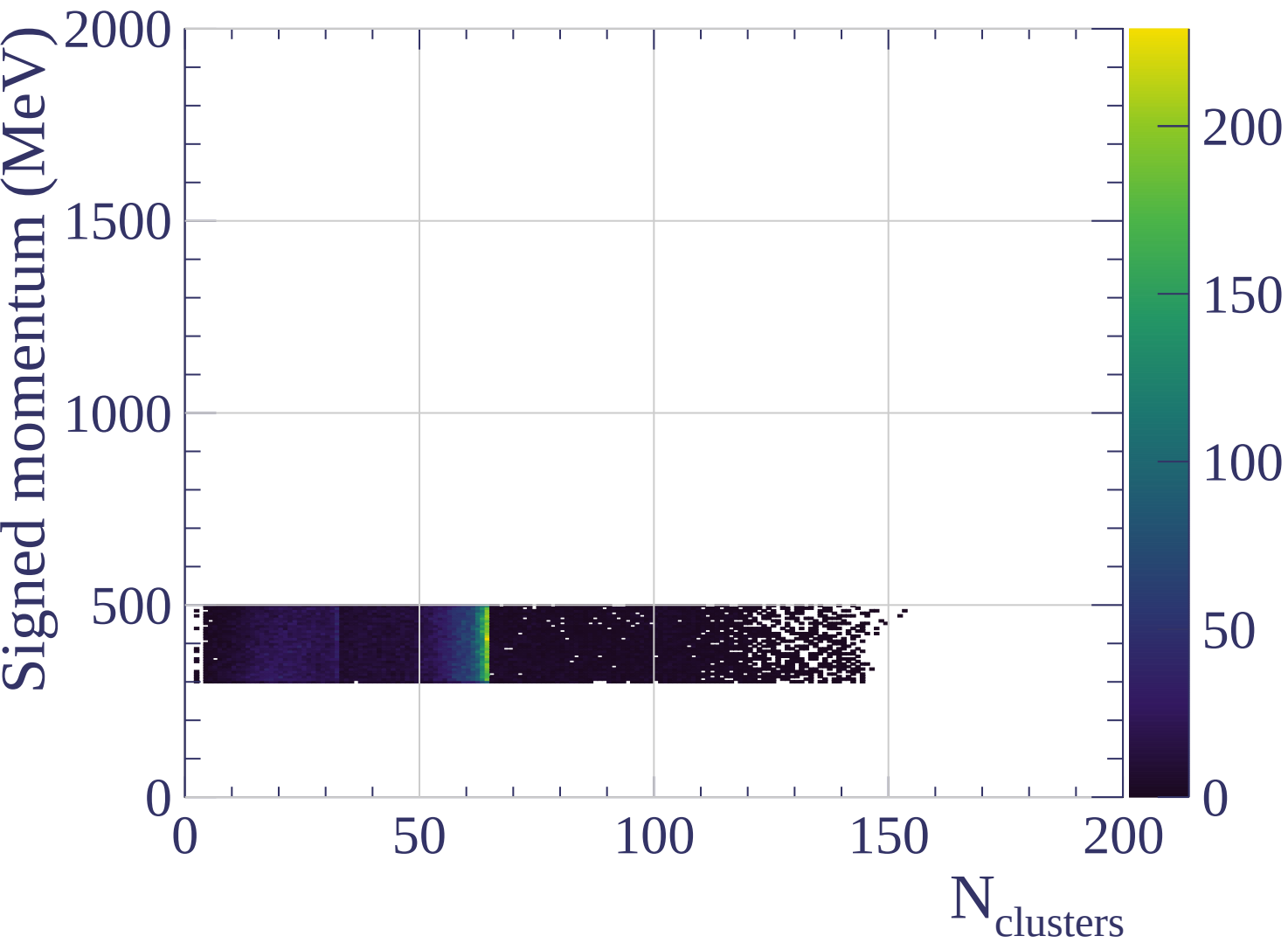


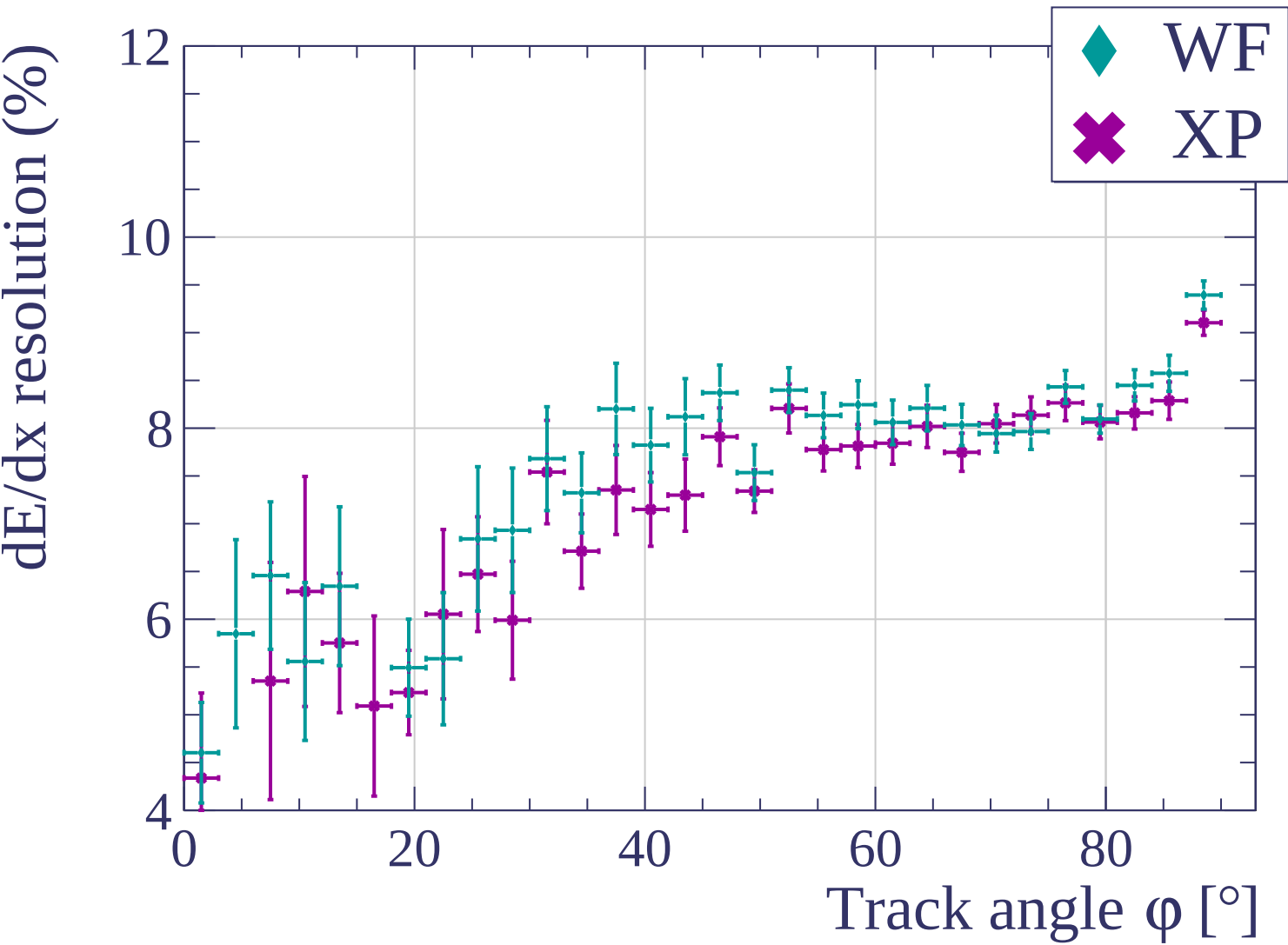


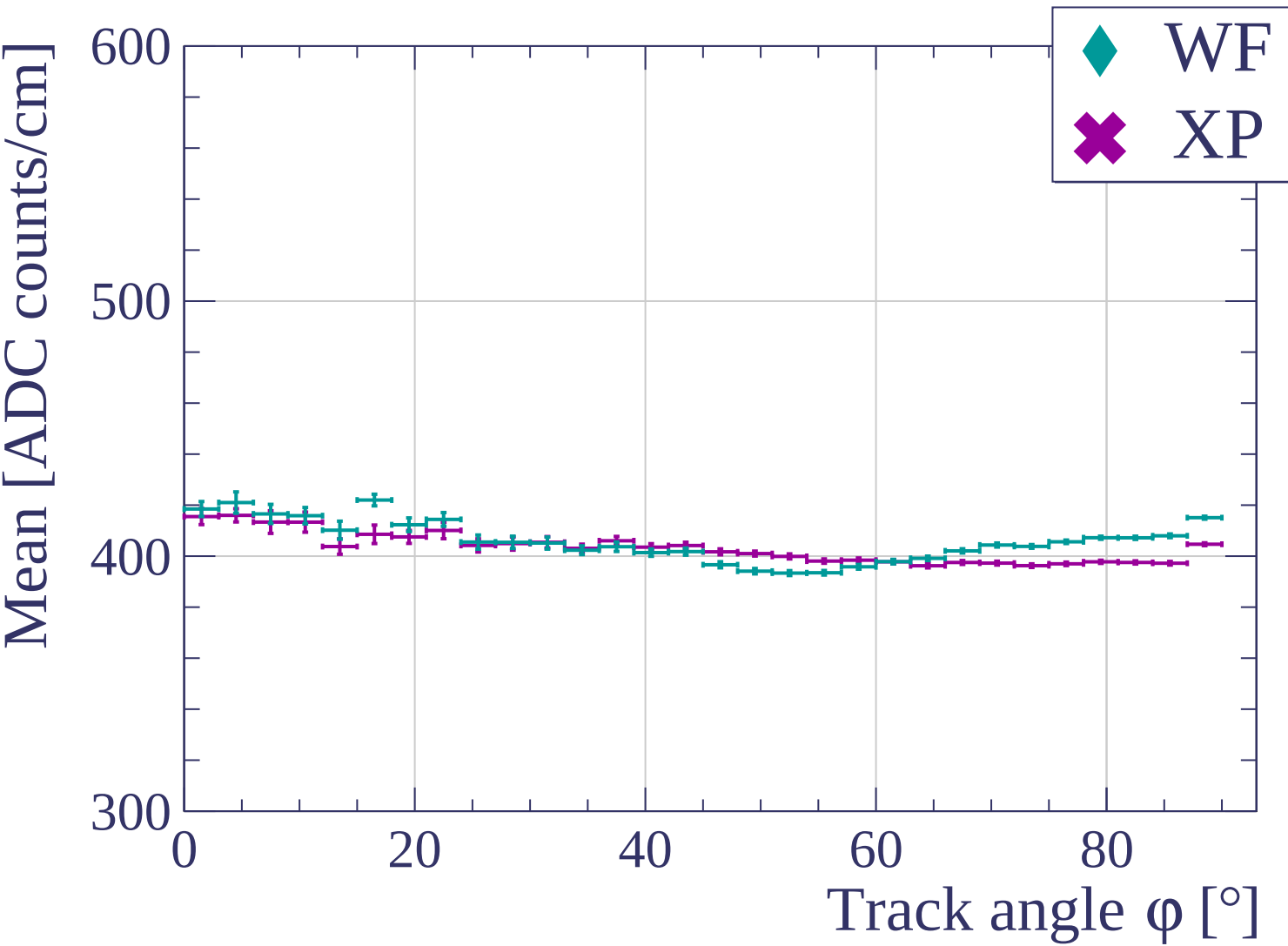


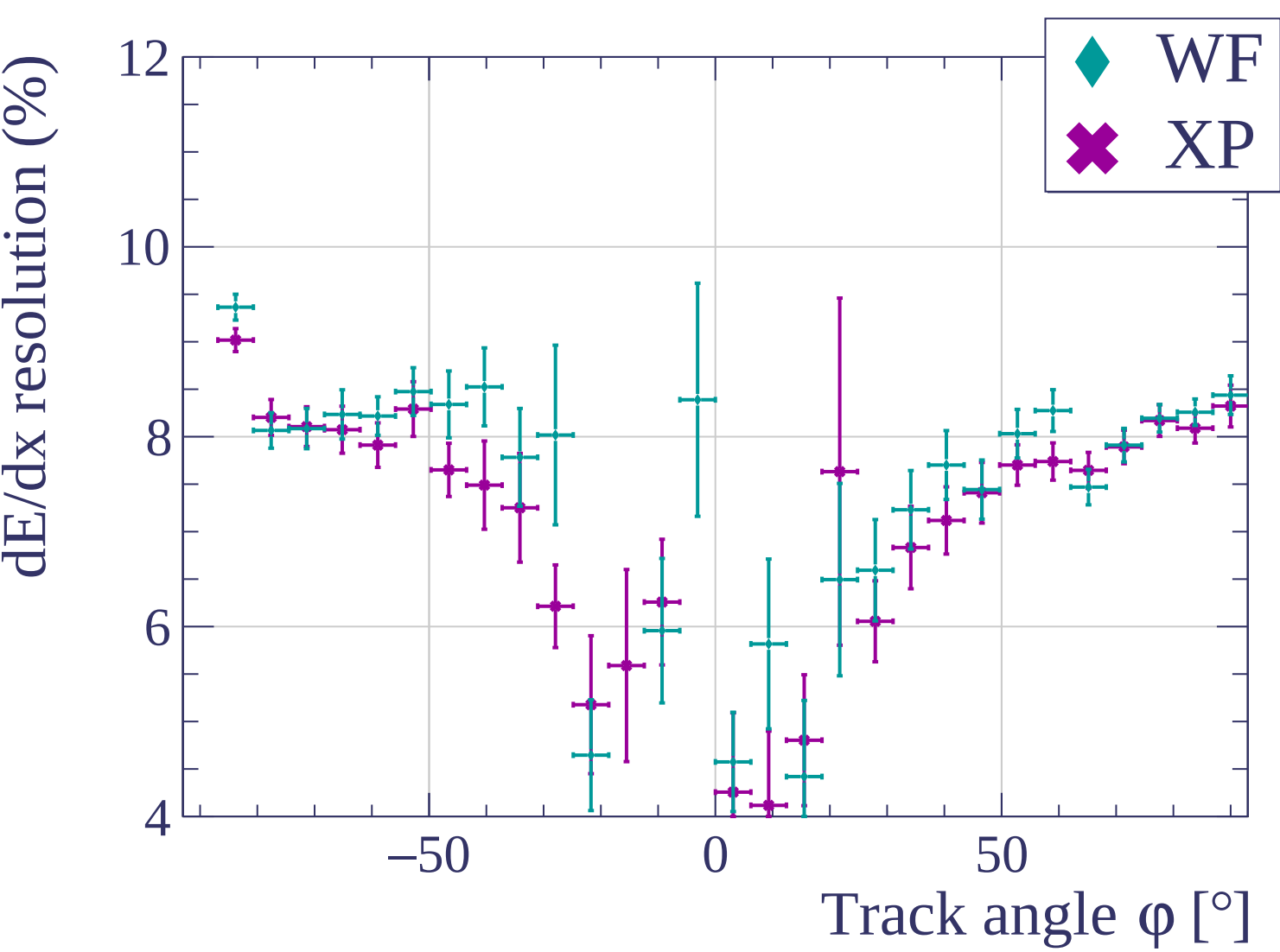


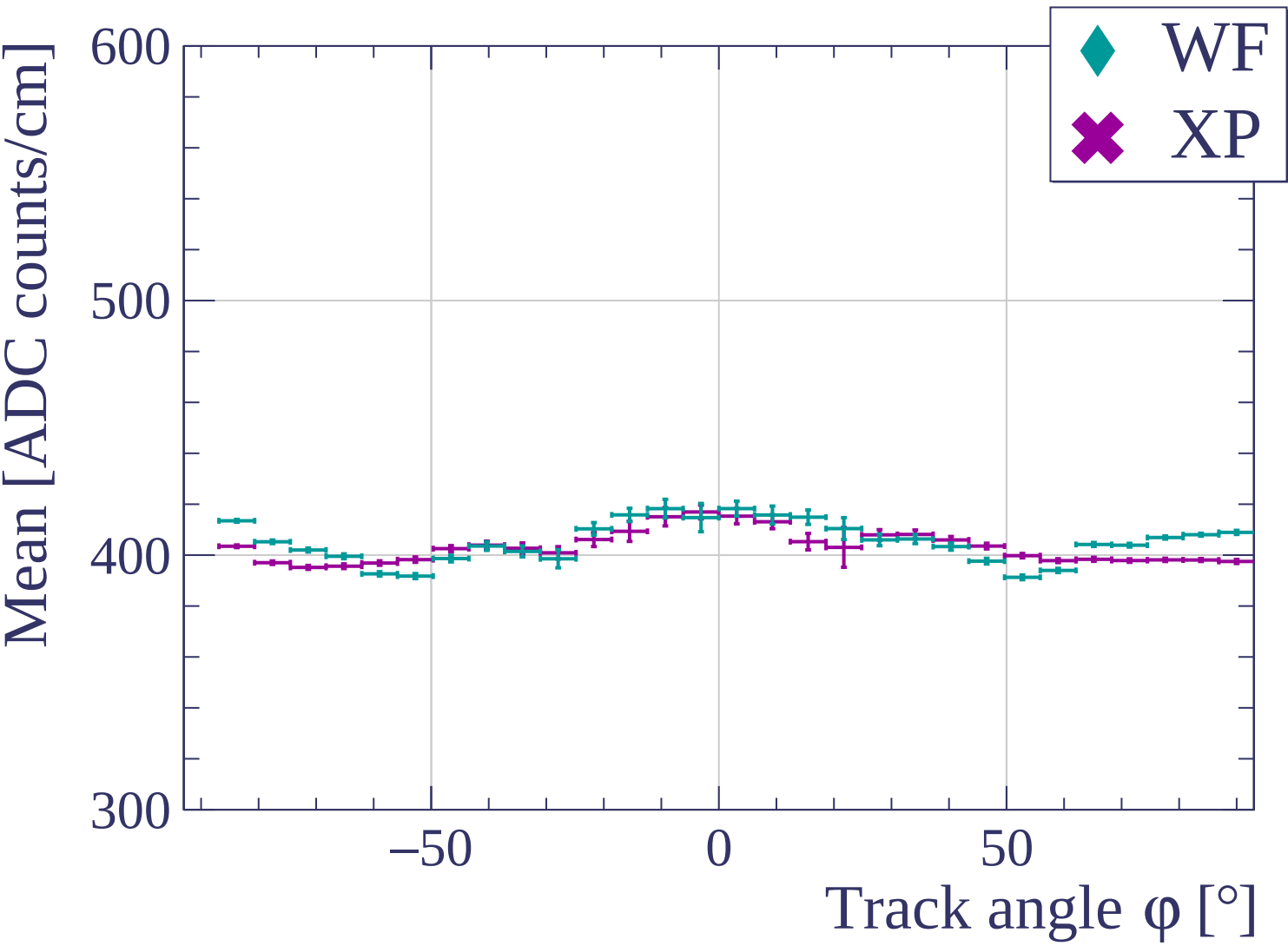


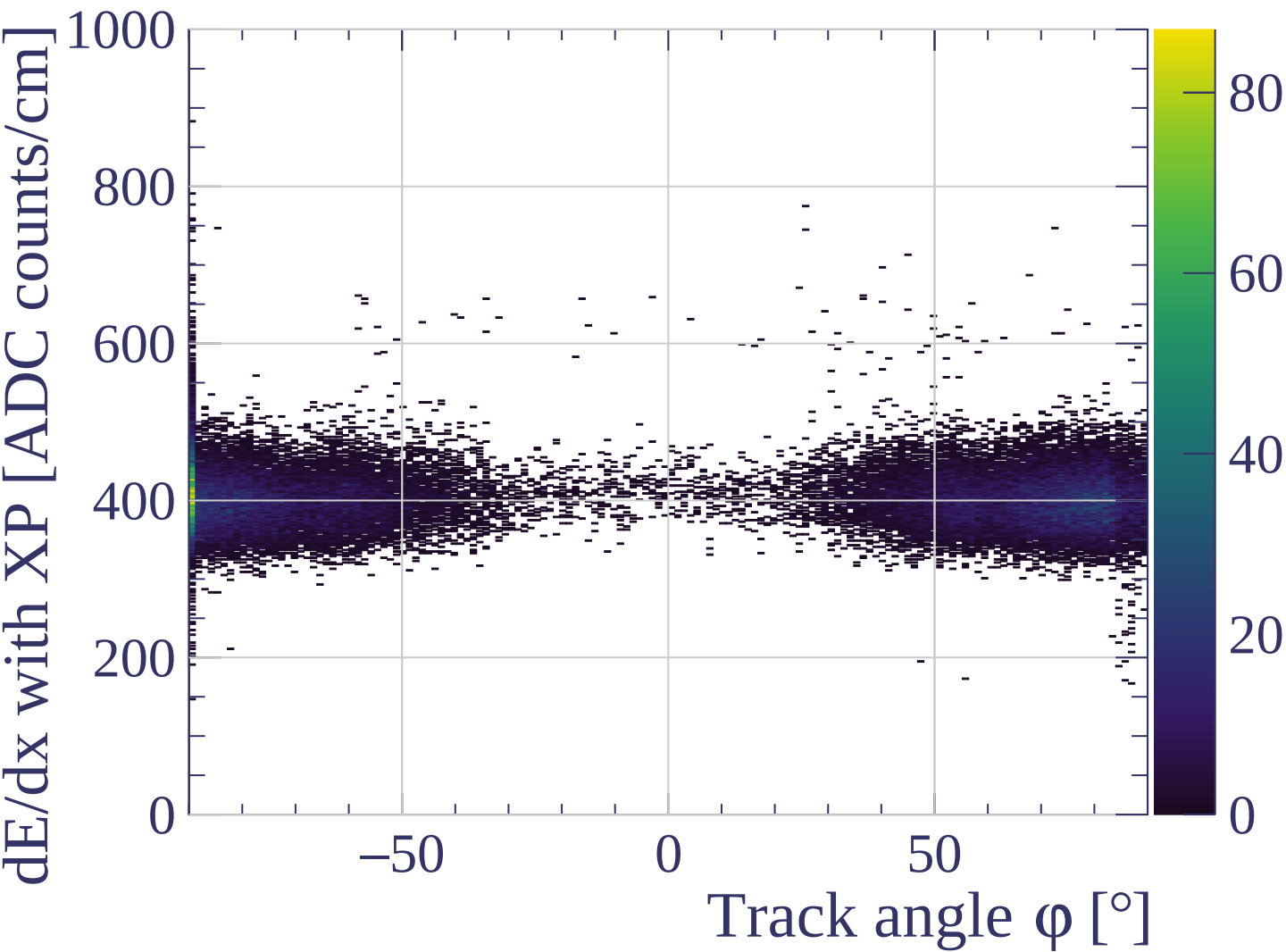




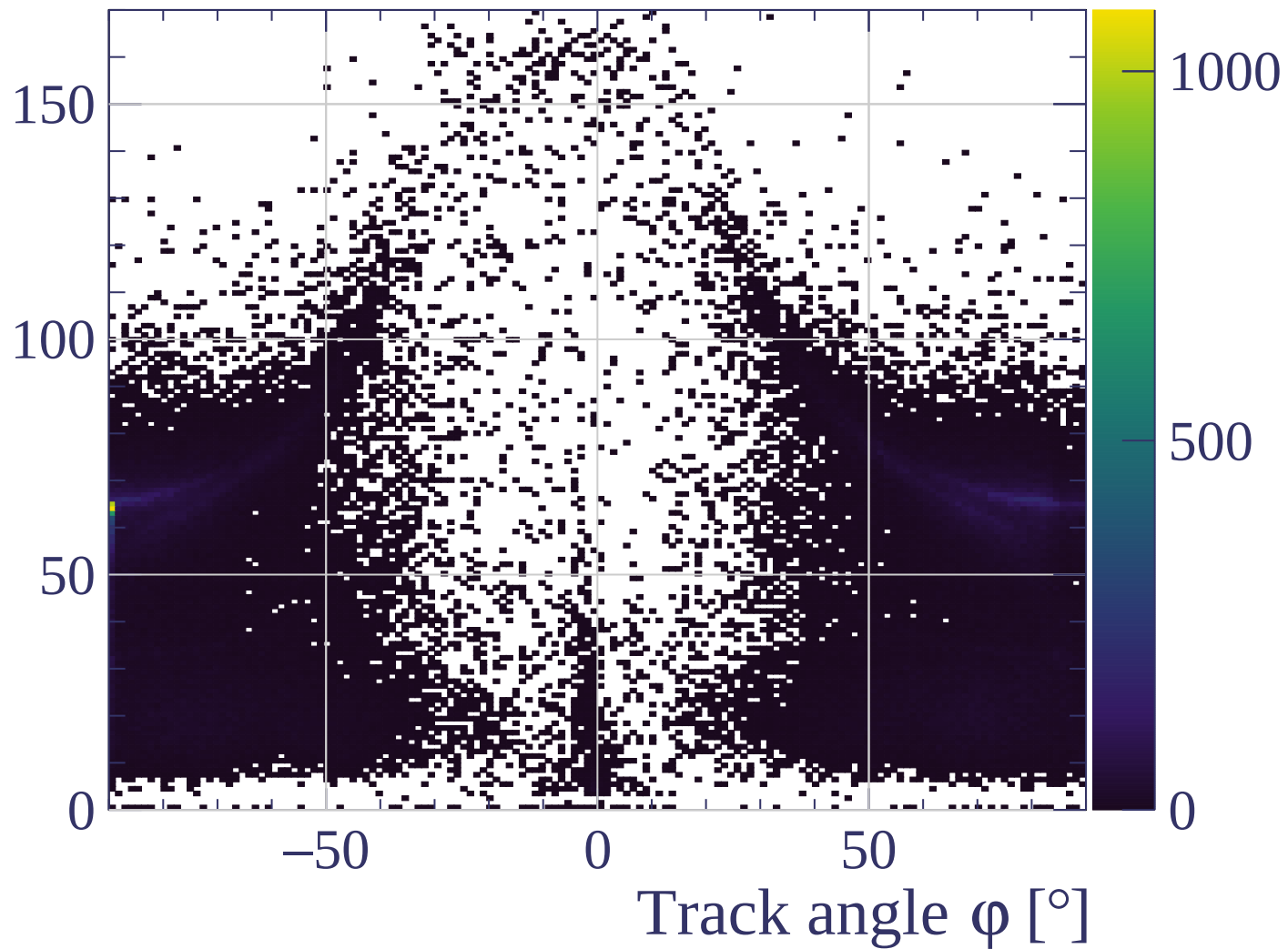


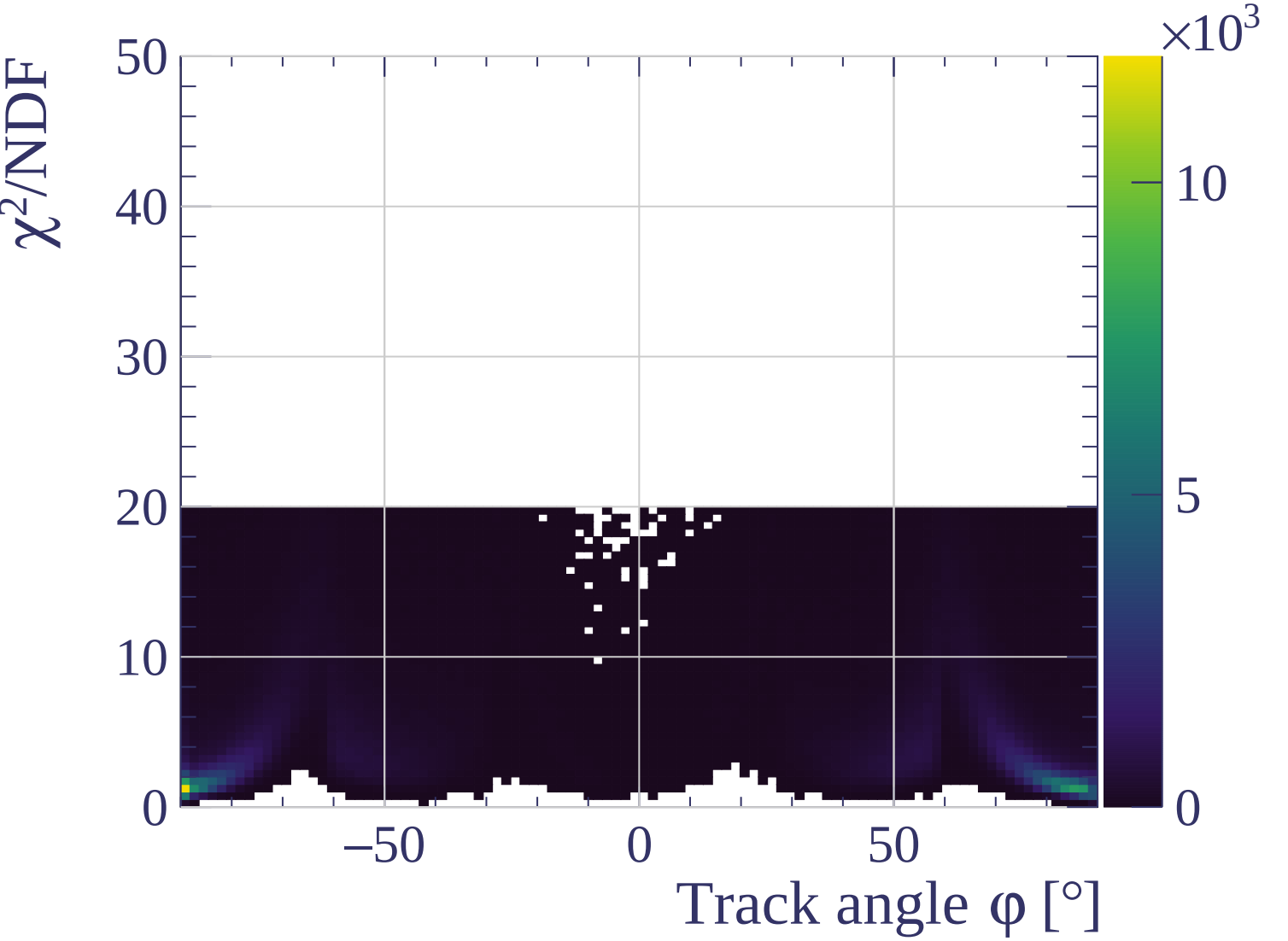


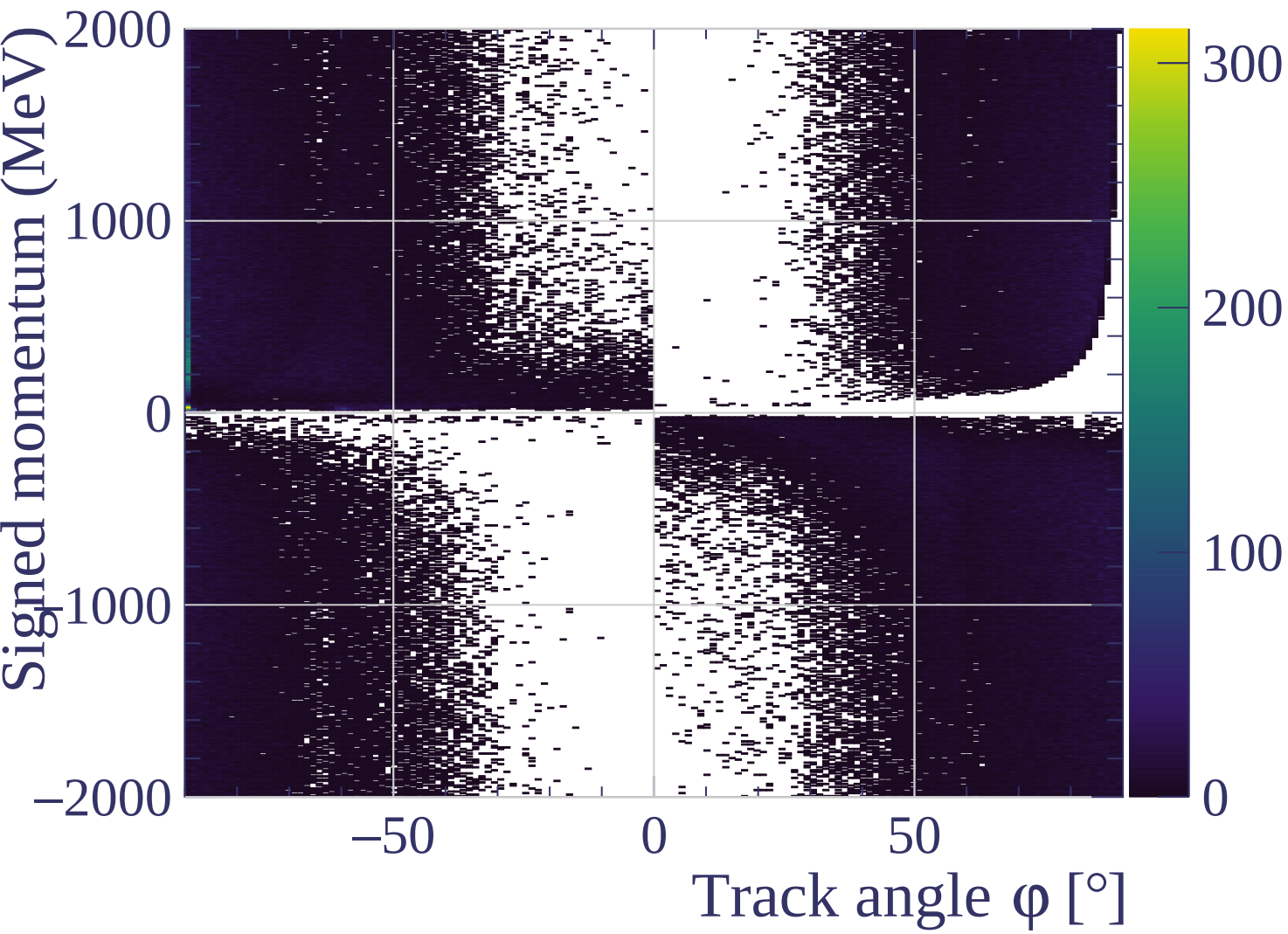


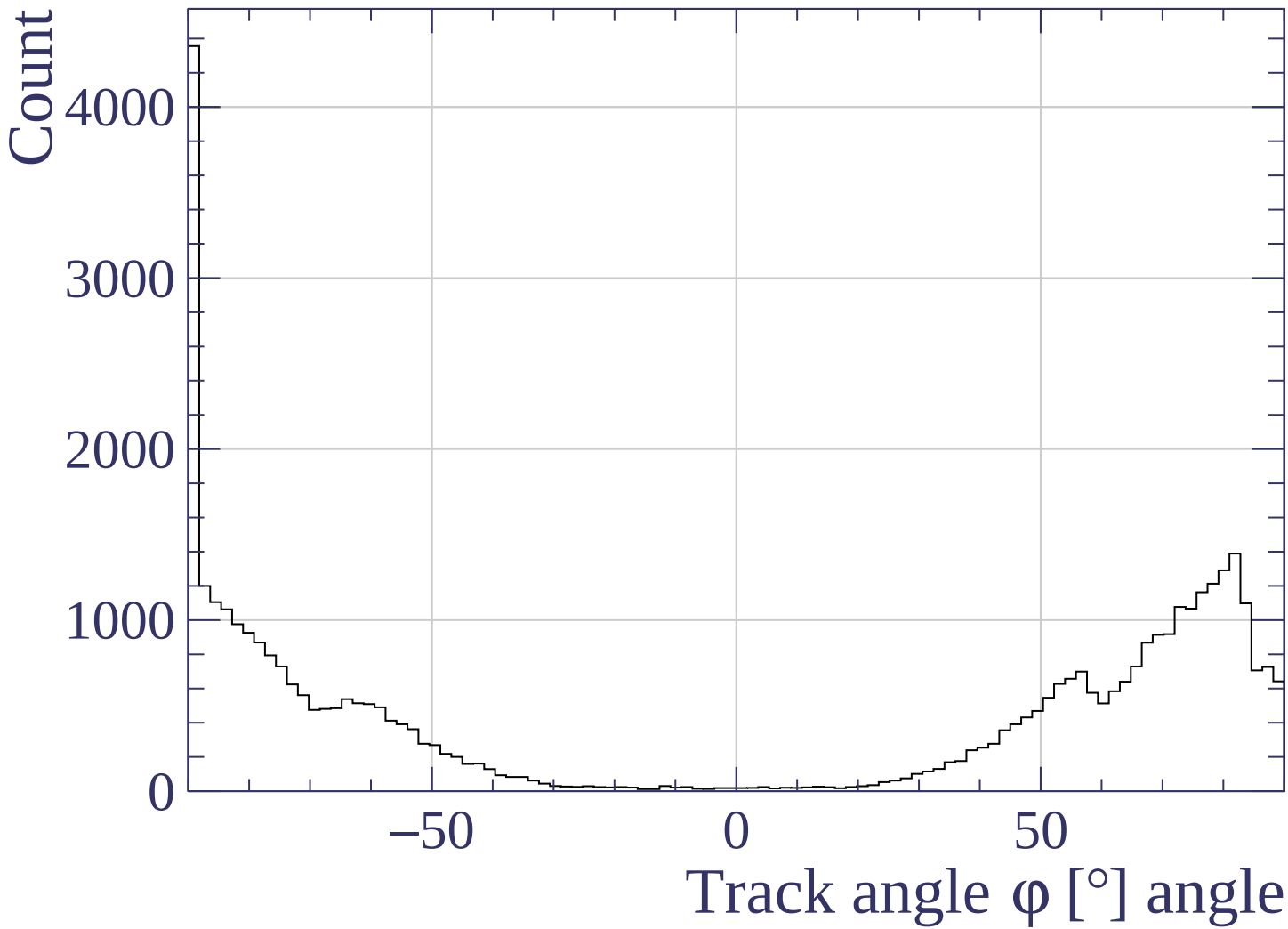


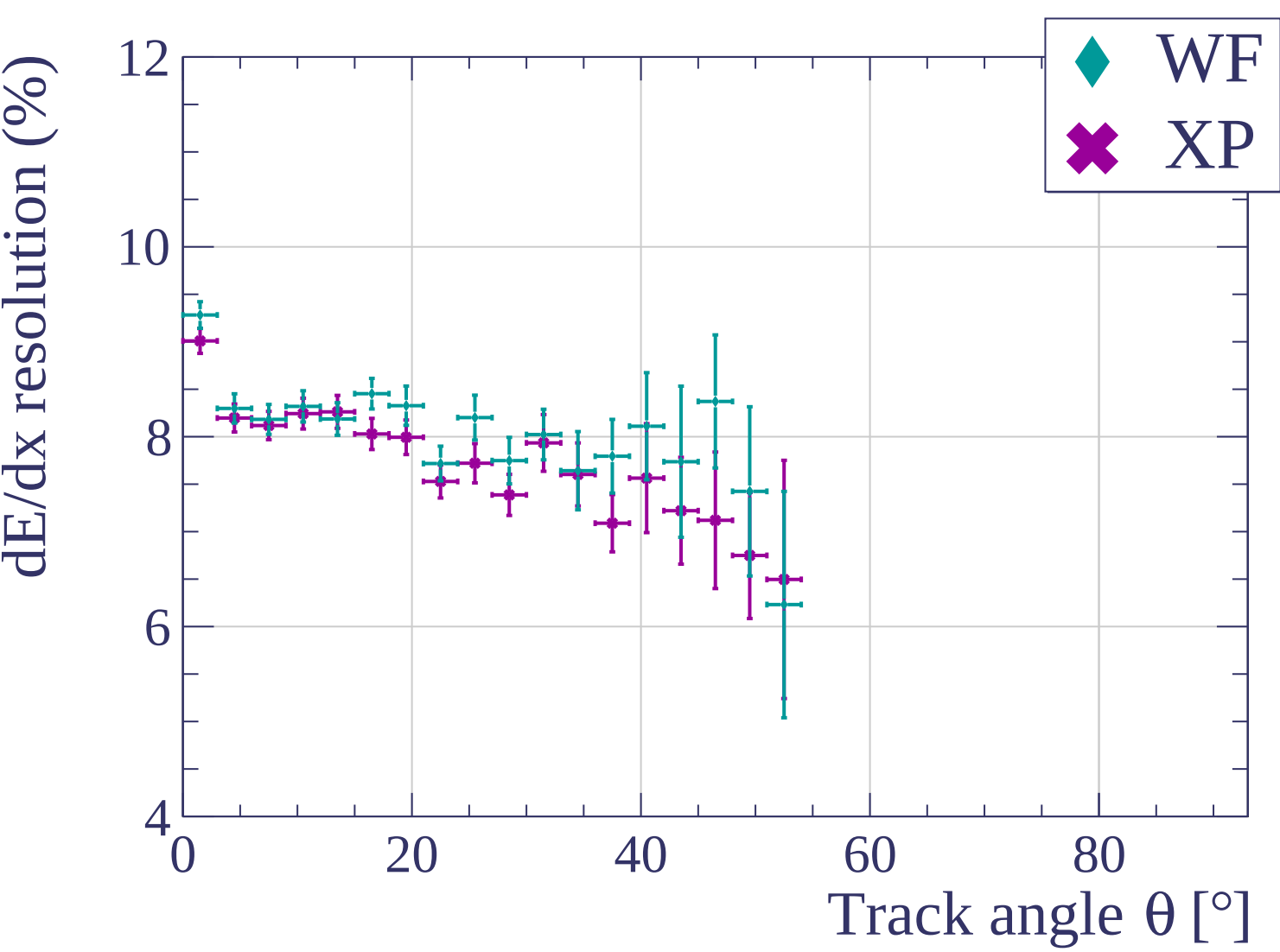
Track length [cm]

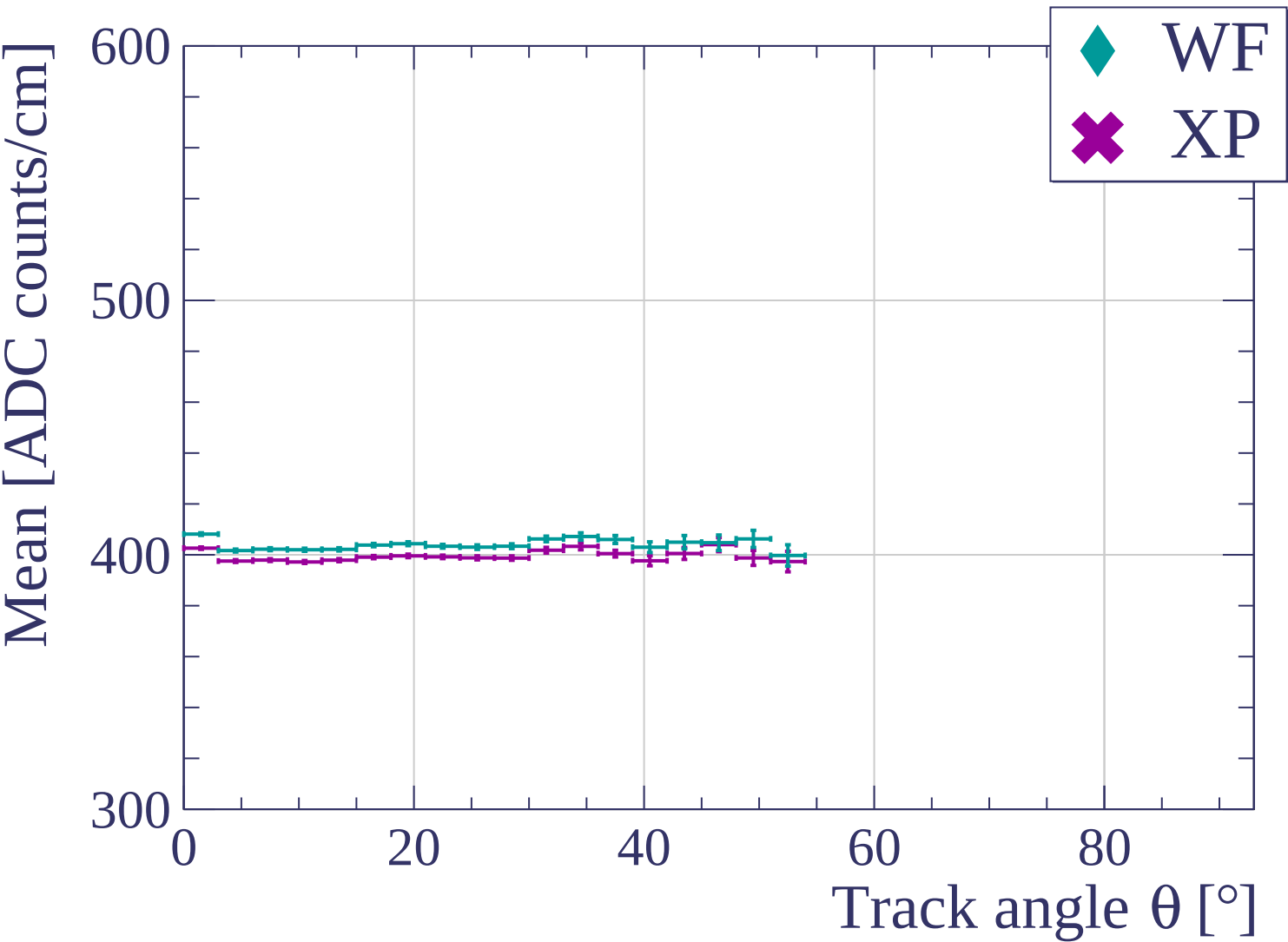


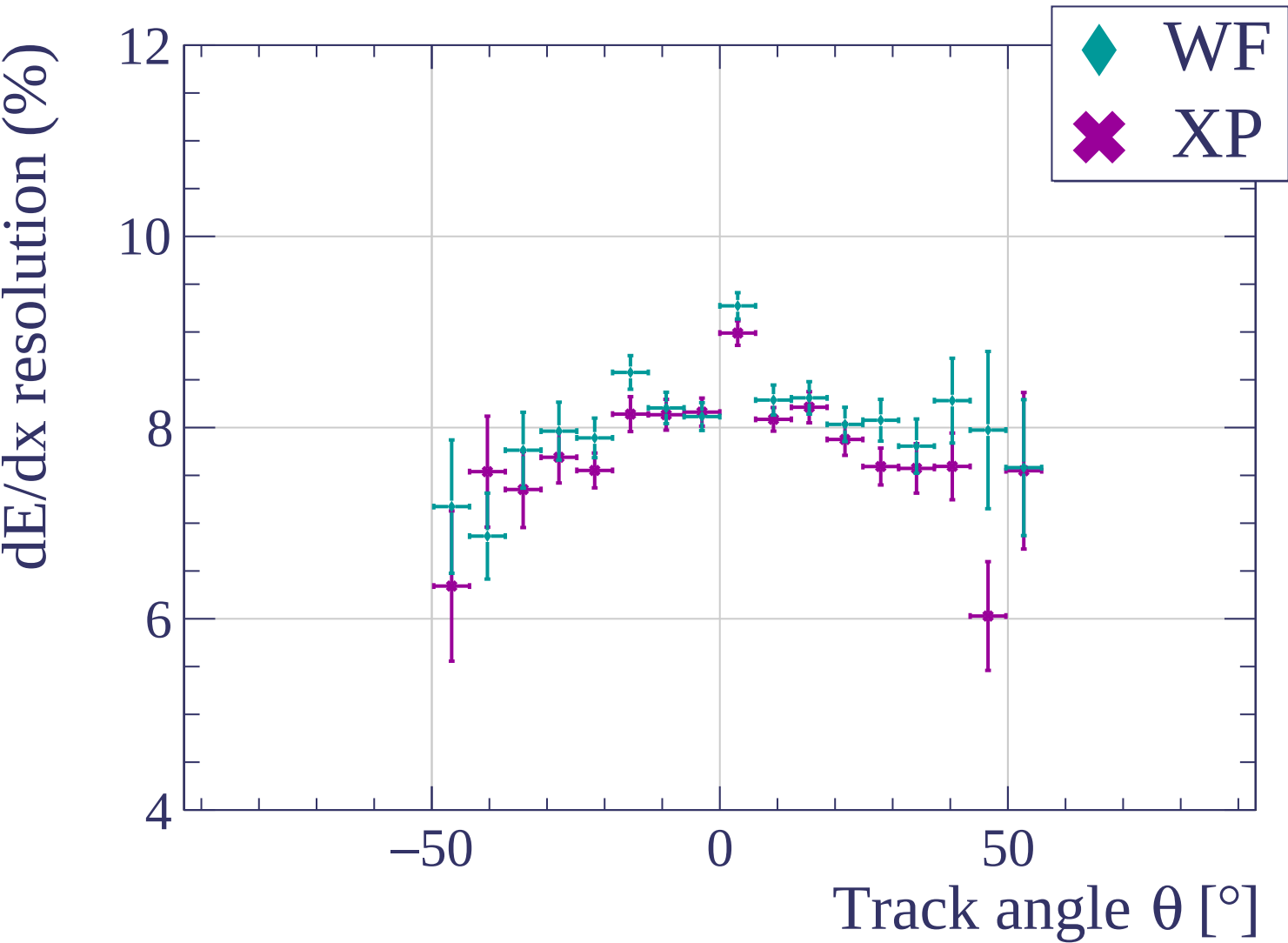


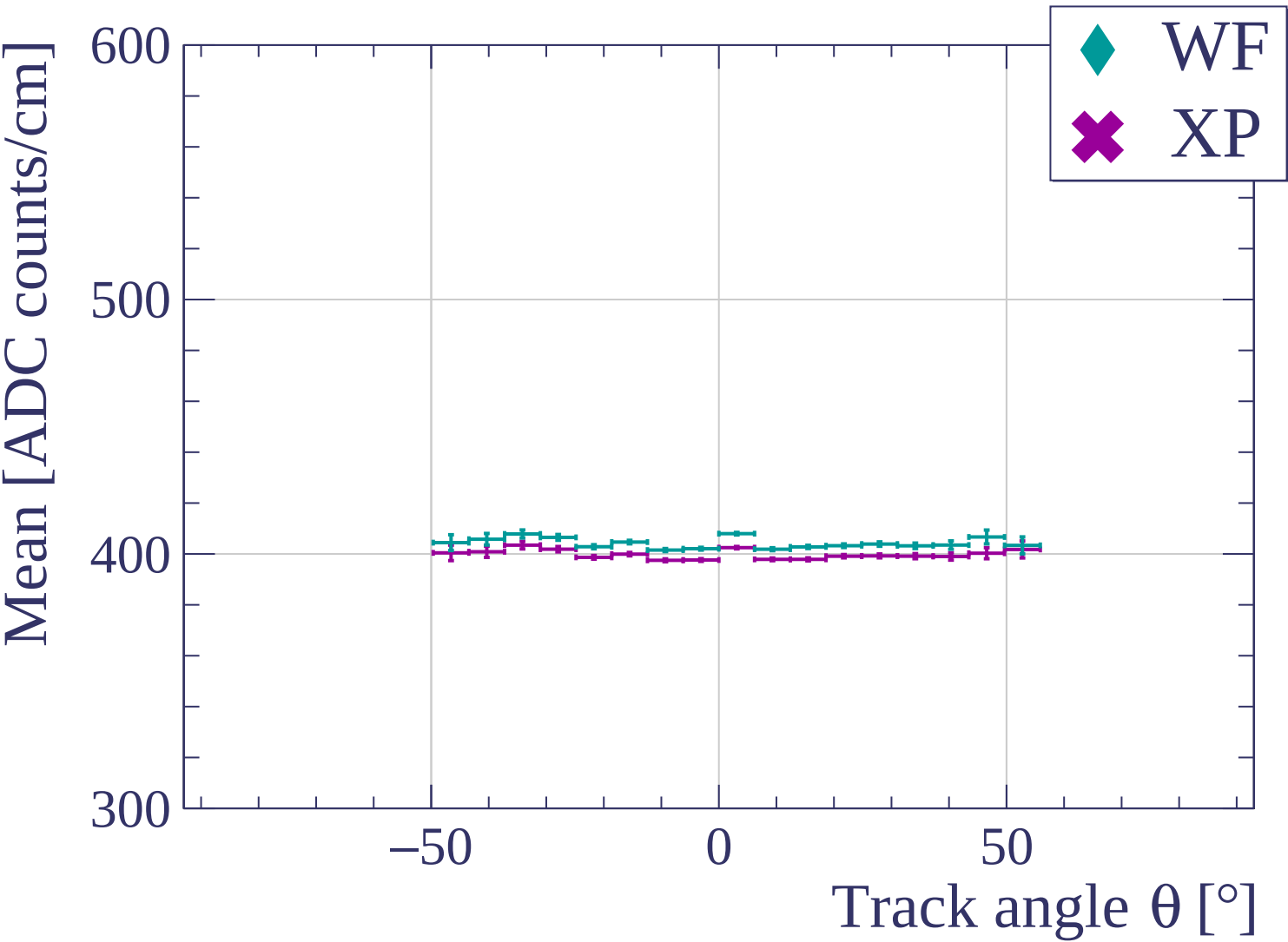


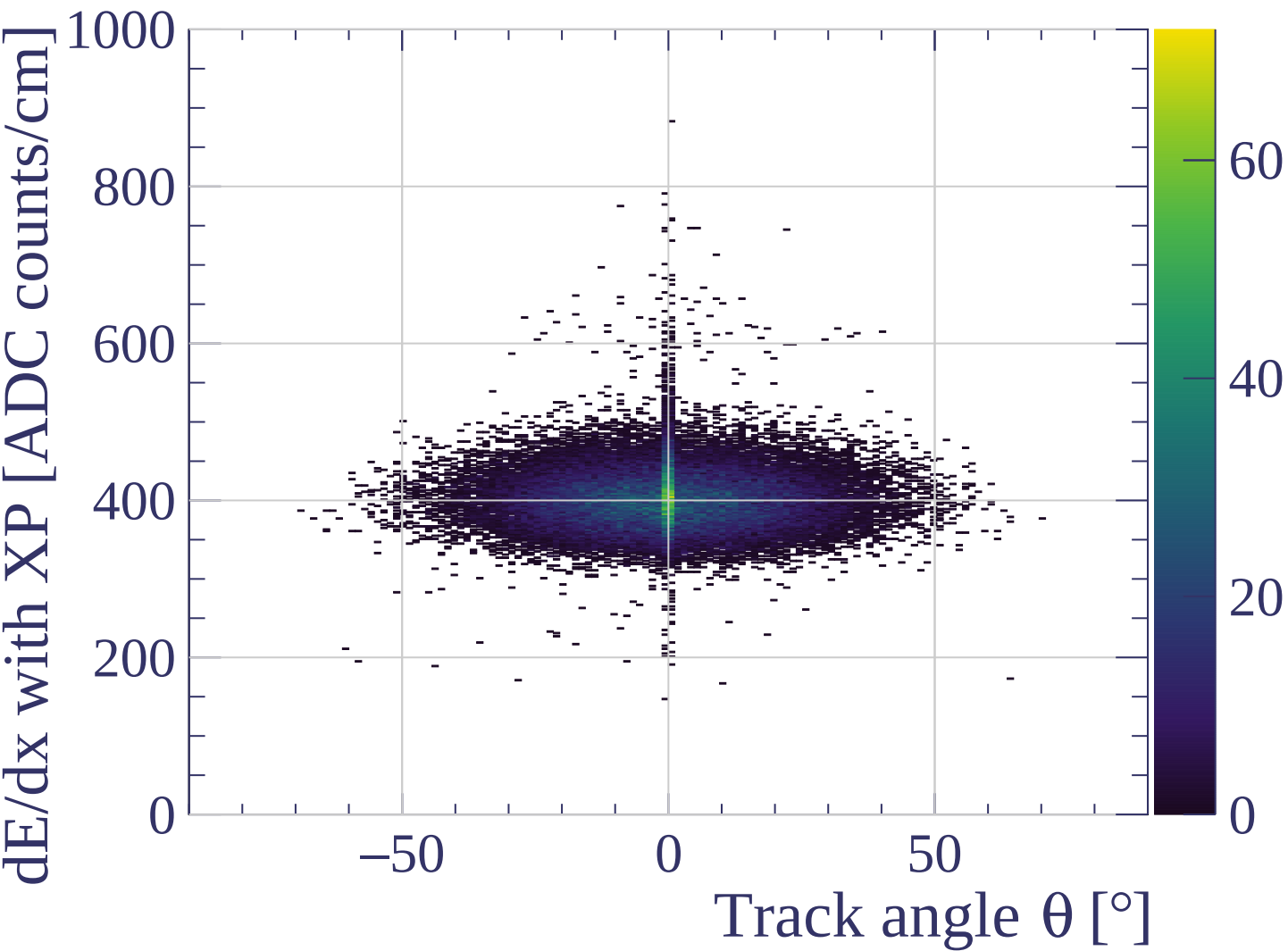




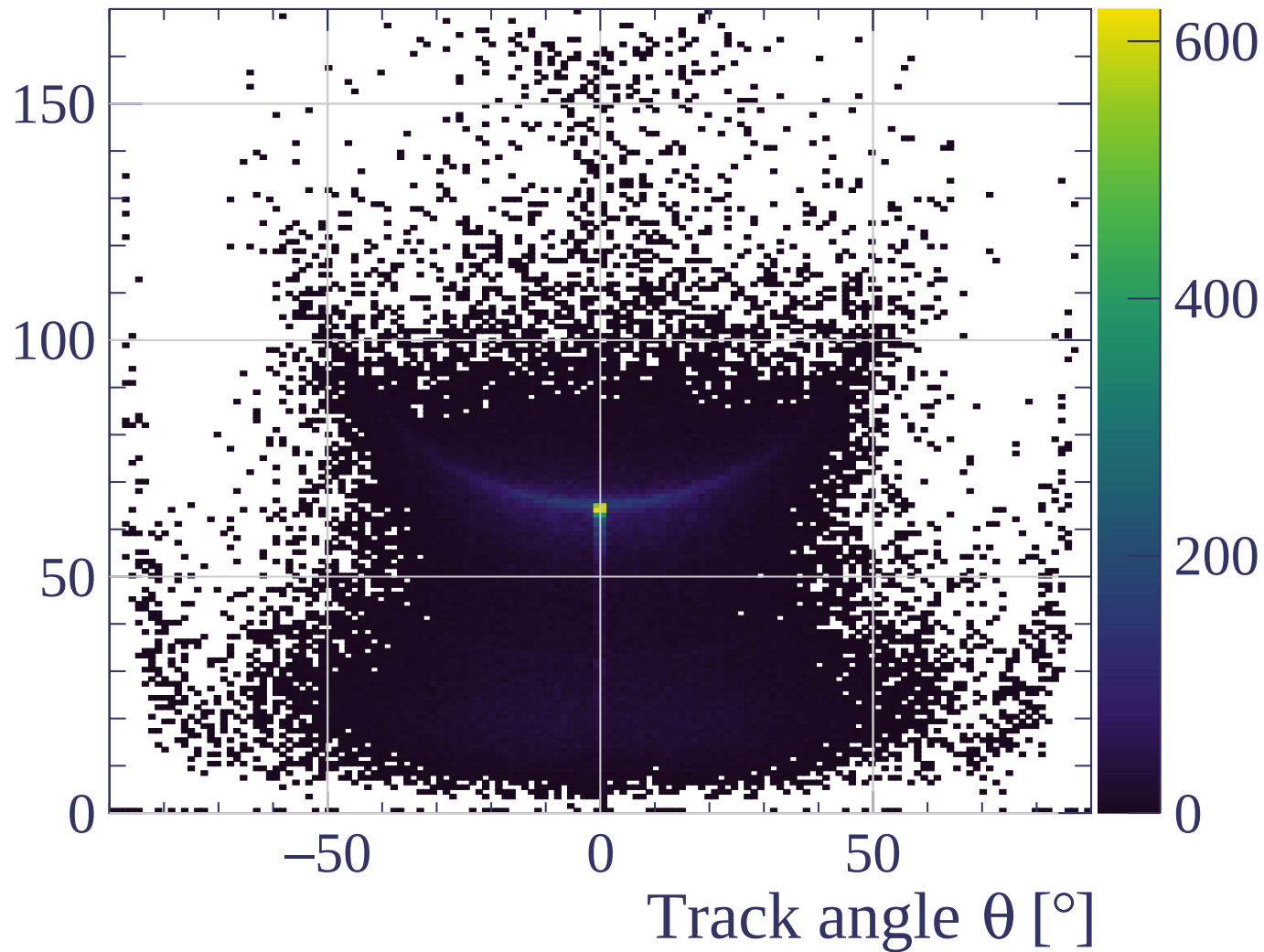


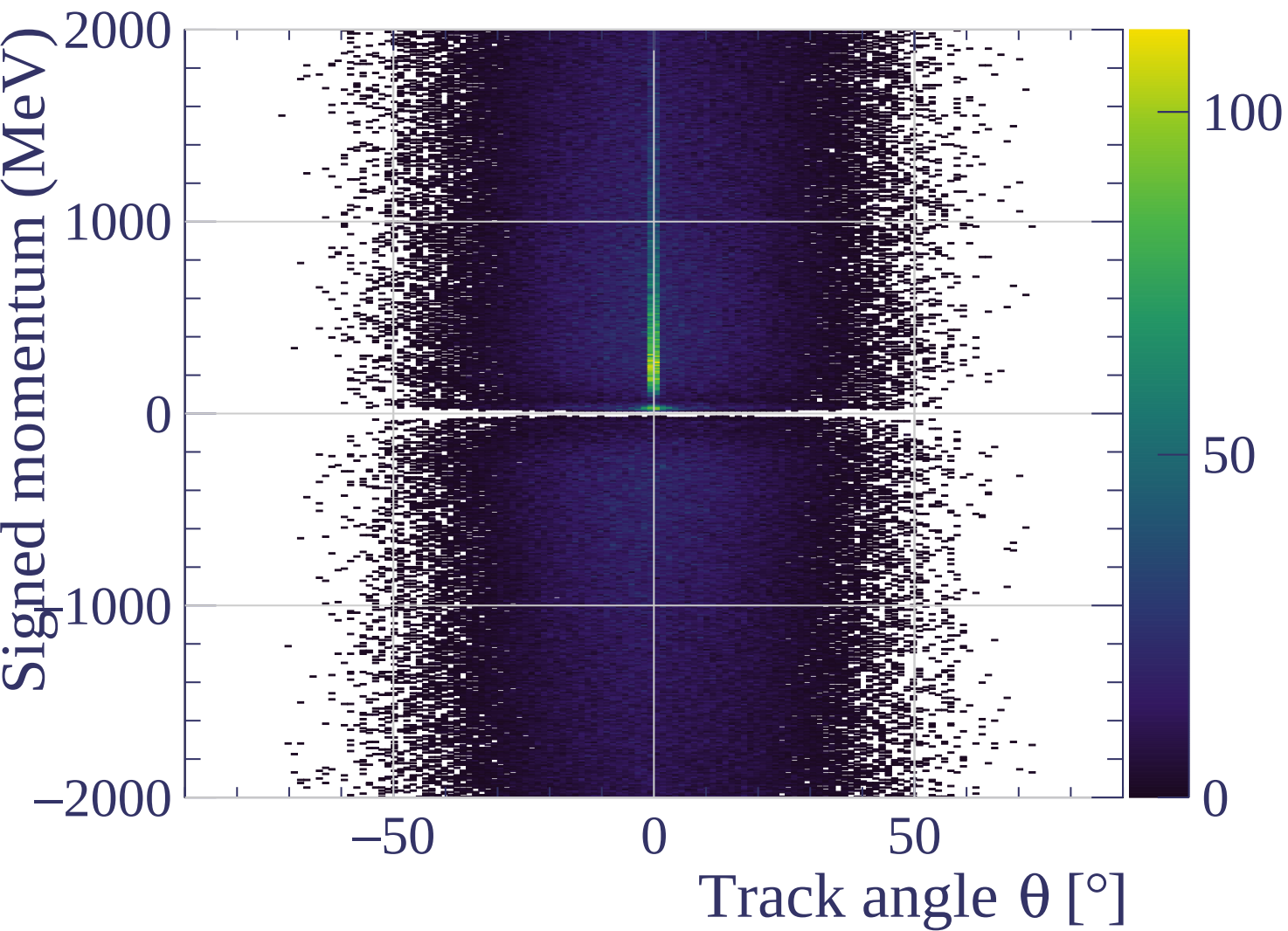


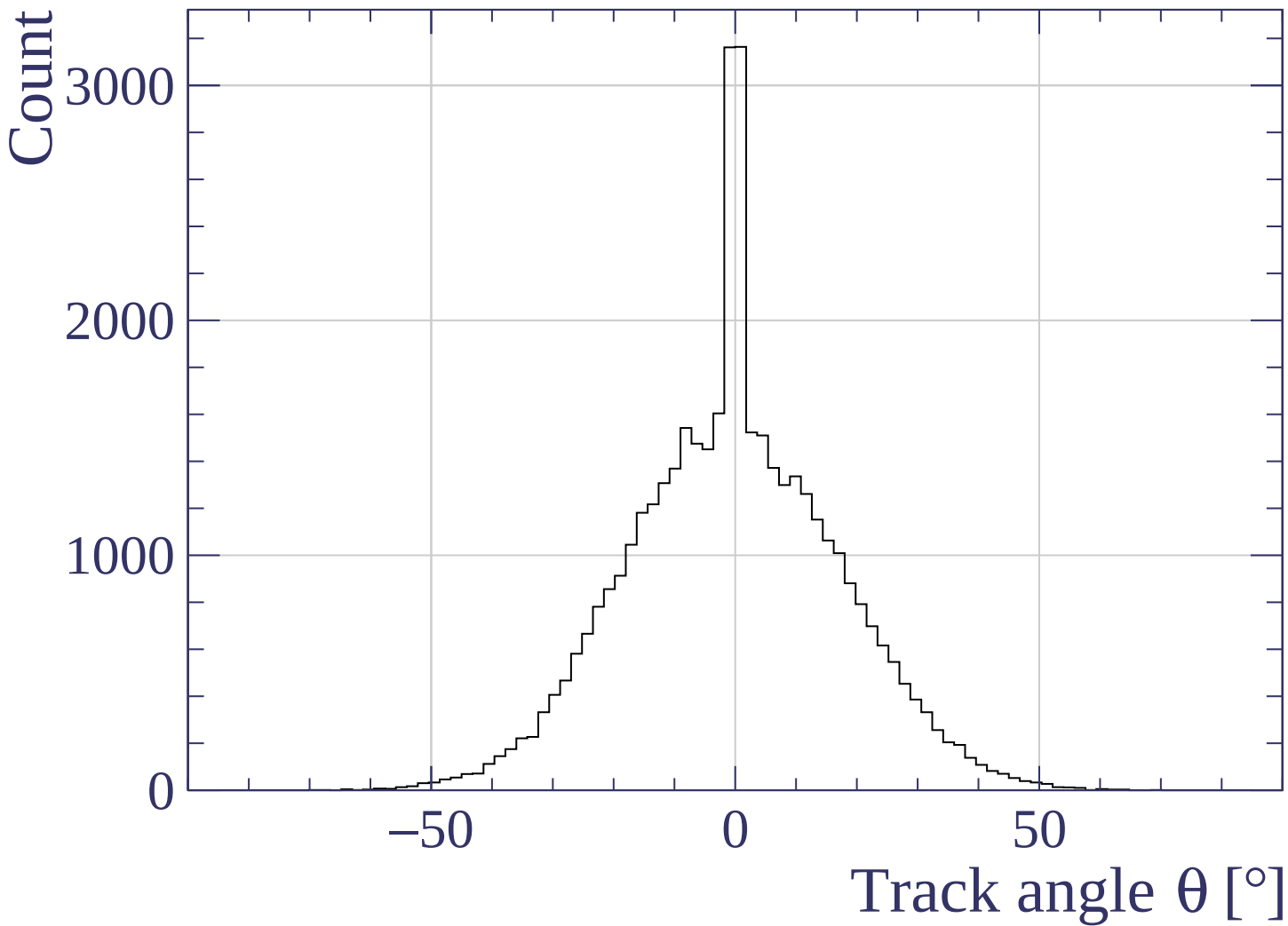




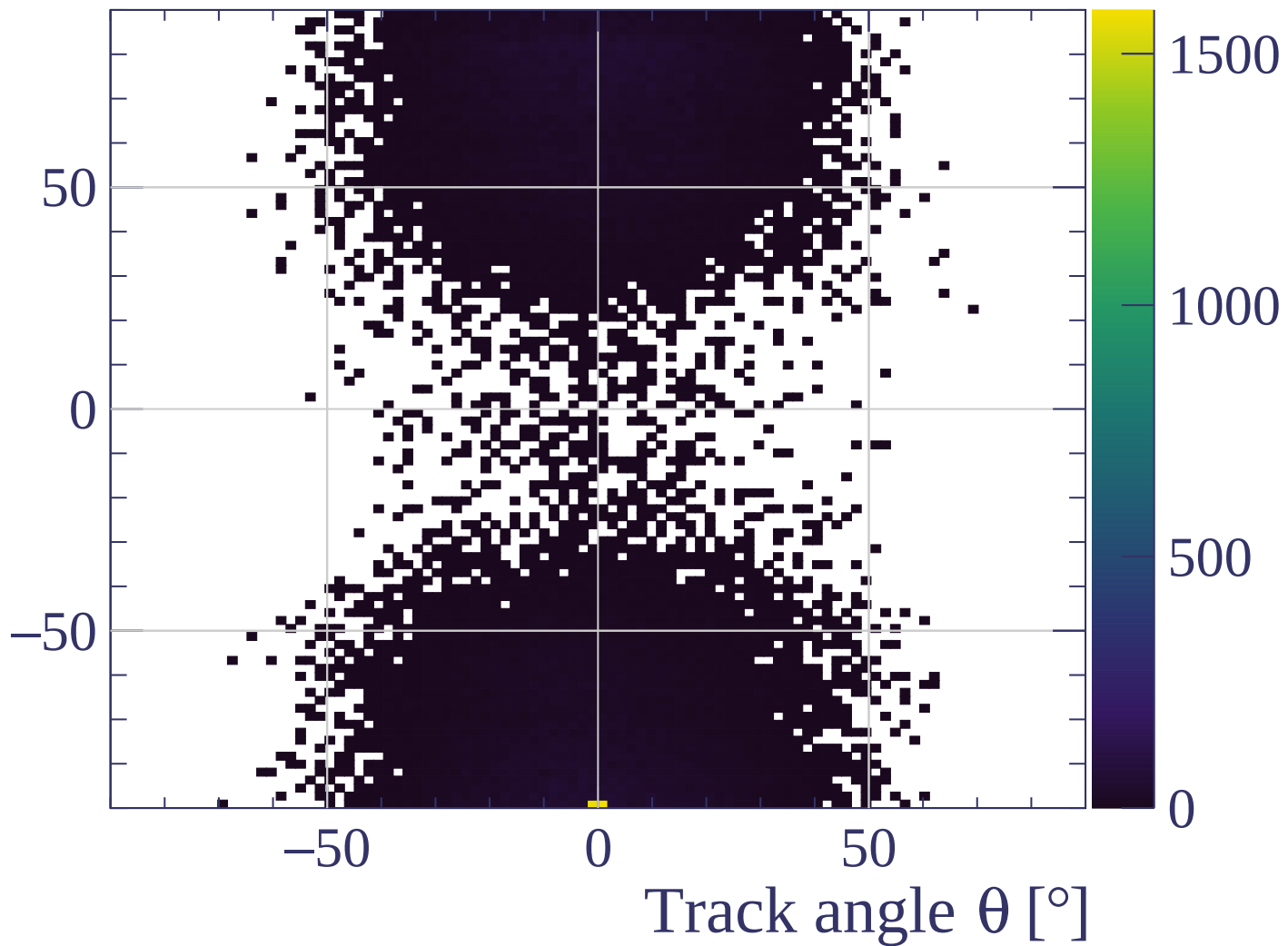
Track length [cm]



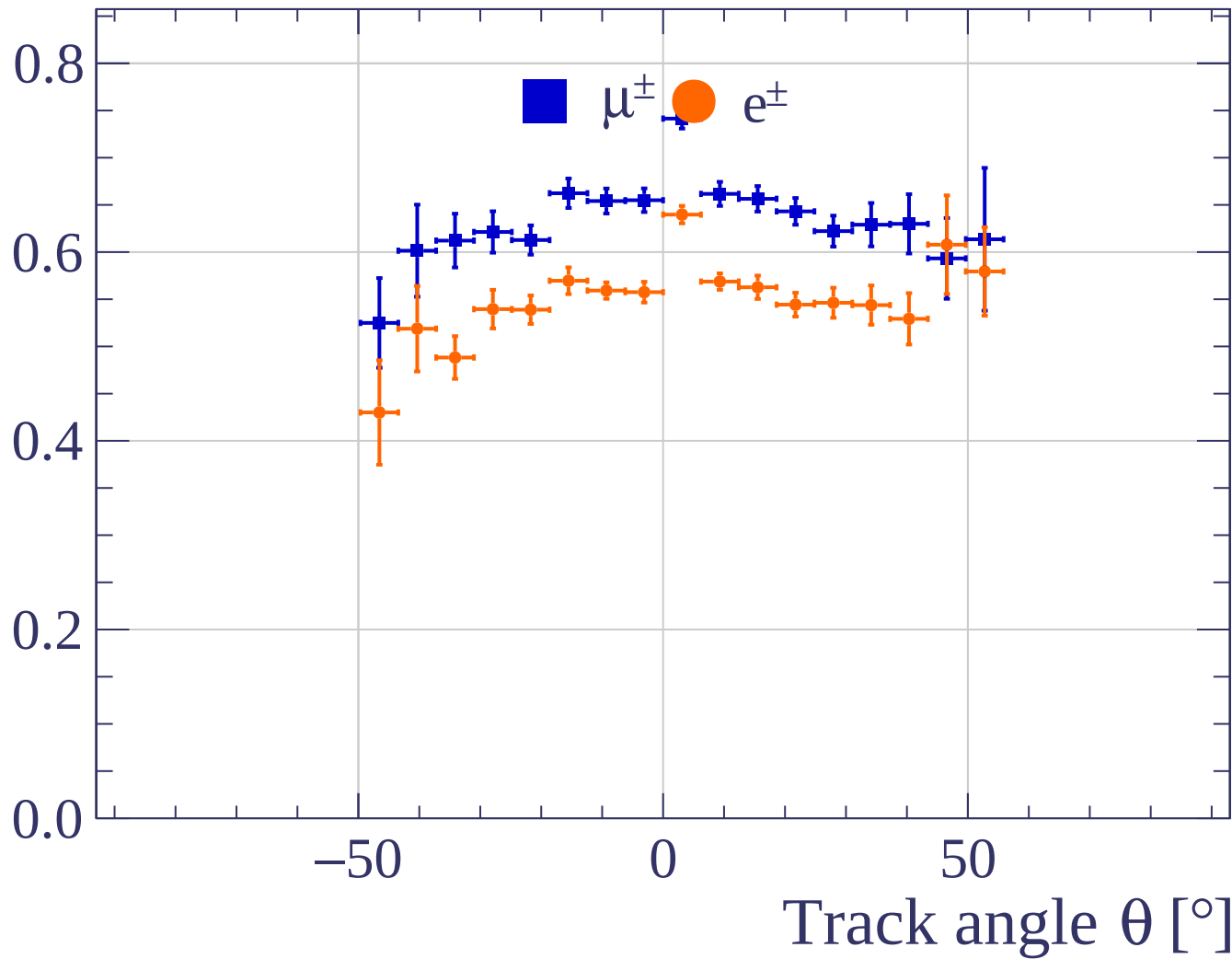




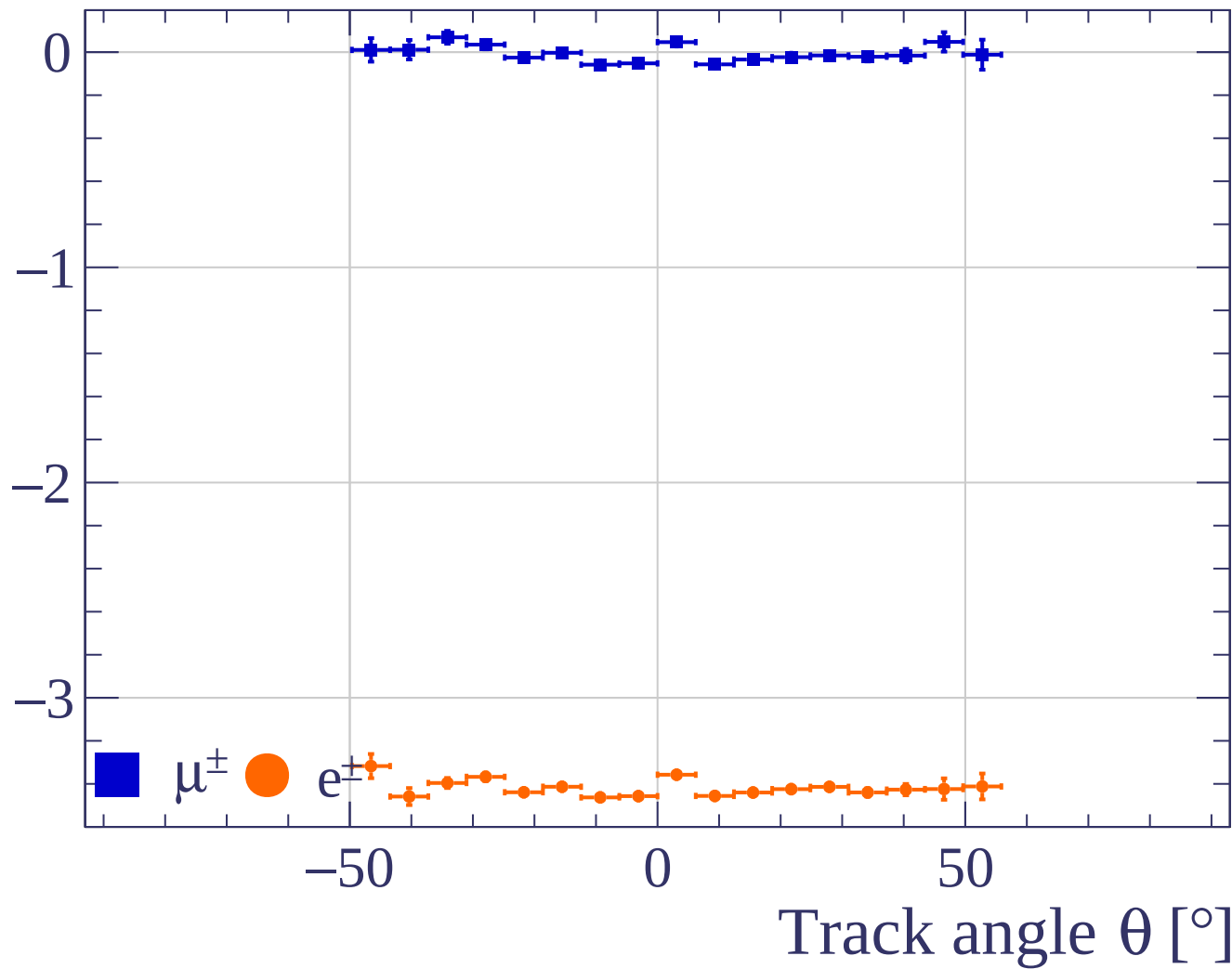
Track angle ϕ [°]



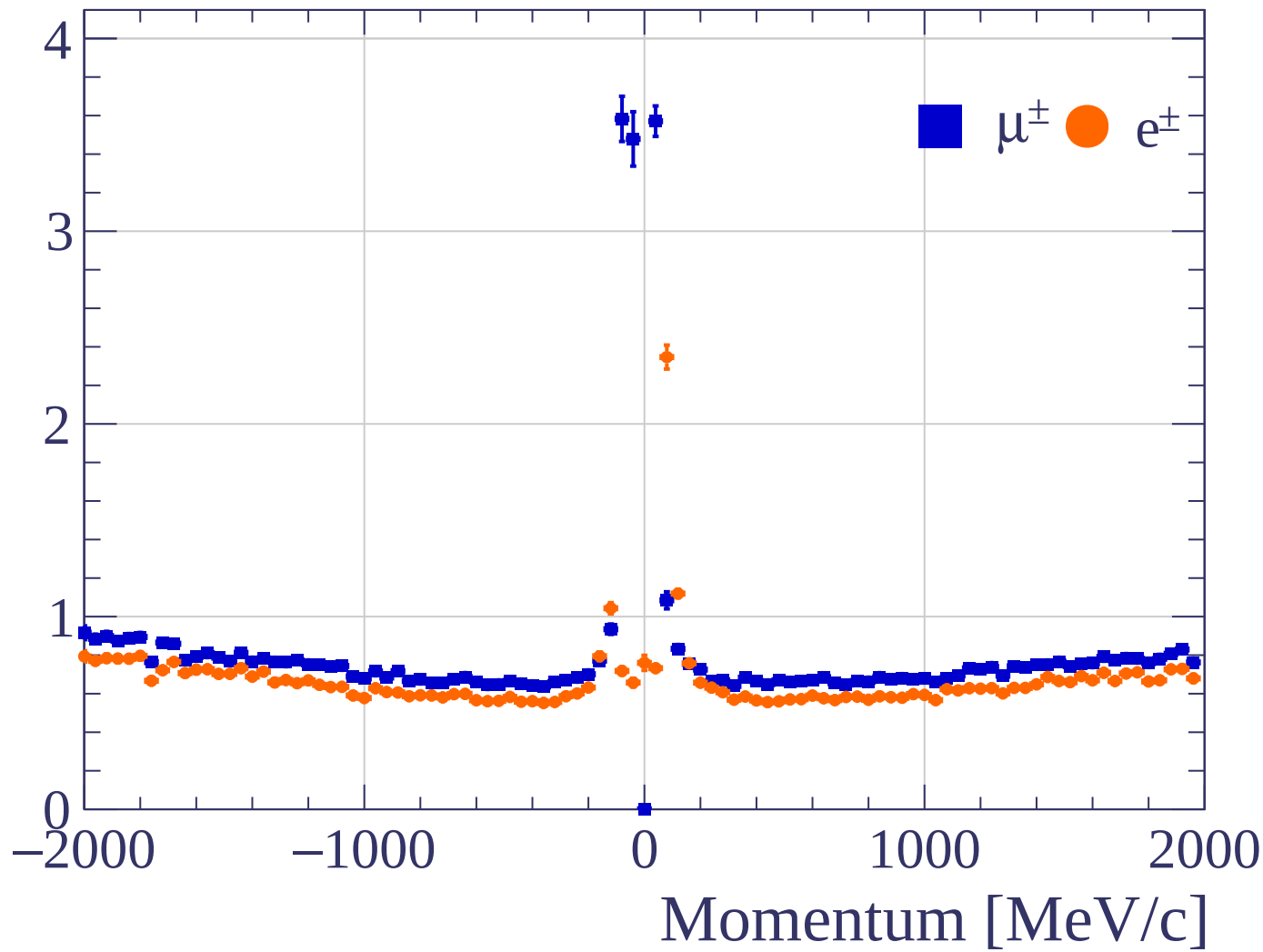
Pulls standard deviation



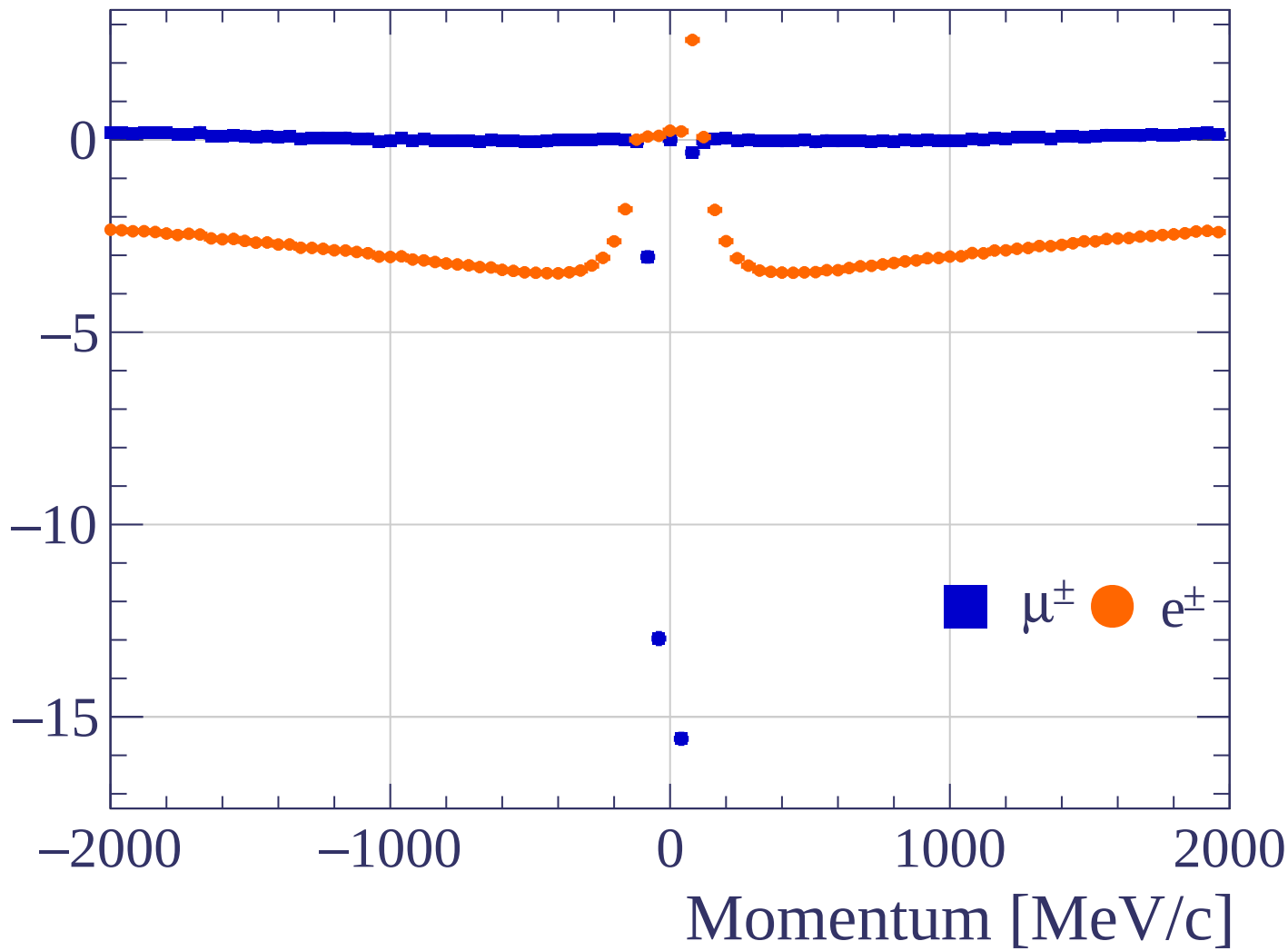
Pulls mean

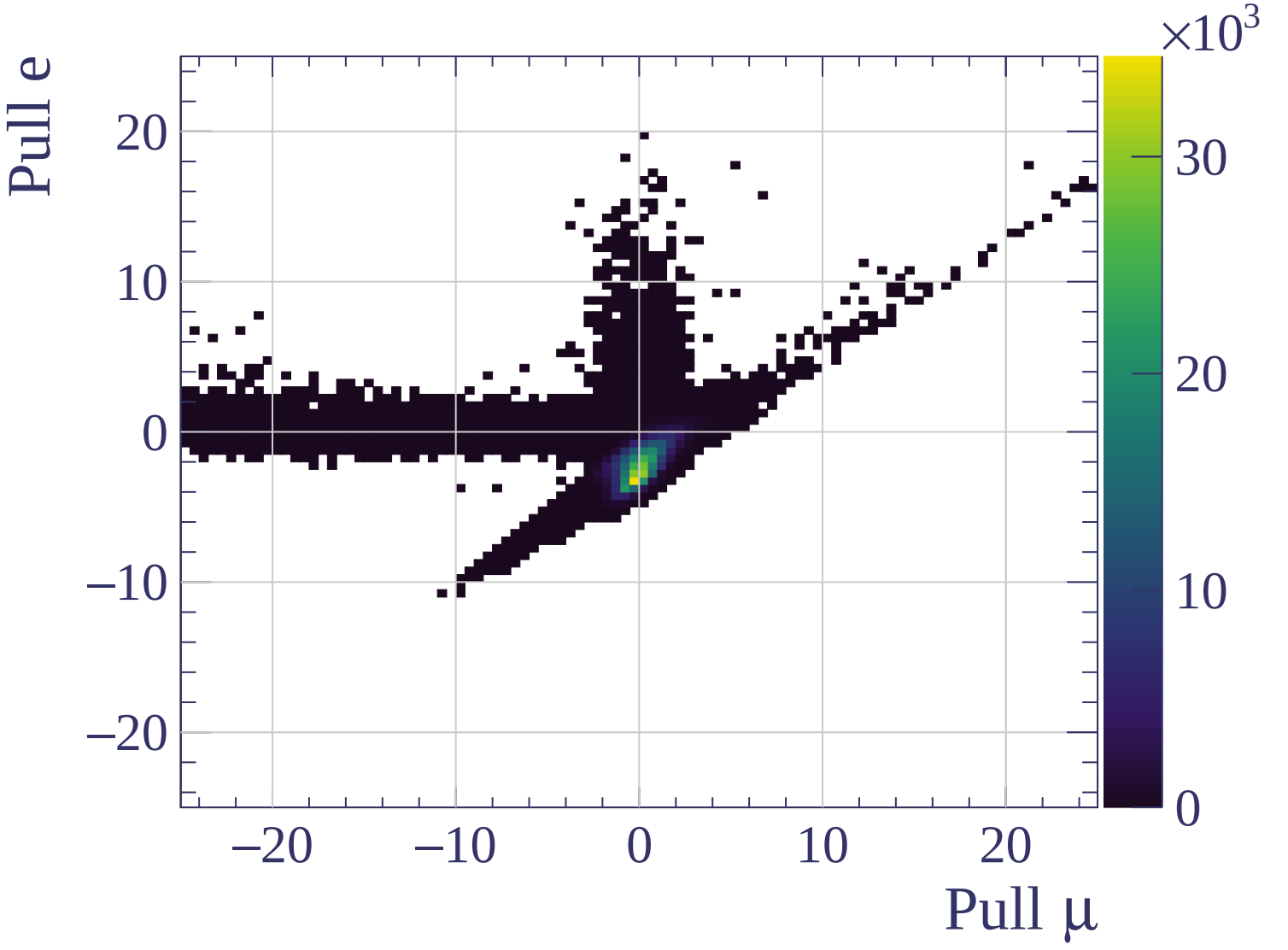


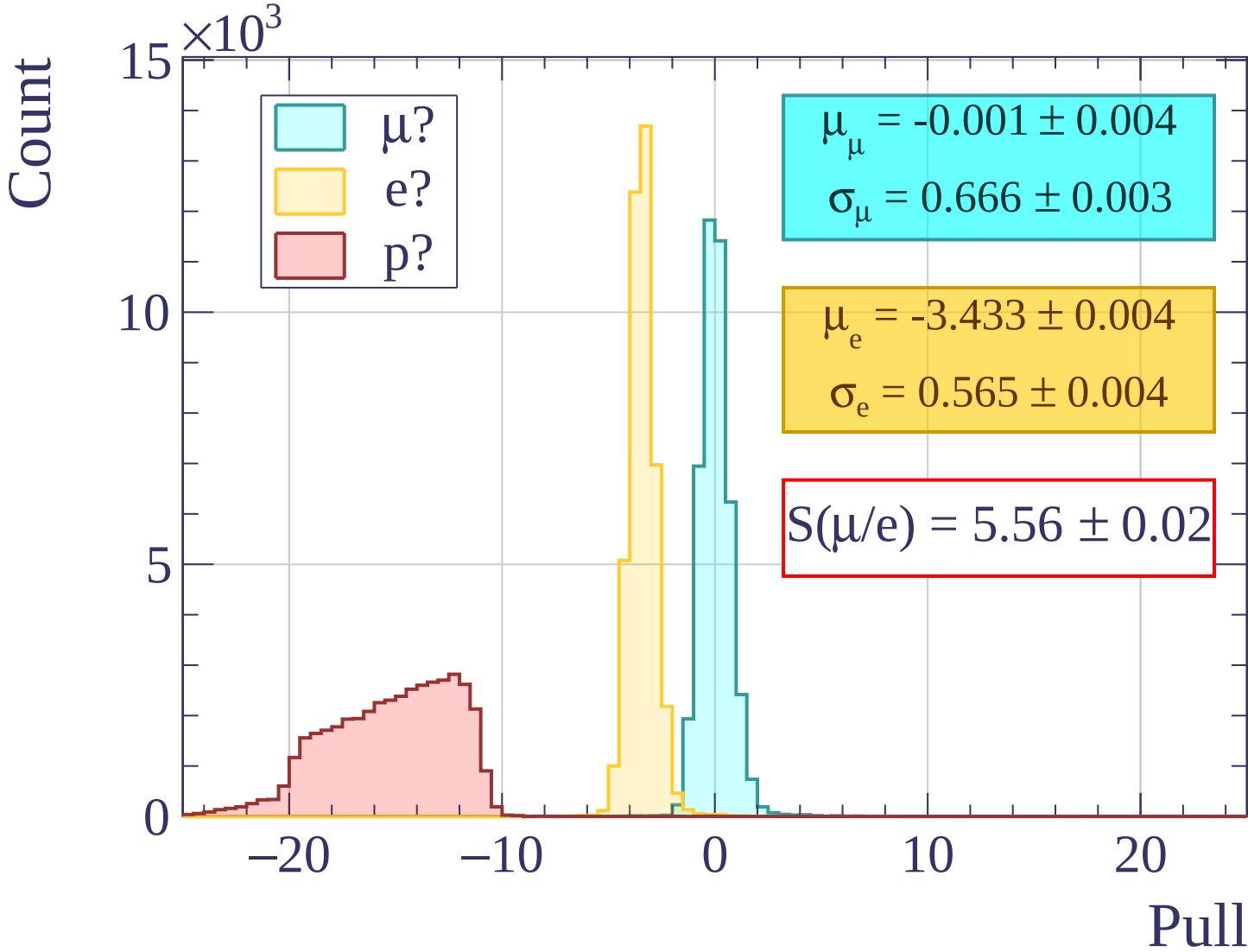
Pulls standard deviation

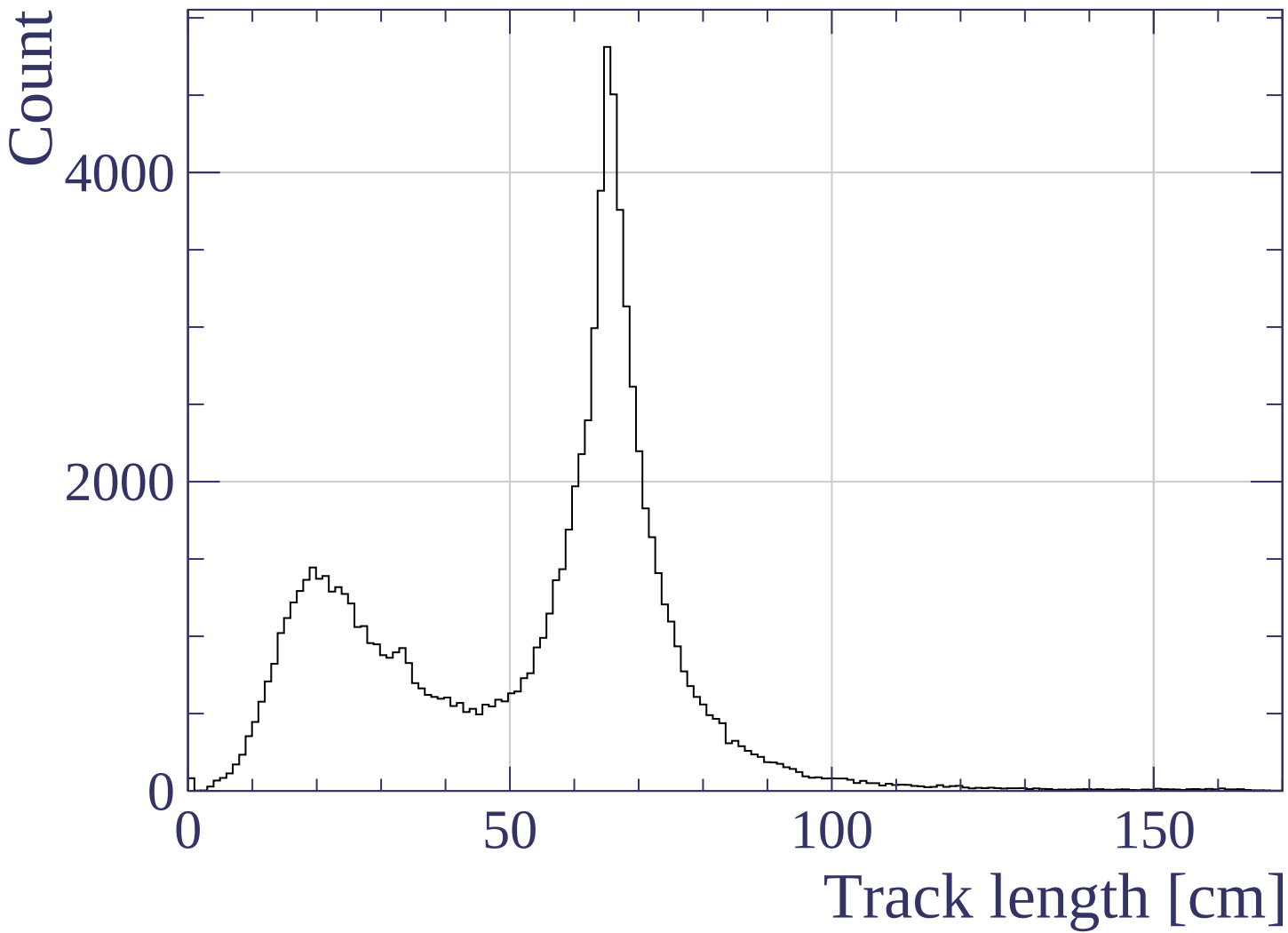


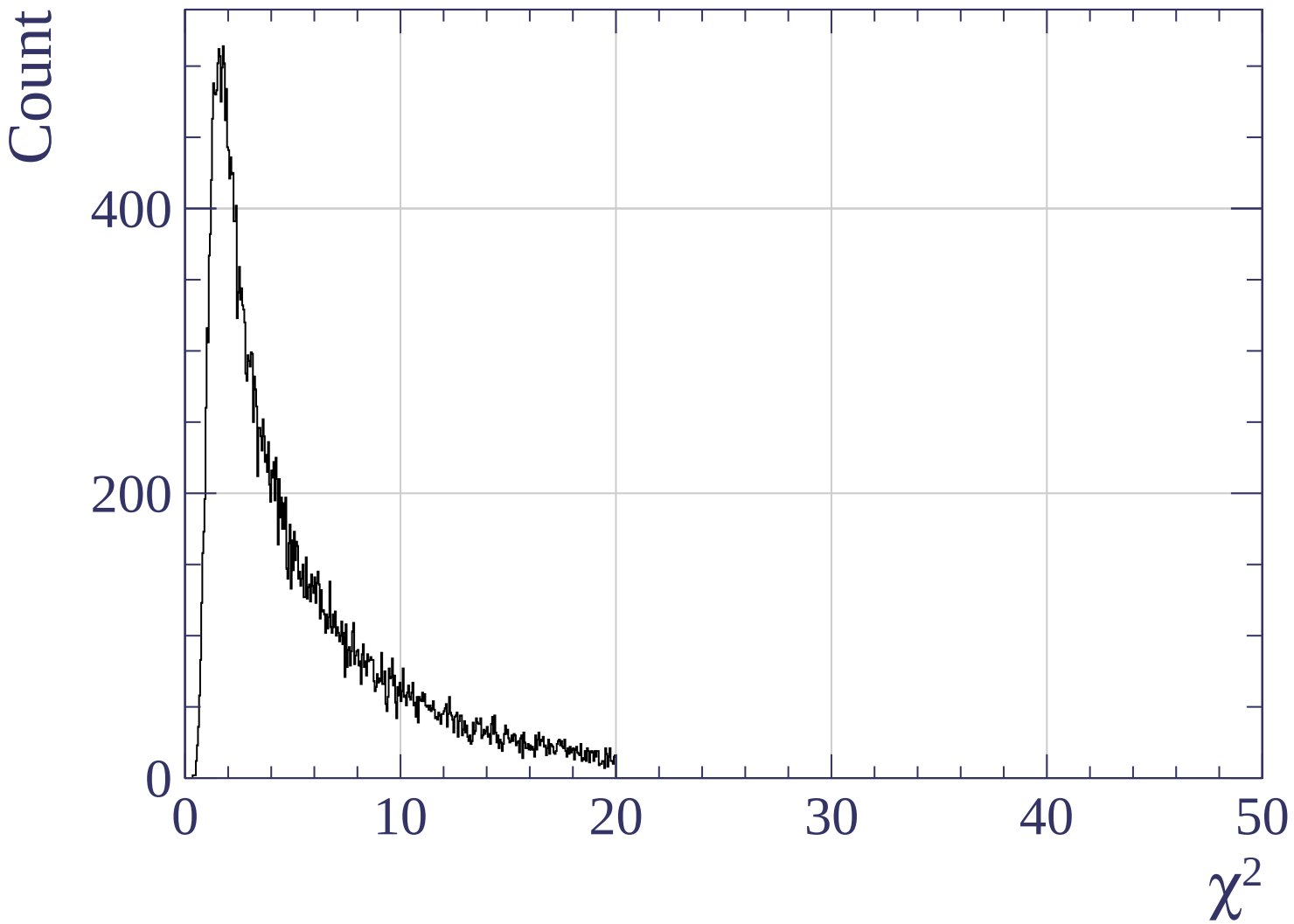
Pulls mean



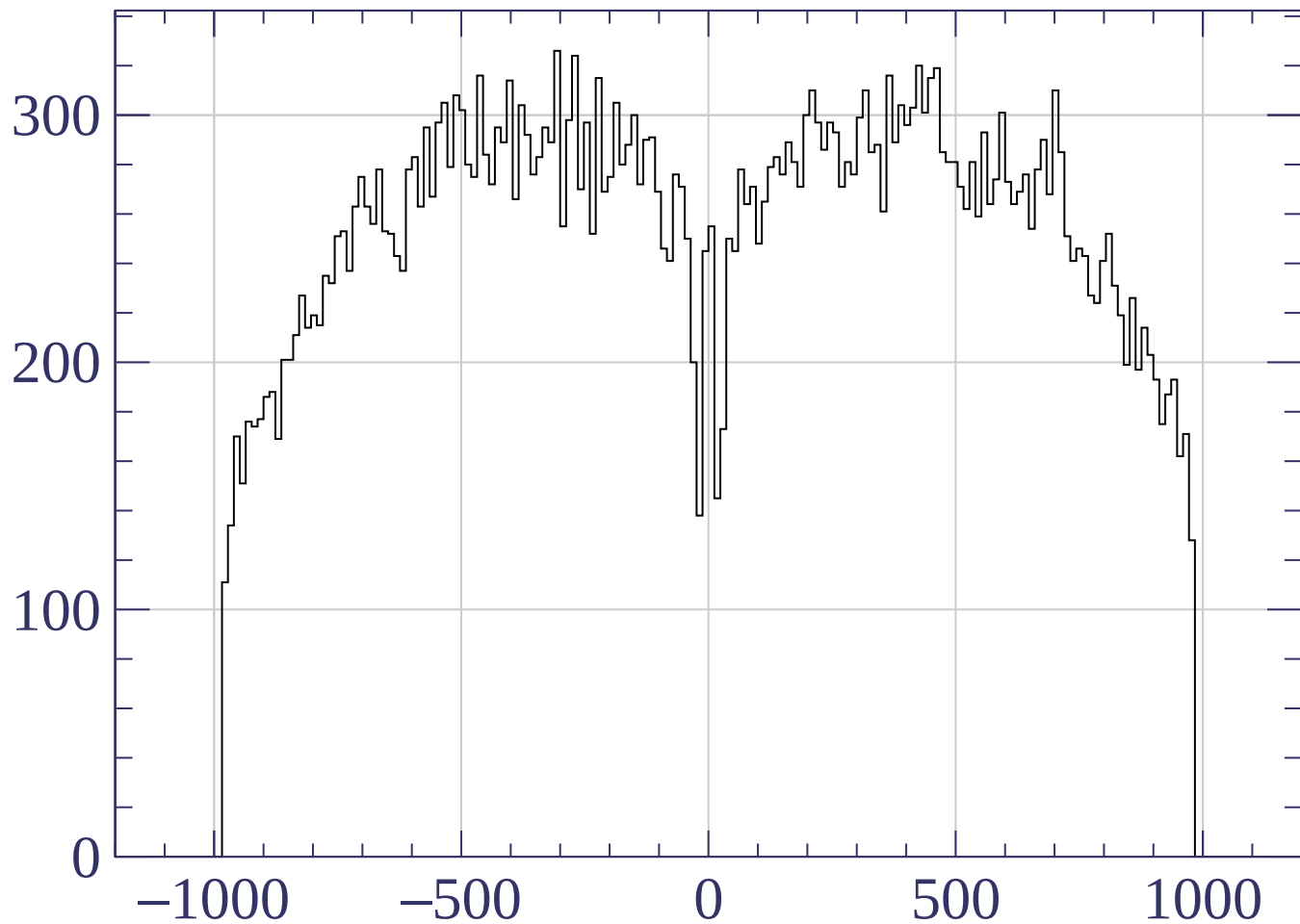




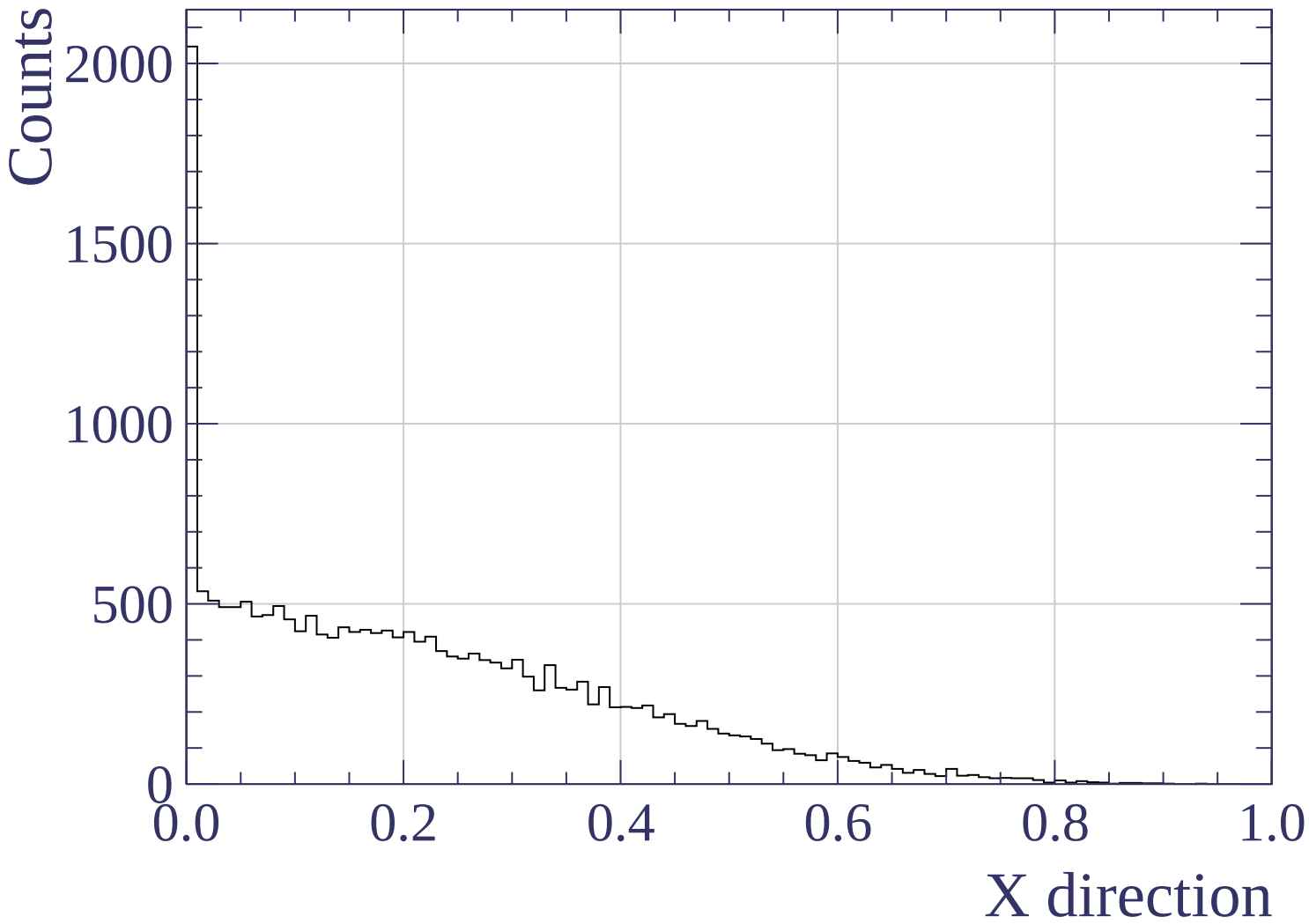


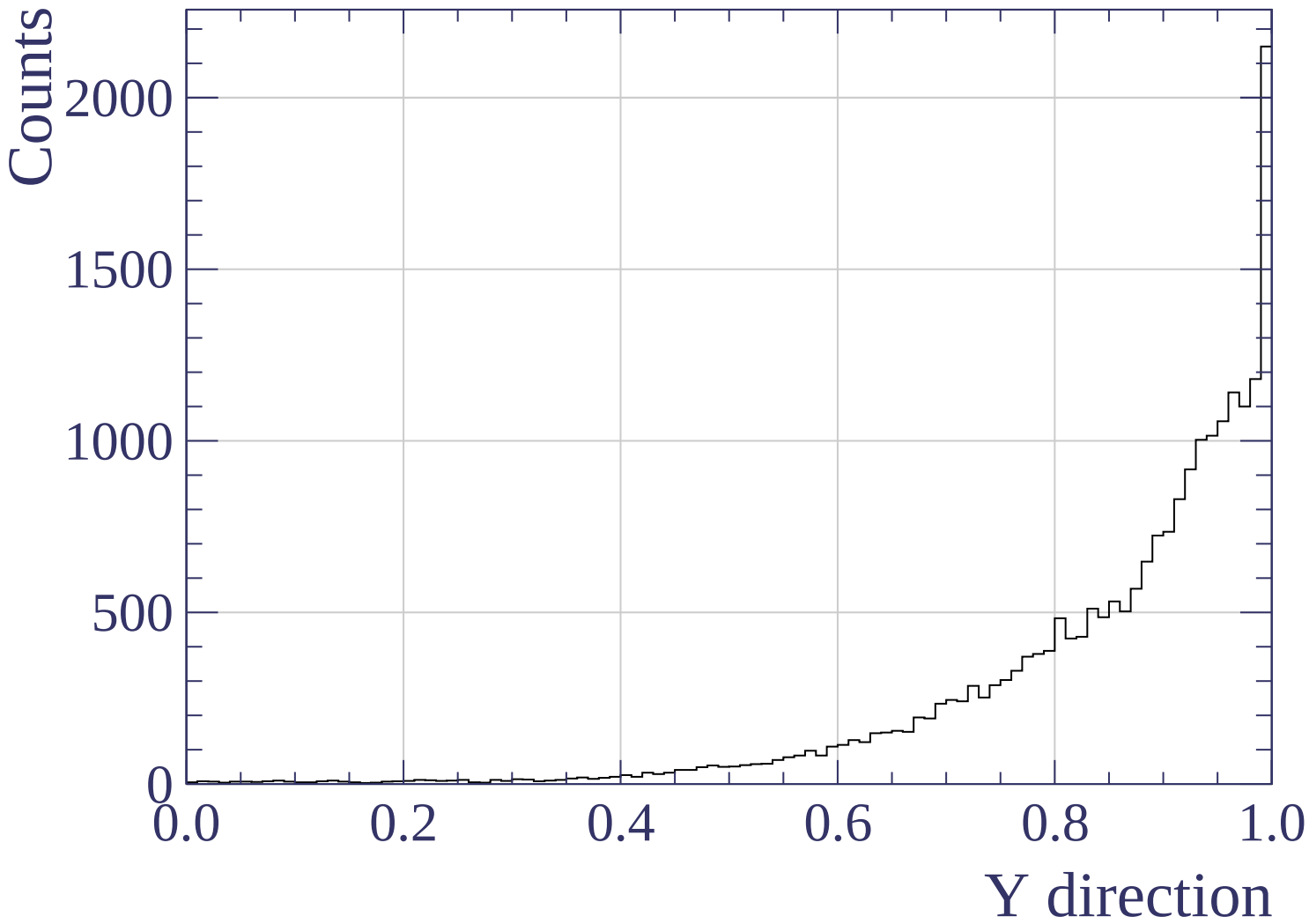


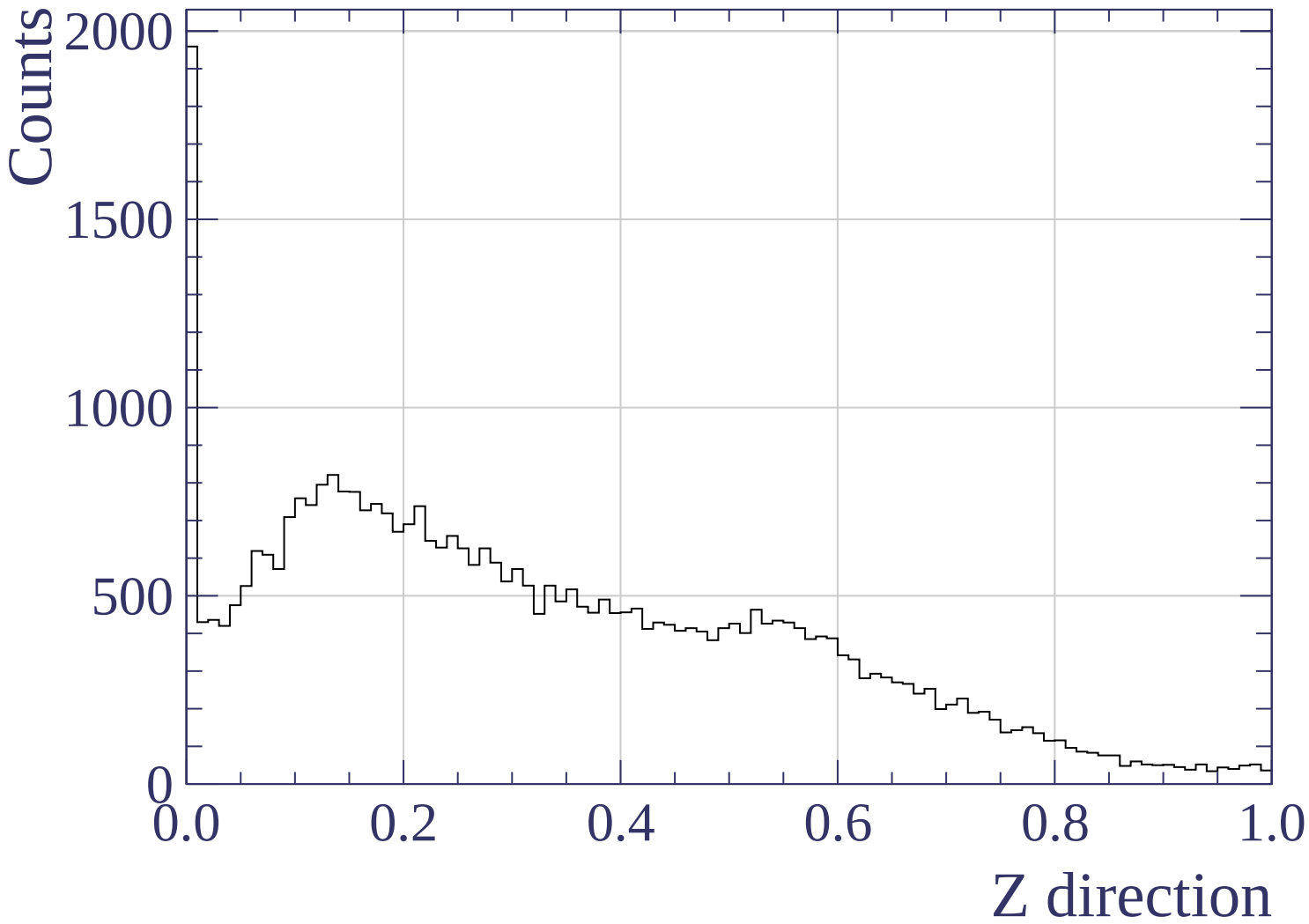
Count

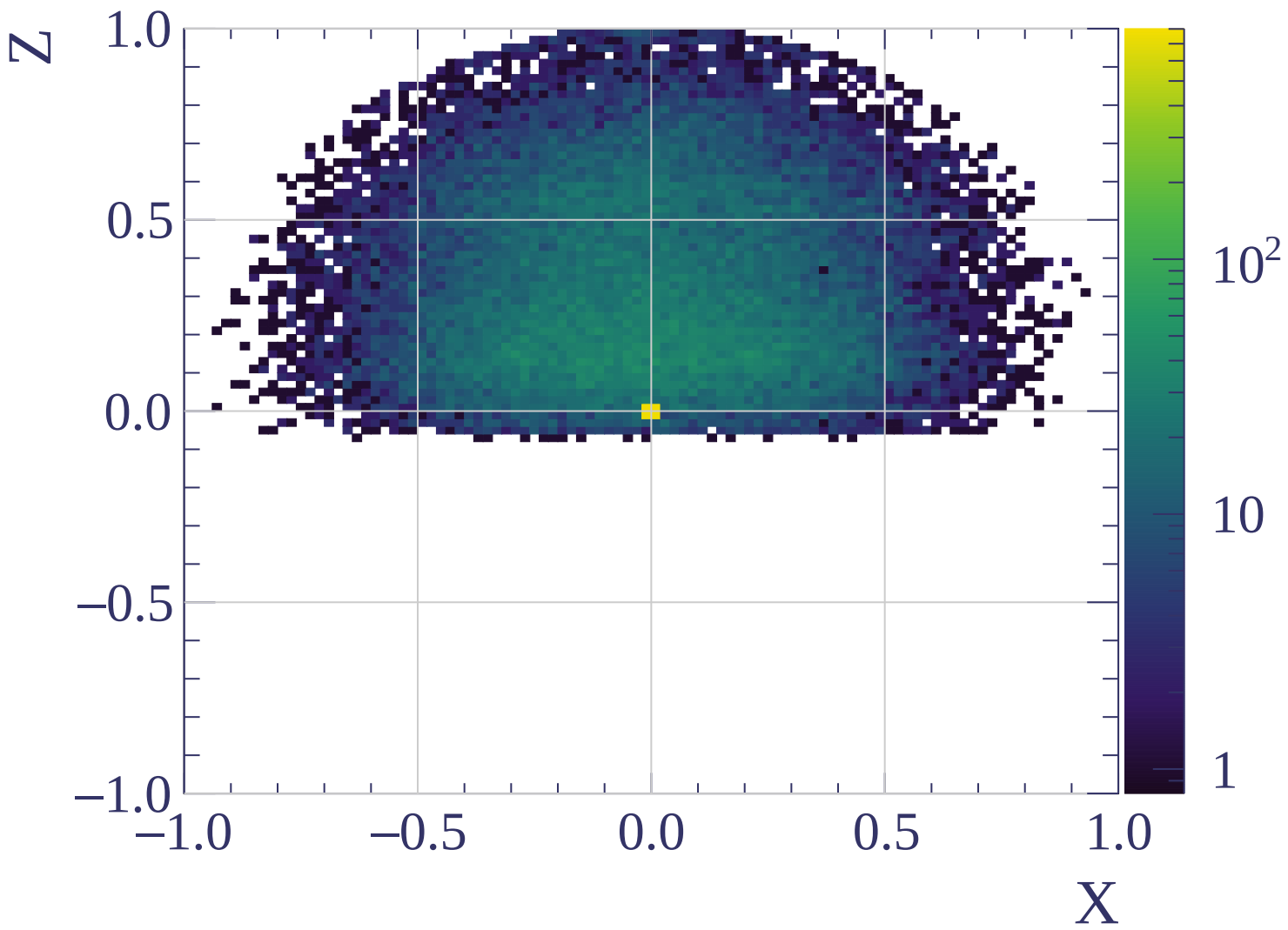


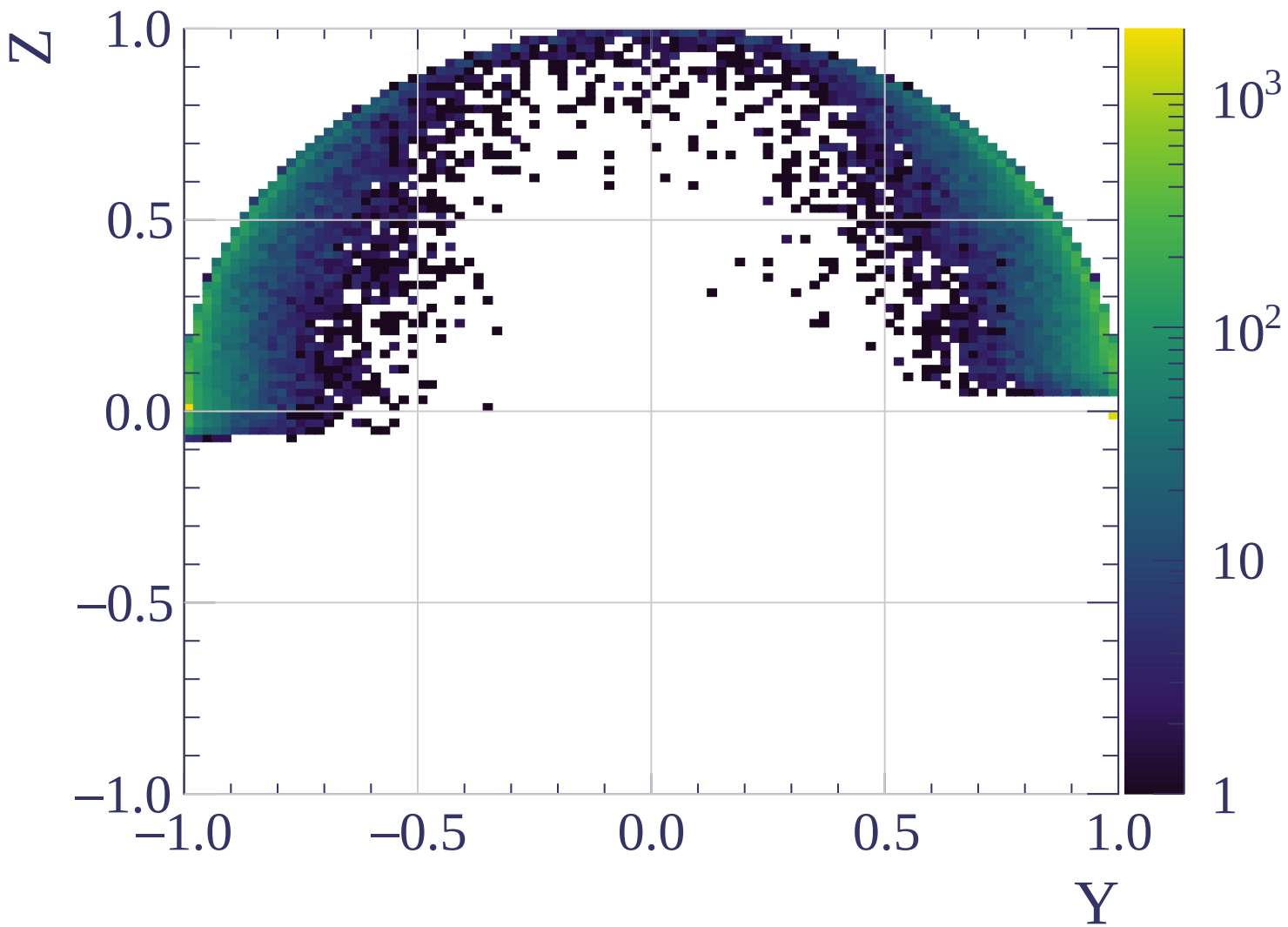
Track X position [mm]

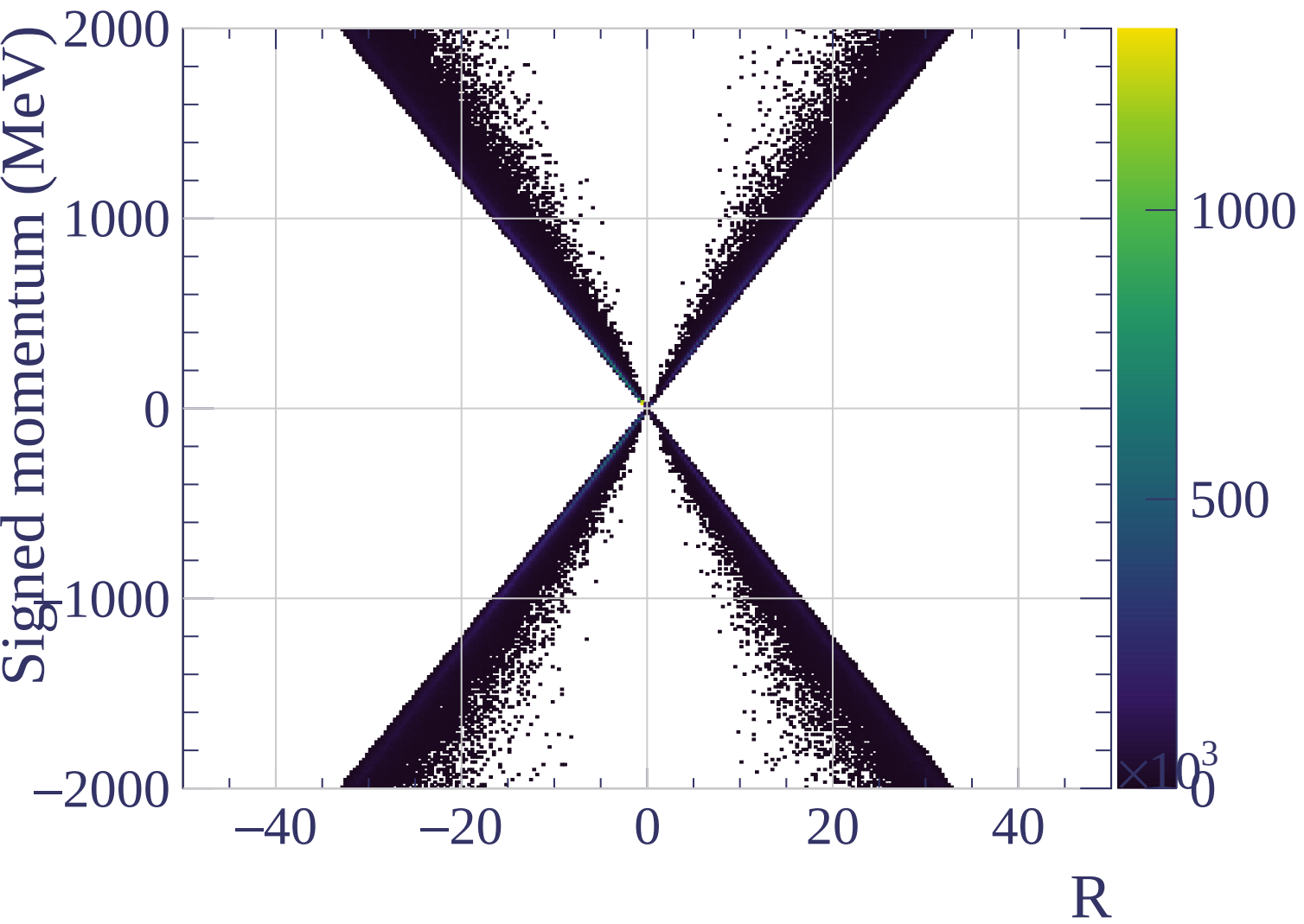


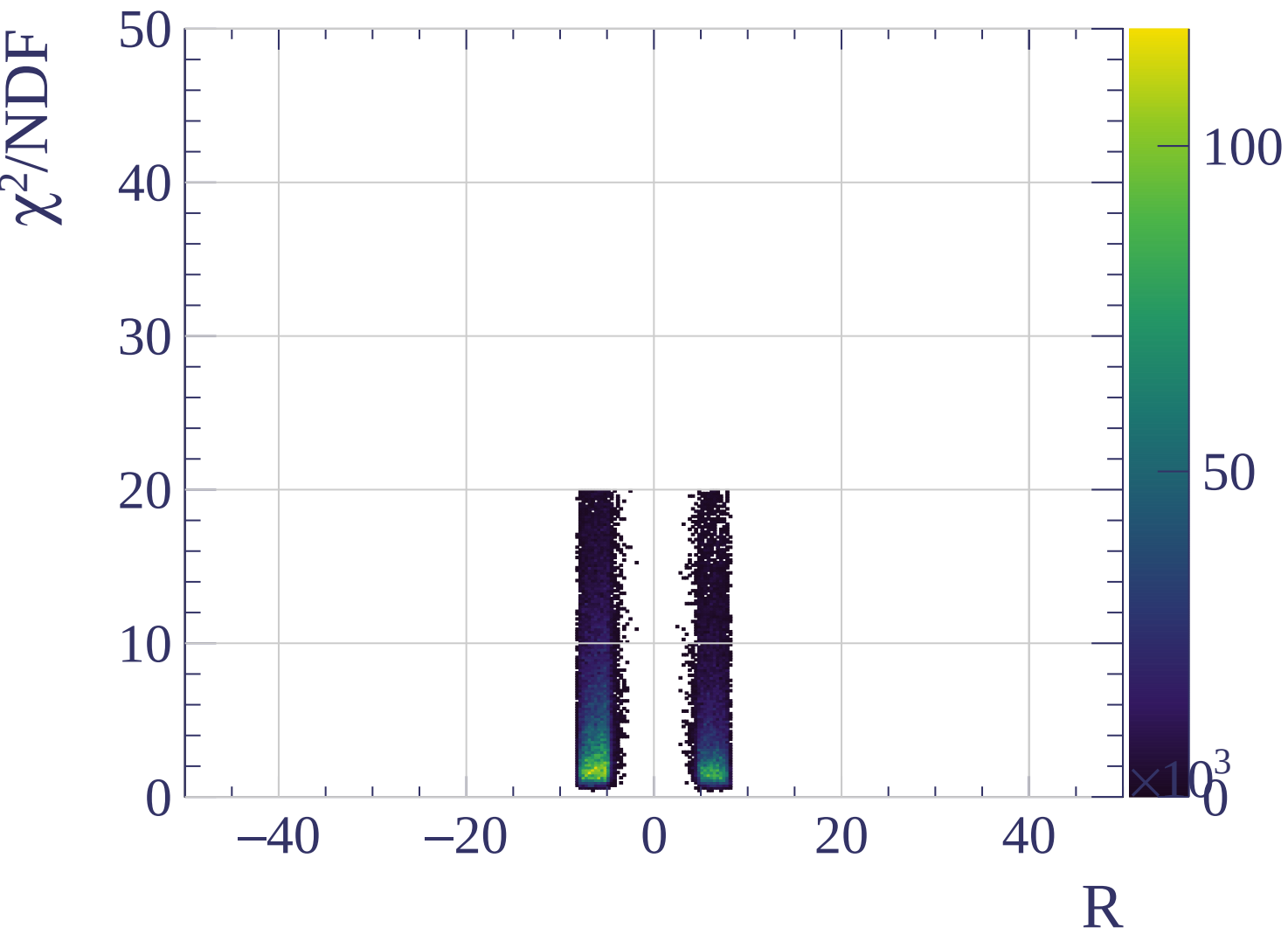




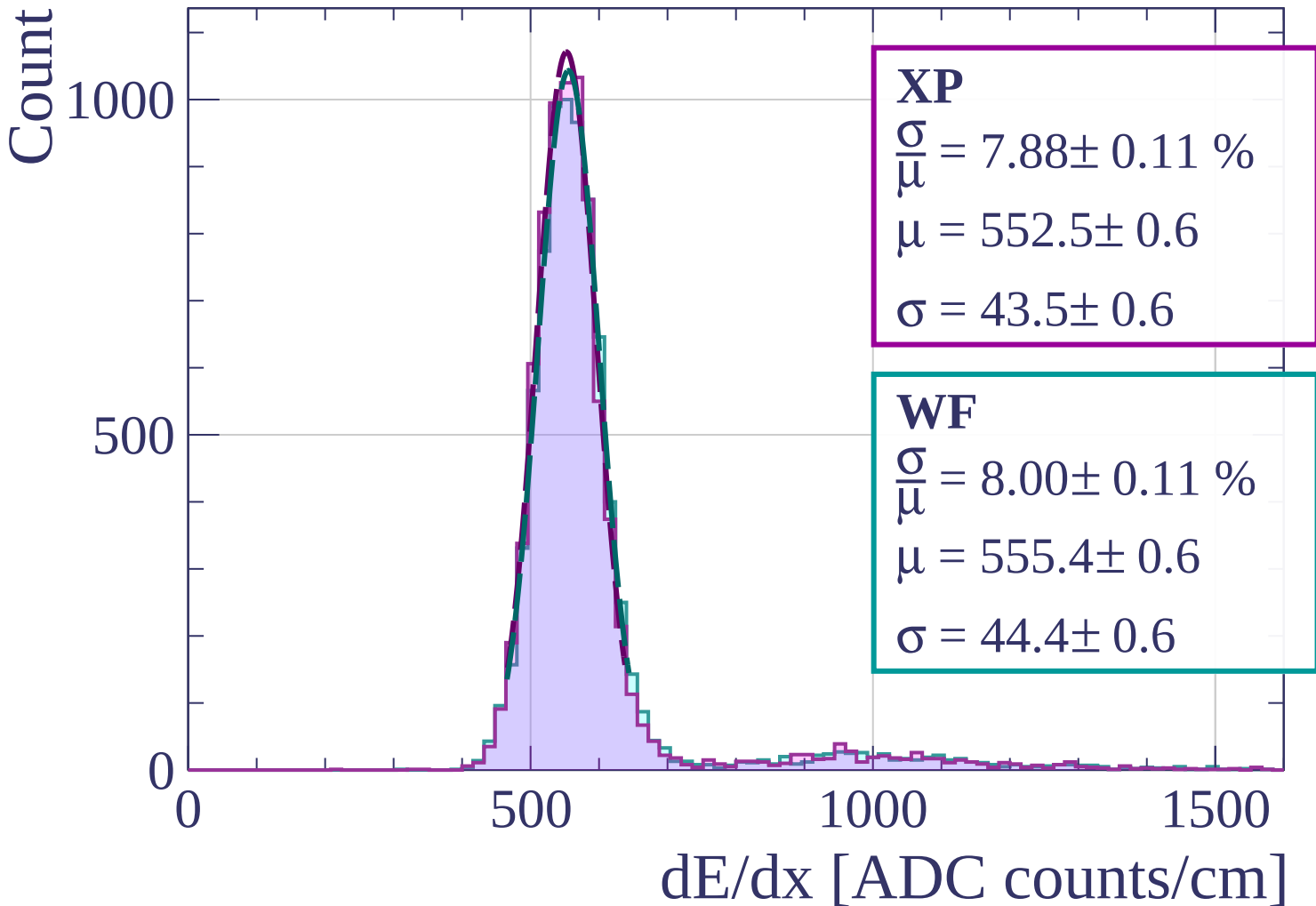






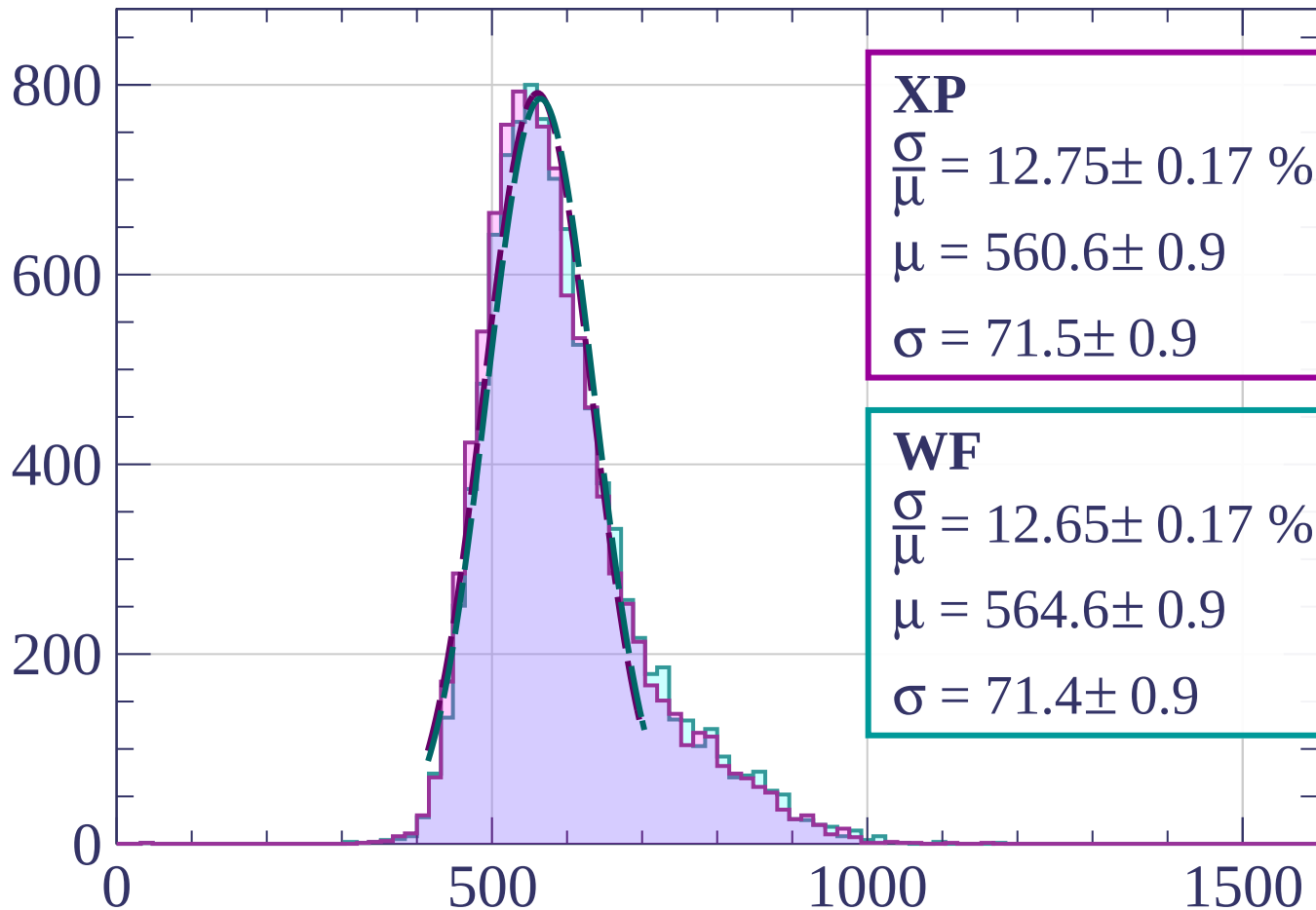


Energy loss | $0 < p < 80$



Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 12.75 \pm 0.17 \%$$

$$\mu = 560.6 \pm 0.9$$

$$\sigma = 71.5 \pm 0.9$$

WF

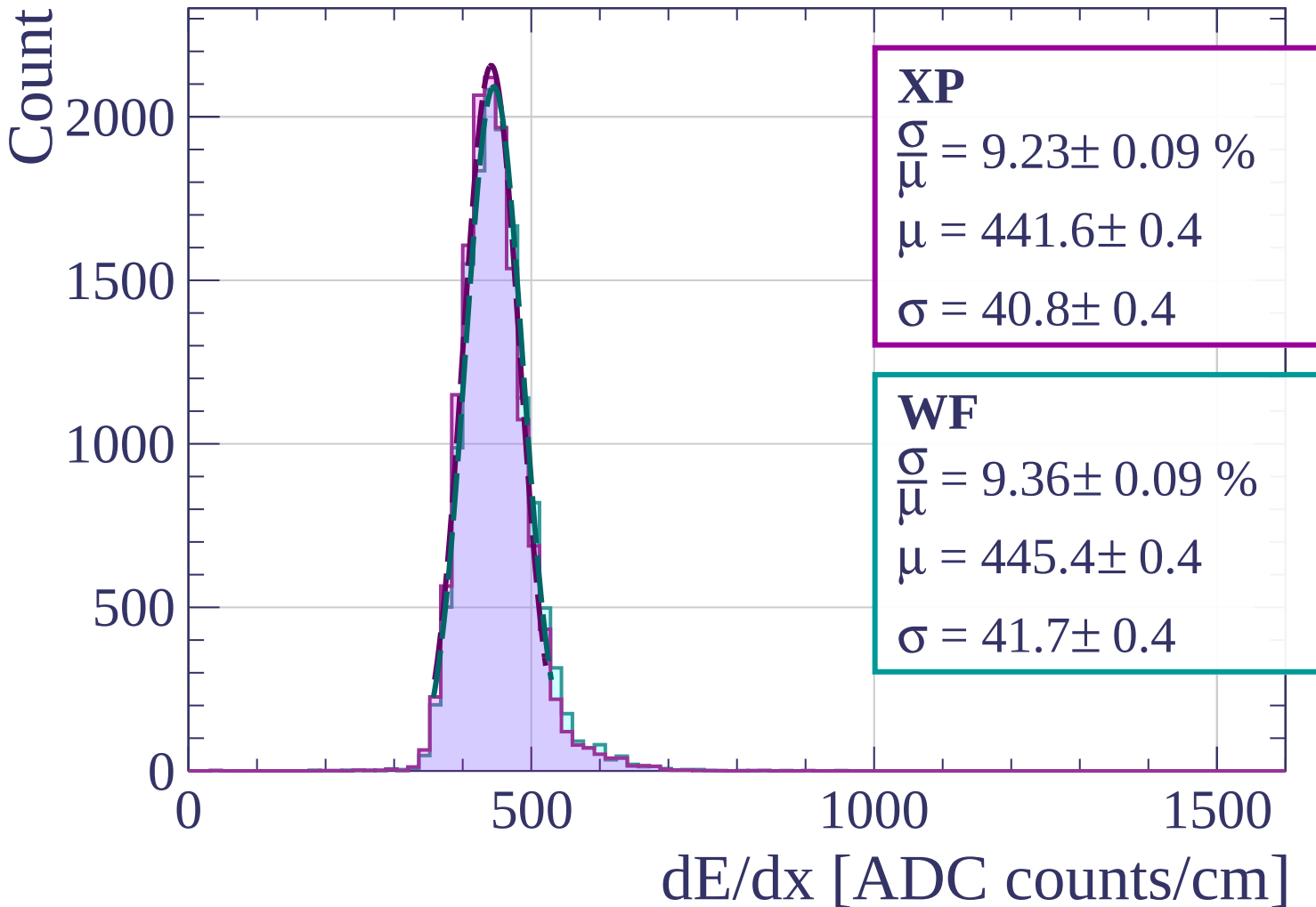
$$\frac{\sigma}{\mu} = 12.65 \pm 0.17 \%$$

$$\mu = 564.6 \pm 0.9$$

$$\sigma = 71.4 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

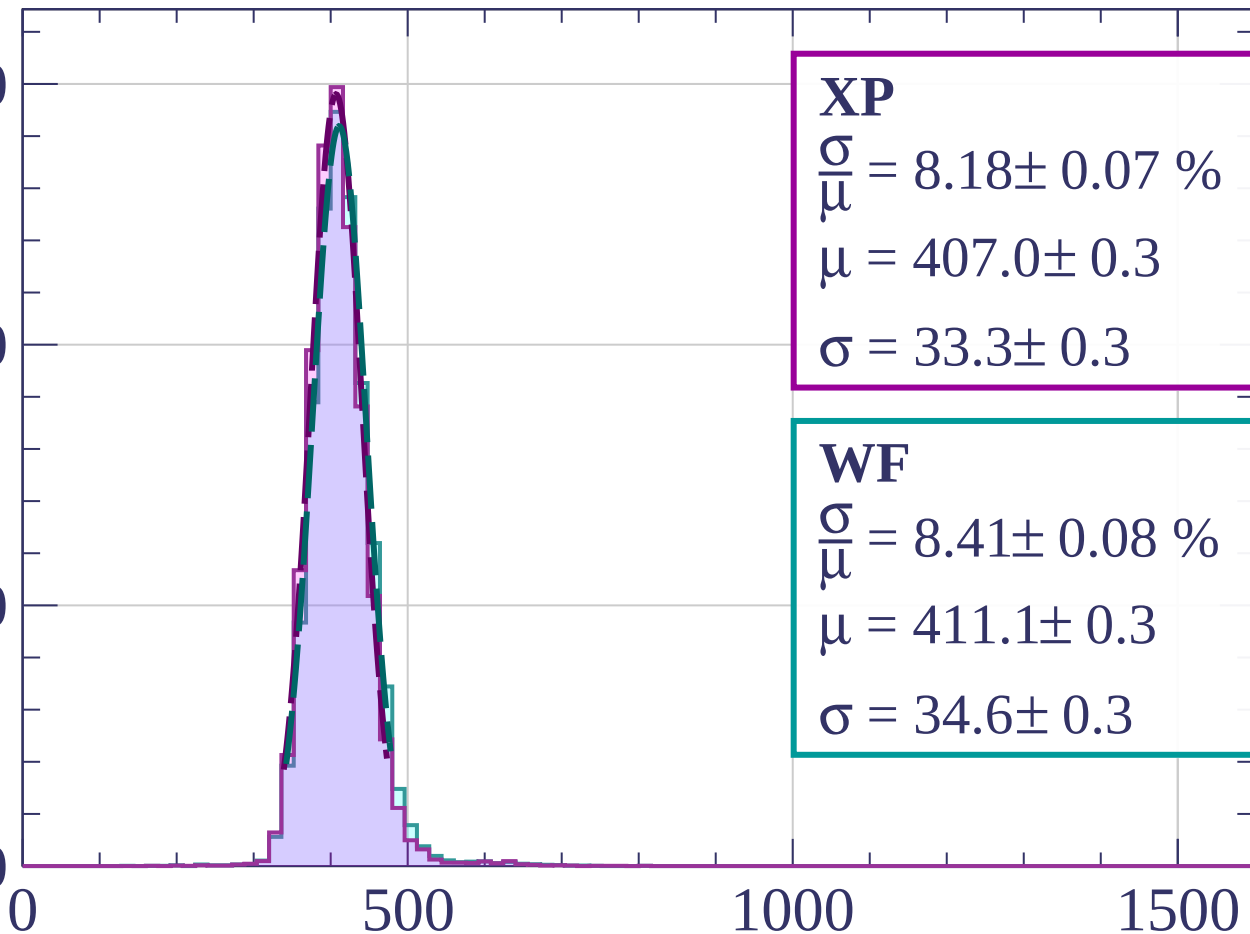
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.18 \pm 0.07 \%$$

$$\mu = 407.0 \pm 0.3$$

$$\sigma = 33.3 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.41 \pm 0.08 \%$$

$$\mu = 411.1 \pm 0.3$$

$$\sigma = 34.6 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $320 < p < 400$

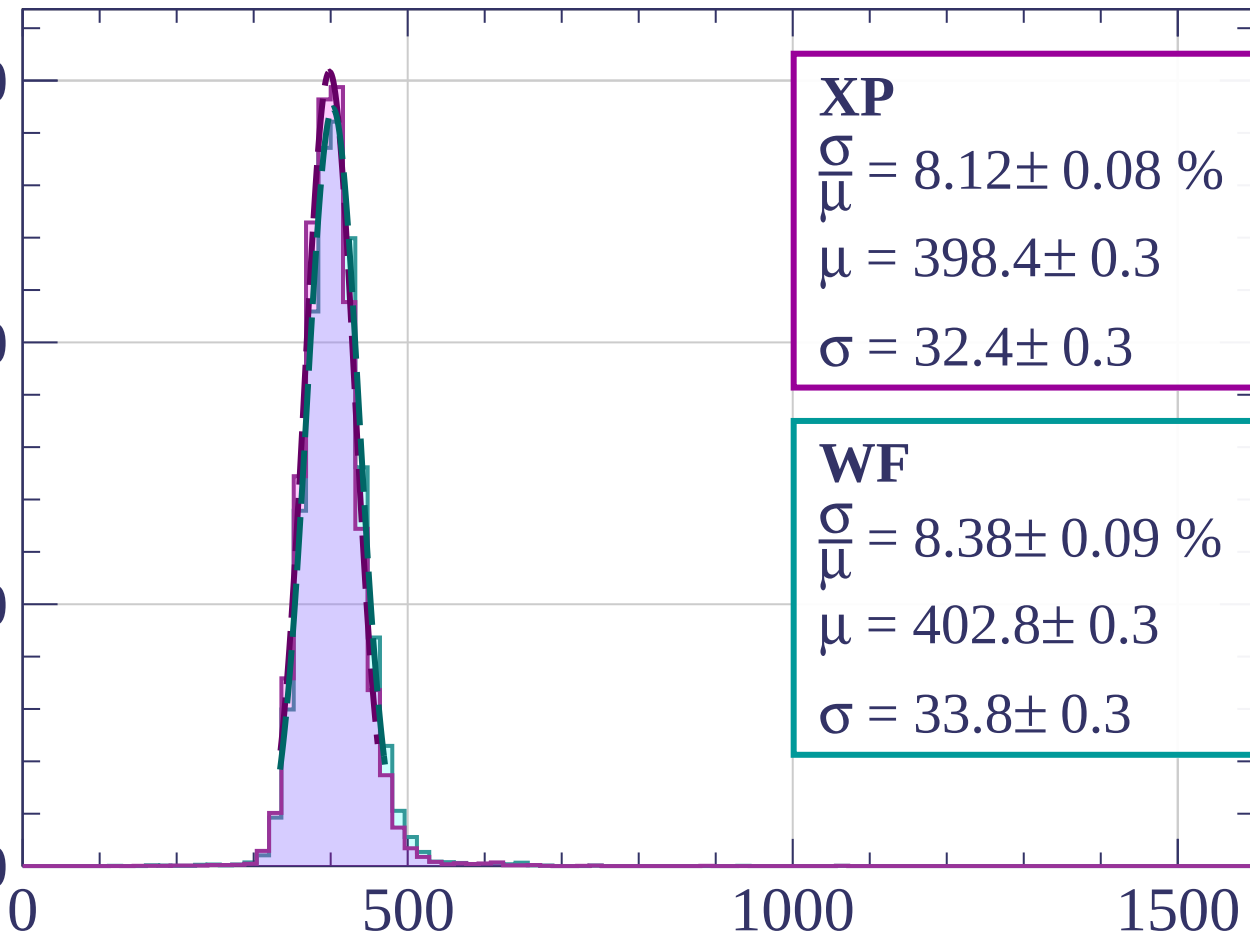
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.12 \pm 0.08 \%$$

$$\mu = 398.4 \pm 0.3$$

$$\sigma = 32.4 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.38 \pm 0.09 \%$$

$$\mu = 402.8 \pm 0.3$$

$$\sigma = 33.8 \pm 0.3$$

dE/dx [ADC counts/cm]

Energy loss | $400 < p < 480$

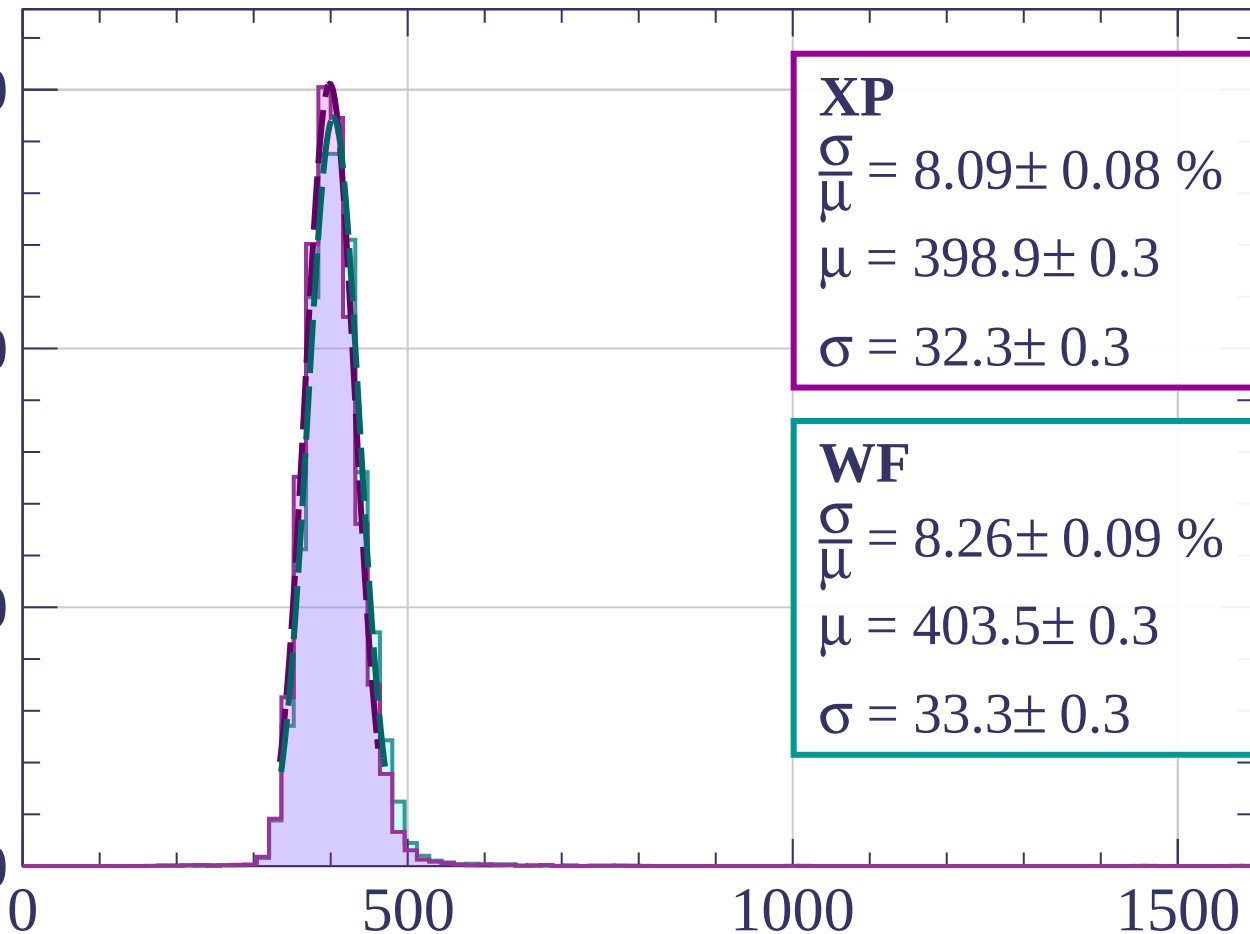
Count

3000

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.09 \pm 0.08 \%$$

$$\mu = 398.9 \pm 0.3$$

$$\sigma = 32.3 \pm 0.3$$

WF

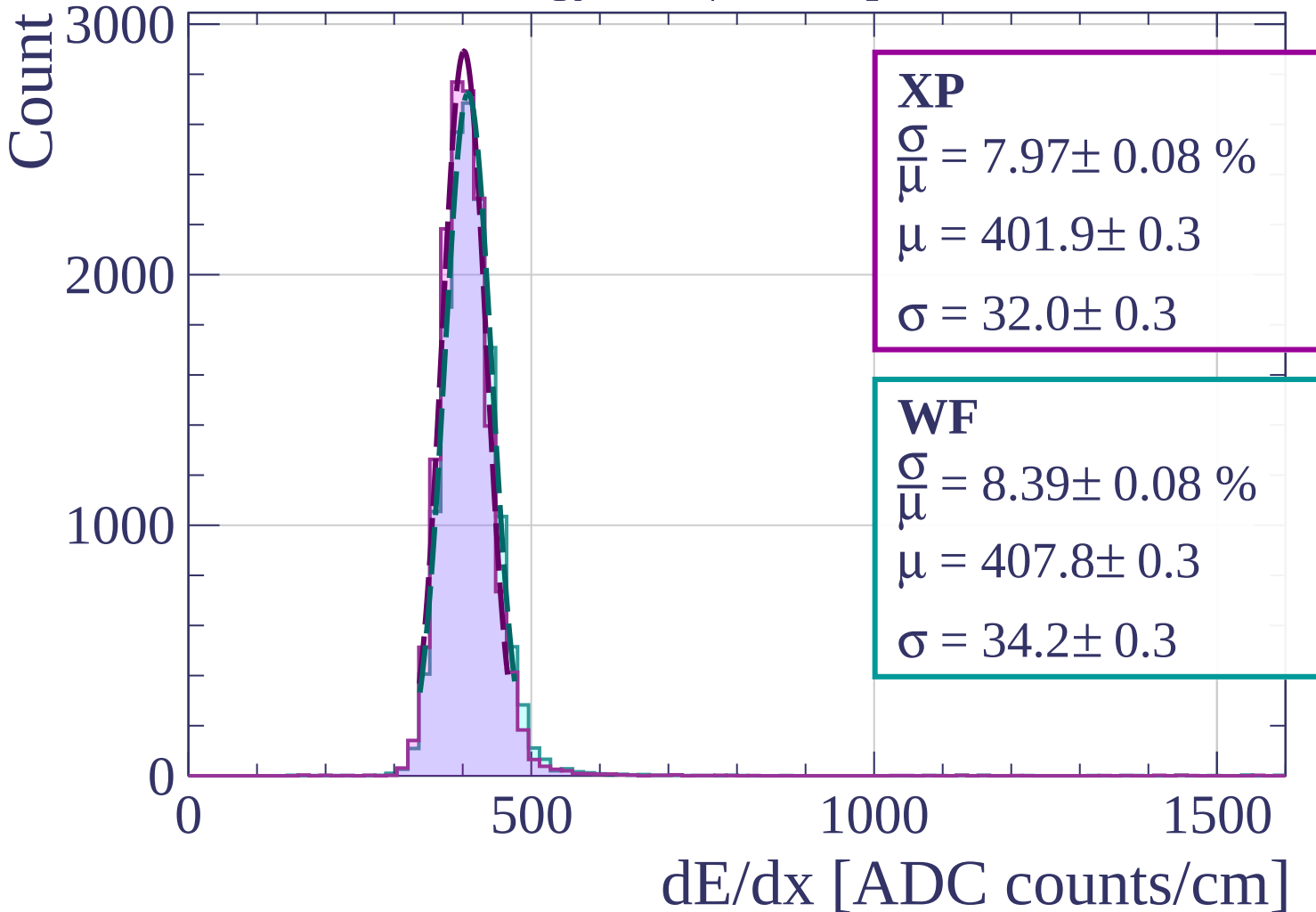
$$\frac{\sigma}{\mu} = 8.26 \pm 0.09 \%$$

$$\mu = 403.5 \pm 0.3$$

$$\sigma = 33.3 \pm 0.3$$

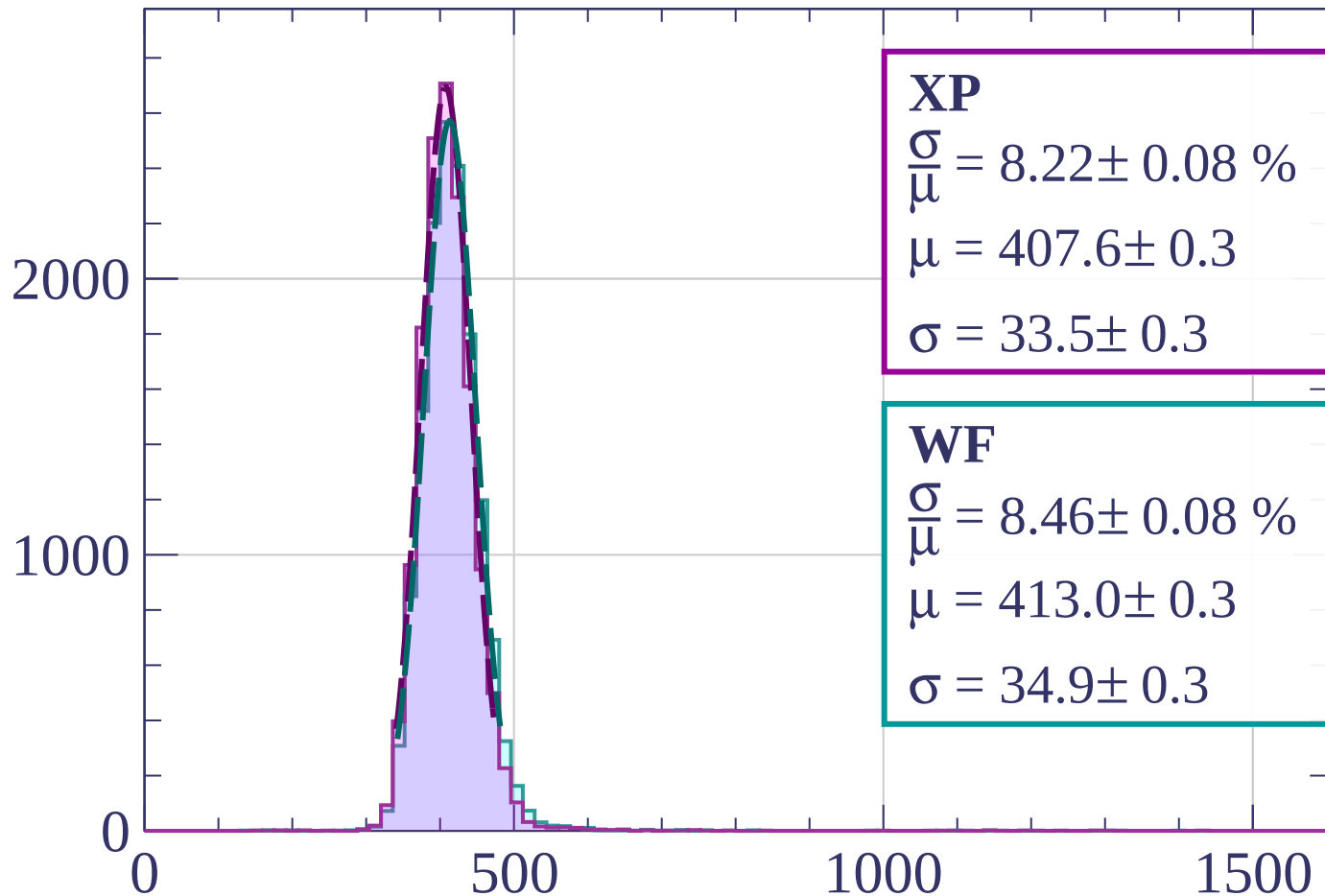
dE/dx [ADC counts/cm]

Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.08 \%$$

$$\mu = 407.6 \pm 0.3$$

$$\sigma = 33.5 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 8.46 \pm 0.08 \%$$

$$\mu = 413.0 \pm 0.3$$

$$\sigma = 34.9 \pm 0.3$$

dE/dx [ADC counts/cm]

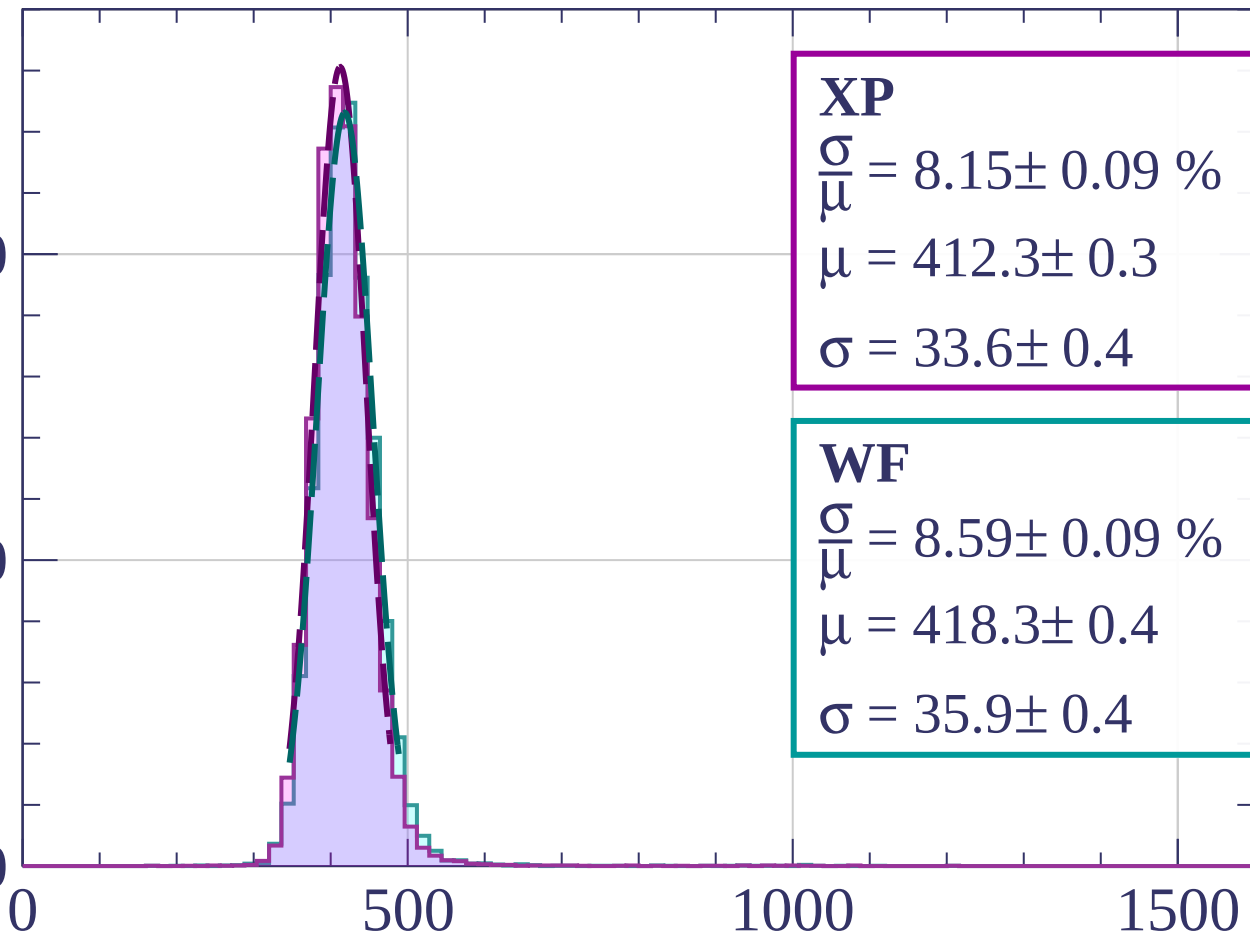
Energy loss | $640 < p < 720$

Count

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.15 \pm 0.09 \%$$

$$\mu = 412.3 \pm 0.3$$

$$\sigma = 33.6 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.59 \pm 0.09 \%$$

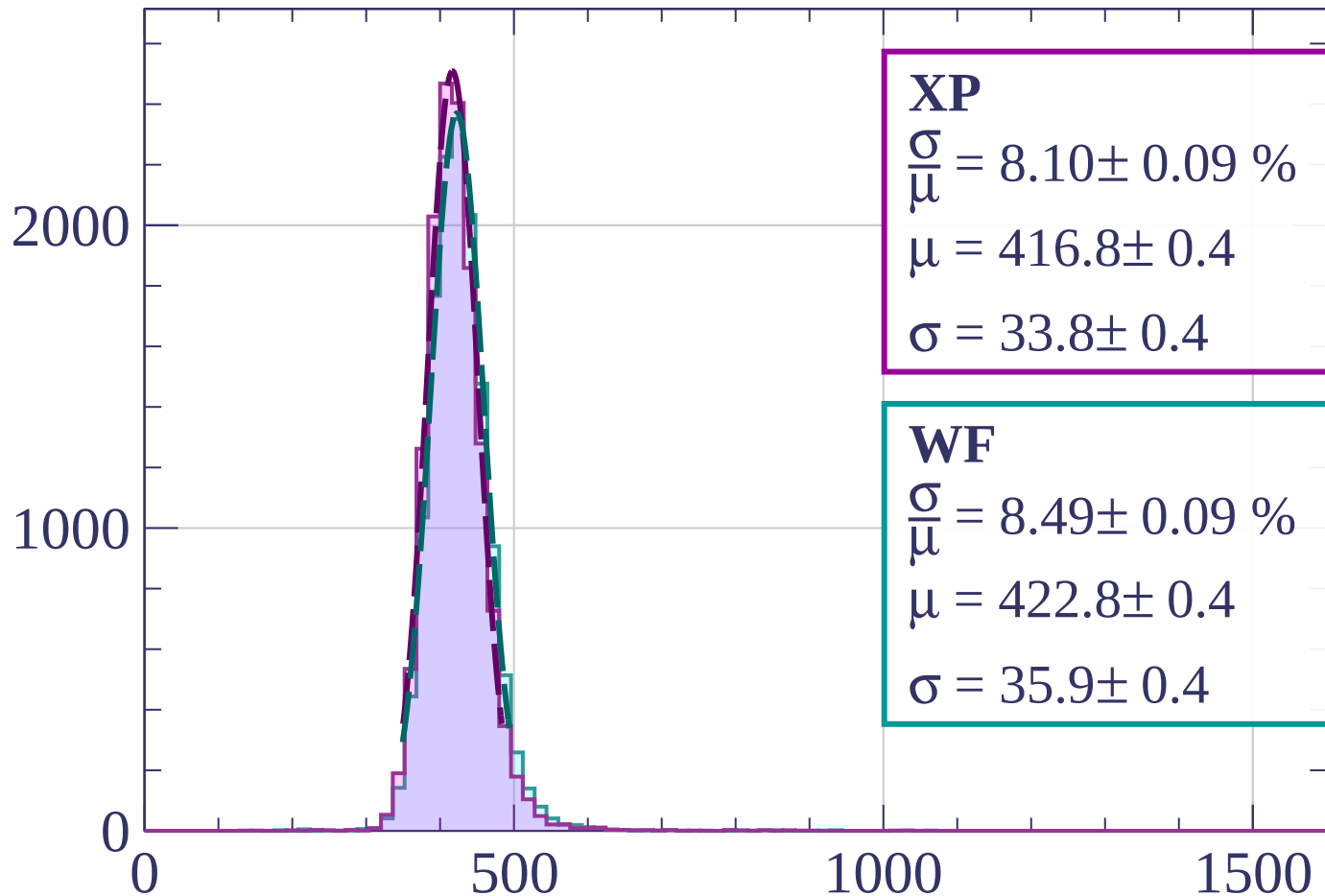
$$\mu = 418.3 \pm 0.4$$

$$\sigma = 35.9 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.10 \pm 0.09 \%$$

$$\mu = 416.8 \pm 0.4$$

$$\sigma = 33.8 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.49 \pm 0.09 \%$$

$$\mu = 422.8 \pm 0.4$$

$$\sigma = 35.9 \pm 0.4$$

dE/dx [ADC counts/cm]

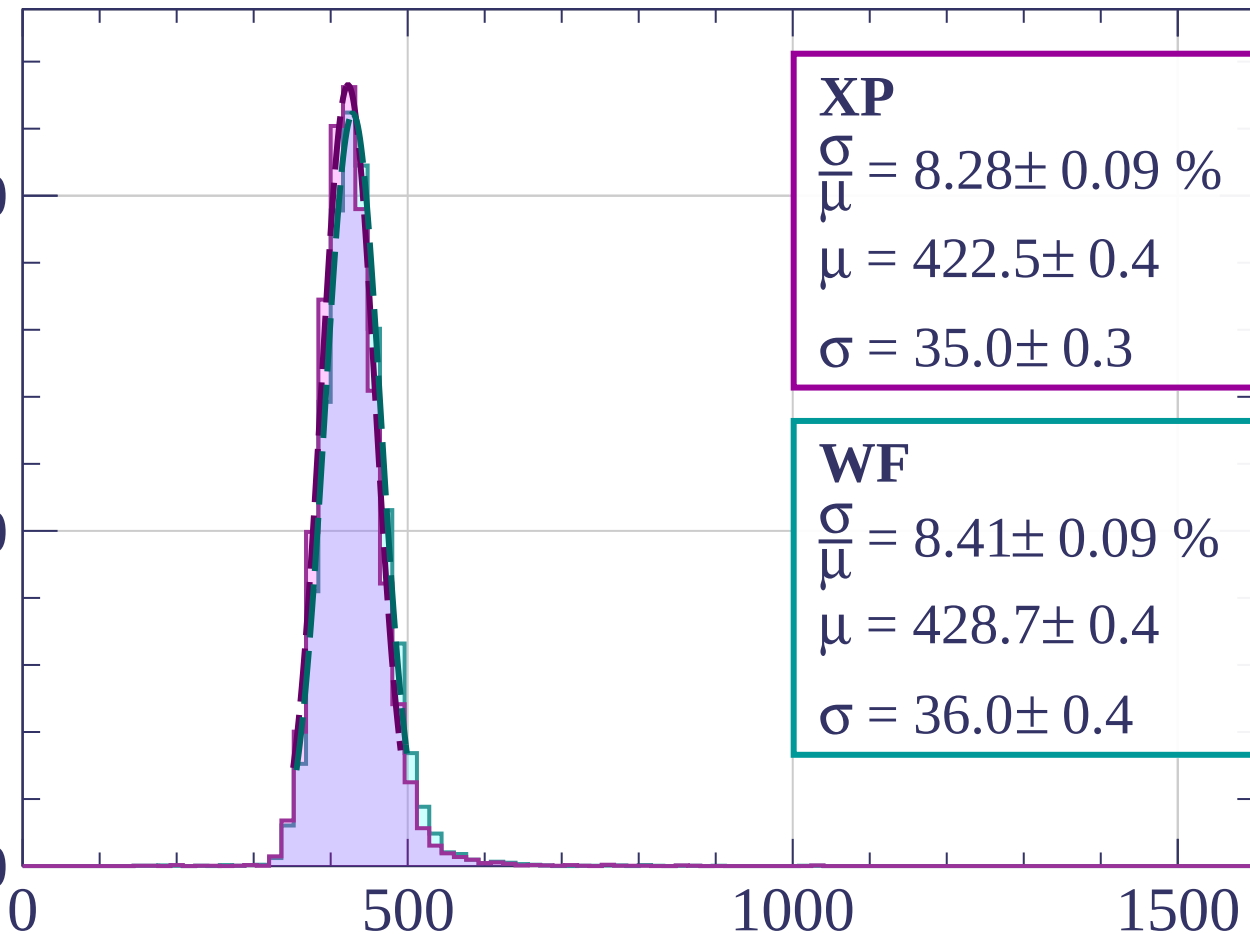
Energy loss | $800 < p < 880$

Count

2000

1000

0



XP

$$\frac{\sigma}{\mu} = 8.28 \pm 0.09 \%$$

$$\mu = 422.5 \pm 0.4$$

$$\sigma = 35.0 \pm 0.3$$

WF

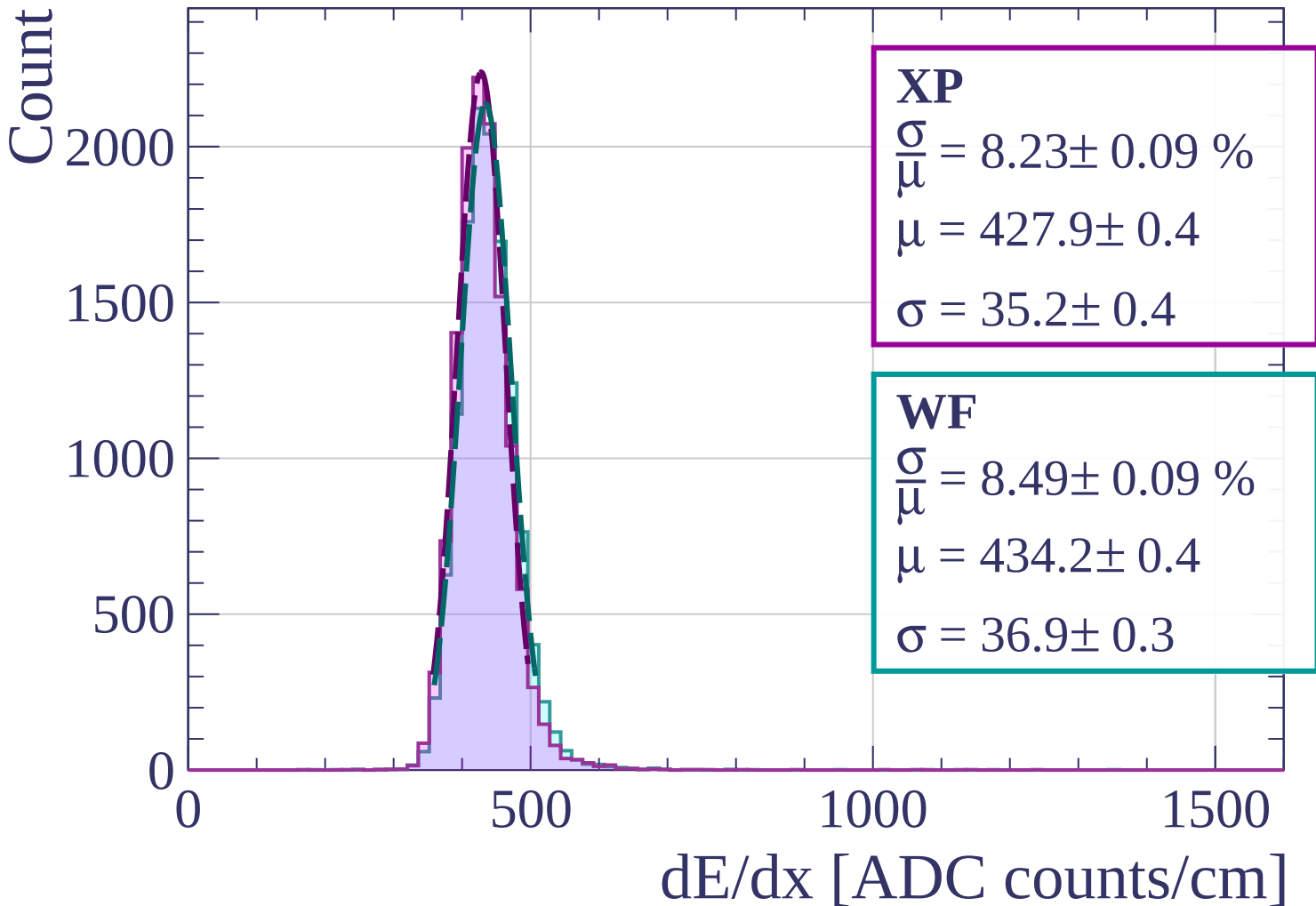
$$\frac{\sigma}{\mu} = 8.41 \pm 0.09 \%$$

$$\mu = 428.7 \pm 0.4$$

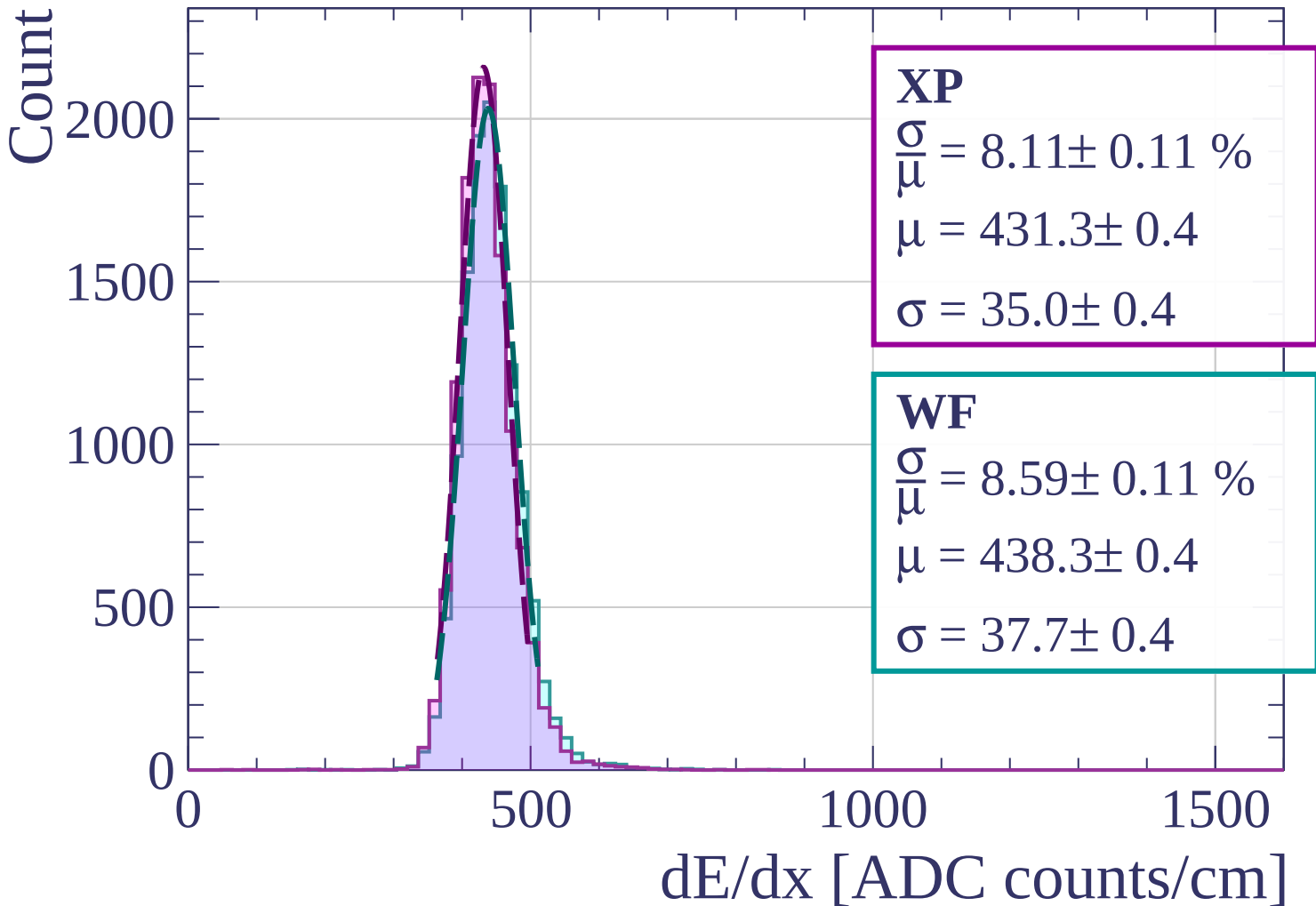
$$\sigma = 36.0 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $880 < p < 960$



Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count

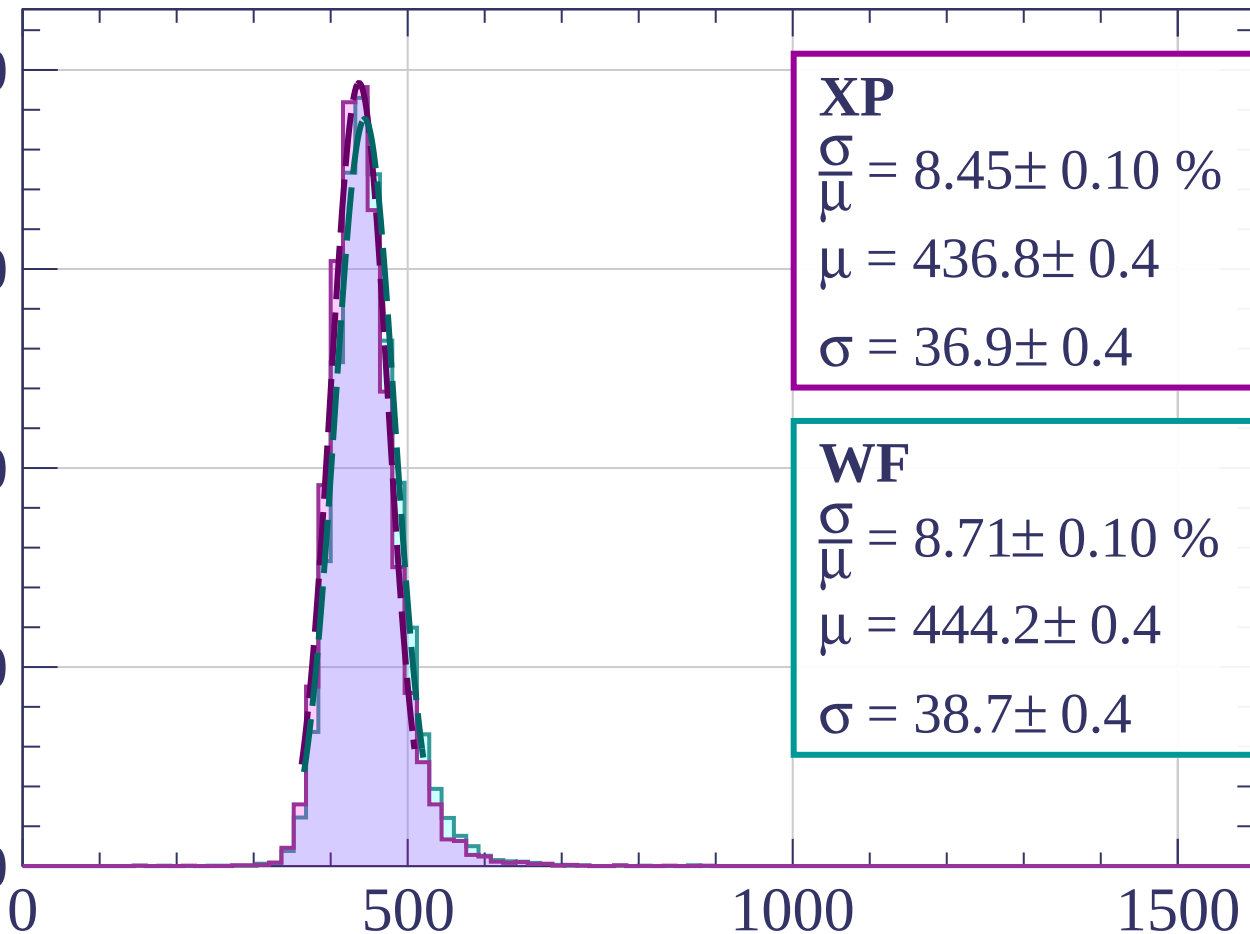
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.45 \pm 0.10 \%$$

$$\mu = 436.8 \pm 0.4$$

$$\sigma = 36.9 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.71 \pm 0.10 \%$$

$$\mu = 444.2 \pm 0.4$$

$$\sigma = 38.7 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1120 < p < 1200$

Count

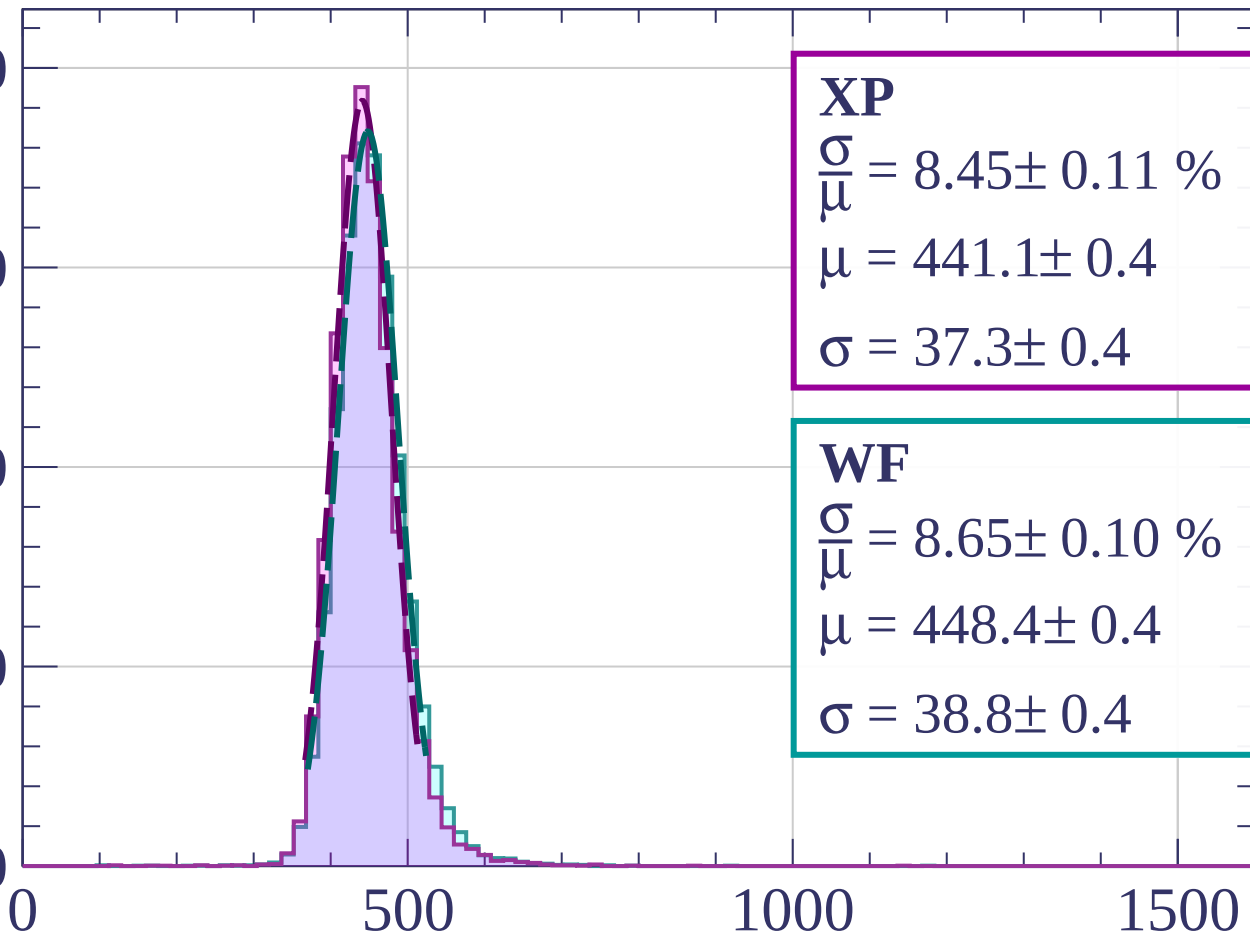
2000

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.45 \pm 0.11 \%$$

$$\mu = 441.1 \pm 0.4$$

$$\sigma = 37.3 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.65 \pm 0.10 \%$$

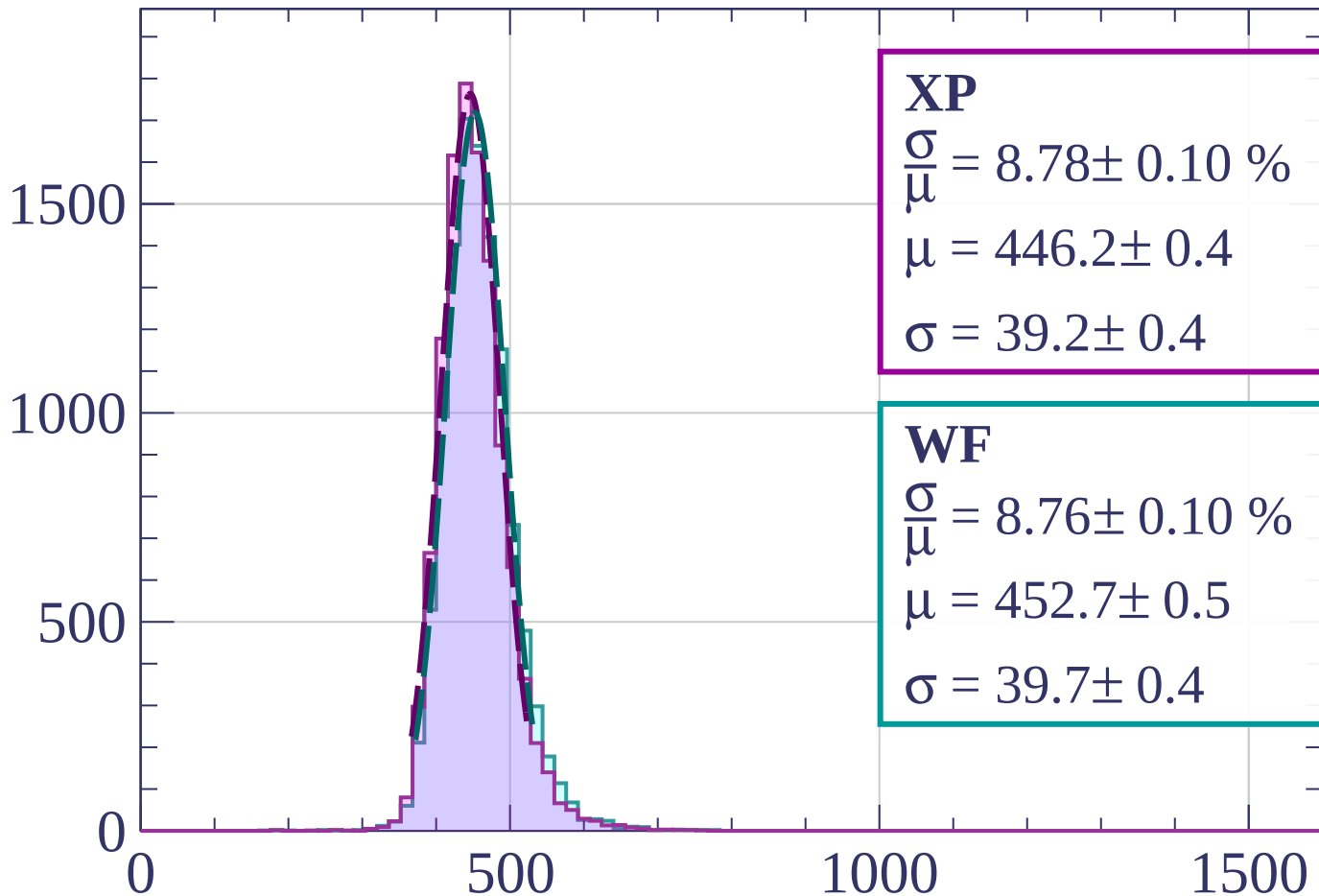
$$\mu = 448.4 \pm 0.4$$

$$\sigma = 38.8 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.10 \%$$

$$\mu = 446.2 \pm 0.4$$

$$\sigma = 39.2 \pm 0.4$$

WF

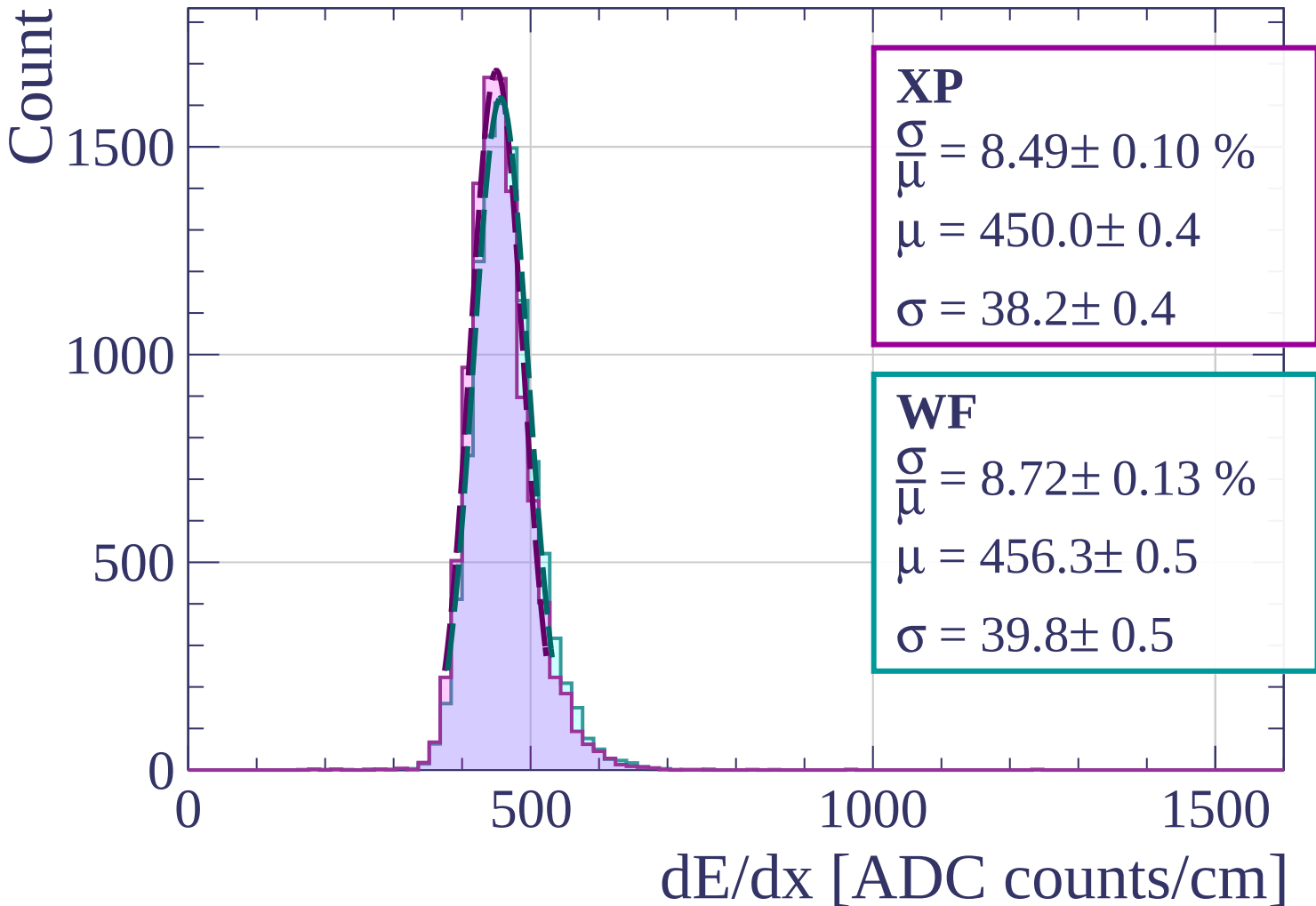
$$\frac{\sigma}{\mu} = 8.76 \pm 0.10 \%$$

$$\mu = 452.7 \pm 0.5$$

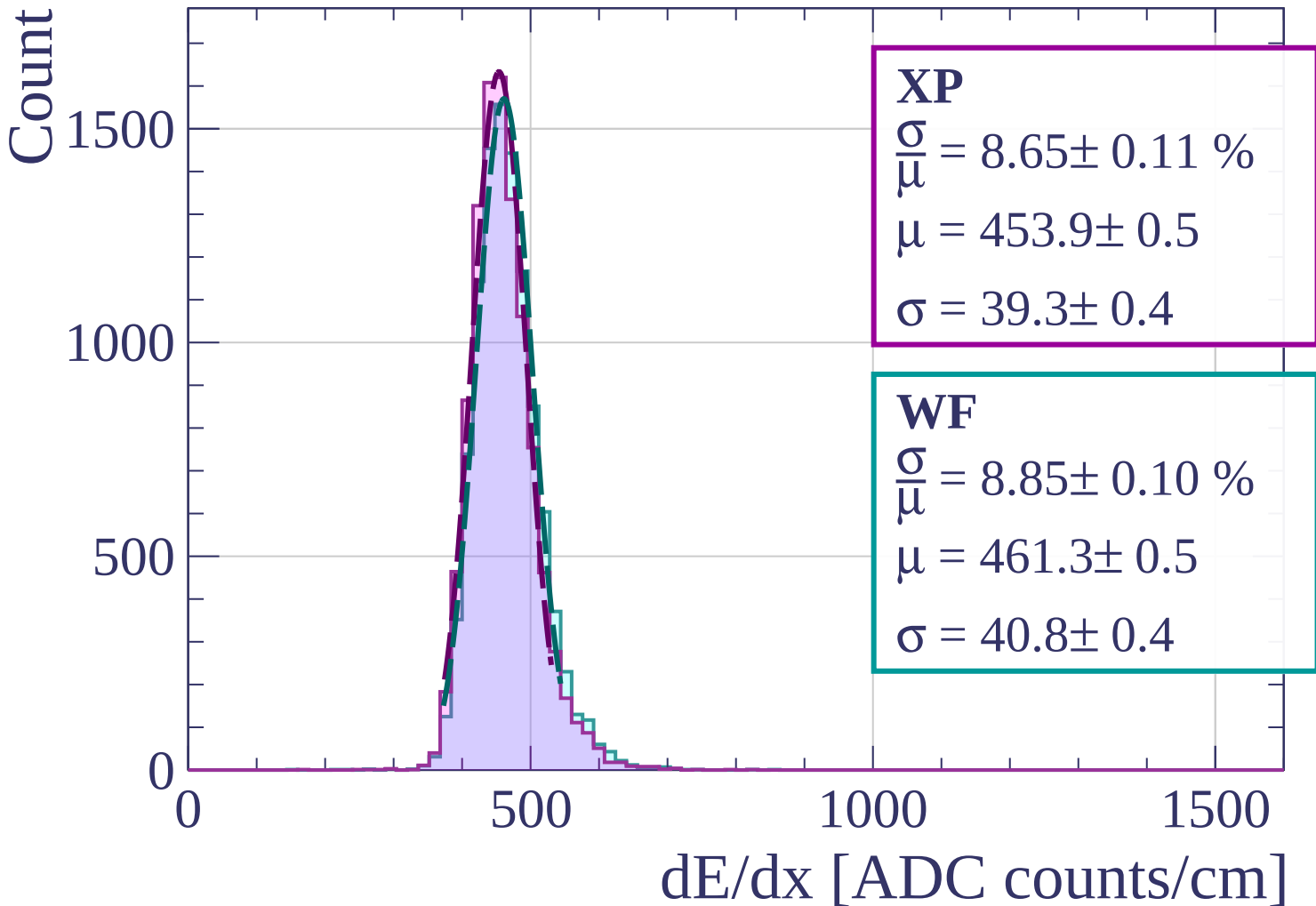
$$\sigma = 39.7 \pm 0.4$$

dE/dx [ADC counts/cm]

Energy loss | $1280 < p < 1360$

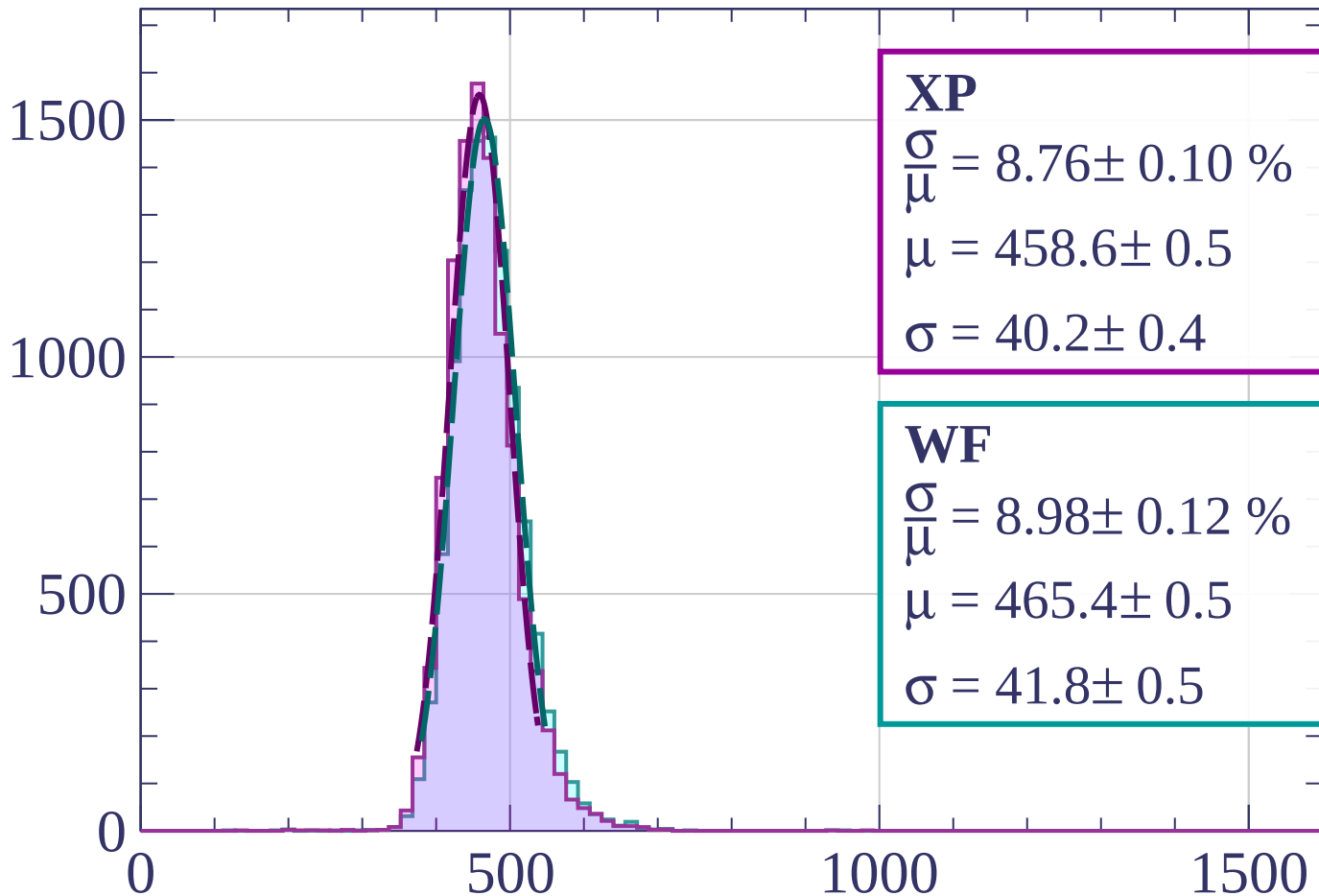


Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.10 \%$$

$$\mu = 458.6 \pm 0.5$$

$$\sigma = 40.2 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.12 \%$$

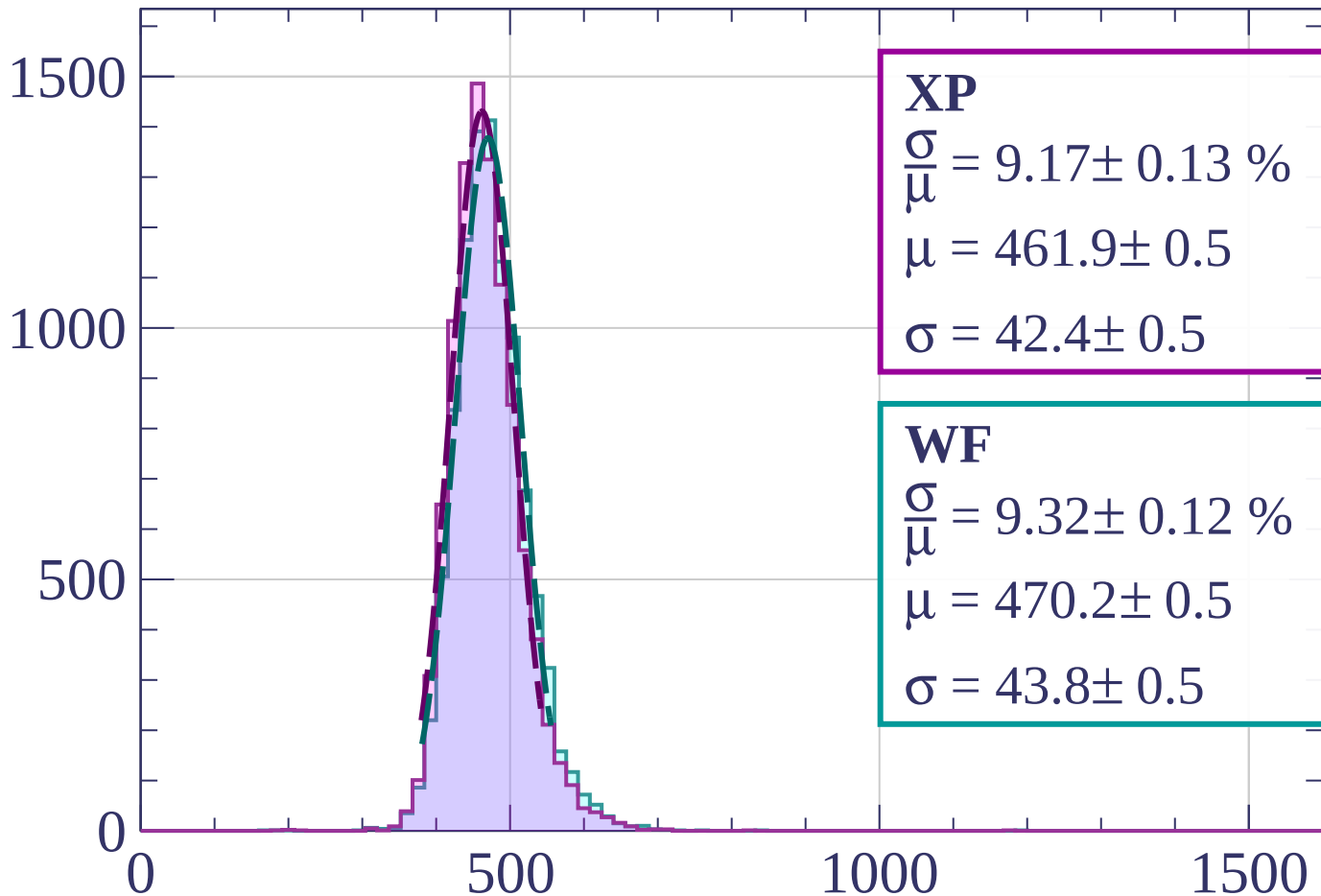
$$\mu = 465.4 \pm 0.5$$

$$\sigma = 41.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 9.17 \pm 0.13 \%$$

$$\mu = 461.9 \pm 0.5$$

$$\sigma = 42.4 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.12 \%$$

$$\mu = 470.2 \pm 0.5$$

$$\sigma = 43.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1600 < p < 1680$

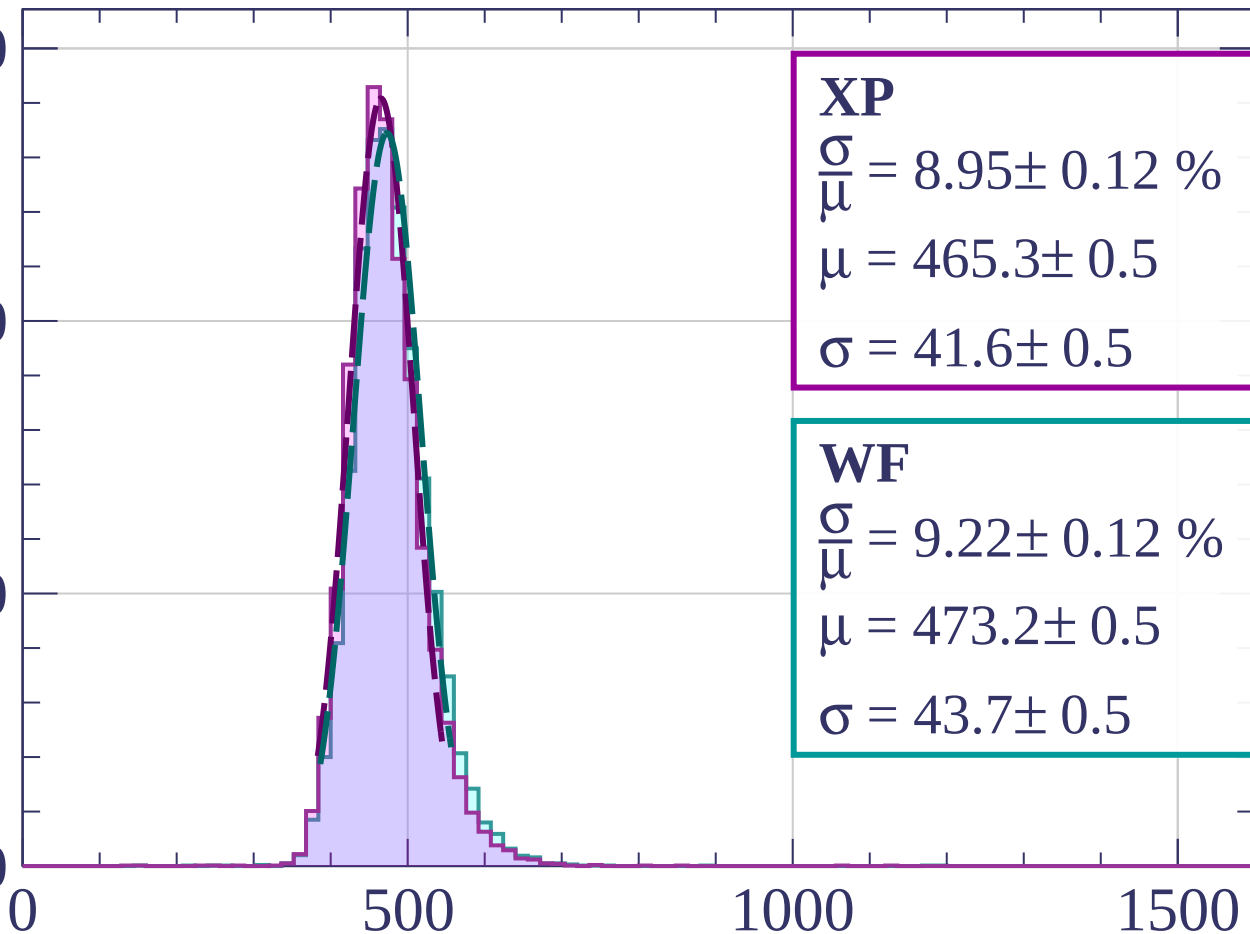
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 8.95 \pm 0.12 \%$$

$$\mu = 465.3 \pm 0.5$$

$$\sigma = 41.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.22 \pm 0.12 \%$$

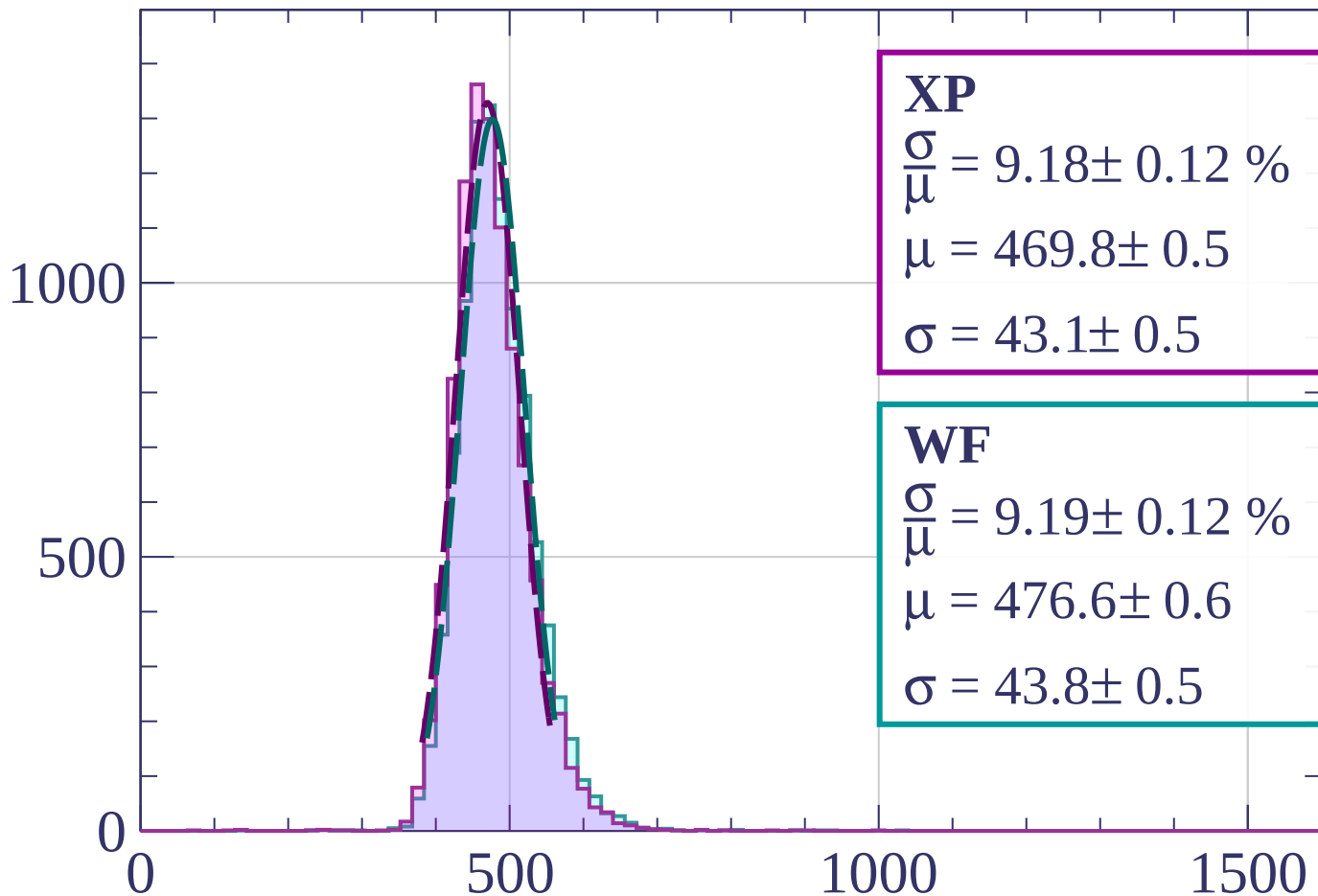
$$\mu = 473.2 \pm 0.5$$

$$\sigma = 43.7 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.18 \pm 0.12 \%$$

$$\mu = 469.8 \pm 0.5$$

$$\sigma = 43.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.12 \%$$

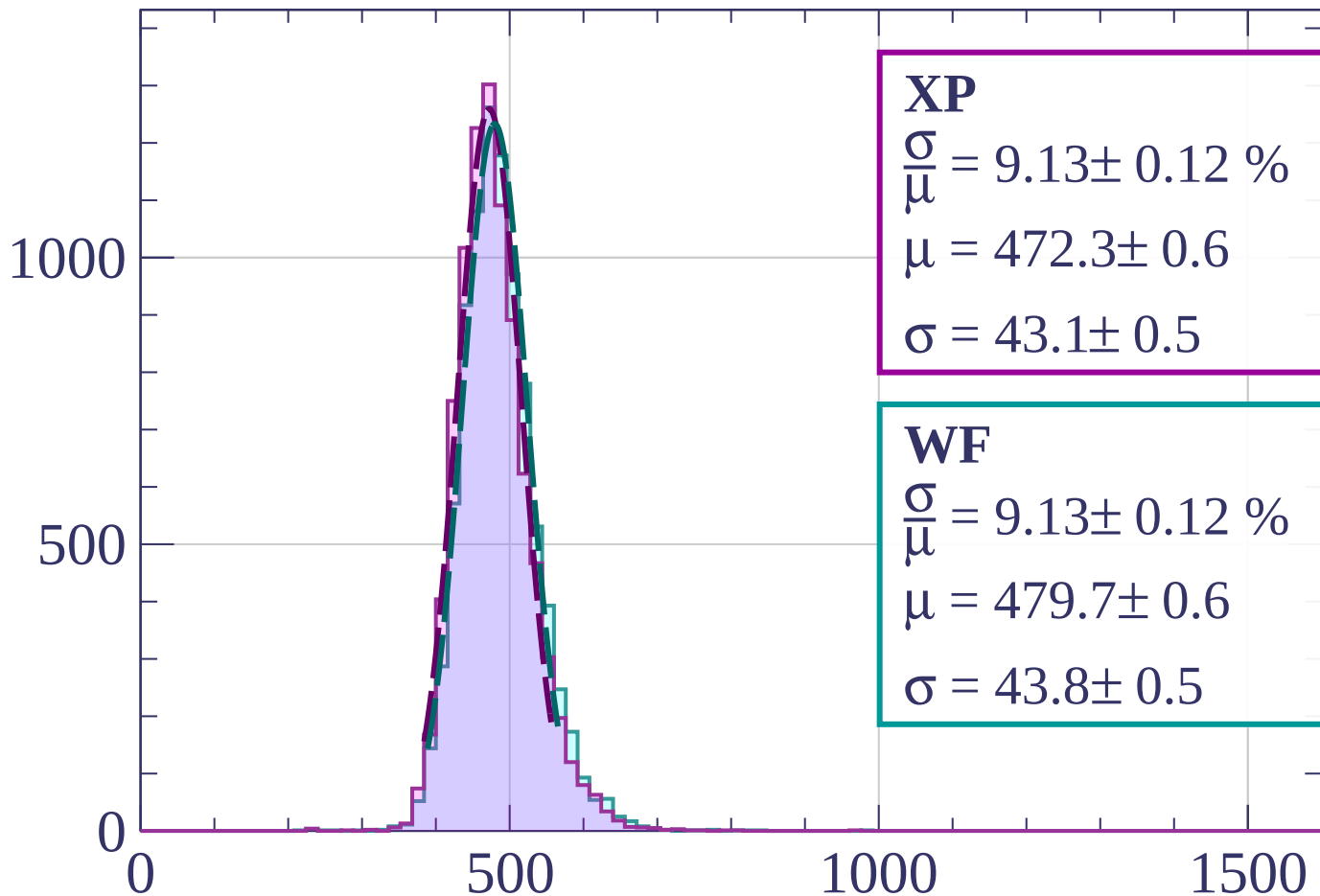
$$\mu = 476.6 \pm 0.6$$

$$\sigma = 43.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.12 \%$$

$$\mu = 472.3 \pm 0.6$$

$$\sigma = 43.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.13 \pm 0.12 \%$$

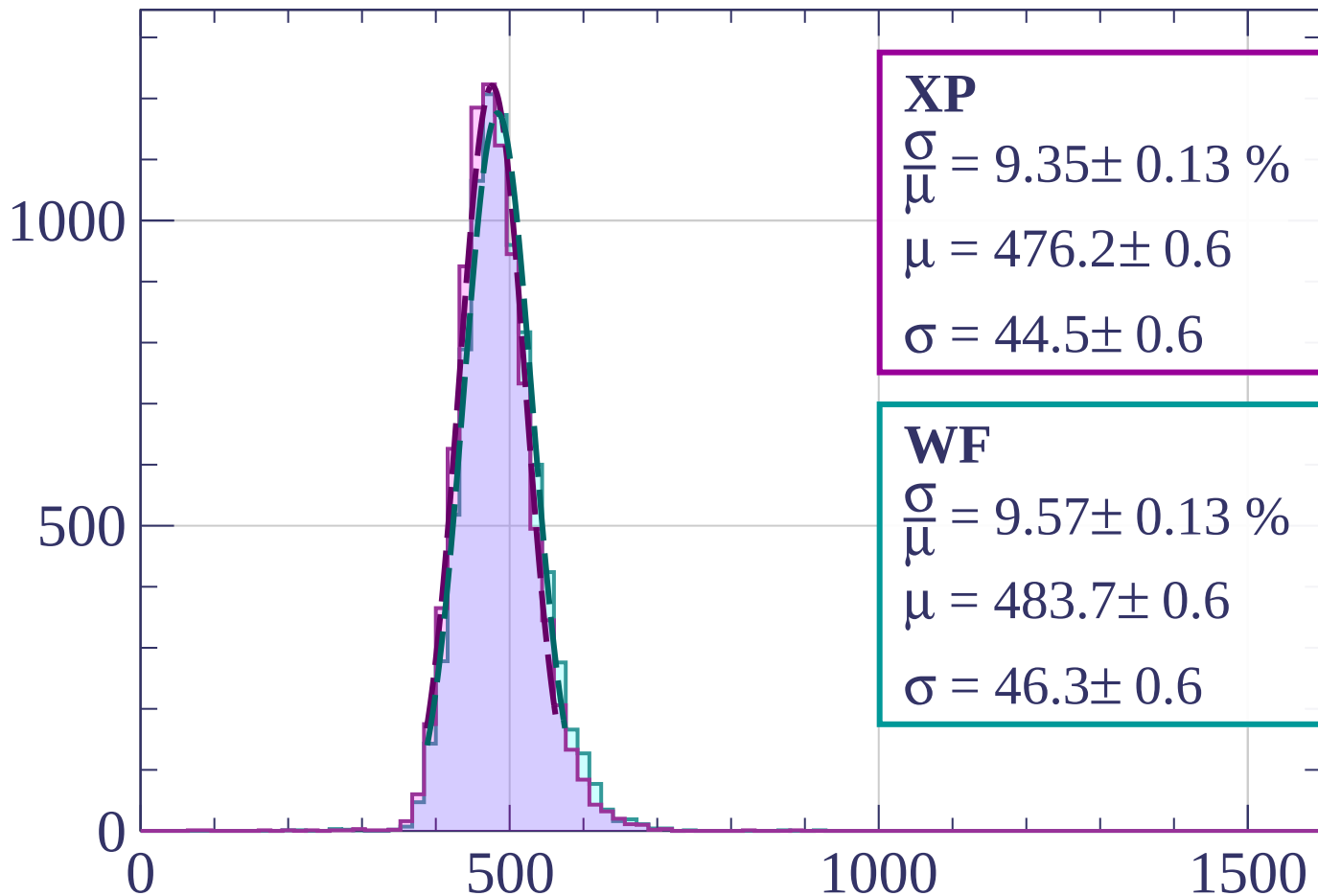
$$\mu = 479.7 \pm 0.6$$

$$\sigma = 43.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.35 \pm 0.13 \%$$

$$\mu = 476.2 \pm 0.6$$

$$\sigma = 44.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.57 \pm 0.13 \%$$

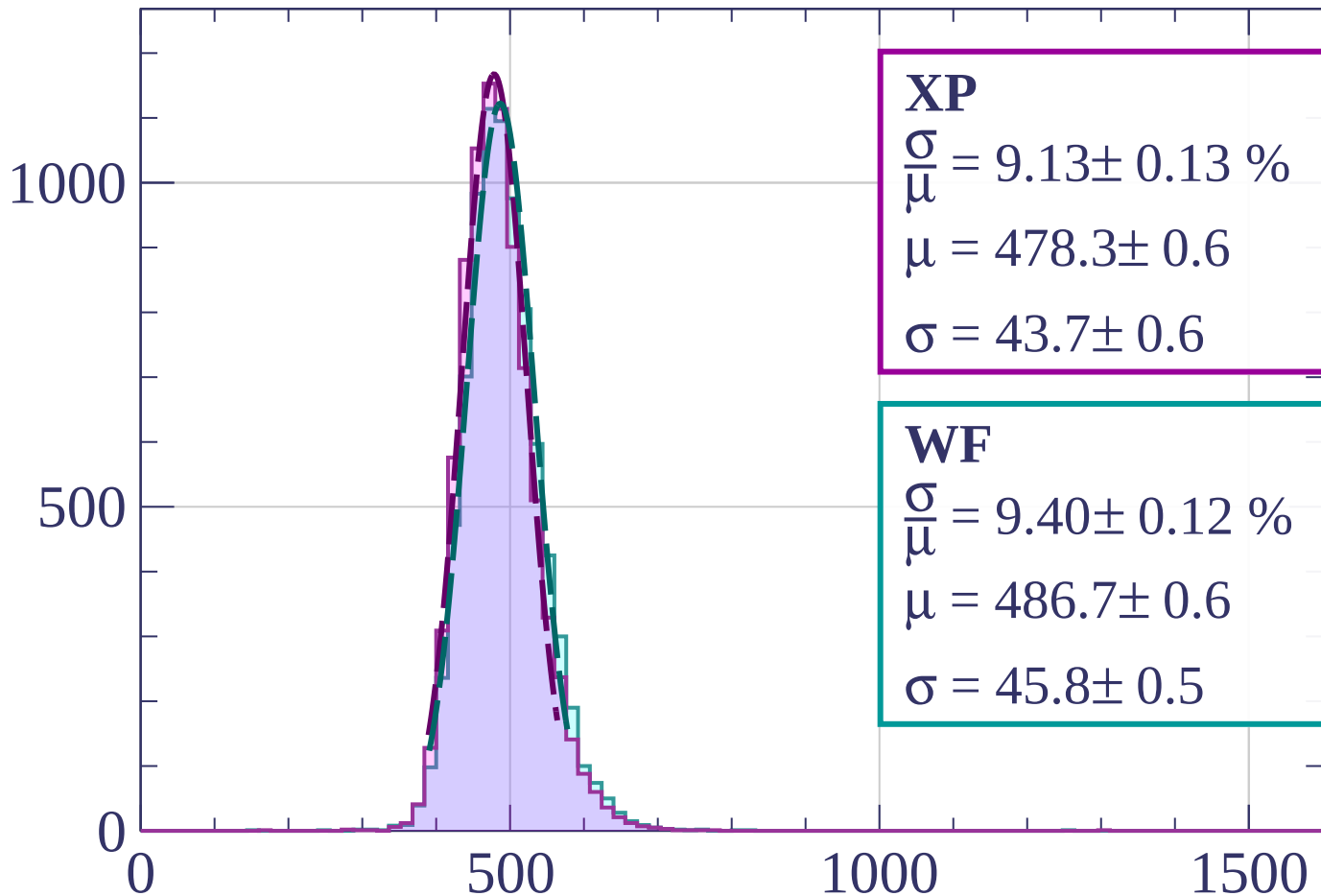
$$\mu = 483.7 \pm 0.6$$

$$\sigma = 46.3 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.13 \%$$

$$\mu = 478.3 \pm 0.6$$

$$\sigma = 43.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.12 \%$$

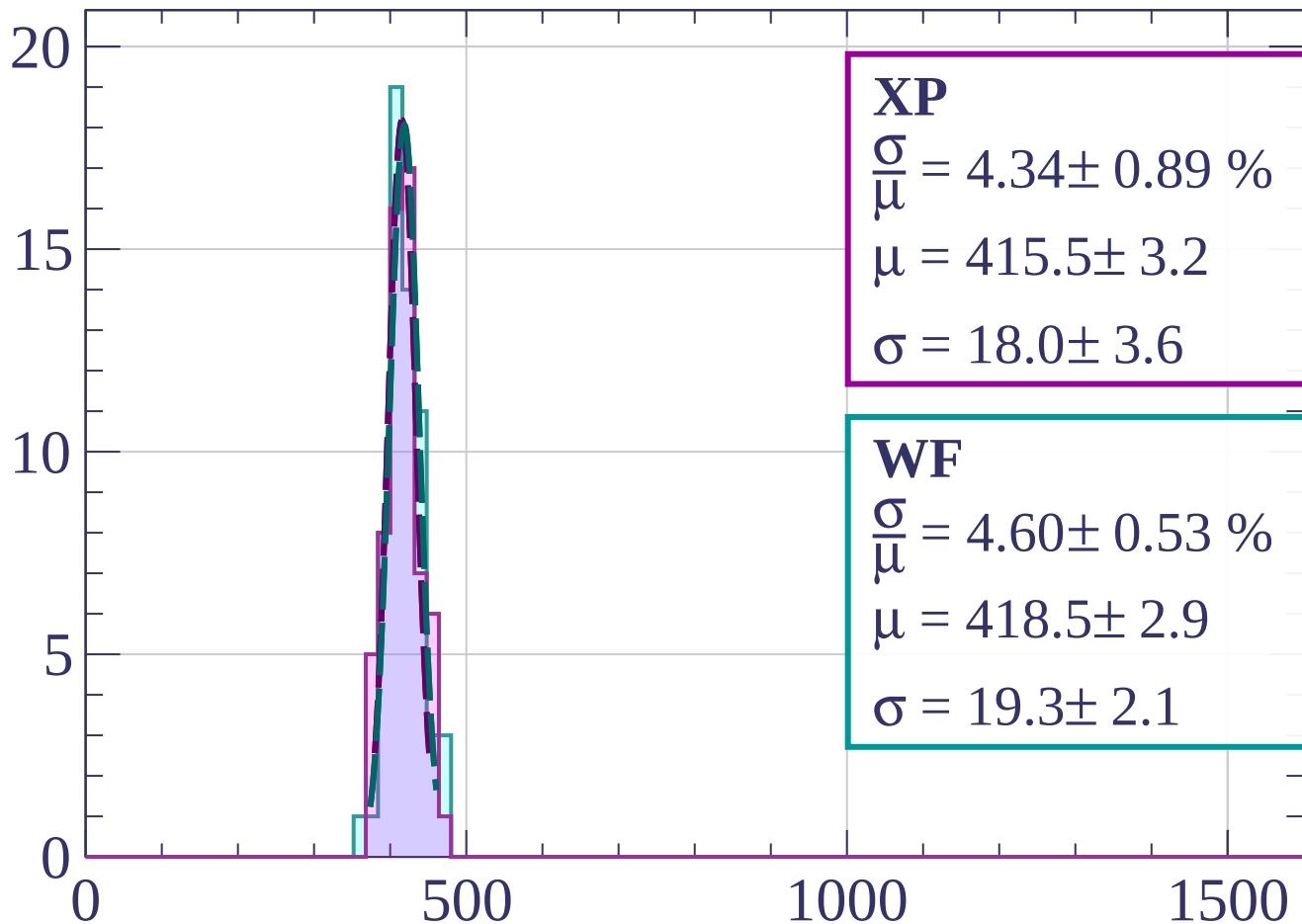
$$\mu = 486.7 \pm 0.6$$

$$\sigma = 45.8 \pm 0.5$$

dE/dx [ADC counts/cm]

Count

Energy loss | $0 < \phi < 3$



XP

$$\frac{\sigma}{\mu} = 4.34 \pm 0.89 \%$$

$$\mu = 415.5 \pm 3.2$$

$$\sigma = 18.0 \pm 3.6$$

WF

$$\frac{\sigma}{\mu} = 4.60 \pm 0.53 \%$$

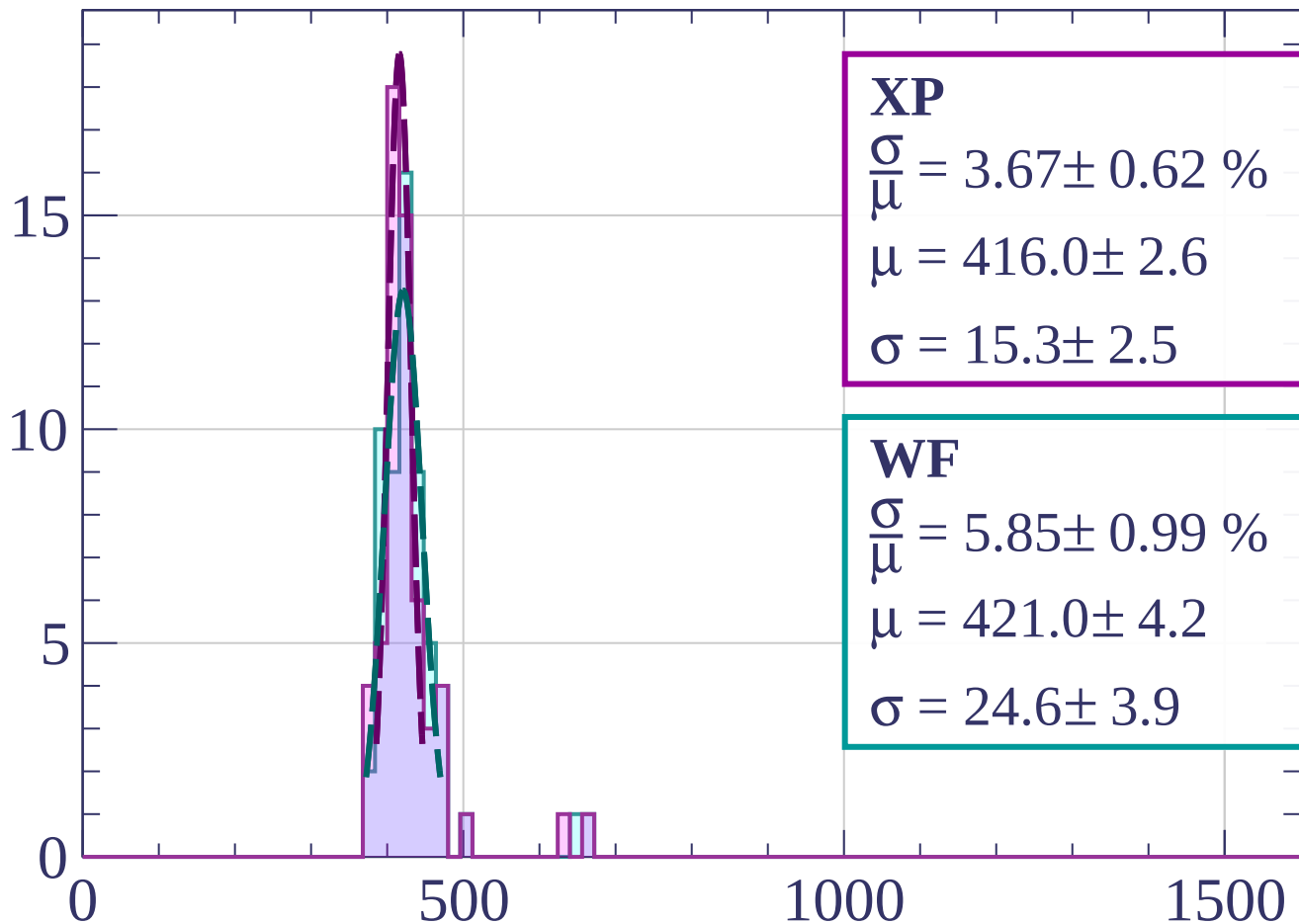
$$\mu = 418.5 \pm 2.9$$

$$\sigma = 19.3 \pm 2.1$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \phi < 6$

Count



XP

$$\frac{\sigma}{\mu} = 3.67 \pm 0.62 \%$$

$$\mu = 416.0 \pm 2.6$$

$$\sigma = 15.3 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 5.85 \pm 0.99 \%$$

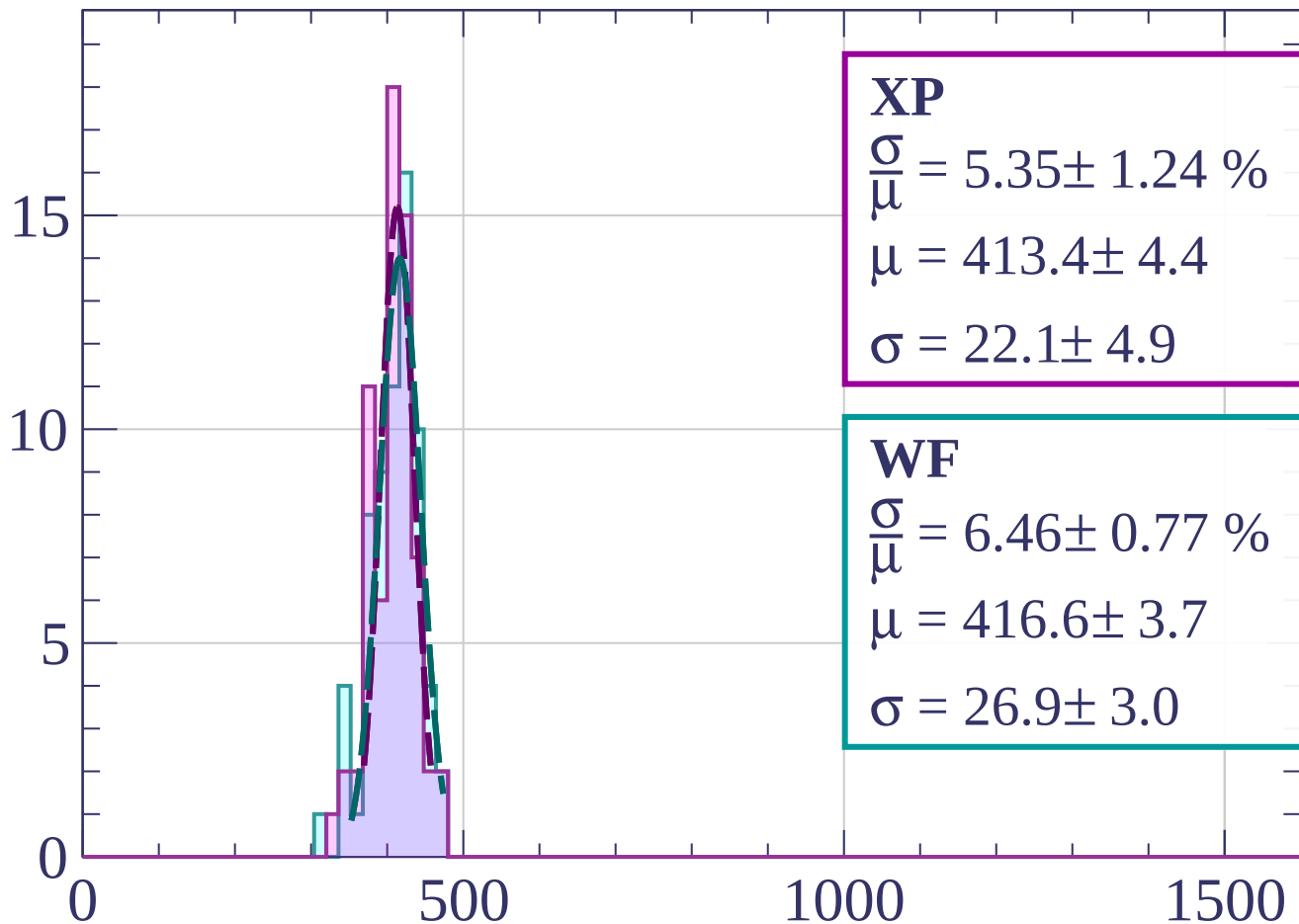
$$\mu = 421.0 \pm 4.2$$

$$\sigma = 24.6 \pm 3.9$$

dE/dx [ADC counts/cm]

Energy loss | $6 < \phi < 9$

Count



XP

$$\frac{\sigma}{\mu} = 5.35 \pm 1.24 \%$$

$$\mu = 413.4 \pm 4.4$$

$$\sigma = 22.1 \pm 4.9$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 0.77 \%$$

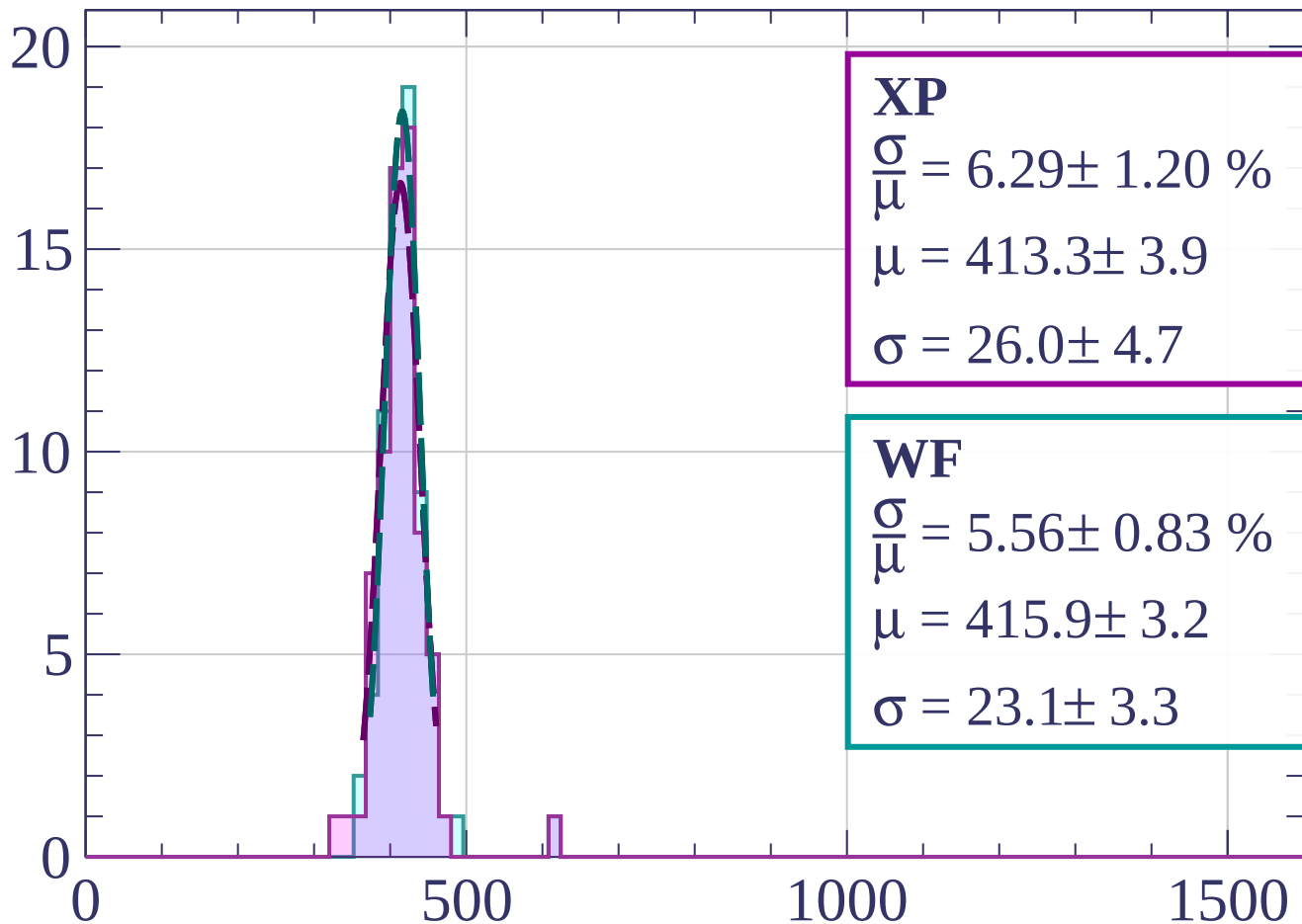
$$\mu = 416.6 \pm 3.7$$

$$\sigma = 26.9 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 6.29 \pm 1.20 \%$$

$$\mu = 413.3 \pm 3.9$$

$$\sigma = 26.0 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 5.56 \pm 0.83 \%$$

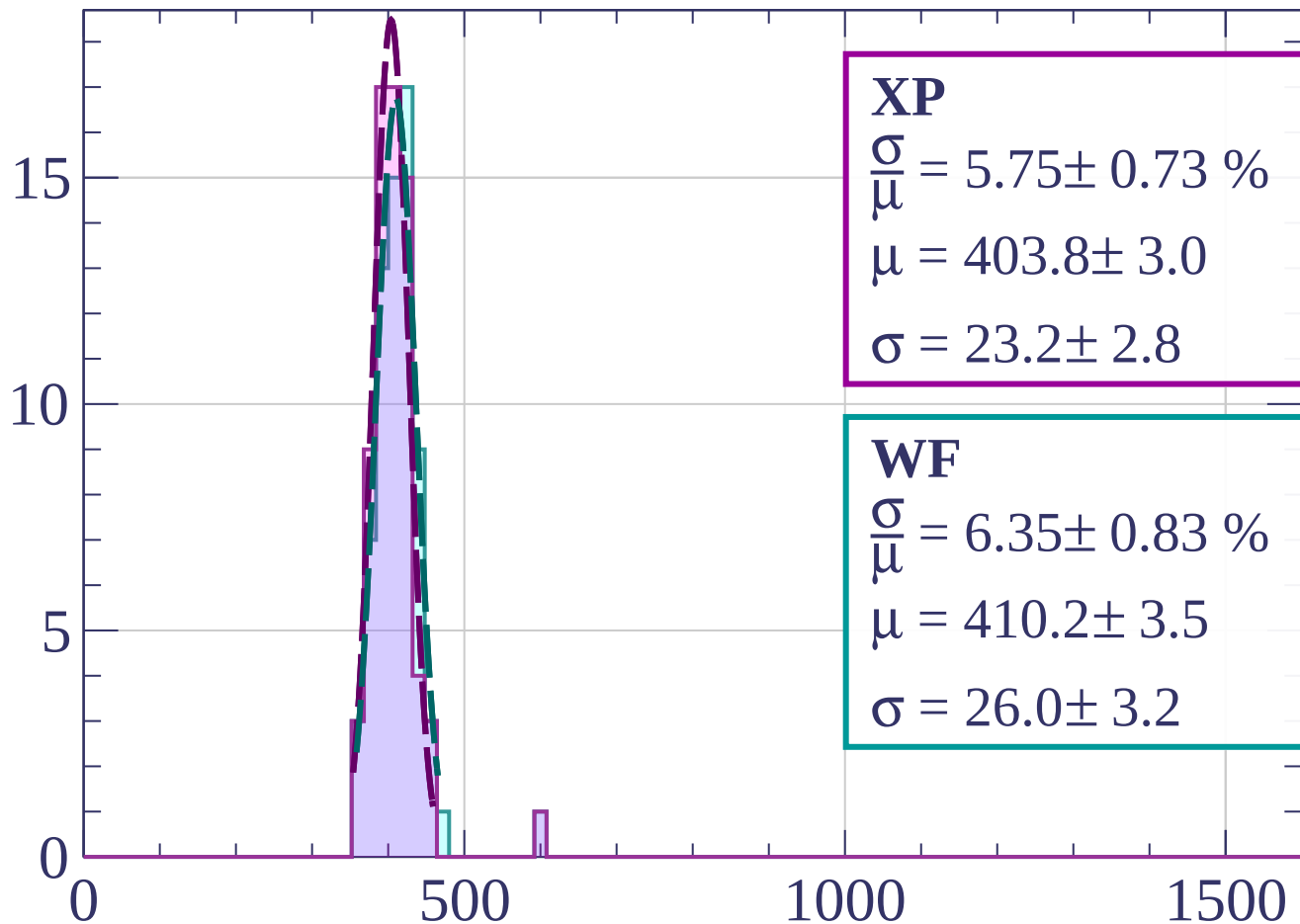
$$\mu = 415.9 \pm 3.2$$

$$\sigma = 23.1 \pm 3.3$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \phi < 15$

Count



XP

$$\frac{\sigma}{\mu} = 5.75 \pm 0.73 \%$$

$$\mu = 403.8 \pm 3.0$$

$$\sigma = 23.2 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 6.35 \pm 0.83 \%$$

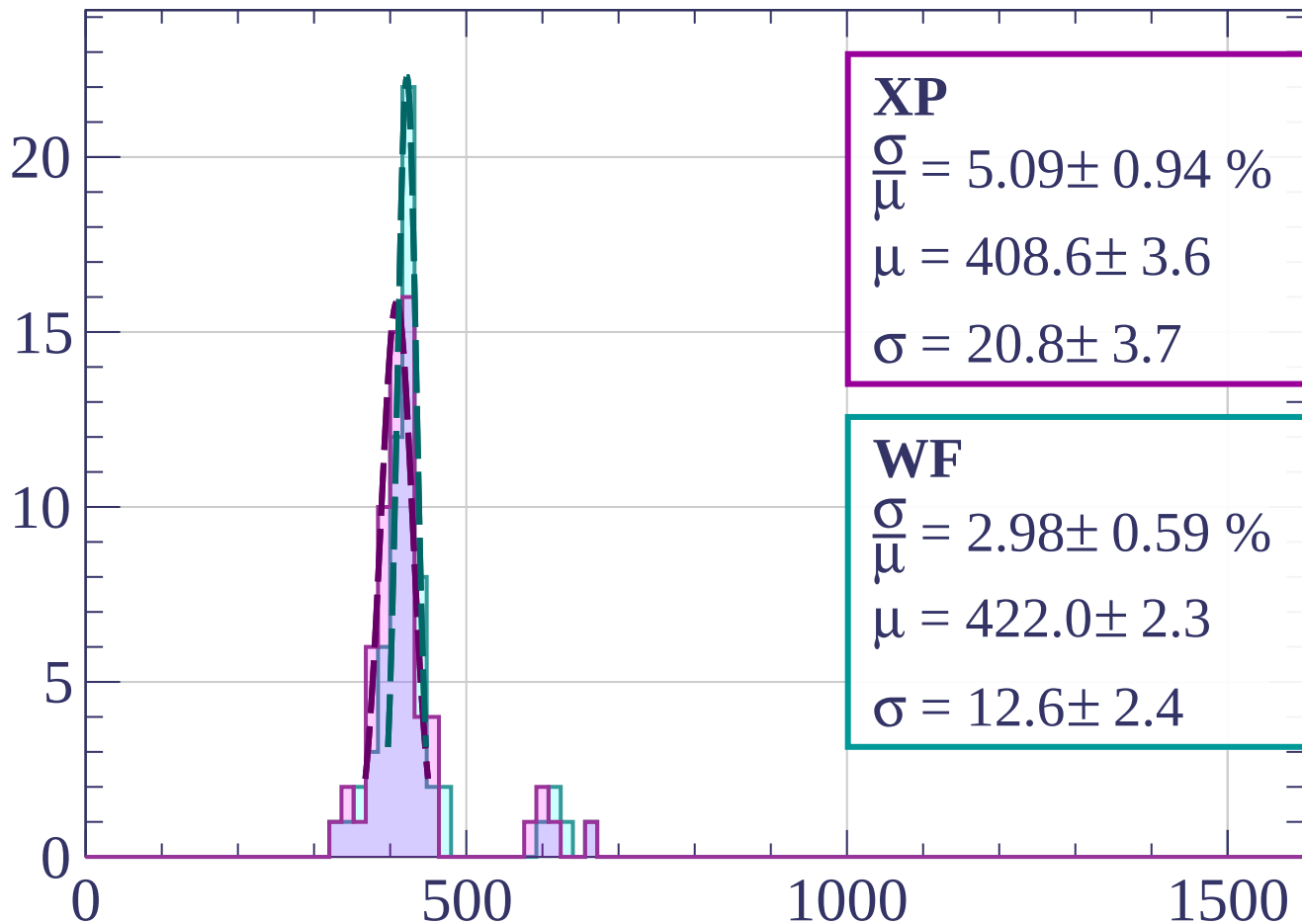
$$\mu = 410.2 \pm 3.5$$

$$\sigma = 26.0 \pm 3.2$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \phi < 18$

Count



XP

$$\frac{\sigma}{\mu} = 5.09 \pm 0.94 \%$$

$$\mu = 408.6 \pm 3.6$$

$$\sigma = 20.8 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 2.98 \pm 0.59 \%$$

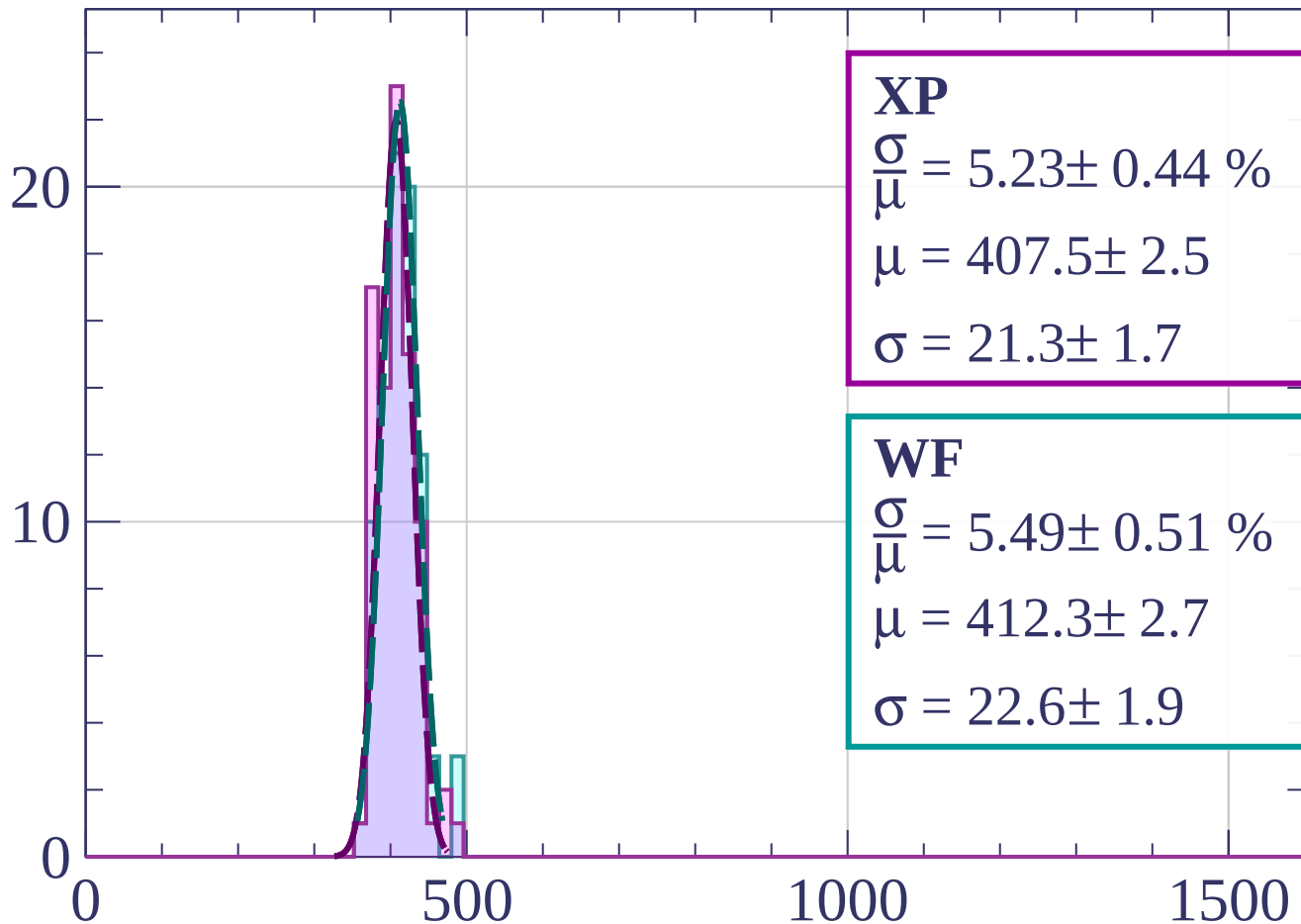
$$\mu = 422.0 \pm 2.3$$

$$\sigma = 12.6 \pm 2.4$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 5.23 \pm 0.44 \%$$

$$\mu = 407.5 \pm 2.5$$

$$\sigma = 21.3 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 5.49 \pm 0.51 \%$$

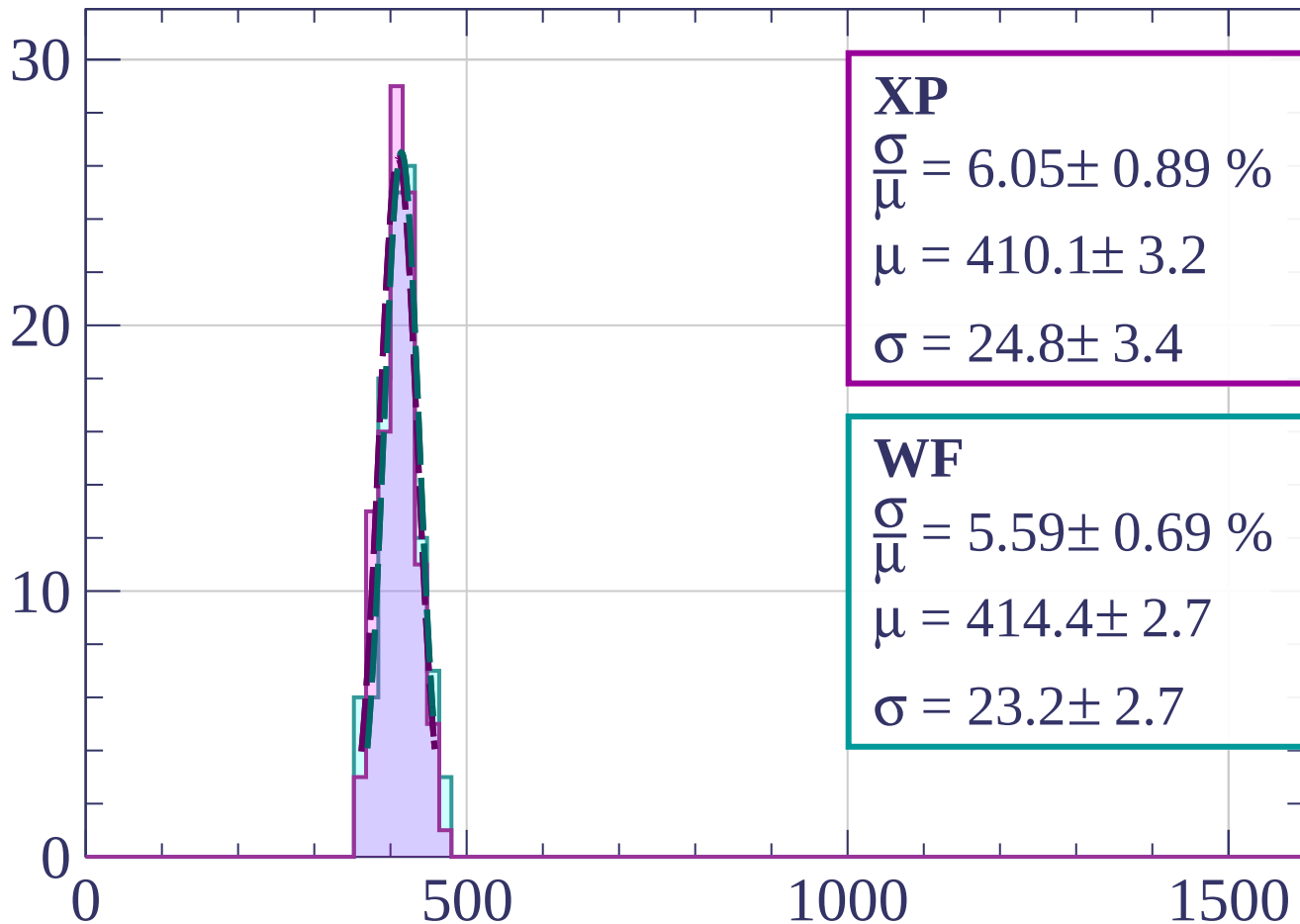
$$\mu = 412.3 \pm 2.7$$

$$\sigma = 22.6 \pm 1.9$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \phi < 24$

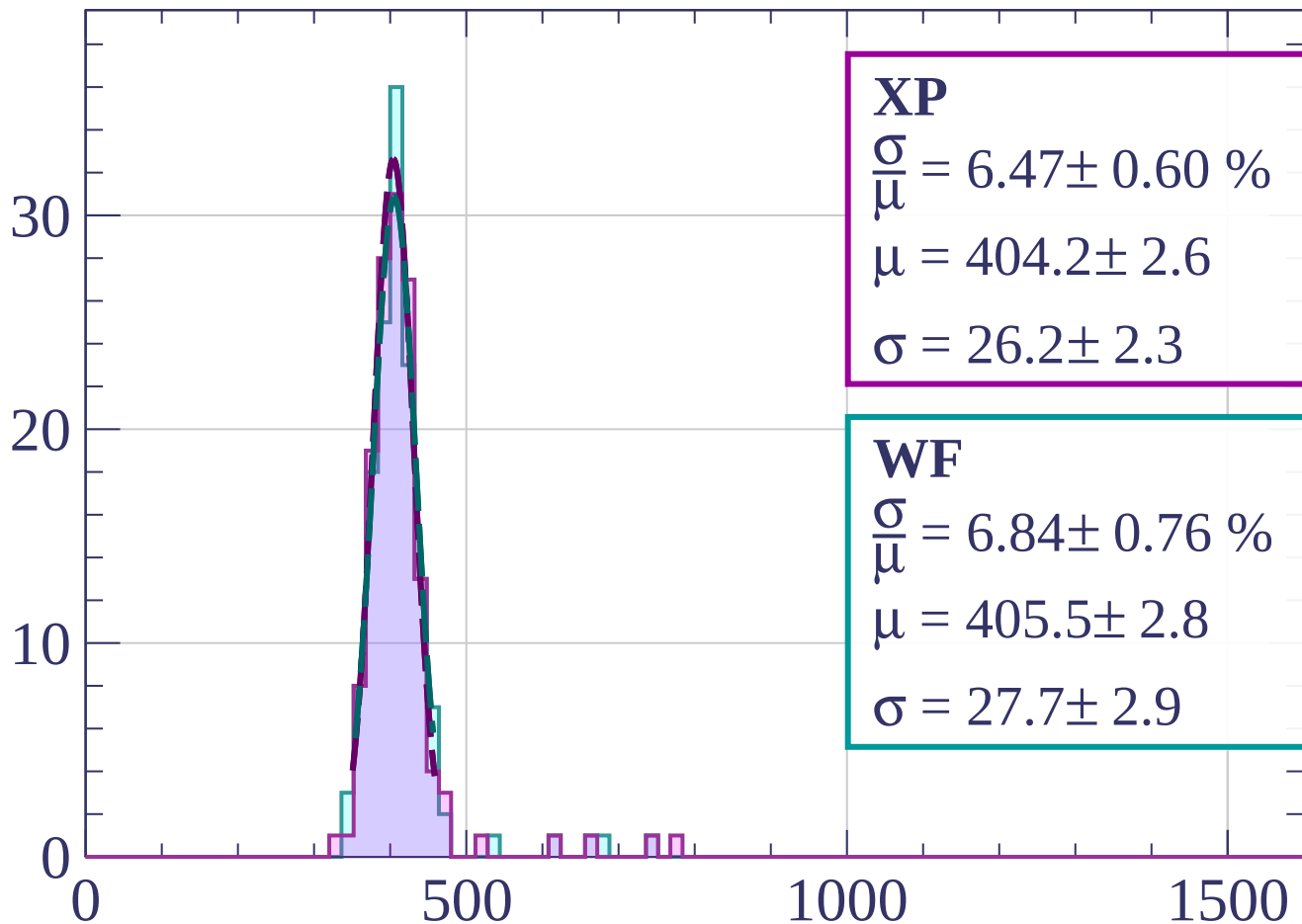
Count



dE/dx [ADC counts/cm]

Energy loss | $24 < \phi < 27$

Count



XP

$$\frac{\sigma}{\mu} = 6.47 \pm 0.60 \%$$

$$\mu = 404.2 \pm 2.6$$

$$\sigma = 26.2 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 6.84 \pm 0.76 \%$$

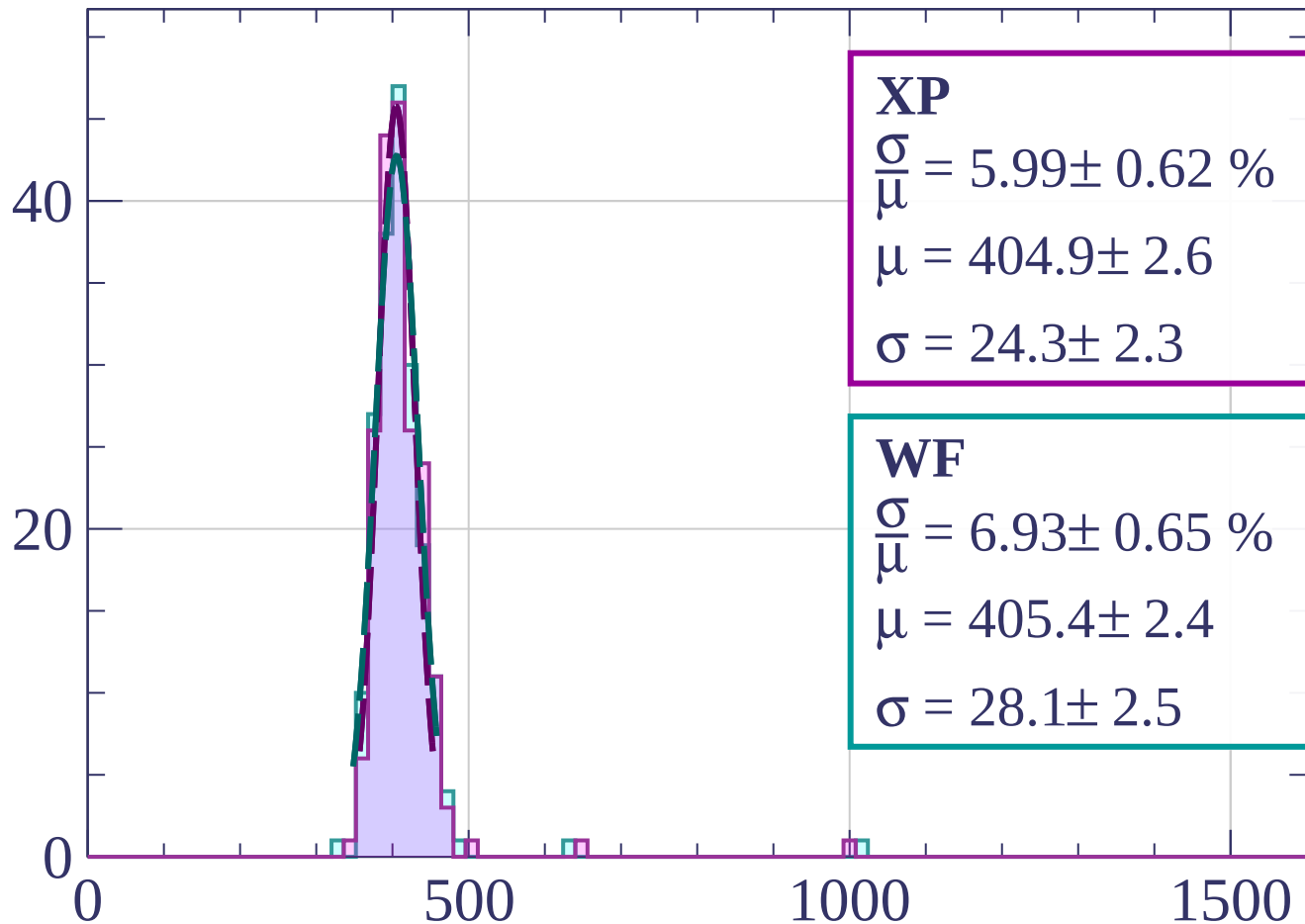
$$\mu = 405.5 \pm 2.8$$

$$\sigma = 27.7 \pm 2.9$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 5.99 \pm 0.62 \%$$

$$\mu = 404.9 \pm 2.6$$

$$\sigma = 24.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 6.93 \pm 0.65 \%$$

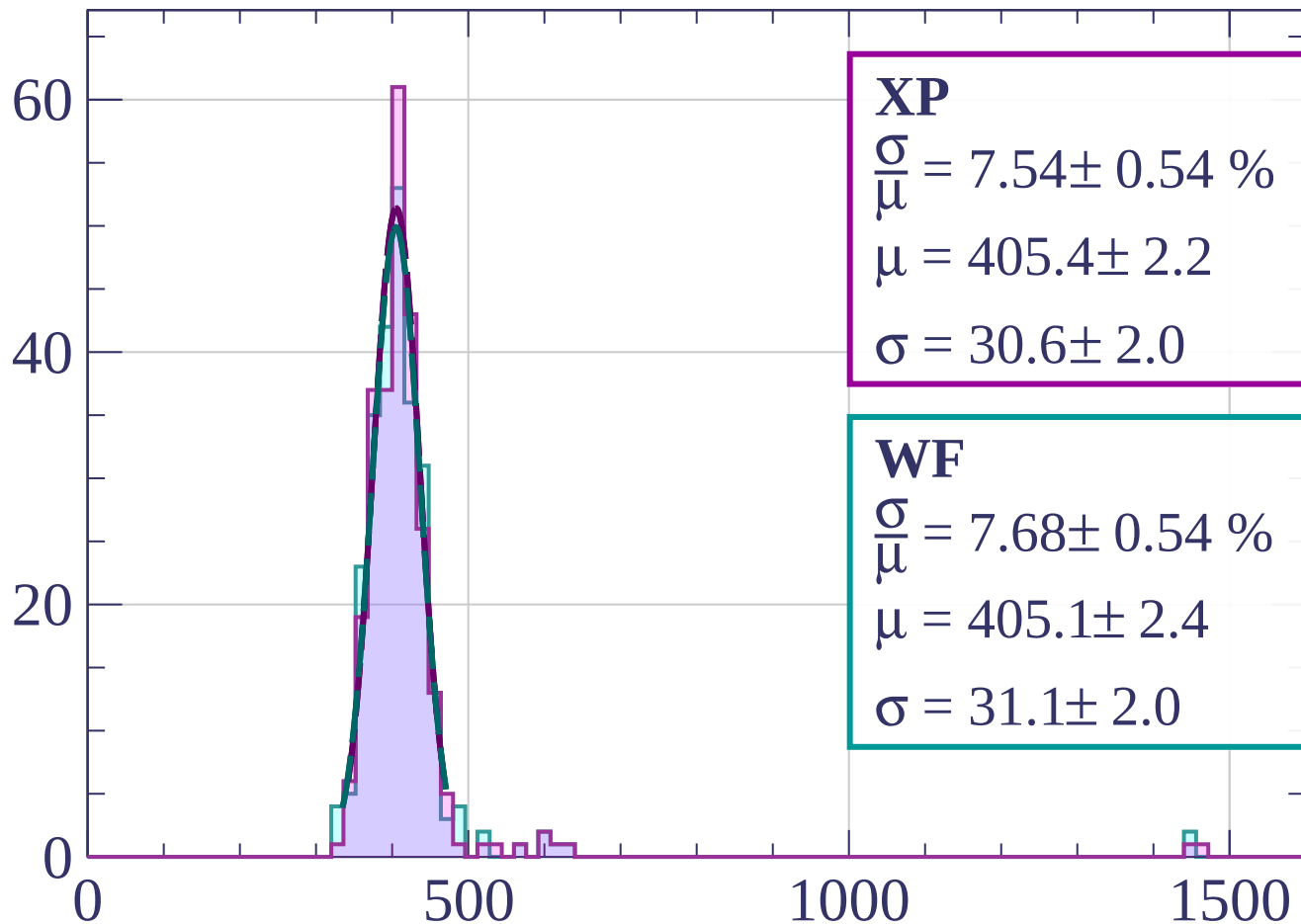
$$\mu = 405.4 \pm 2.4$$

$$\sigma = 28.1 \pm 2.5$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.54 \pm 0.54 \%$$

$$\mu = 405.4 \pm 2.2$$

$$\sigma = 30.6 \pm 2.0$$

WF

$$\frac{\sigma}{\mu} = 7.68 \pm 0.54 \%$$

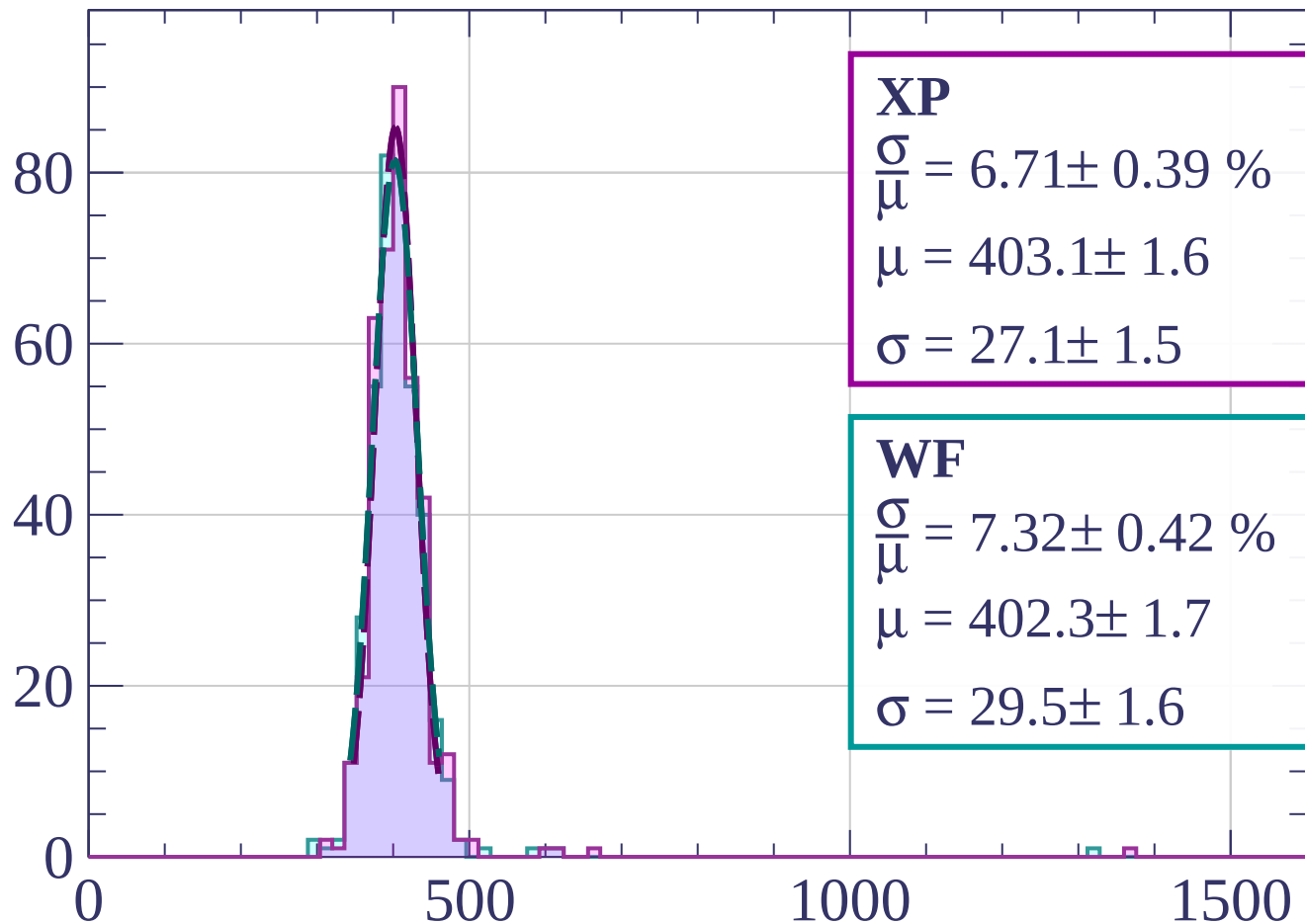
$$\mu = 405.1 \pm 2.4$$

$$\sigma = 31.1 \pm 2.0$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 6.71 \pm 0.39 \%$$

$$\mu = 403.1 \pm 1.6$$

$$\sigma = 27.1 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.32 \pm 0.42 \%$$

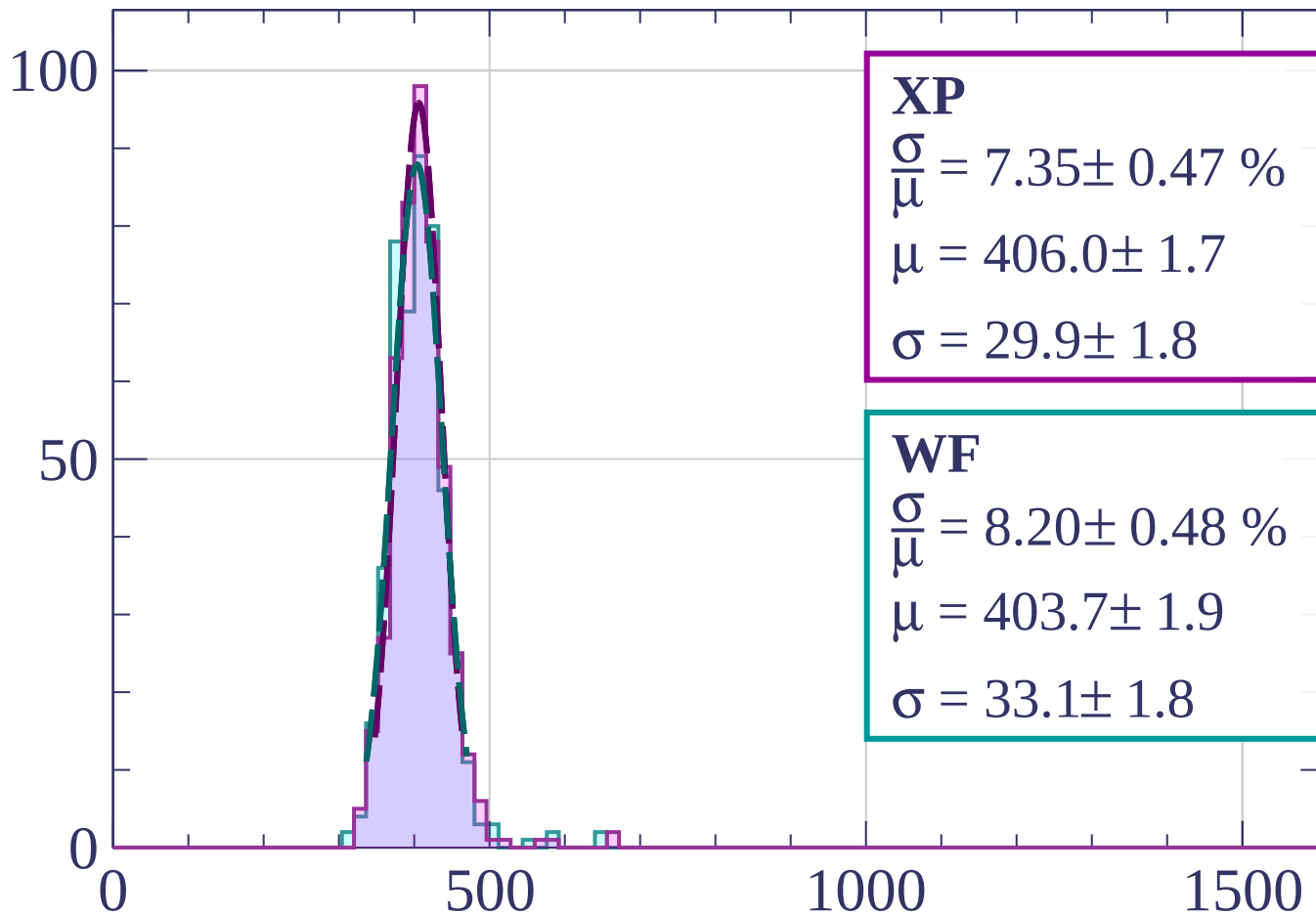
$$\mu = 402.3 \pm 1.7$$

$$\sigma = 29.5 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \phi < 39$

Count



XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.47 \%$$

$$\mu = 406.0 \pm 1.7$$

$$\sigma = 29.9 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.48 \%$$

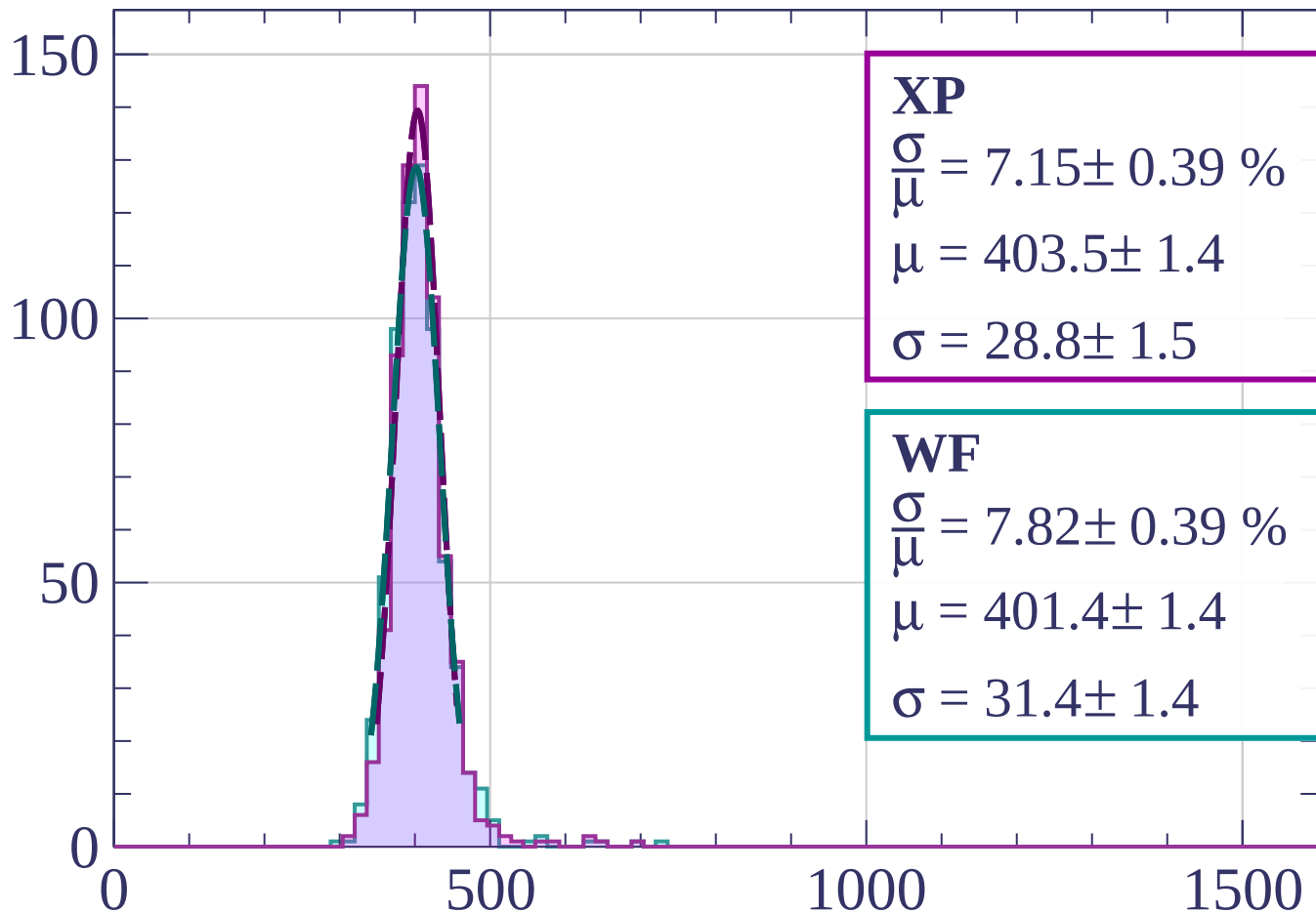
$$\mu = 403.7 \pm 1.9$$

$$\sigma = 33.1 \pm 1.8$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 7.15 \pm 0.39 \%$$

$$\mu = 403.5 \pm 1.4$$

$$\sigma = 28.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 7.82 \pm 0.39 \%$$

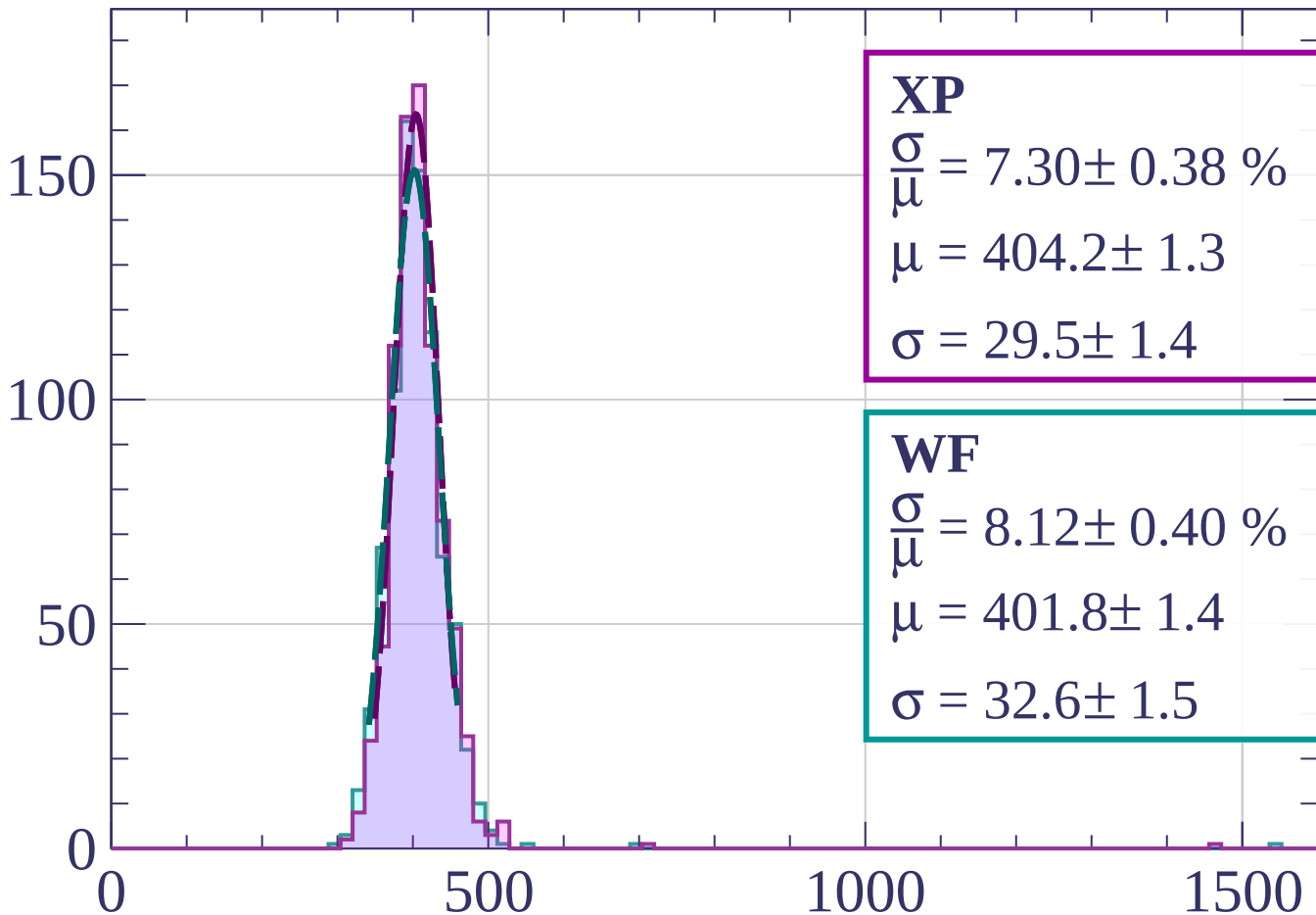
$$\mu = 401.4 \pm 1.4$$

$$\sigma = 31.4 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss | $42 < \phi < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.30 \pm 0.38 \%$$

$$\mu = 404.2 \pm 1.3$$

$$\sigma = 29.5 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.12 \pm 0.40 \%$$

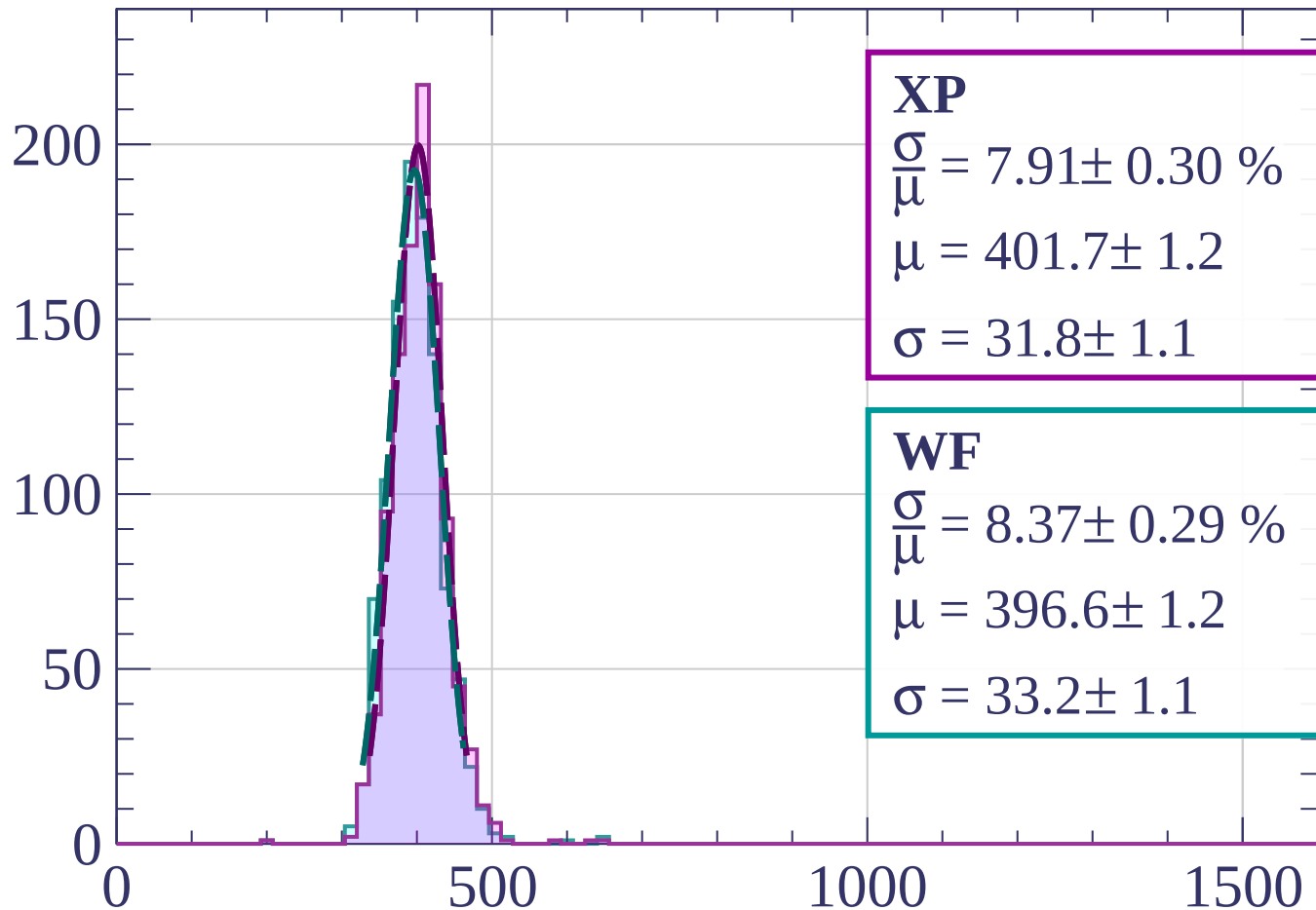
$$\mu = 401.8 \pm 1.4$$

$$\sigma = 32.6 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \phi < 48$

Count



XP

$$\frac{\sigma}{\mu} = 7.91 \pm 0.30 \%$$

$$\mu = 401.7 \pm 1.2$$

$$\sigma = 31.8 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.29 \%$$

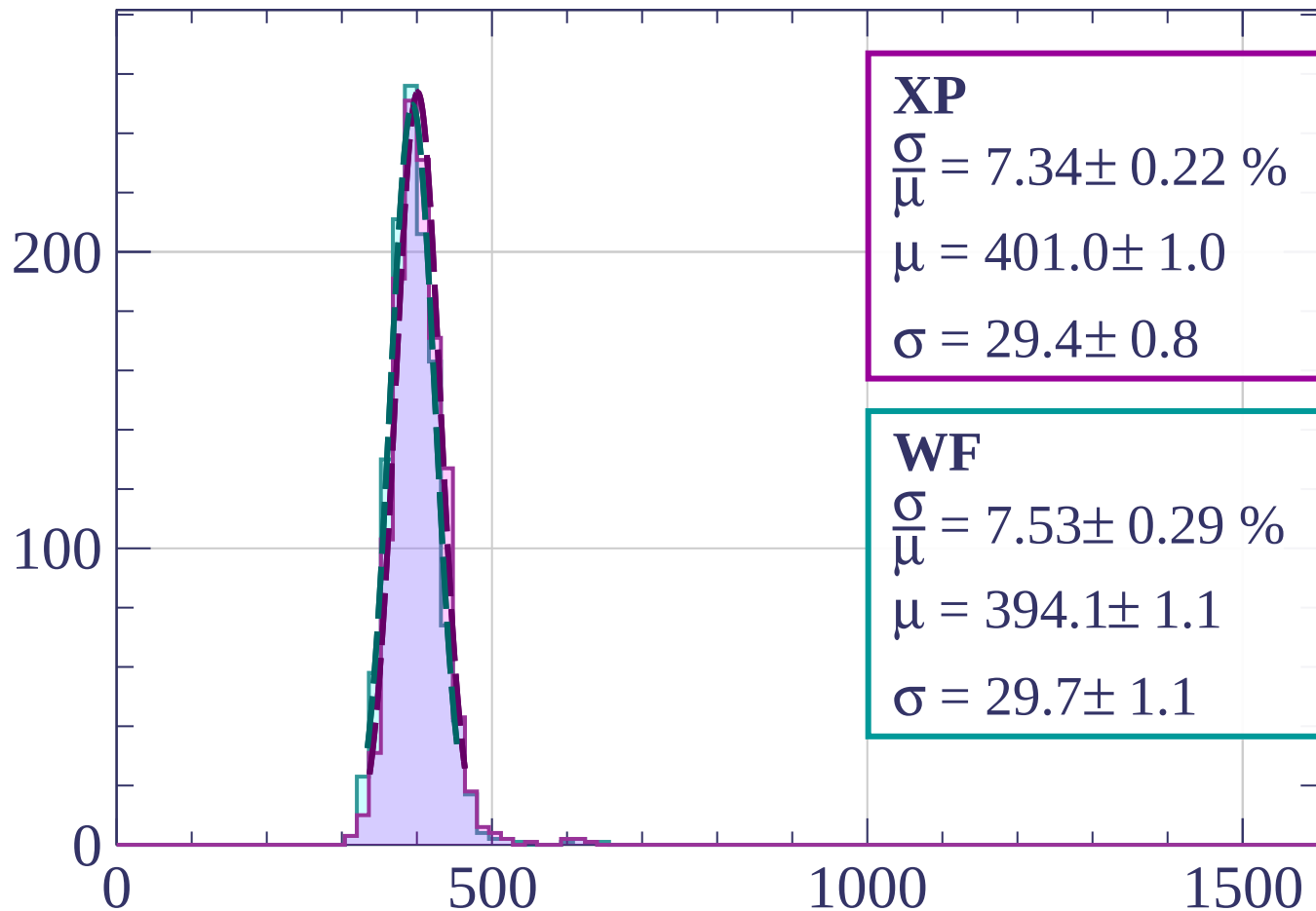
$$\mu = 396.6 \pm 1.2$$

$$\sigma = 33.2 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \phi < 51$

Count



XP

$$\frac{\sigma}{\mu} = 7.34 \pm 0.22 \%$$

$$\mu = 401.0 \pm 1.0$$

$$\sigma = 29.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.53 \pm 0.29 \%$$

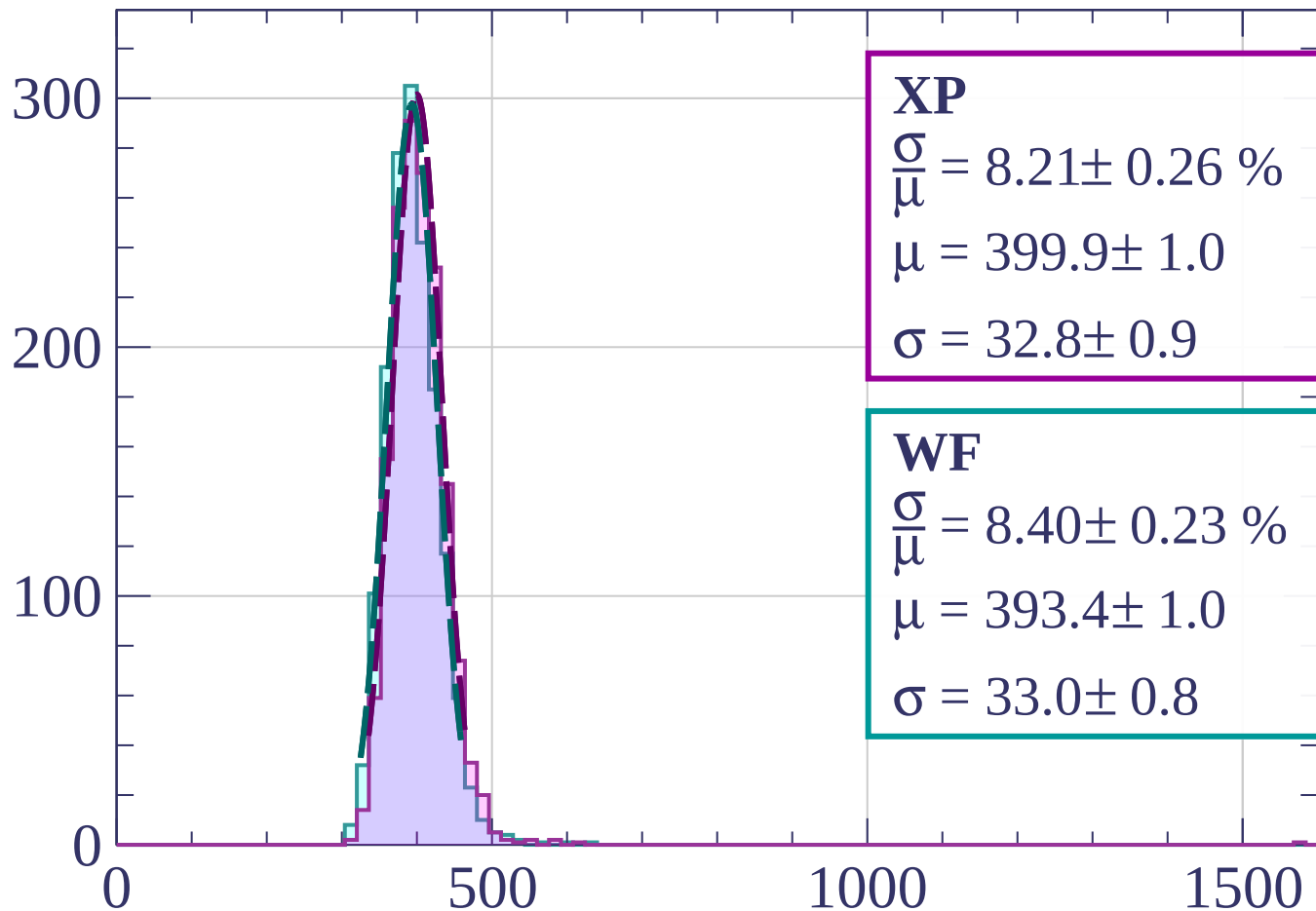
$$\mu = 394.1 \pm 1.1$$

$$\sigma = 29.7 \pm 1.1$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \phi < 54$

Count



XP

$$\frac{\sigma}{\mu} = 8.21 \pm 0.26 \%$$

$$\mu = 399.9 \pm 1.0$$

$$\sigma = 32.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.40 \pm 0.23 \%$$

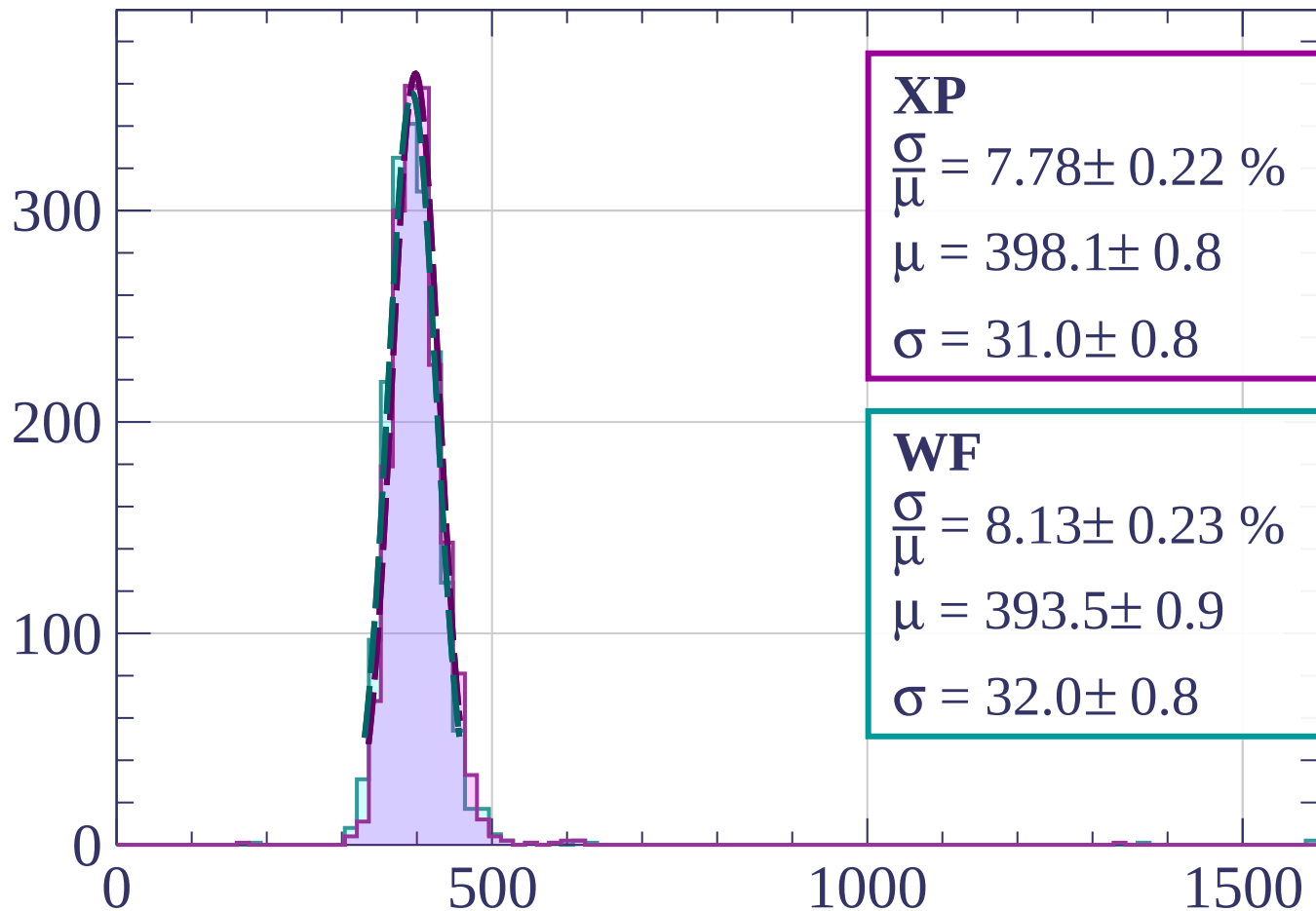
$$\mu = 393.4 \pm 1.0$$

$$\sigma = 33.0 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \phi < 57$

Count



XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.22 \%$$

$$\mu = 398.1 \pm 0.8$$

$$\sigma = 31.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.13 \pm 0.23 \%$$

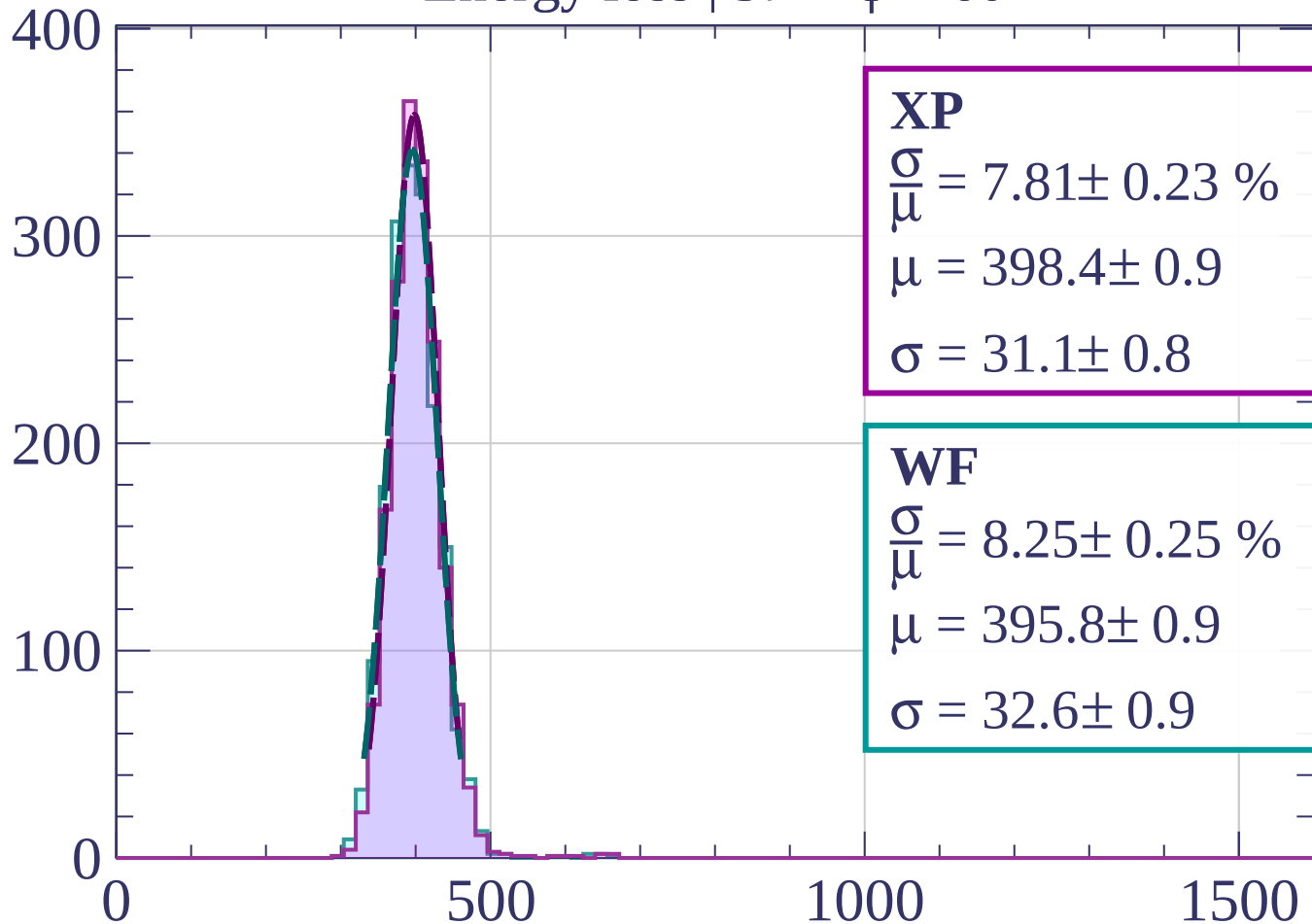
$$\mu = 393.5 \pm 0.9$$

$$\sigma = 32.0 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 7.81 \pm 0.23 \%$$

$$\mu = 398.4 \pm 0.9$$

$$\sigma = 31.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.25 \pm 0.25 \%$$

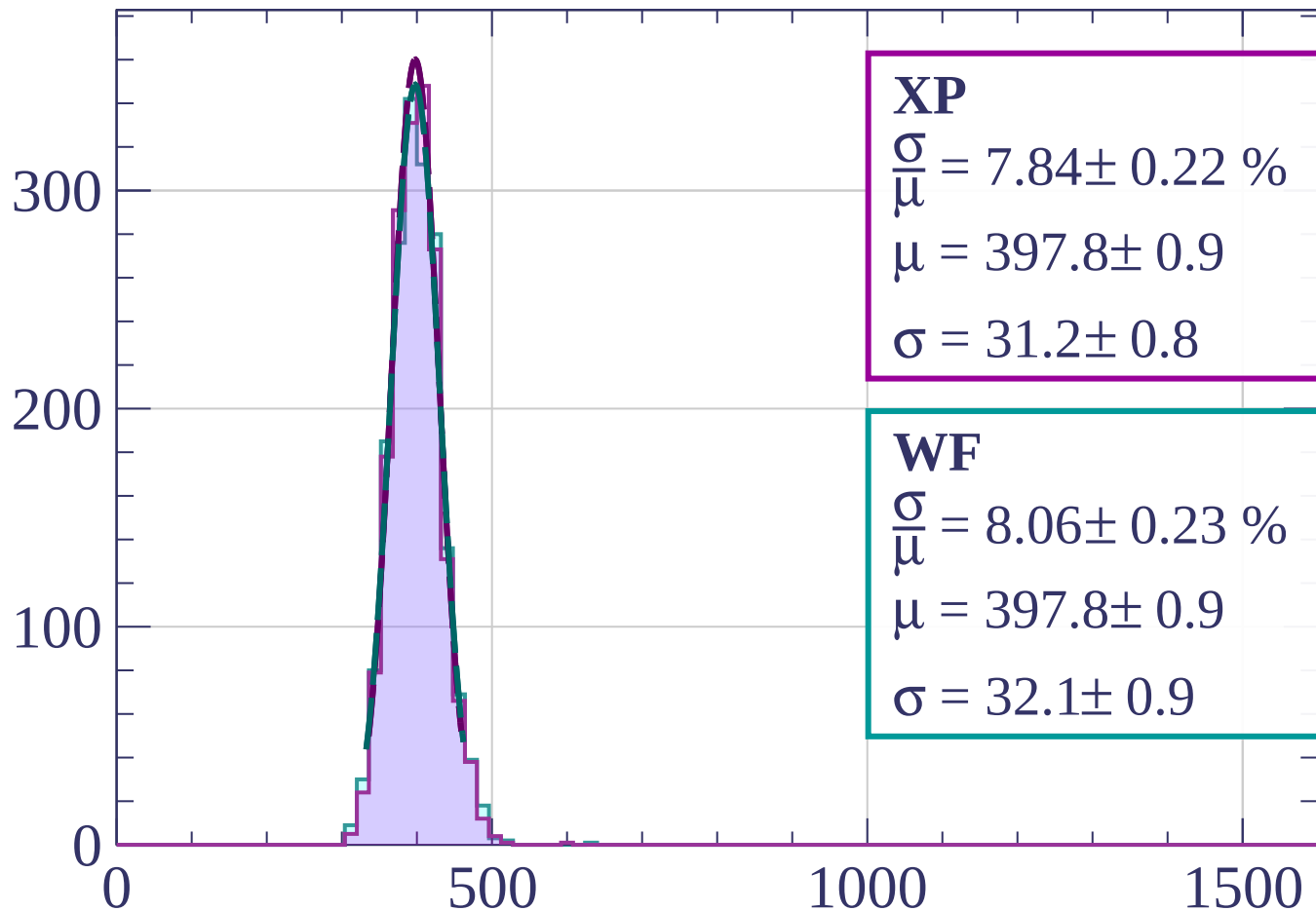
$$\mu = 395.8 \pm 0.9$$

$$\sigma = 32.6 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \phi < 63$

Count



XP

$$\frac{\sigma}{\mu} = 7.84 \pm 0.22 \%$$

$$\mu = 397.8 \pm 0.9$$

$$\sigma = 31.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.06 \pm 0.23 \%$$

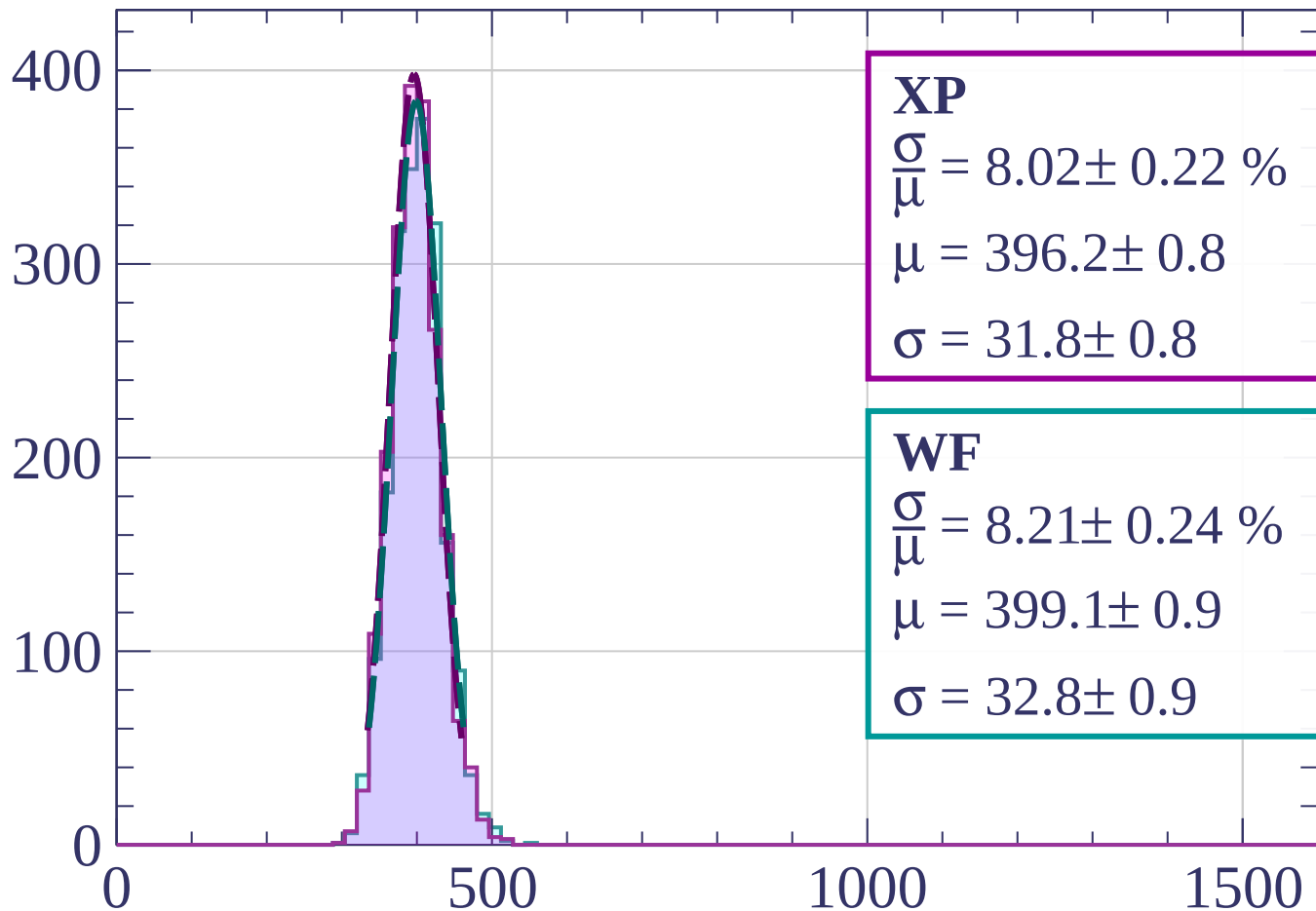
$$\mu = 397.8 \pm 0.9$$

$$\sigma = 32.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \phi < 66$

Count



XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.22 \%$$

$$\mu = 396.2 \pm 0.8$$

$$\sigma = 31.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.21 \pm 0.24 \%$$

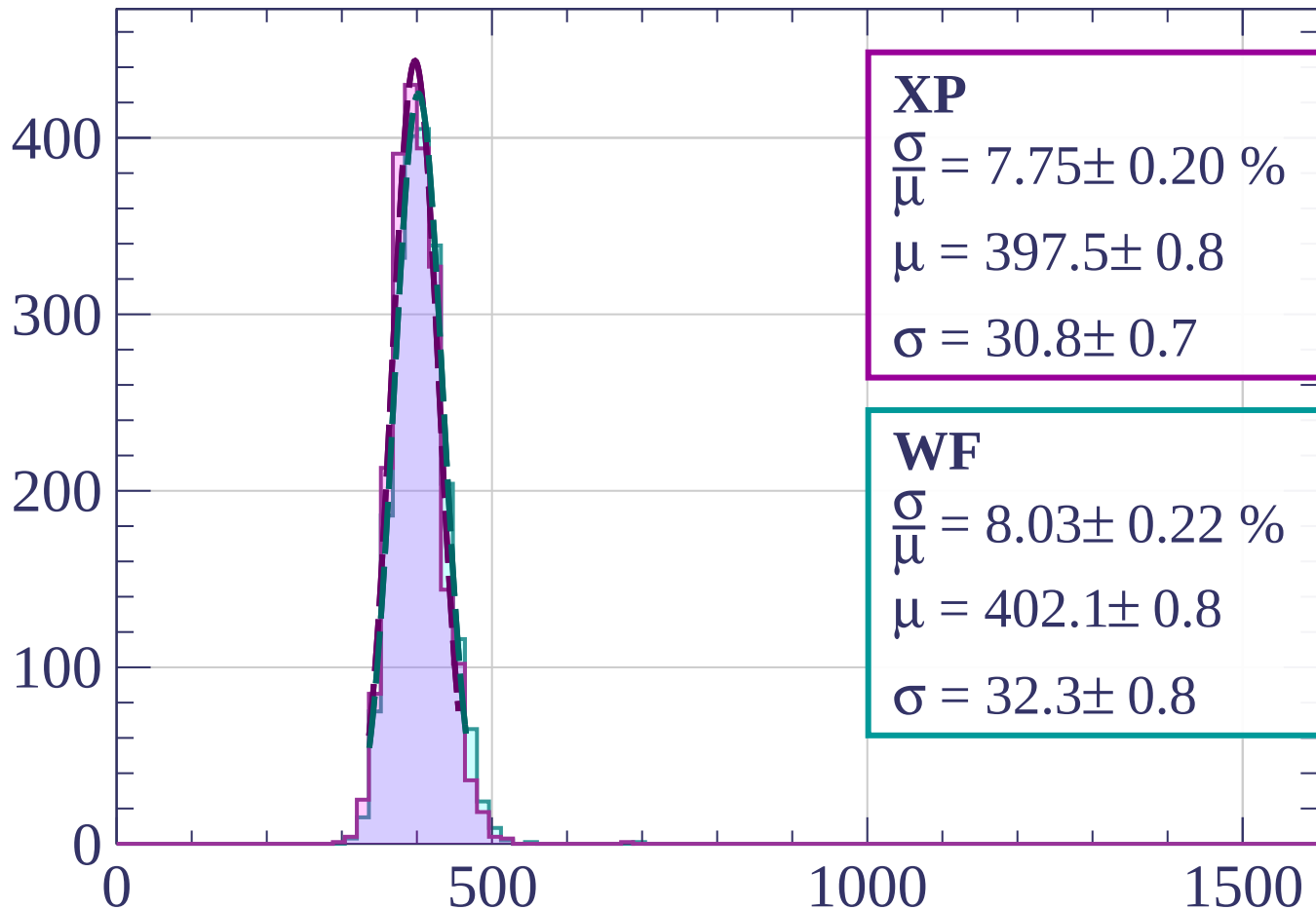
$$\mu = 399.1 \pm 0.9$$

$$\sigma = 32.8 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \phi < 69$

Count



XP

$$\frac{\sigma}{\mu} = 7.75 \pm 0.20 \%$$

$$\mu = 397.5 \pm 0.8$$

$$\sigma = 30.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.03 \pm 0.22 \%$$

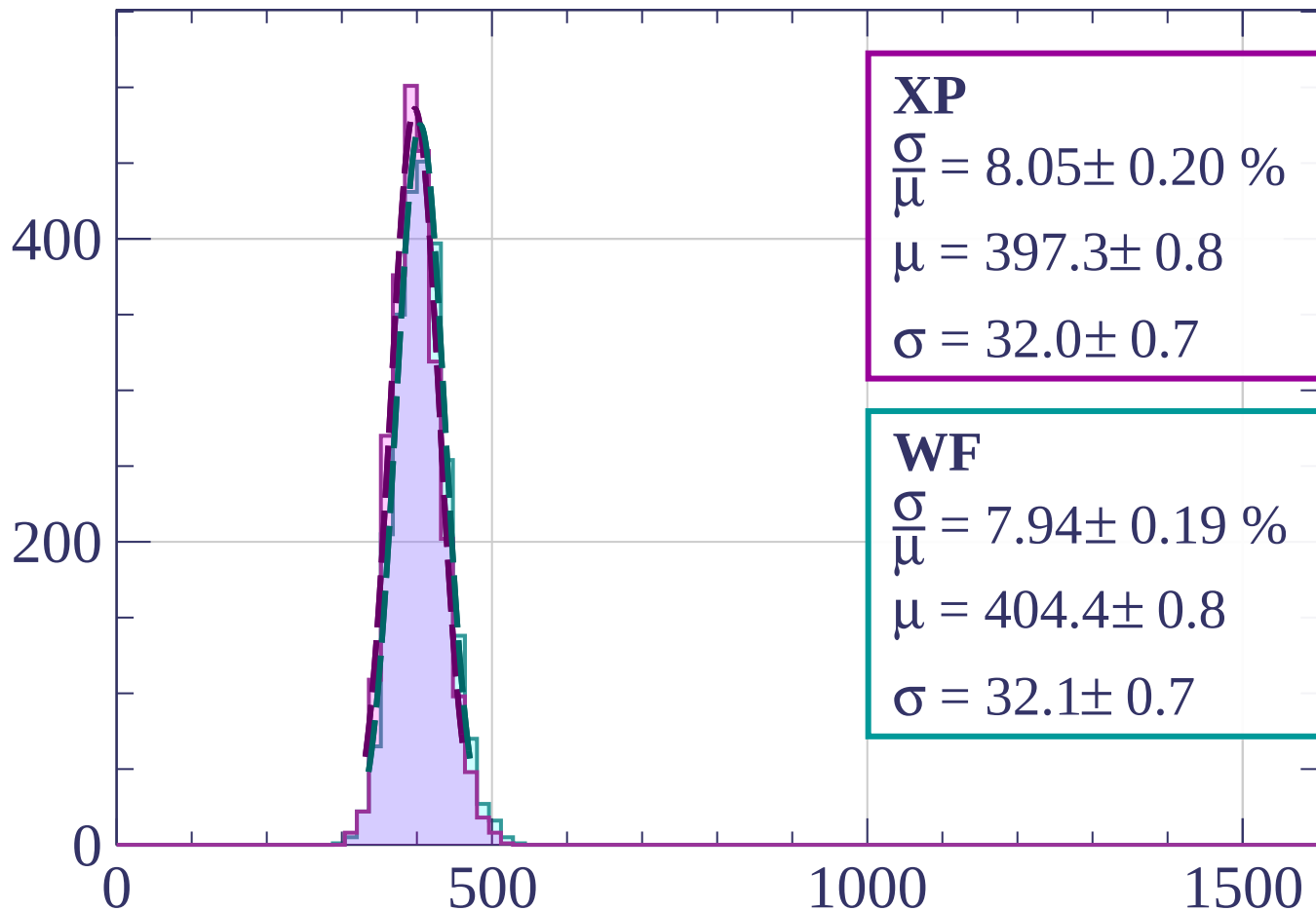
$$\mu = 402.1 \pm 0.8$$

$$\sigma = 32.3 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $69 < \phi < 72$

Count



XP

$$\frac{\sigma}{\mu} = 8.05 \pm 0.20 \%$$

$$\mu = 397.3 \pm 0.8$$

$$\sigma = 32.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.94 \pm 0.19 \%$$

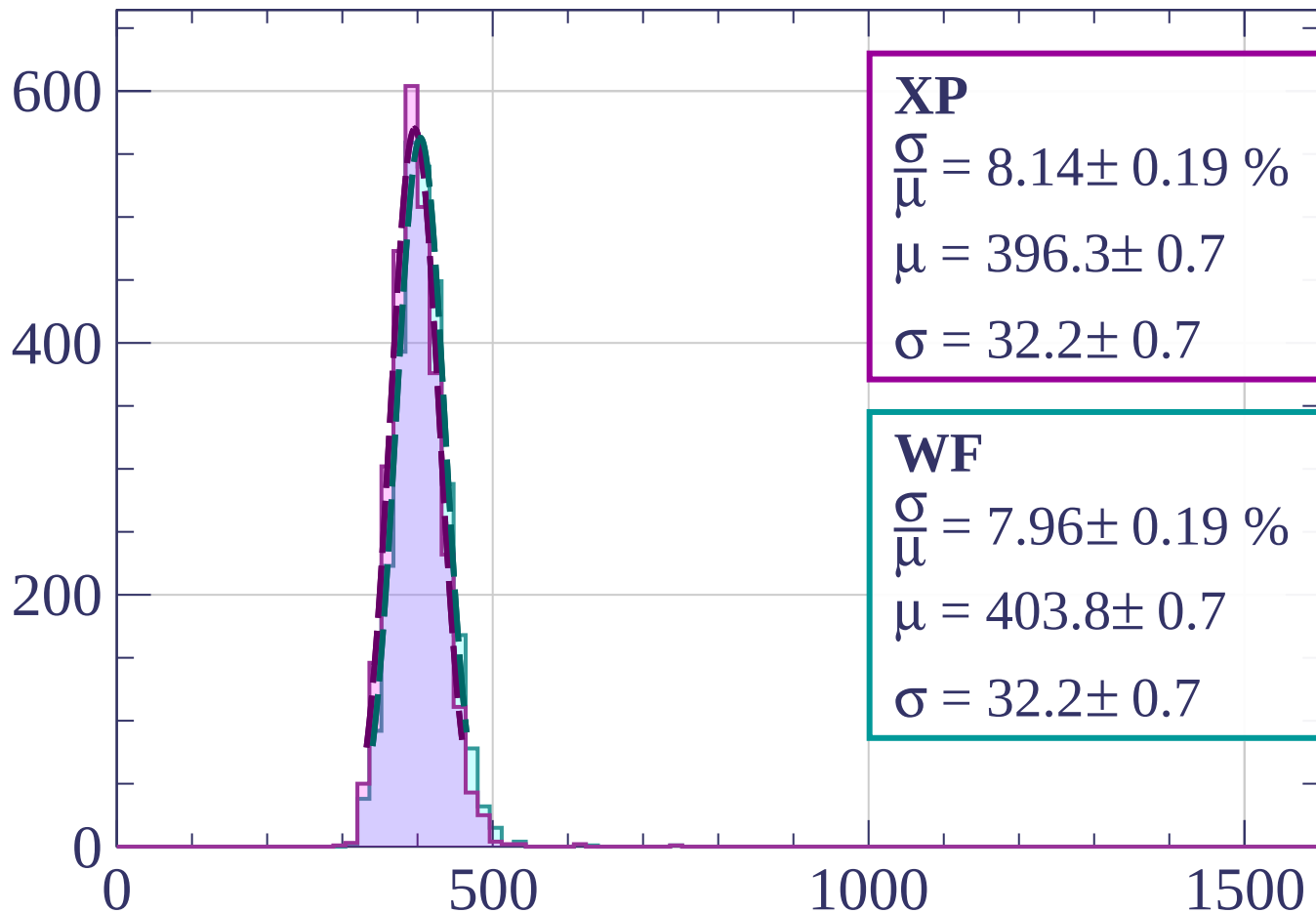
$$\mu = 404.4 \pm 0.8$$

$$\sigma = 32.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $72 < \phi < 75$

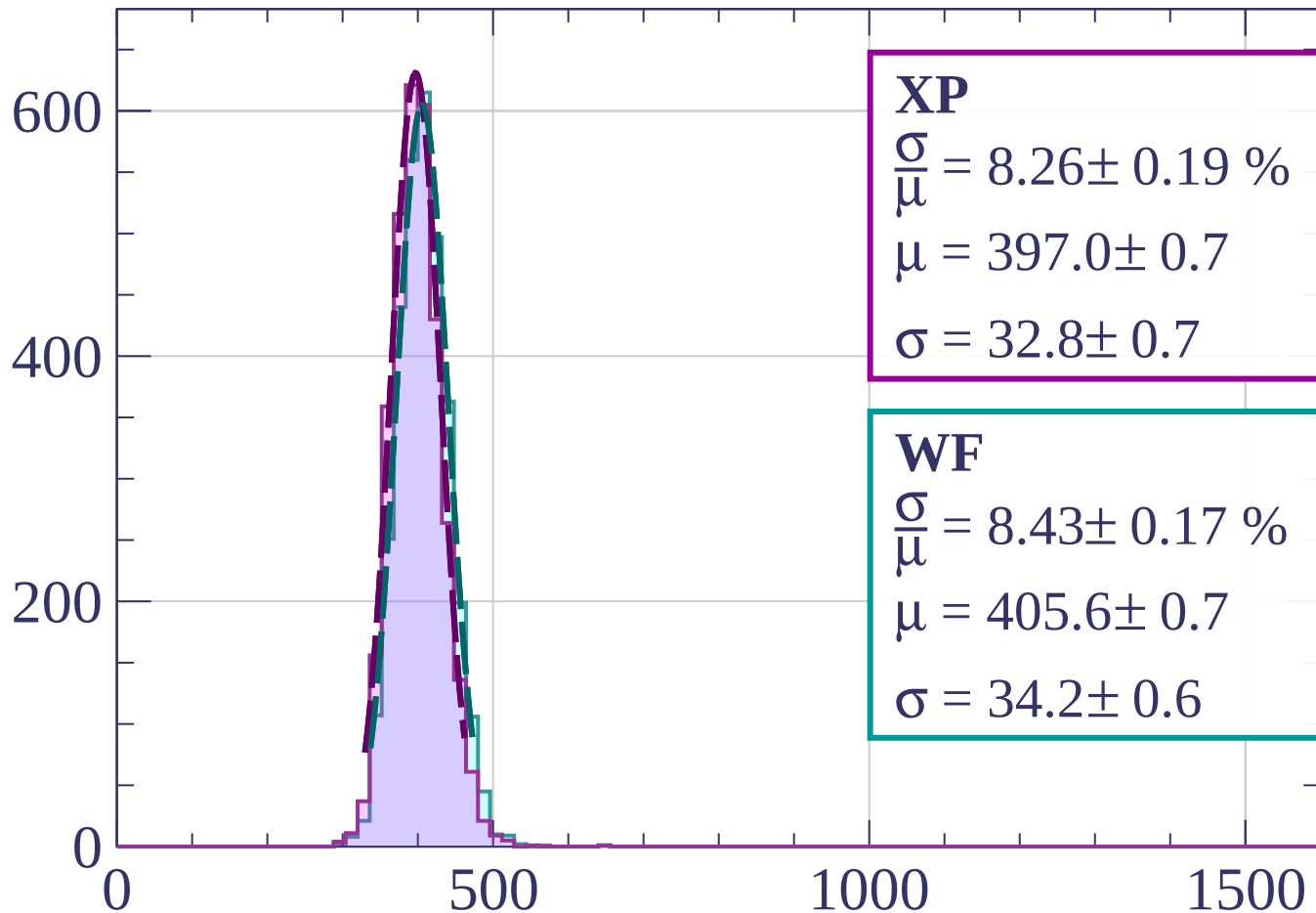
Count



dE/dx [ADC counts/cm]

Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.19 \%$$

$$\mu = 397.0 \pm 0.7$$

$$\sigma = 32.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.43 \pm 0.17 \%$$

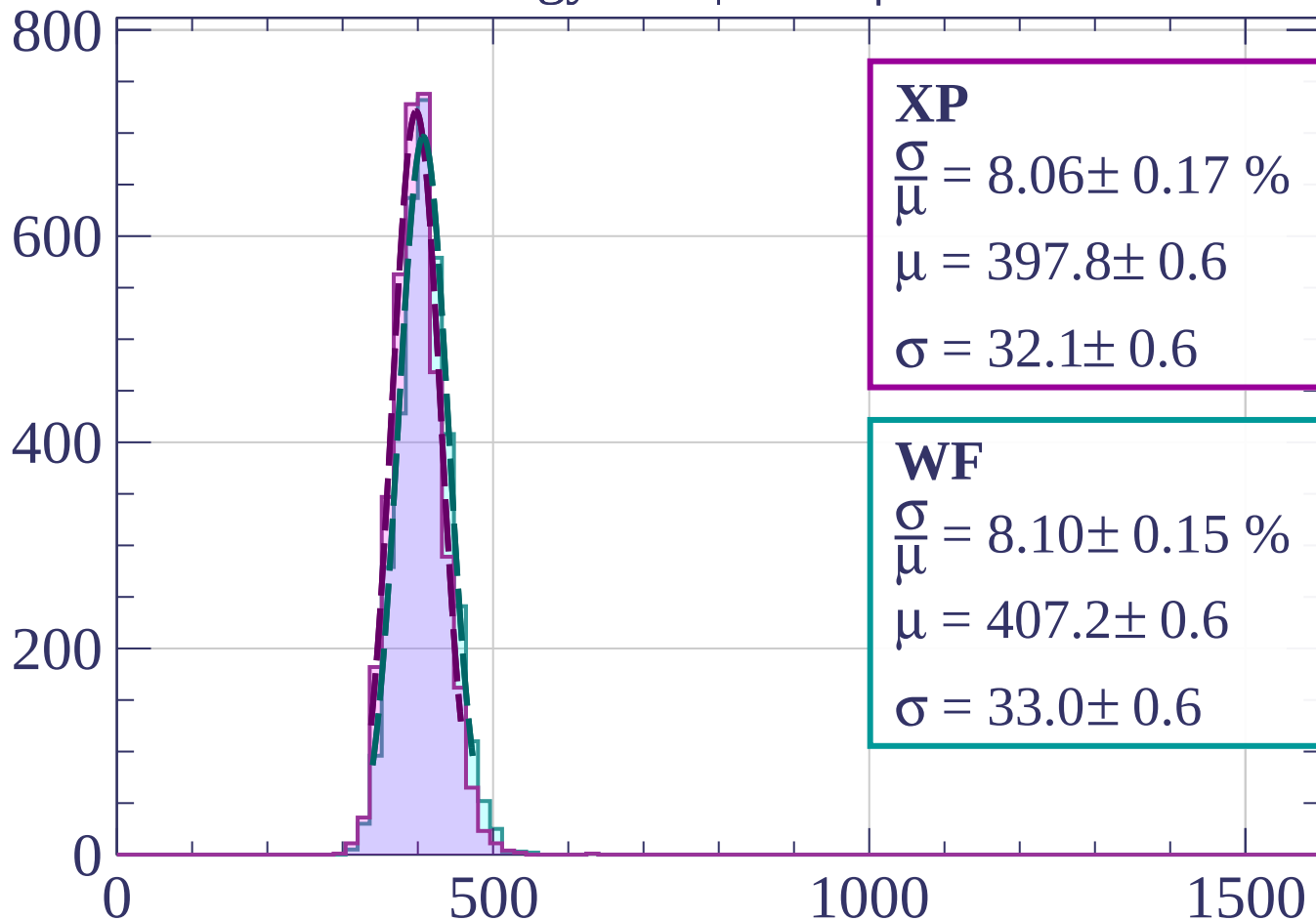
$$\mu = 405.6 \pm 0.7$$

$$\sigma = 34.2 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $78 < \phi < 81$

Count



XP

$$\frac{\sigma}{\mu} = 8.06 \pm 0.17 \%$$

$$\mu = 397.8 \pm 0.6$$

$$\sigma = 32.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.10 \pm 0.15 \%$$

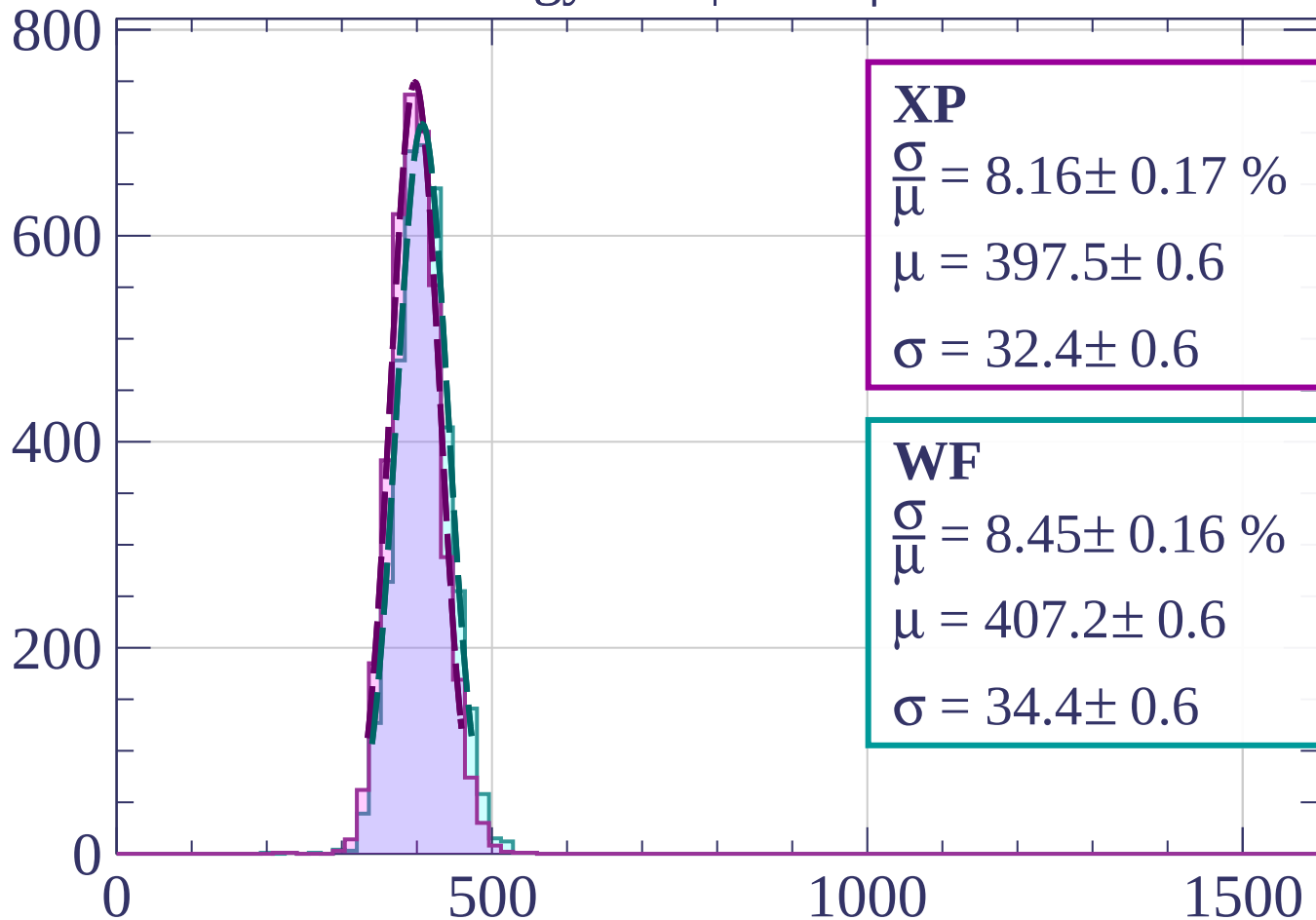
$$\mu = 407.2 \pm 0.6$$

$$\sigma = 33.0 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $81 < \phi < 84$

Count



XP

$$\frac{\sigma}{\mu} = 8.16 \pm 0.17 \%$$

$$\mu = 397.5 \pm 0.6$$

$$\sigma = 32.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.45 \pm 0.16 \%$$

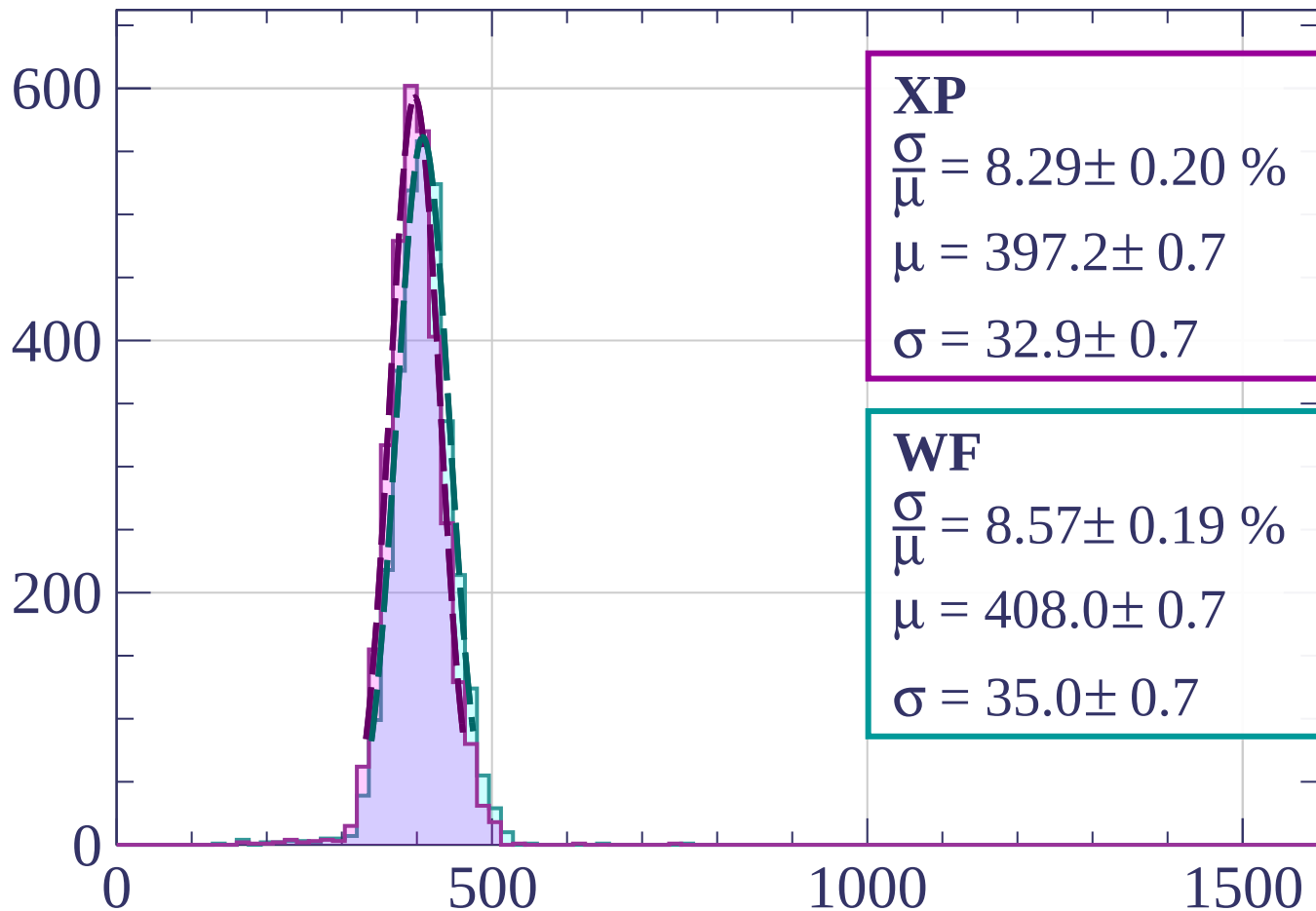
$$\mu = 407.2 \pm 0.6$$

$$\sigma = 34.4 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $84 < \phi < 87$

Count



XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.20 \%$$

$$\mu = 397.2 \pm 0.7$$

$$\sigma = 32.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.57 \pm 0.19 \%$$

$$\mu = 408.0 \pm 0.7$$

$$\sigma = 35.0 \pm 0.7$$

dE/dx [ADC counts/cm]

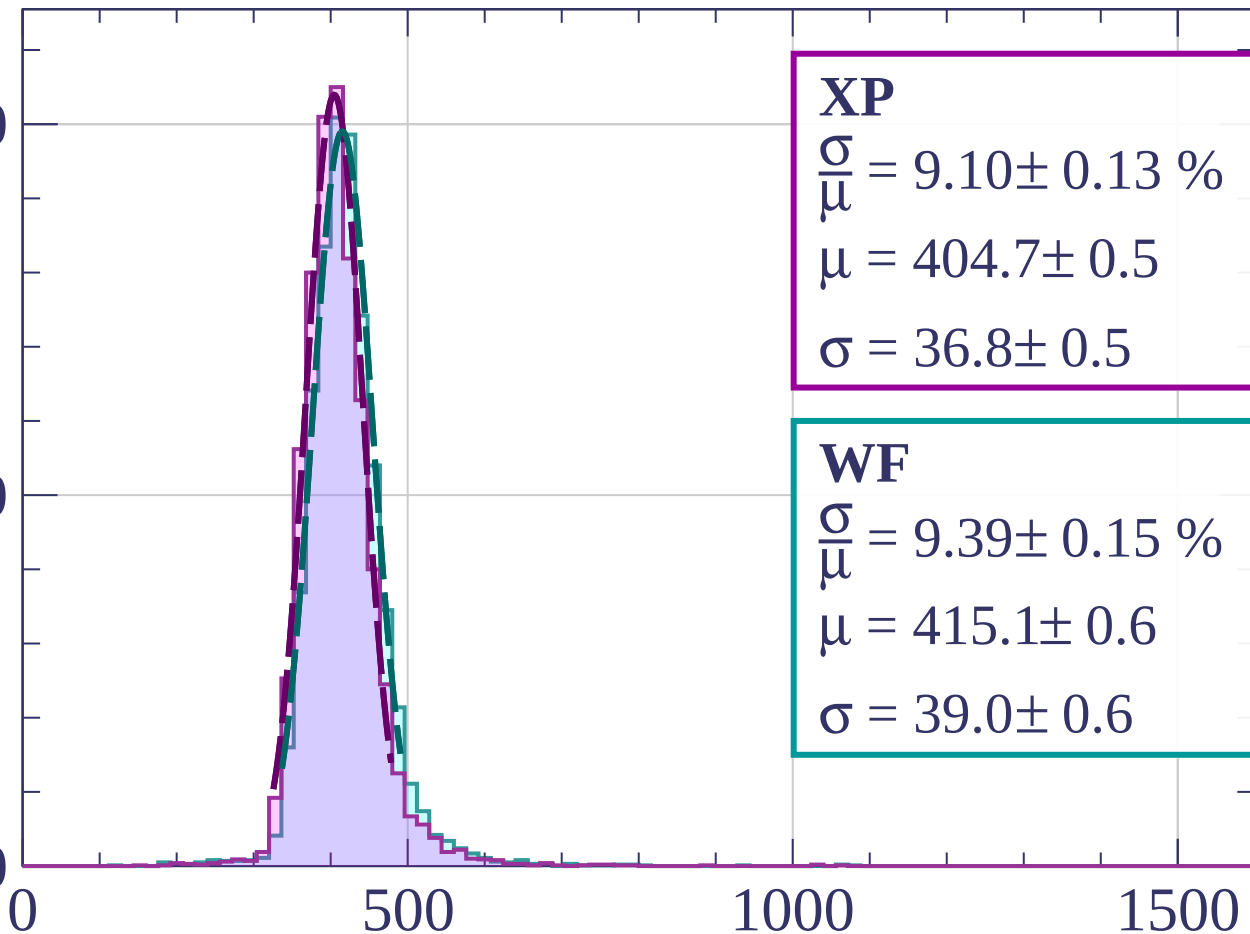
Energy loss | $87 < \phi < 90$

Count

1000

500

0



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.13 \%$$

$$\mu = 404.7 \pm 0.5$$

$$\sigma = 36.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 9.39 \pm 0.15 \%$$

$$\mu = 415.1 \pm 0.6$$

$$\sigma = 39.0 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $0 < \theta < 3$

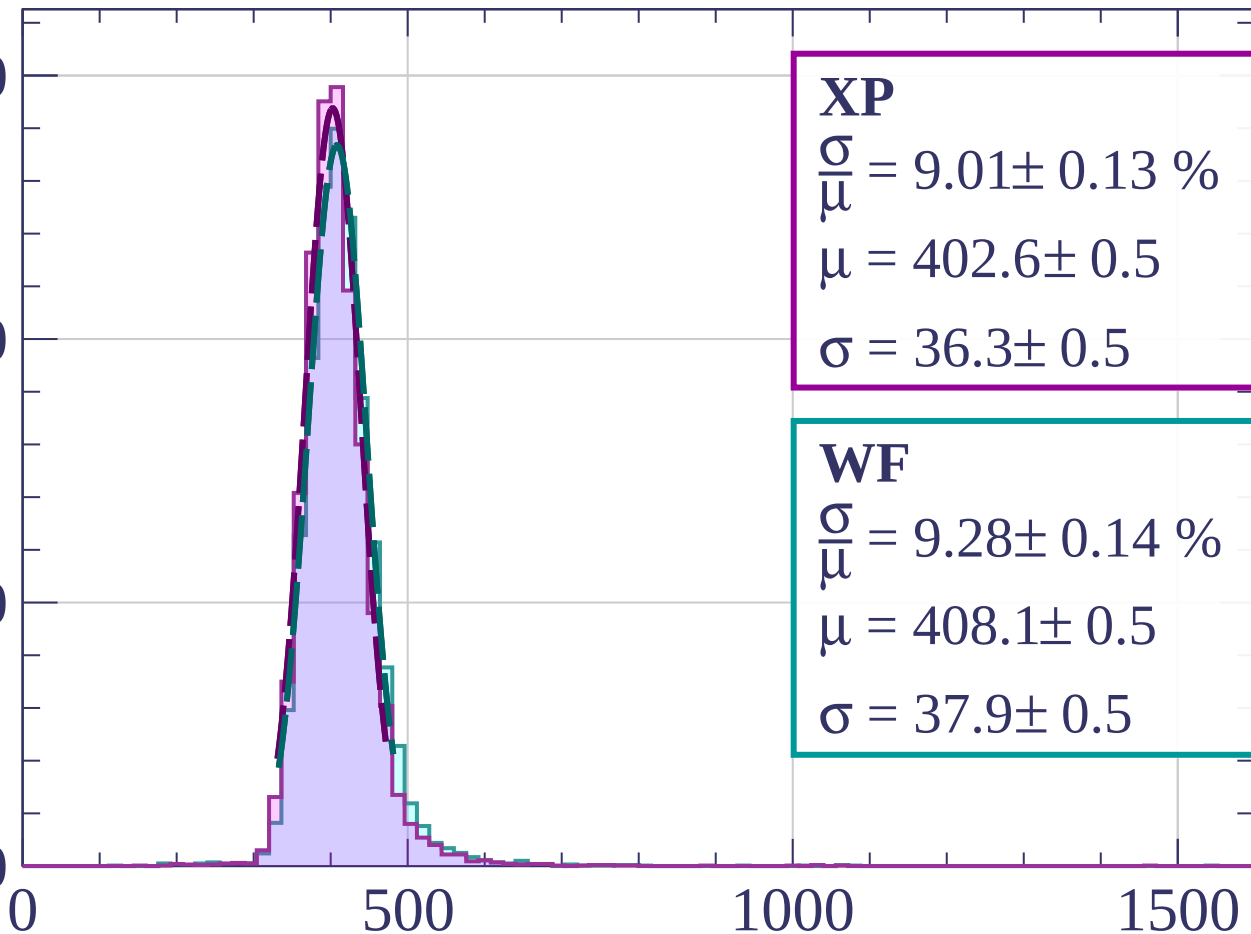
Count

1500

1000

500

0



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.13 \%$$

$$\mu = 402.6 \pm 0.5$$

$$\sigma = 36.3 \pm 0.5$$

WF

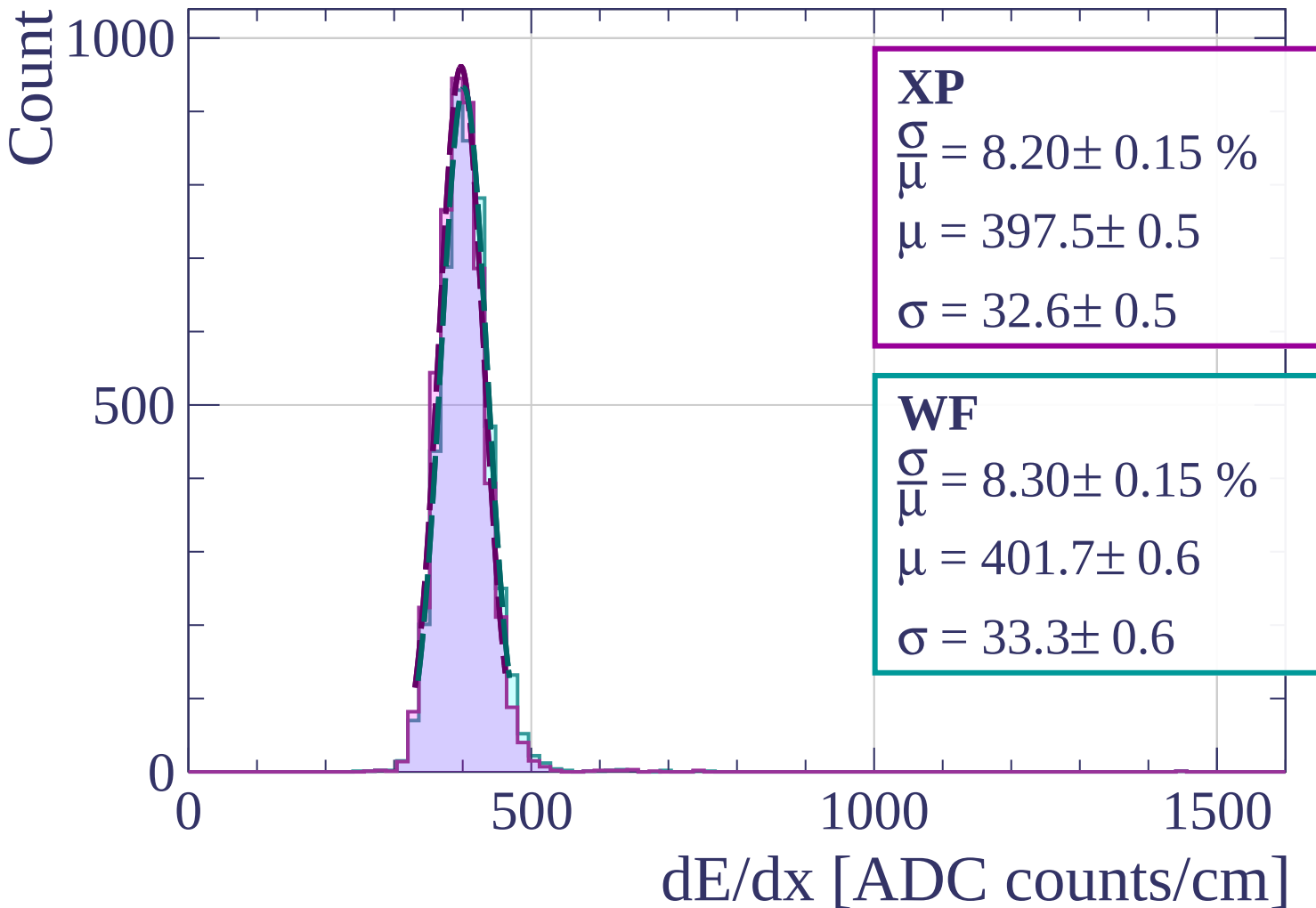
$$\frac{\sigma}{\mu} = 9.28 \pm 0.14 \%$$

$$\mu = 408.1 \pm 0.5$$

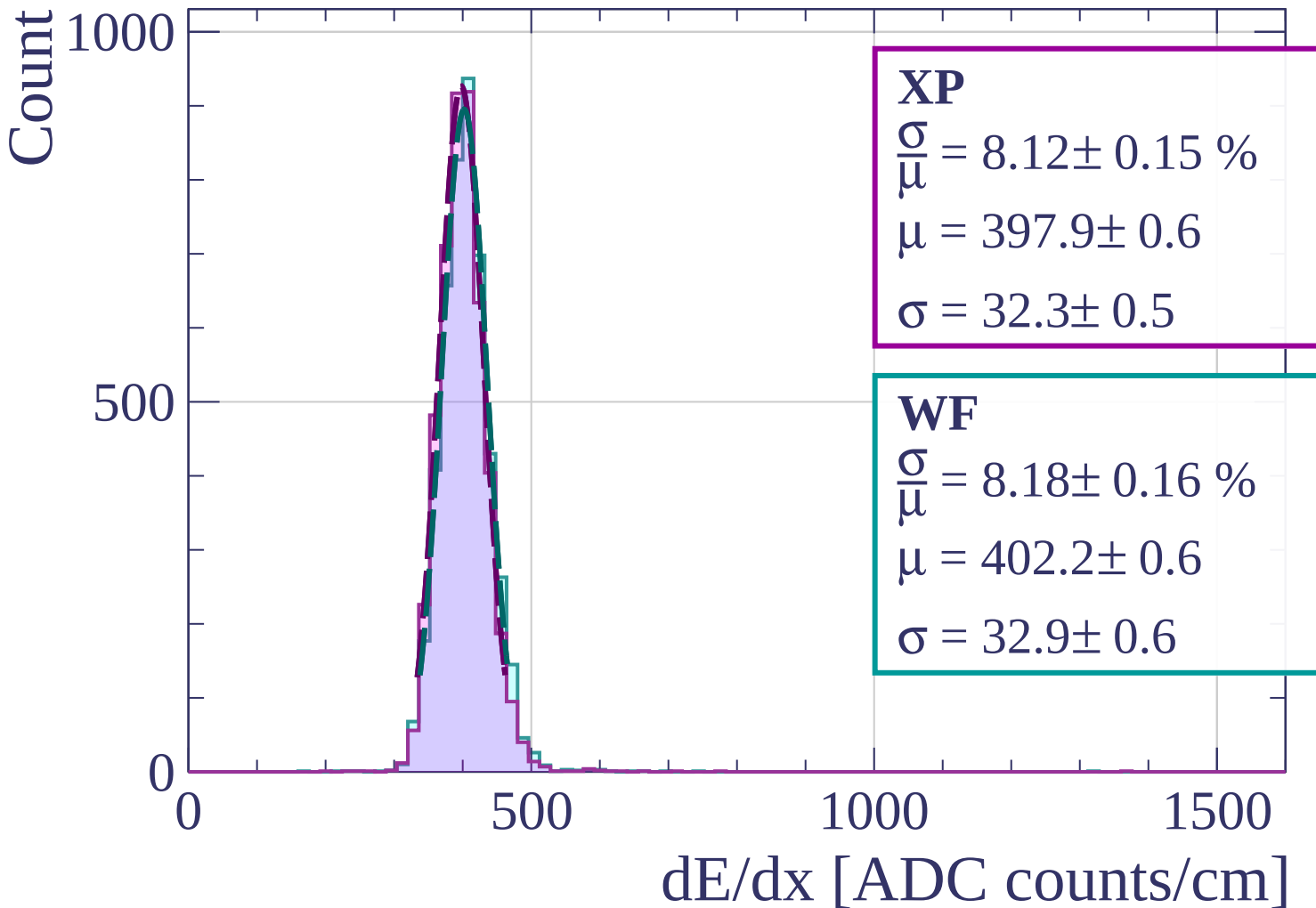
$$\sigma = 37.9 \pm 0.5$$

dE/dx [ADC counts/cm]

Energy loss | $3 < \theta < 6$

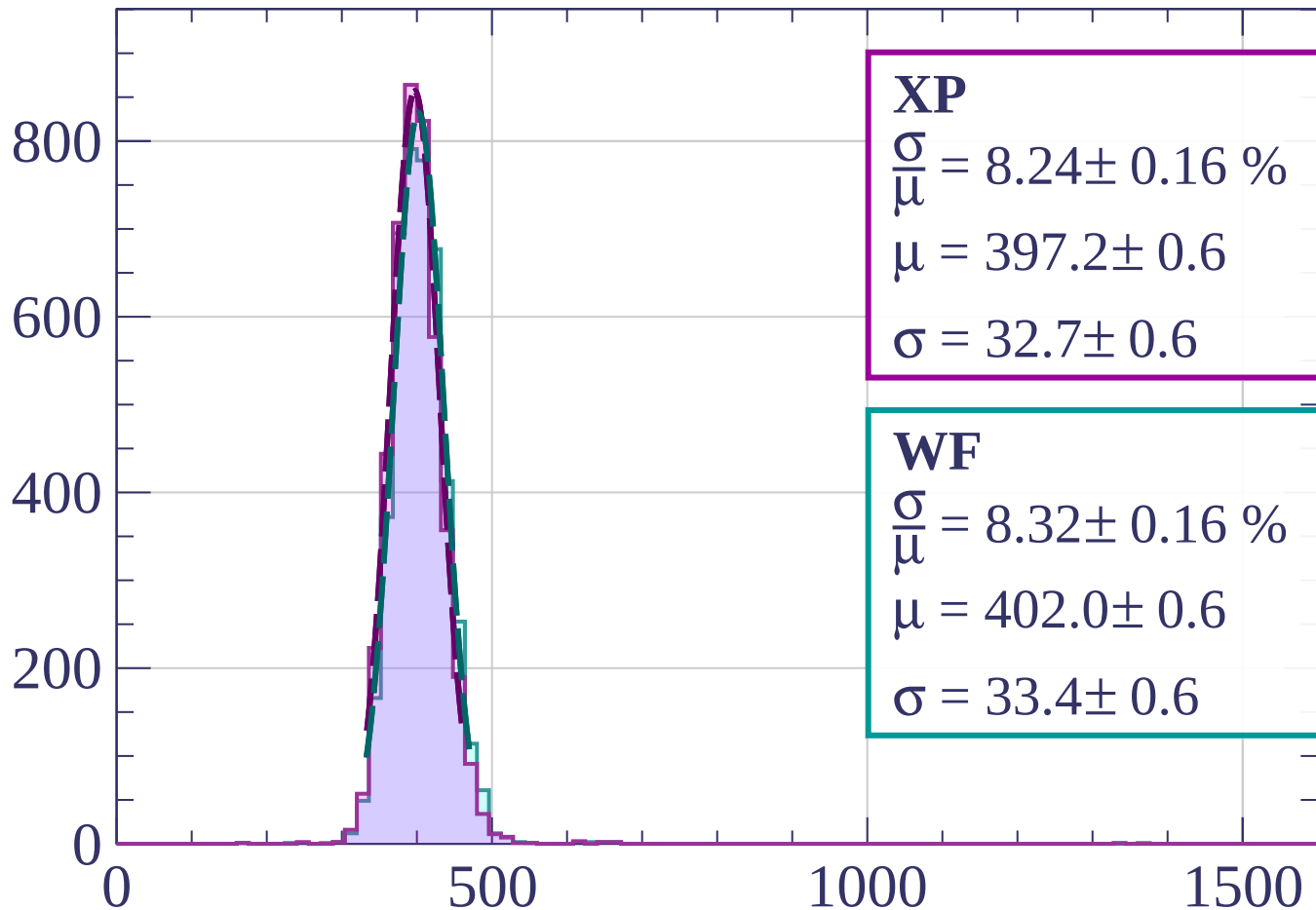


Energy loss | $6 < \theta < 9$



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 8.24 \pm 0.16 \%$$

$$\mu = 397.2 \pm 0.6$$

$$\sigma = 32.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.32 \pm 0.16 \%$$

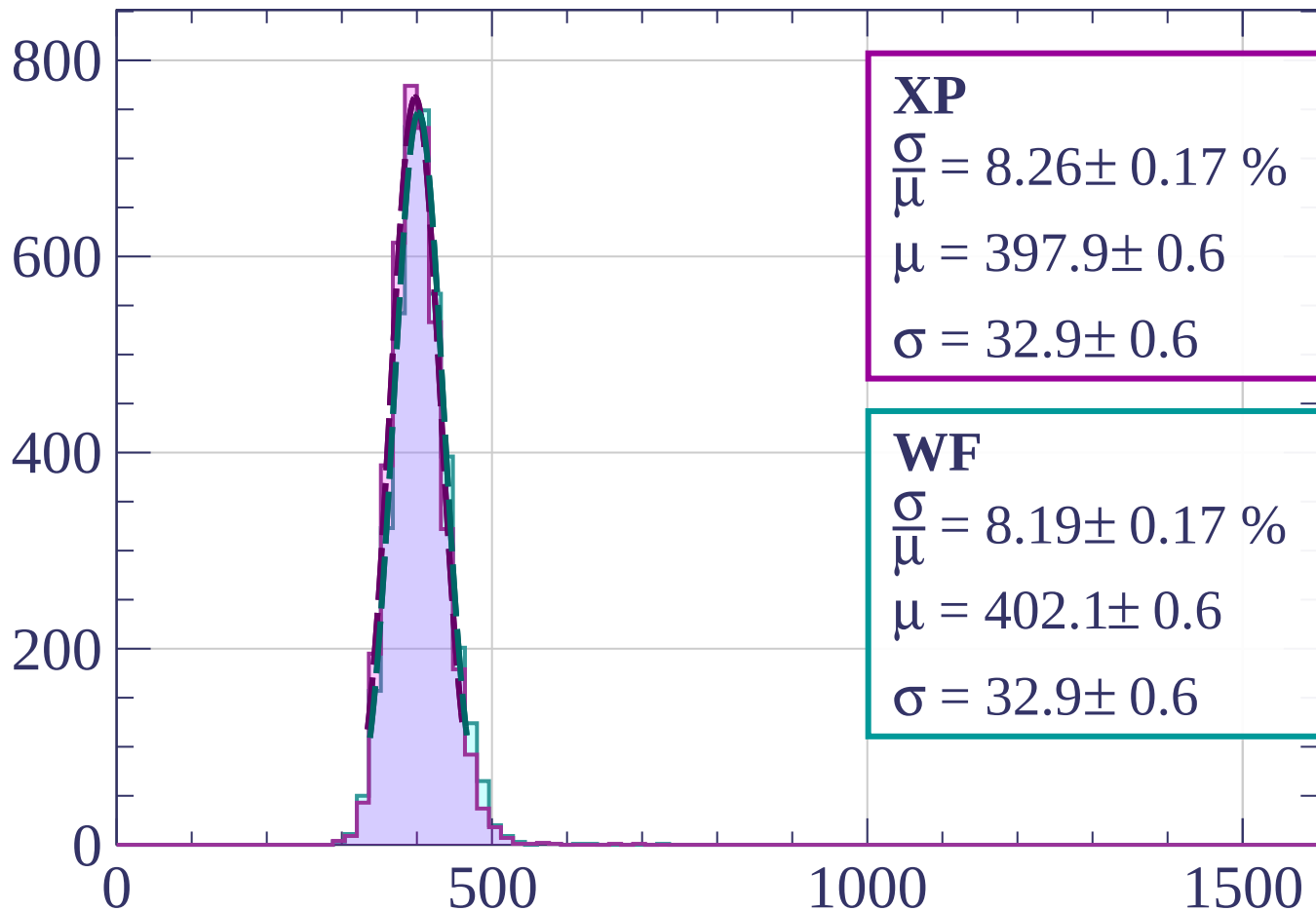
$$\mu = 402.0 \pm 0.6$$

$$\sigma = 33.4 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $12 < \theta < 15$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.17 \%$$

$$\mu = 397.9 \pm 0.6$$

$$\sigma = 32.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.19 \pm 0.17 \%$$

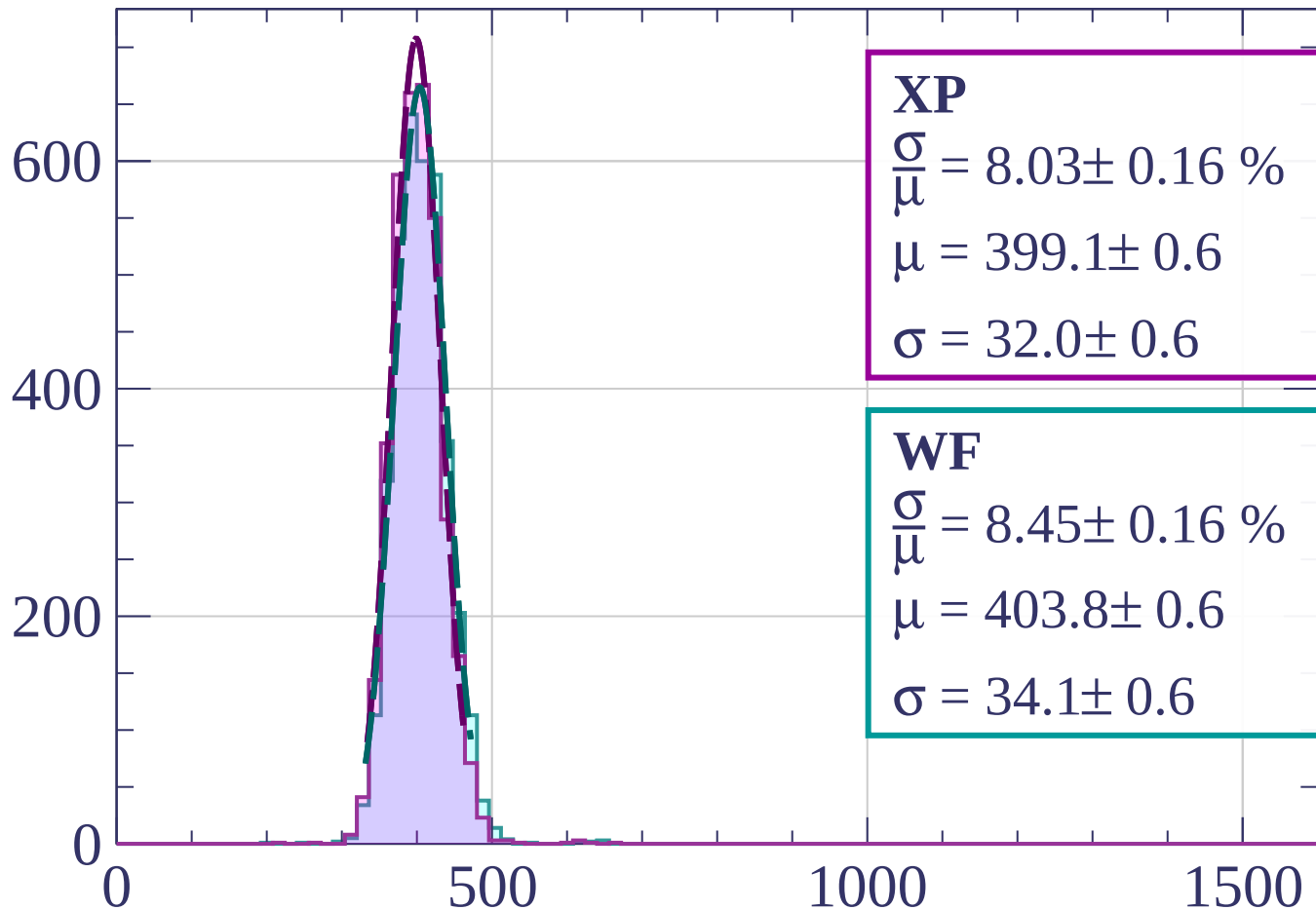
$$\mu = 402.1 \pm 0.6$$

$$\sigma = 32.9 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 8.03 \pm 0.16 \%$$

$$\mu = 399.1 \pm 0.6$$

$$\sigma = 32.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.45 \pm 0.16 \%$$

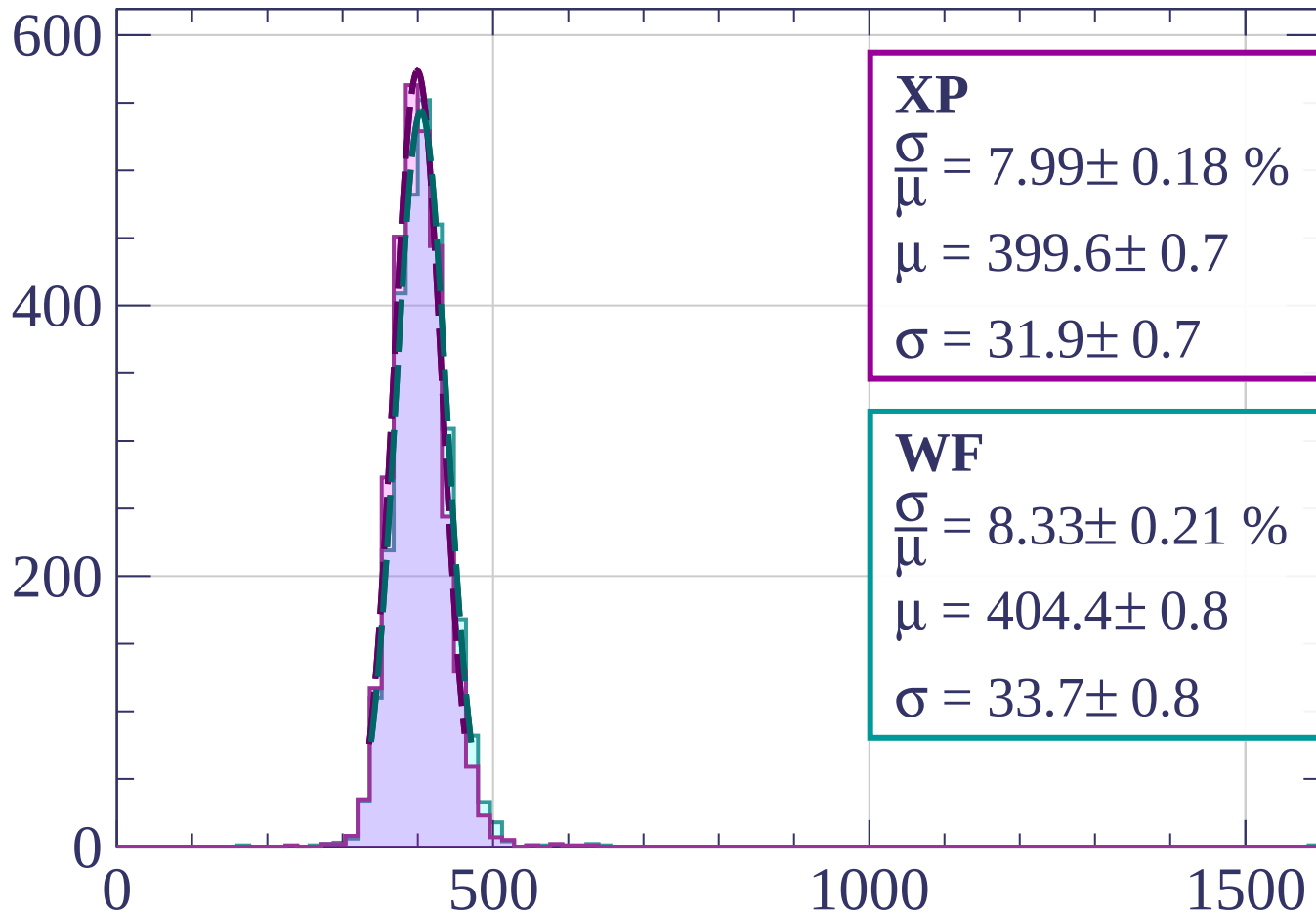
$$\mu = 403.8 \pm 0.6$$

$$\sigma = 34.1 \pm 0.6$$

dE/dx [ADC counts/cm]

Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.18 \%$$

$$\mu = 399.6 \pm 0.7$$

$$\sigma = 31.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.33 \pm 0.21 \%$$

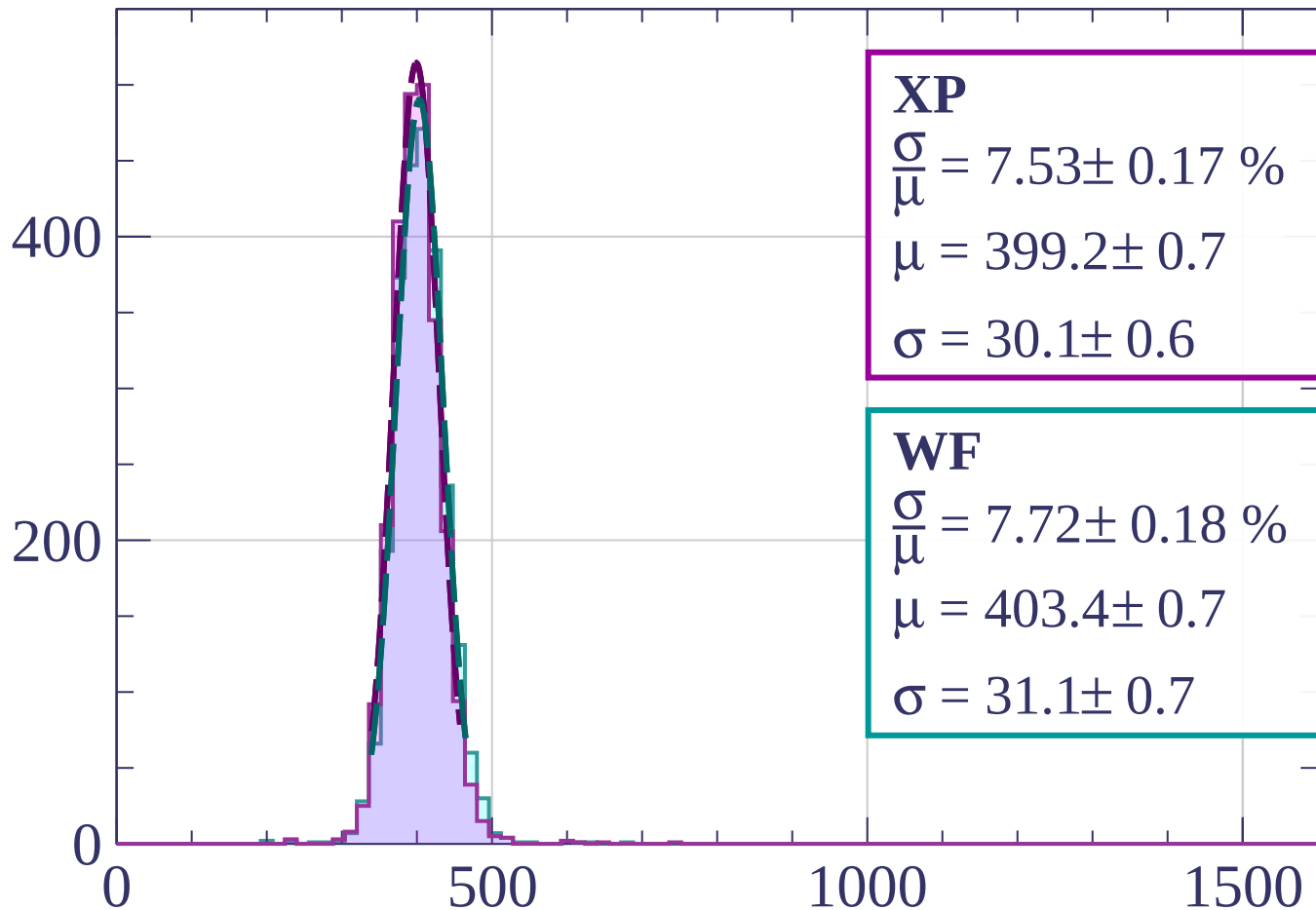
$$\mu = 404.4 \pm 0.8$$

$$\sigma = 33.7 \pm 0.8$$

dE/dx [ADC counts/cm]

Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.53 \pm 0.17 \%$$

$$\mu = 399.2 \pm 0.7$$

$$\sigma = 30.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.72 \pm 0.18 \%$$

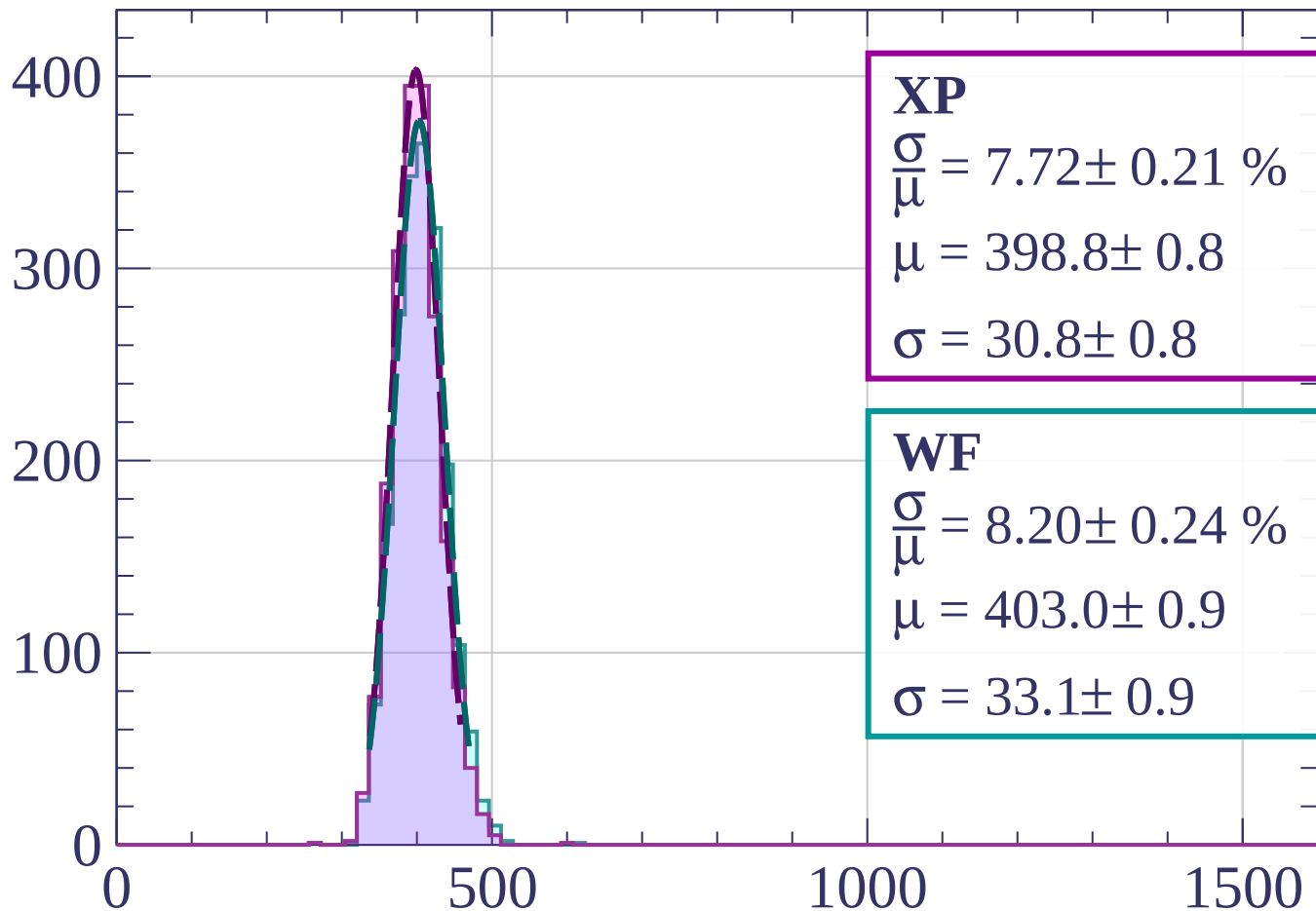
$$\mu = 403.4 \pm 0.7$$

$$\sigma = 31.1 \pm 0.7$$

dE/dx [ADC counts/cm]

Energy loss | $24 < \theta < 27$

Count



XP

$$\frac{\sigma}{\mu} = 7.72 \pm 0.21 \%$$

$$\mu = 398.8 \pm 0.8$$

$$\sigma = 30.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.20 \pm 0.24 \%$$

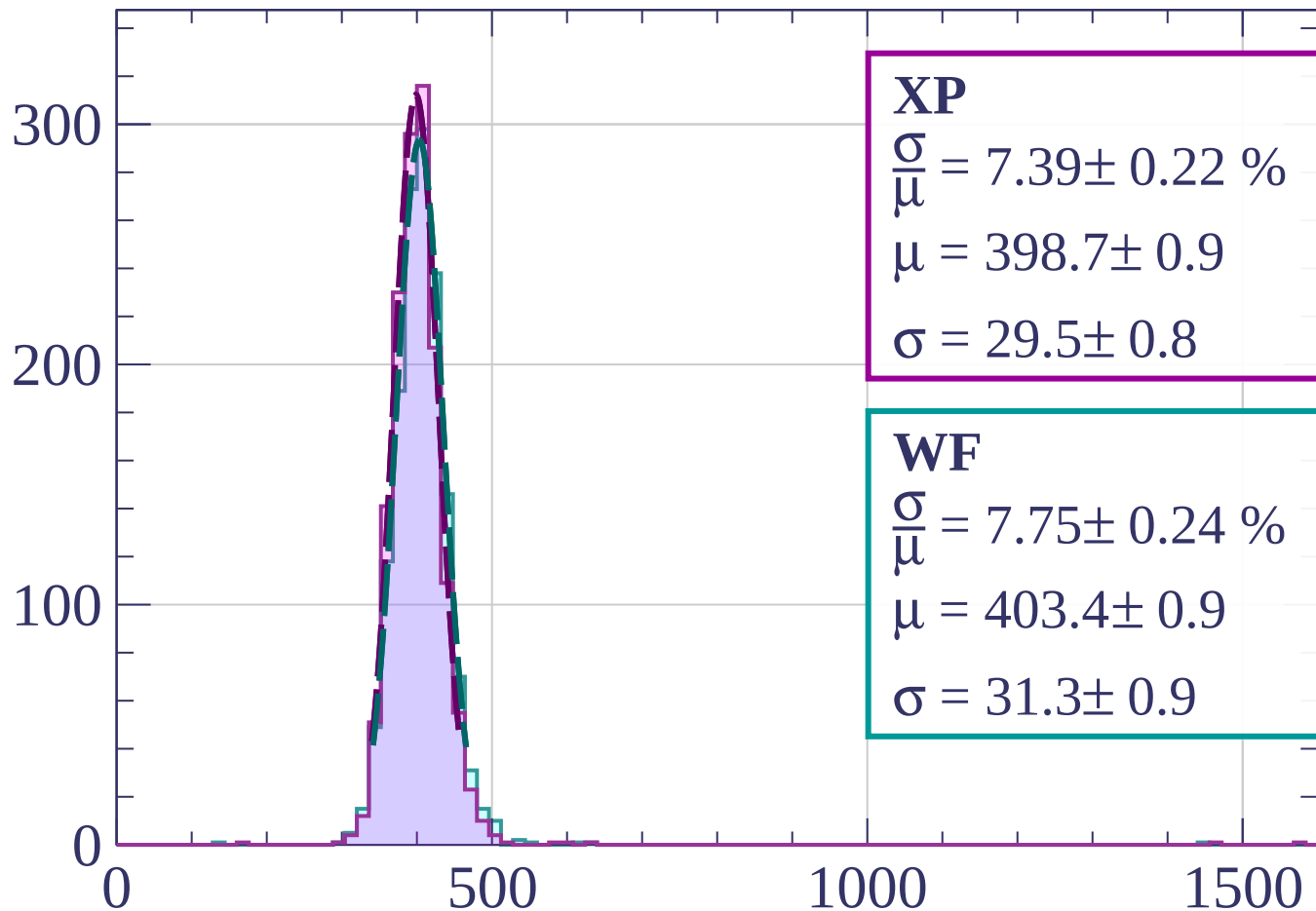
$$\mu = 403.0 \pm 0.9$$

$$\sigma = 33.1 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 7.39 \pm 0.22 \%$$

$$\mu = 398.7 \pm 0.9$$

$$\sigma = 29.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.75 \pm 0.24 \%$$

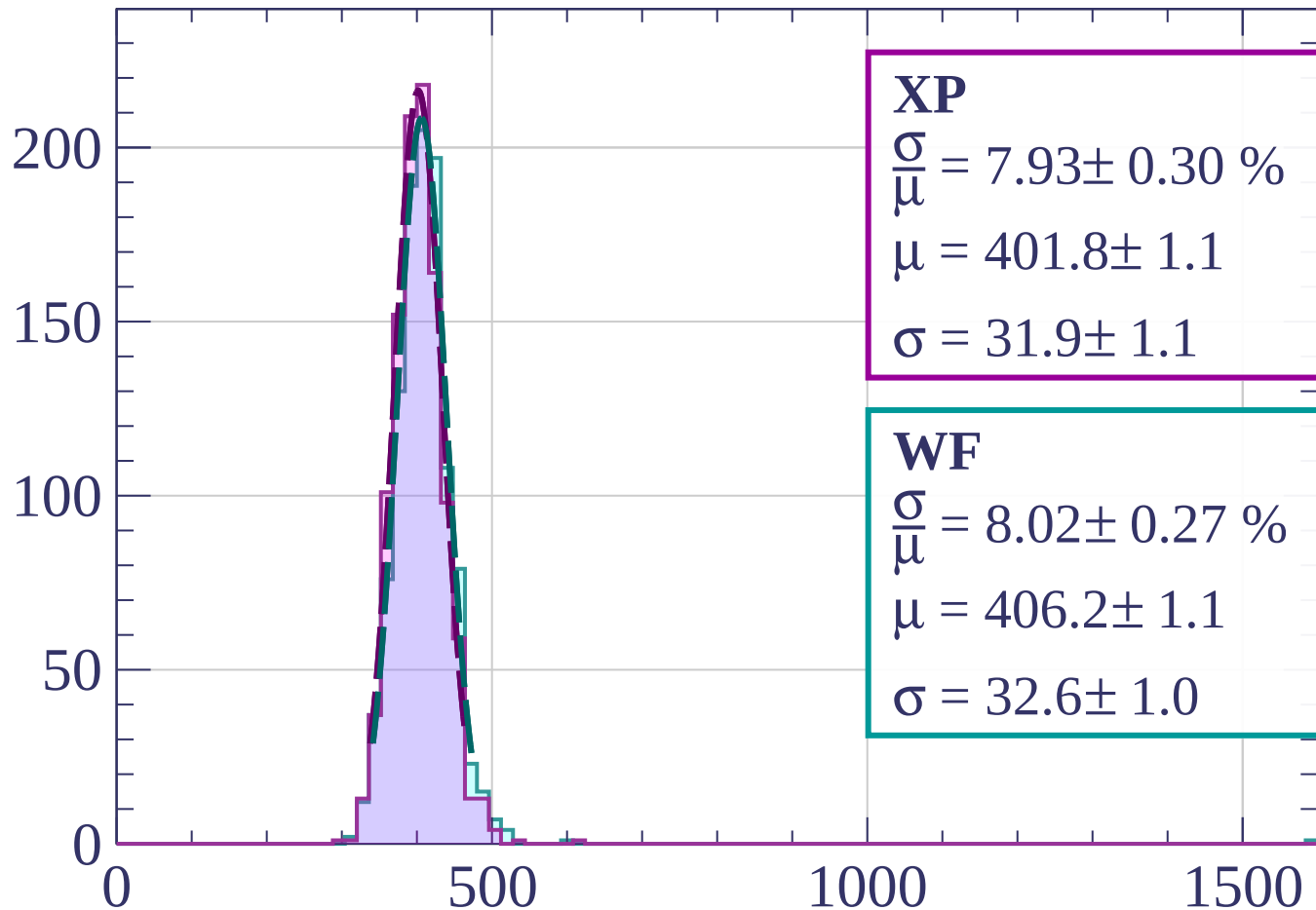
$$\mu = 403.4 \pm 0.9$$

$$\sigma = 31.3 \pm 0.9$$

dE/dx [ADC counts/cm]

Energy loss | $30 < \theta < 33$

Count



XP

$$\frac{\sigma}{\mu} = 7.93 \pm 0.30 \%$$

$$\mu = 401.8 \pm 1.1$$

$$\sigma = 31.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.02 \pm 0.27 \%$$

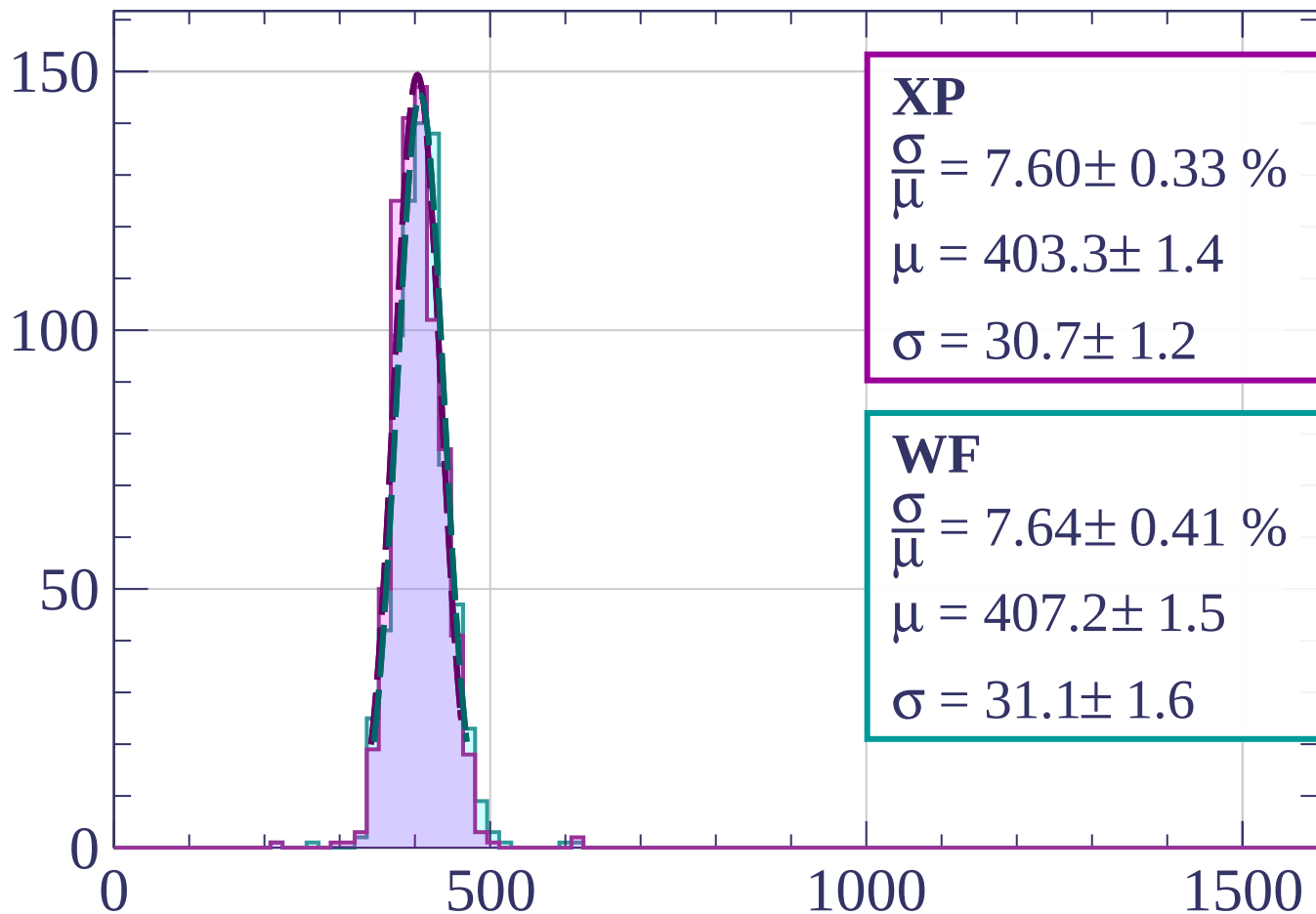
$$\mu = 406.2 \pm 1.1$$

$$\sigma = 32.6 \pm 1.0$$

dE/dx [ADC counts/cm]

Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 7.60 \pm 0.33 \%$$

$$\mu = 403.3 \pm 1.4$$

$$\sigma = 30.7 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 7.64 \pm 0.41 \%$$

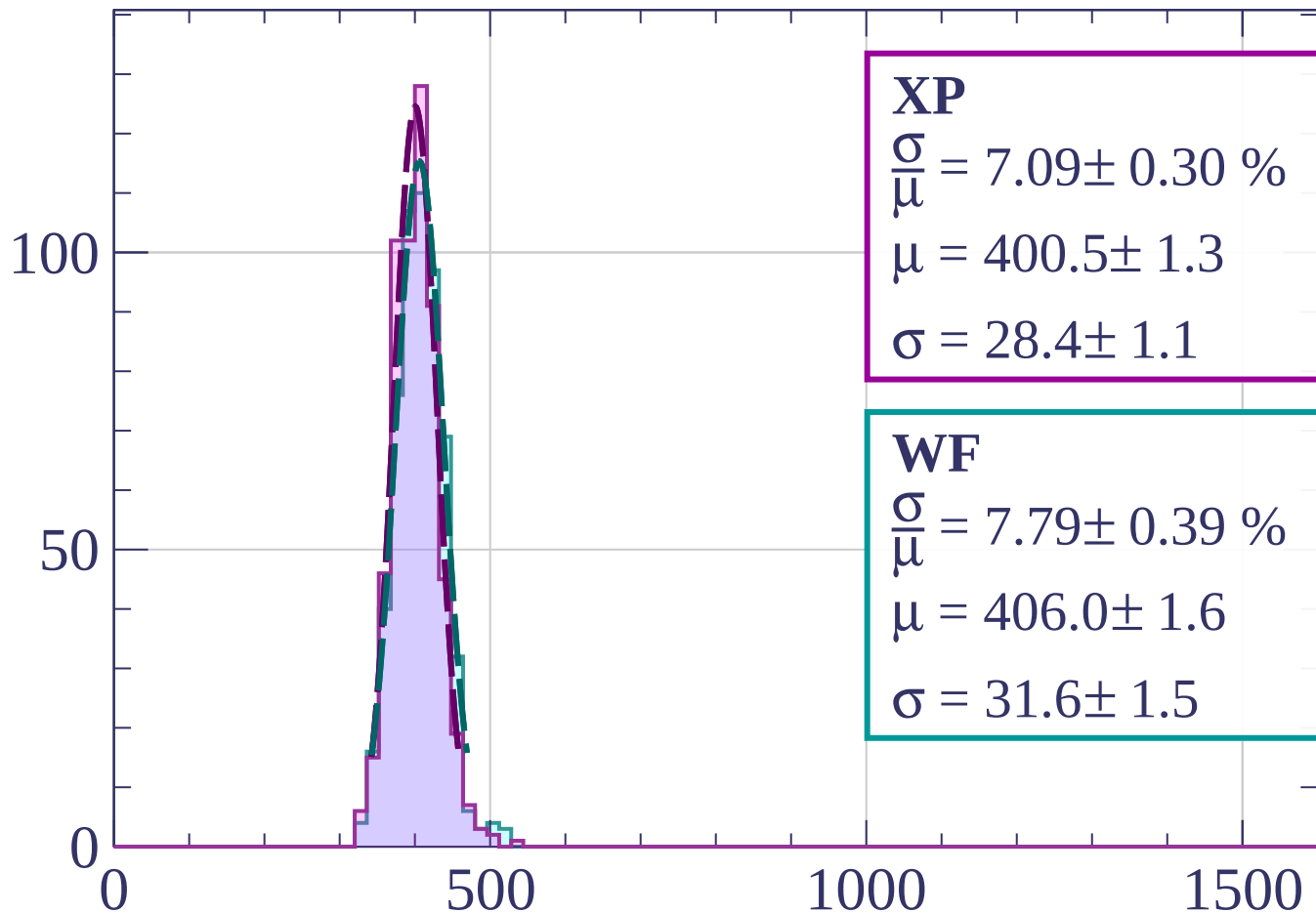
$$\mu = 407.2 \pm 1.5$$

$$\sigma = 31.1 \pm 1.6$$

dE/dx [ADC counts/cm]

Energy loss | $36 < \theta < 39$

Count



XP

$$\frac{\sigma}{\mu} = 7.09 \pm 0.30 \%$$

$$\mu = 400.5 \pm 1.3$$

$$\sigma = 28.4 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.79 \pm 0.39 \%$$

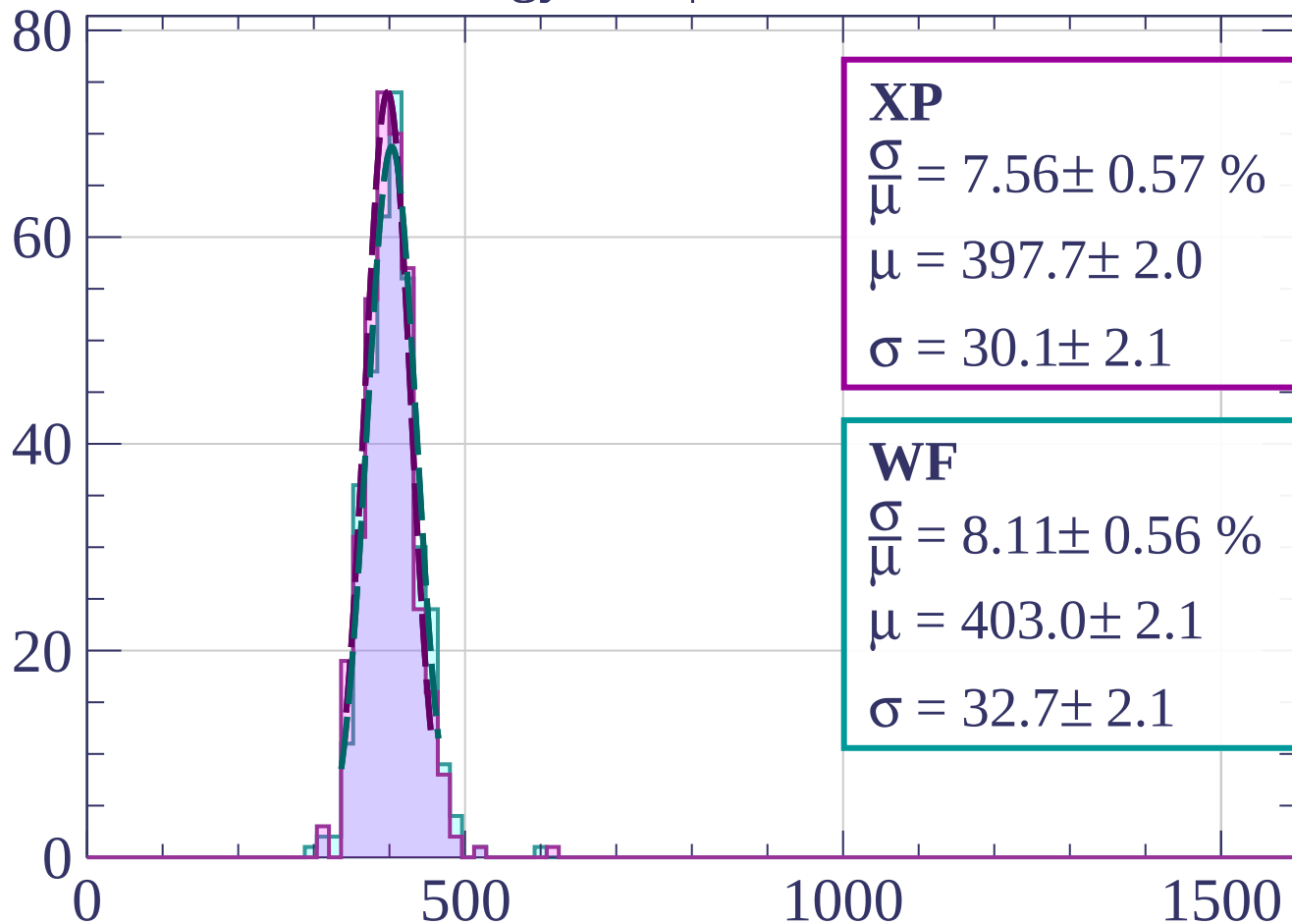
$$\mu = 406.0 \pm 1.6$$

$$\sigma = 31.6 \pm 1.5$$

dE/dx [ADC counts/cm]

Energy loss | $39 < \theta < 42$

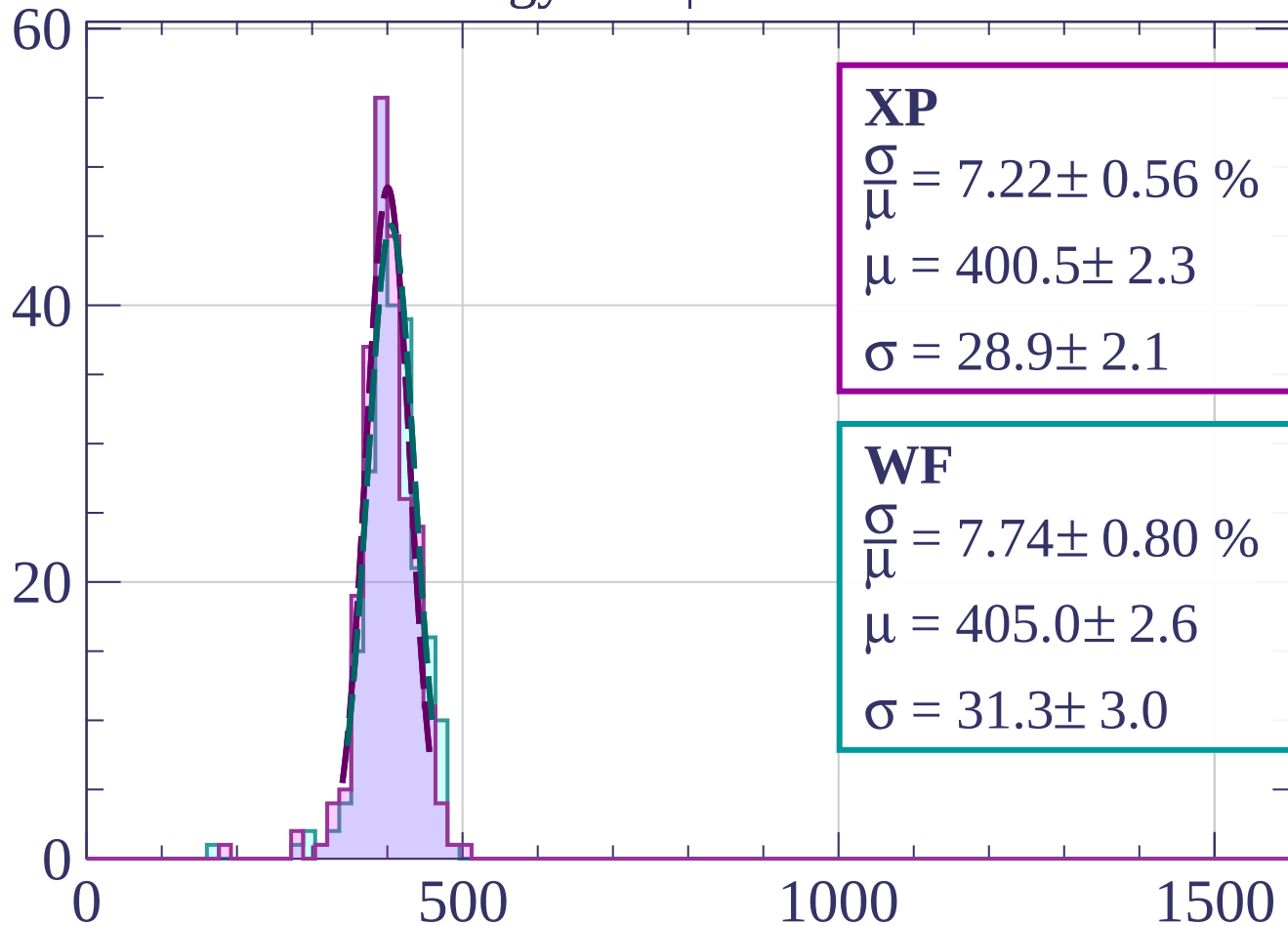
Count



dE/dx [ADC counts/cm]

Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 7.22 \pm 0.56 \%$$

$$\mu = 400.5 \pm 2.3$$

$$\sigma = 28.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 7.74 \pm 0.80 \%$$

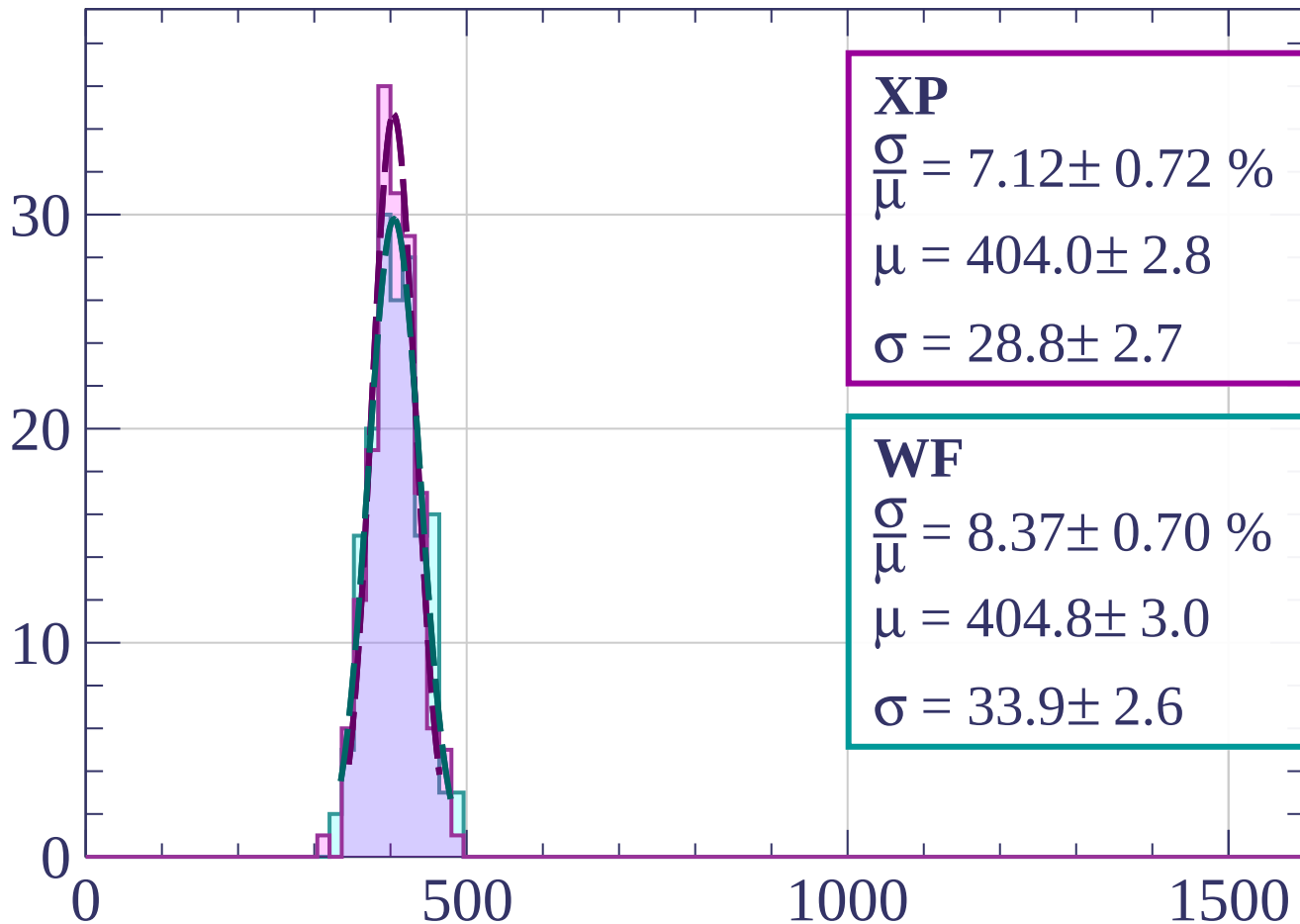
$$\mu = 405.0 \pm 2.6$$

$$\sigma = 31.3 \pm 3.0$$

dE/dx [ADC counts/cm]

Energy loss | $45 < \theta < 48$

Count



XP

$$\frac{\sigma}{\mu} = 7.12 \pm 0.72 \%$$

$$\mu = 404.0 \pm 2.8$$

$$\sigma = 28.8 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 8.37 \pm 0.70 \%$$

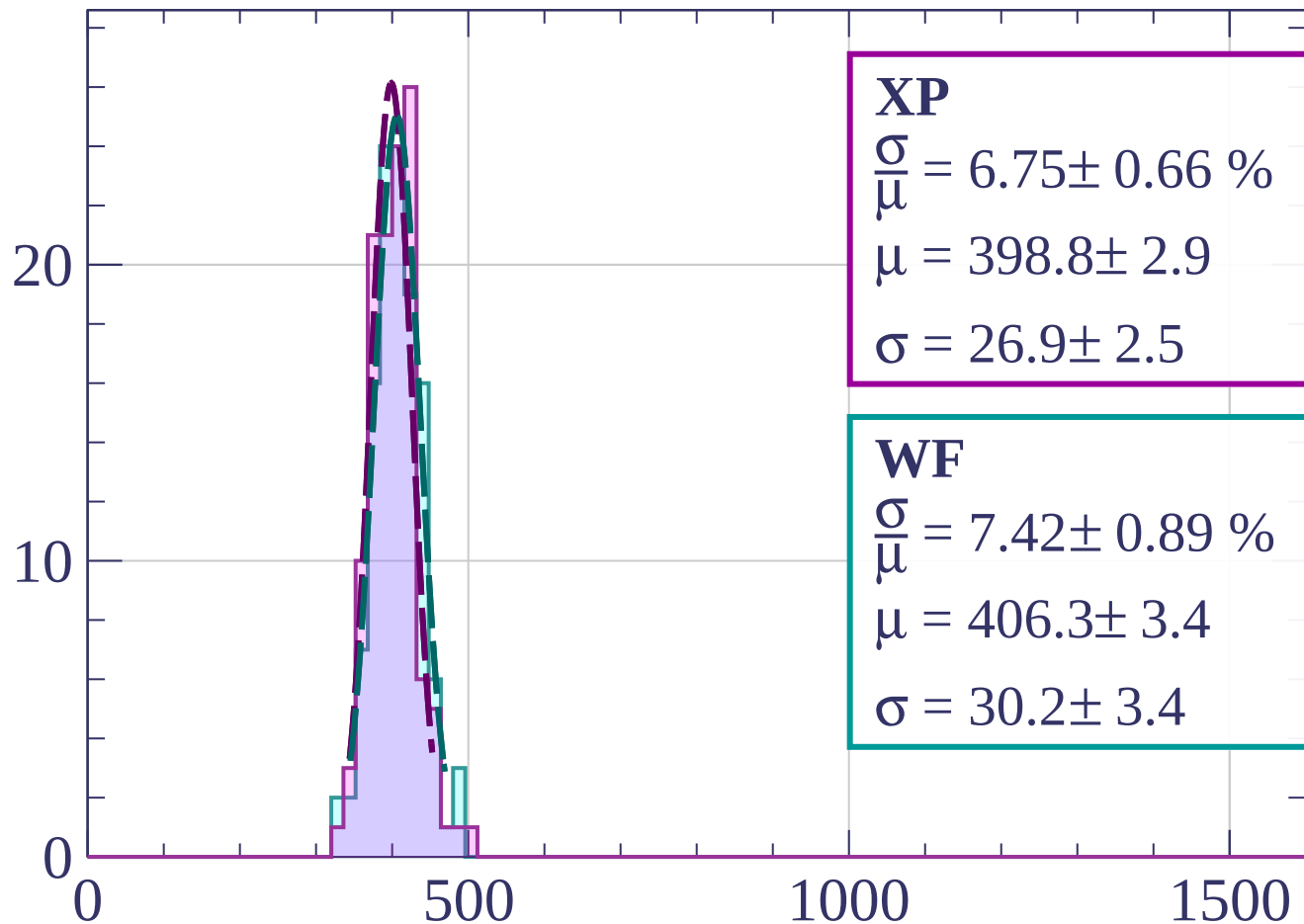
$$\mu = 404.8 \pm 3.0$$

$$\sigma = 33.9 \pm 2.6$$

dE/dx [ADC counts/cm]

Energy loss | $48 < \theta < 51$

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.66 \%$$

$$\mu = 398.8 \pm 2.9$$

$$\sigma = 26.9 \pm 2.5$$

WF

$$\frac{\sigma}{\mu} = 7.42 \pm 0.89 \%$$

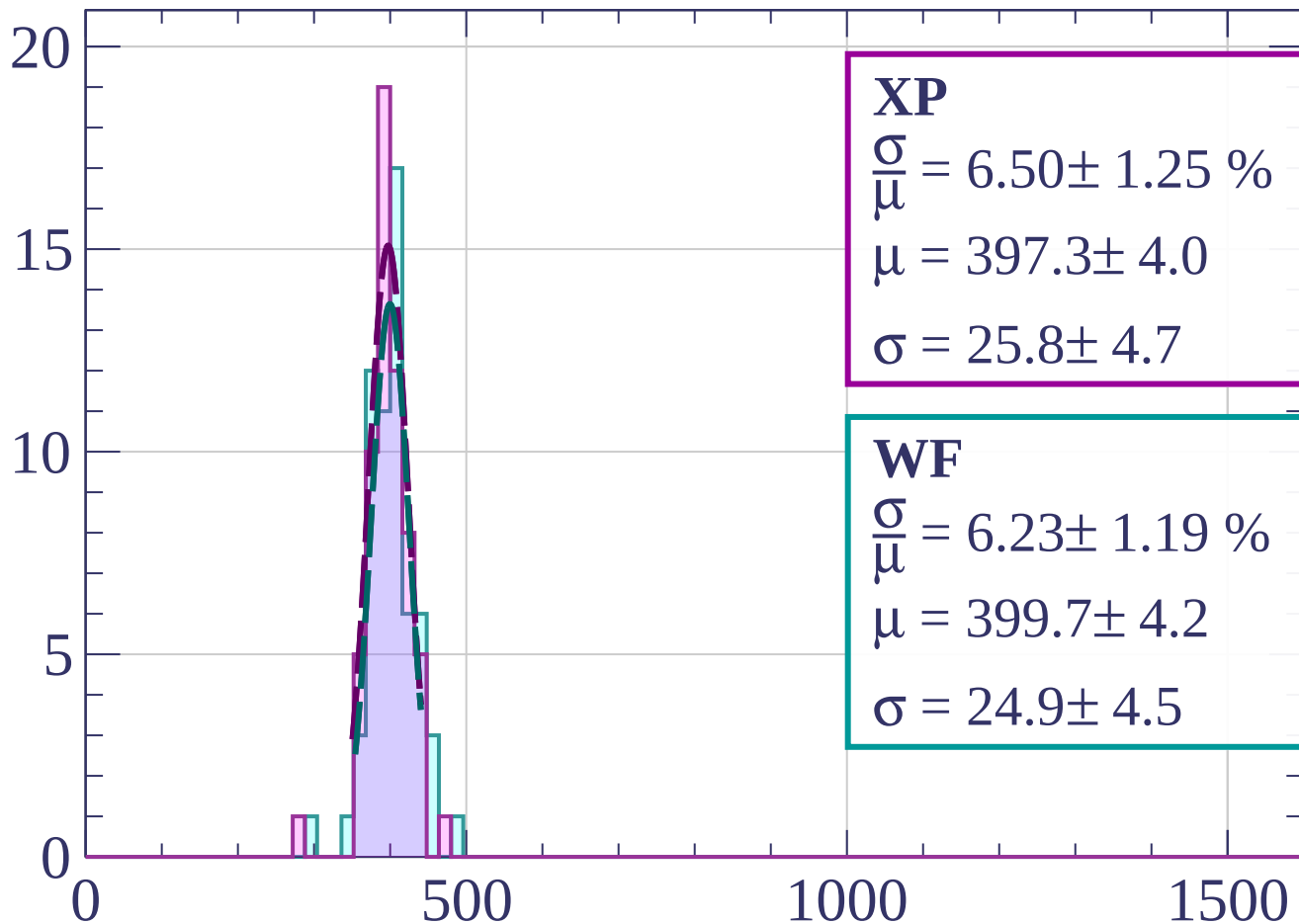
$$\mu = 406.3 \pm 3.4$$

$$\sigma = 30.2 \pm 3.4$$

dE/dx [ADC counts/cm]

Energy loss | $51 < \theta < 54$

Count



XP

$$\frac{\sigma}{\mu} = 6.50 \pm 1.25 \%$$

$$\mu = 397.3 \pm 4.0$$

$$\sigma = 25.8 \pm 4.7$$

WF

$$\frac{\sigma}{\mu} = 6.23 \pm 1.19 \%$$

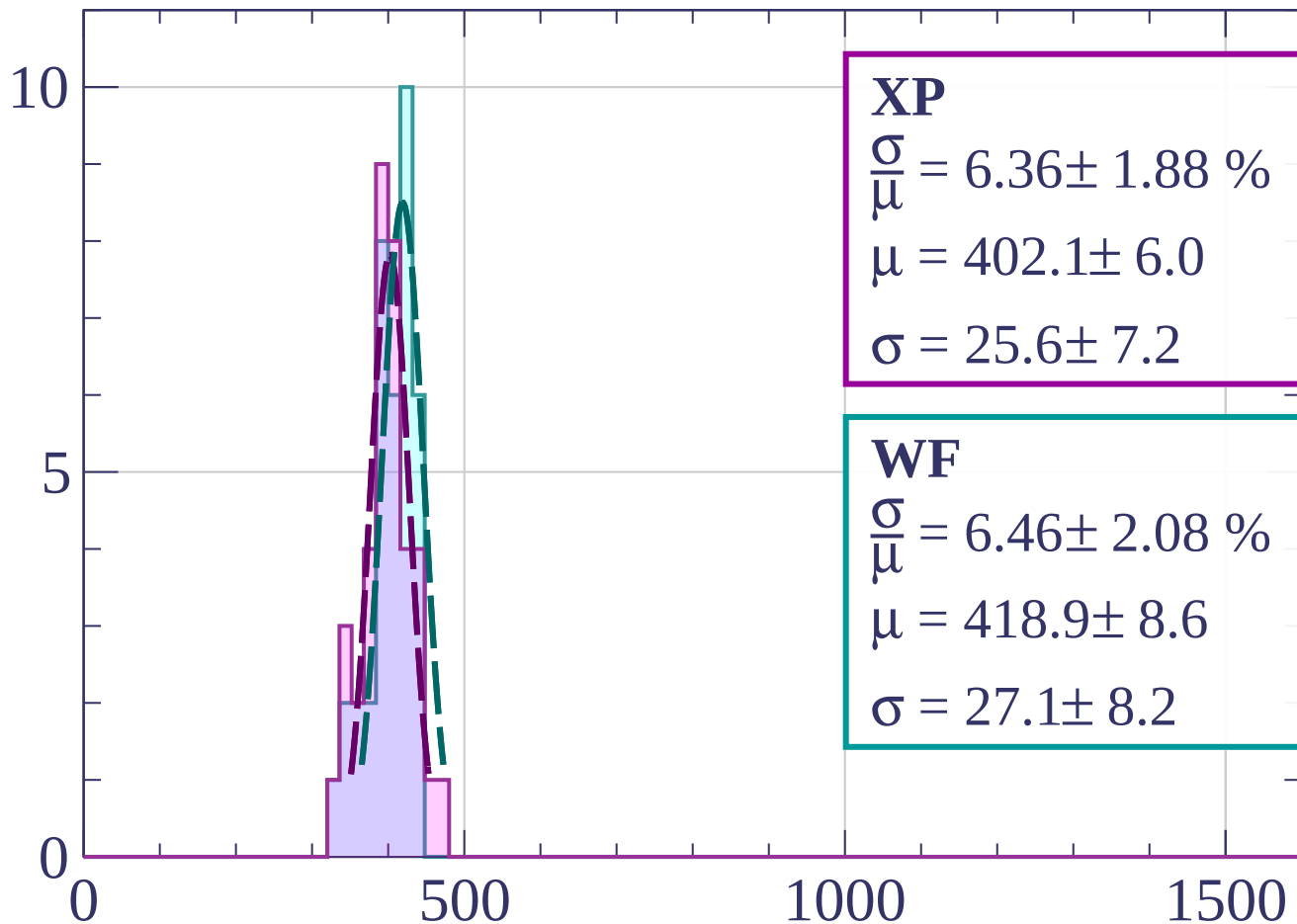
$$\mu = 399.7 \pm 4.2$$

$$\sigma = 24.9 \pm 4.5$$

dE/dx [ADC counts/cm]

Energy loss | $54 < \theta < 57$

Count



XP

$$\frac{\sigma}{\mu} = 6.36 \pm 1.88 \%$$

$$\mu = 402.1 \pm 6.0$$

$$\sigma = 25.6 \pm 7.2$$

WF

$$\frac{\sigma}{\mu} = 6.46 \pm 2.08 \%$$

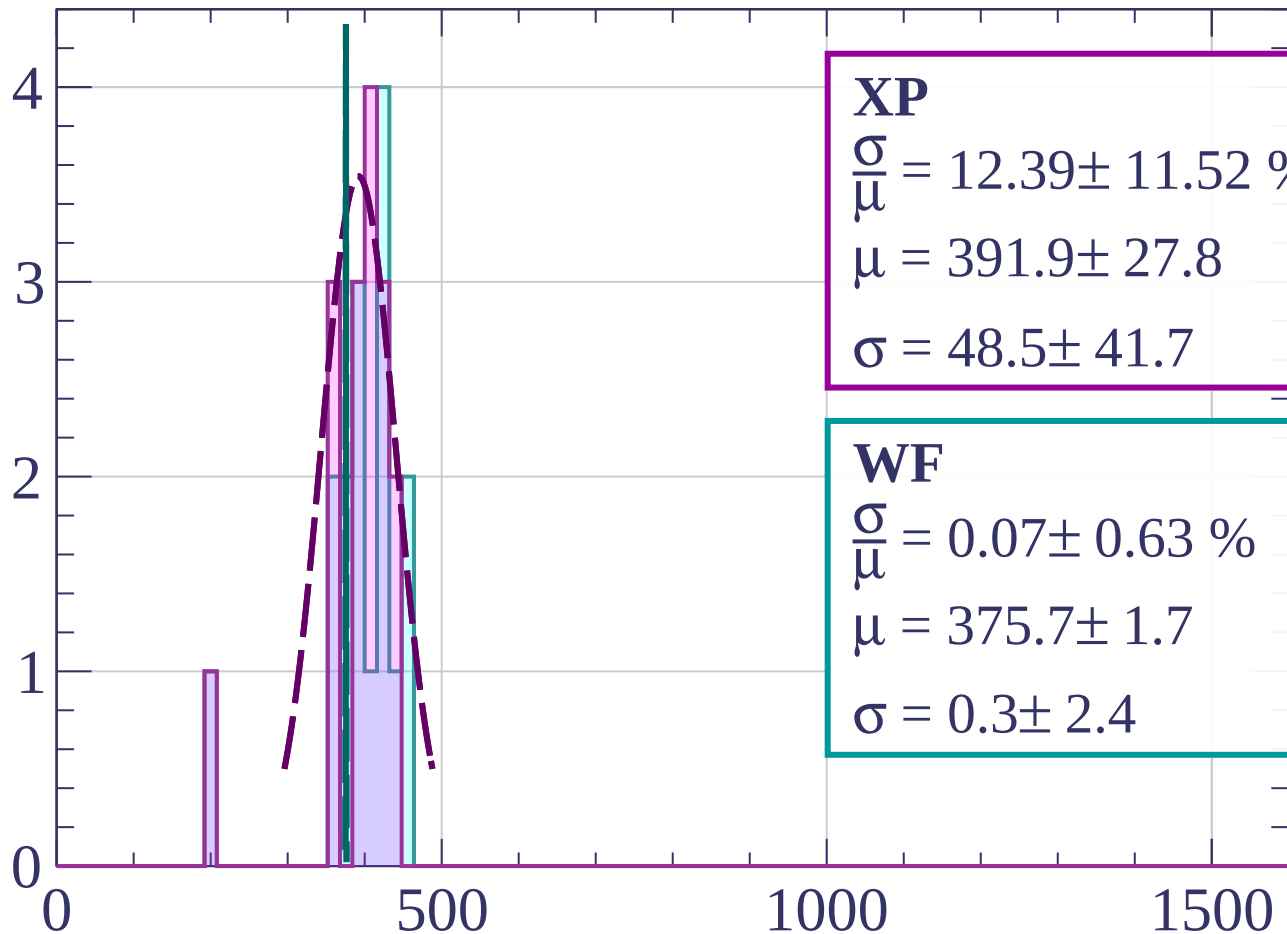
$$\mu = 418.9 \pm 8.6$$

$$\sigma = 27.1 \pm 8.2$$

dE/dx [ADC counts/cm]

Energy loss | $57 < \theta < 60$

Count



XP

$$\frac{\sigma}{\mu} = 12.39 \pm 11.52 \%$$

$$\mu = 391.9 \pm 27.8$$

$$\sigma = 48.5 \pm 41.7$$

WF

$$\frac{\sigma}{\mu} = 0.07 \pm 0.63 \%$$

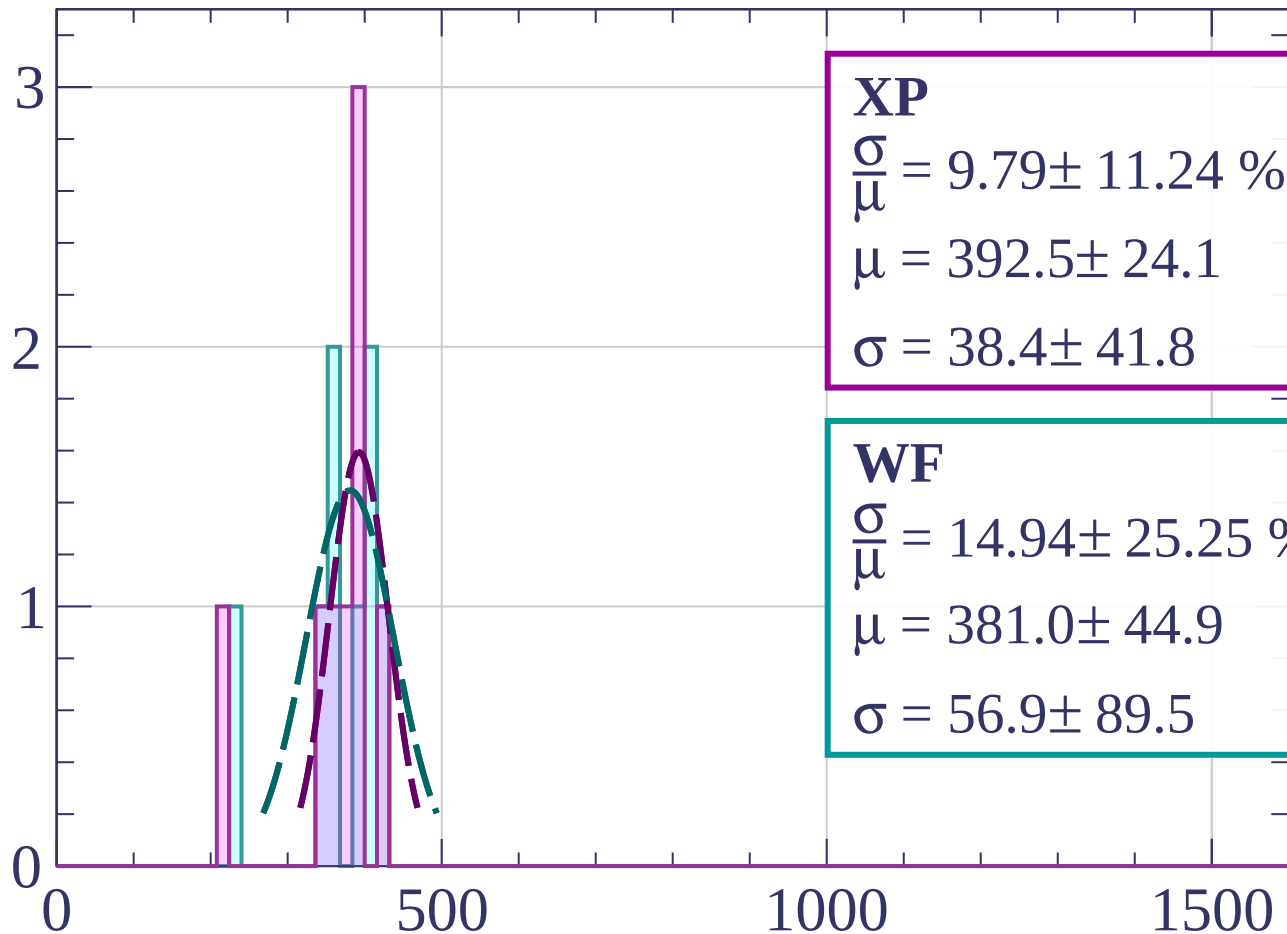
$$\mu = 375.7 \pm 1.7$$

$$\sigma = 0.3 \pm 2.4$$

dE/dx [ADC counts/cm]

Energy loss | $60 < \theta < 63$

Count



XP

$$\frac{\sigma}{\mu} = 9.79 \pm 11.24 \%$$

$$\mu = 392.5 \pm 24.1$$

$$\sigma = 38.4 \pm 41.8$$

WF

$$\frac{\sigma}{\mu} = 14.94 \pm 25.25 \%$$

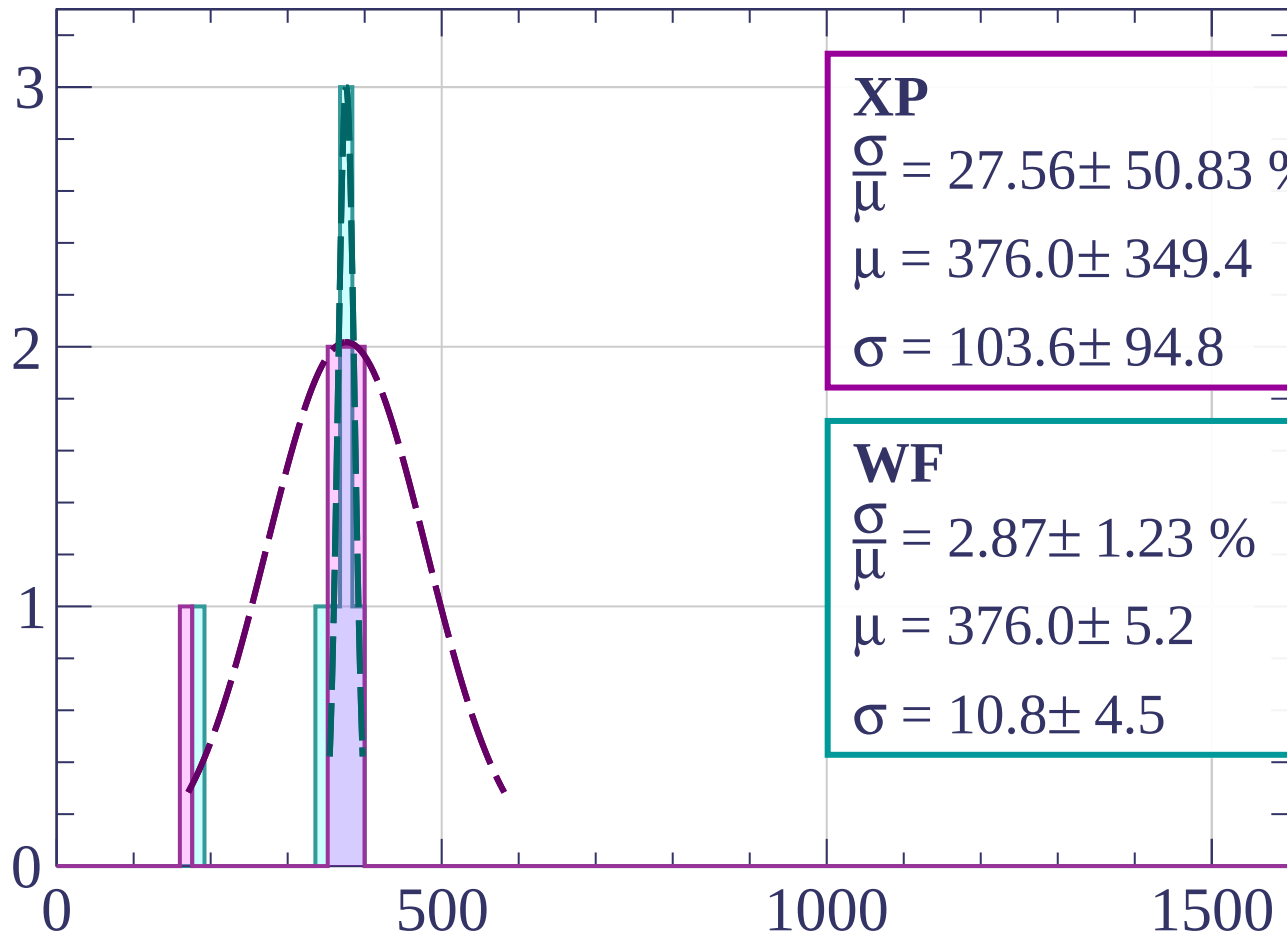
$$\mu = 381.0 \pm 44.9$$

$$\sigma = 56.9 \pm 89.5$$

dE/dx [ADC counts/cm]

Energy loss | $63 < \theta < 66$

Count



XP

$$\frac{\sigma}{\mu} = 27.56 \pm 50.83 \%$$

$$\mu = 376.0 \pm 349.4$$

$$\sigma = 103.6 \pm 94.8$$

WF

$$\frac{\sigma}{\mu} = 2.87 \pm 1.23 \%$$

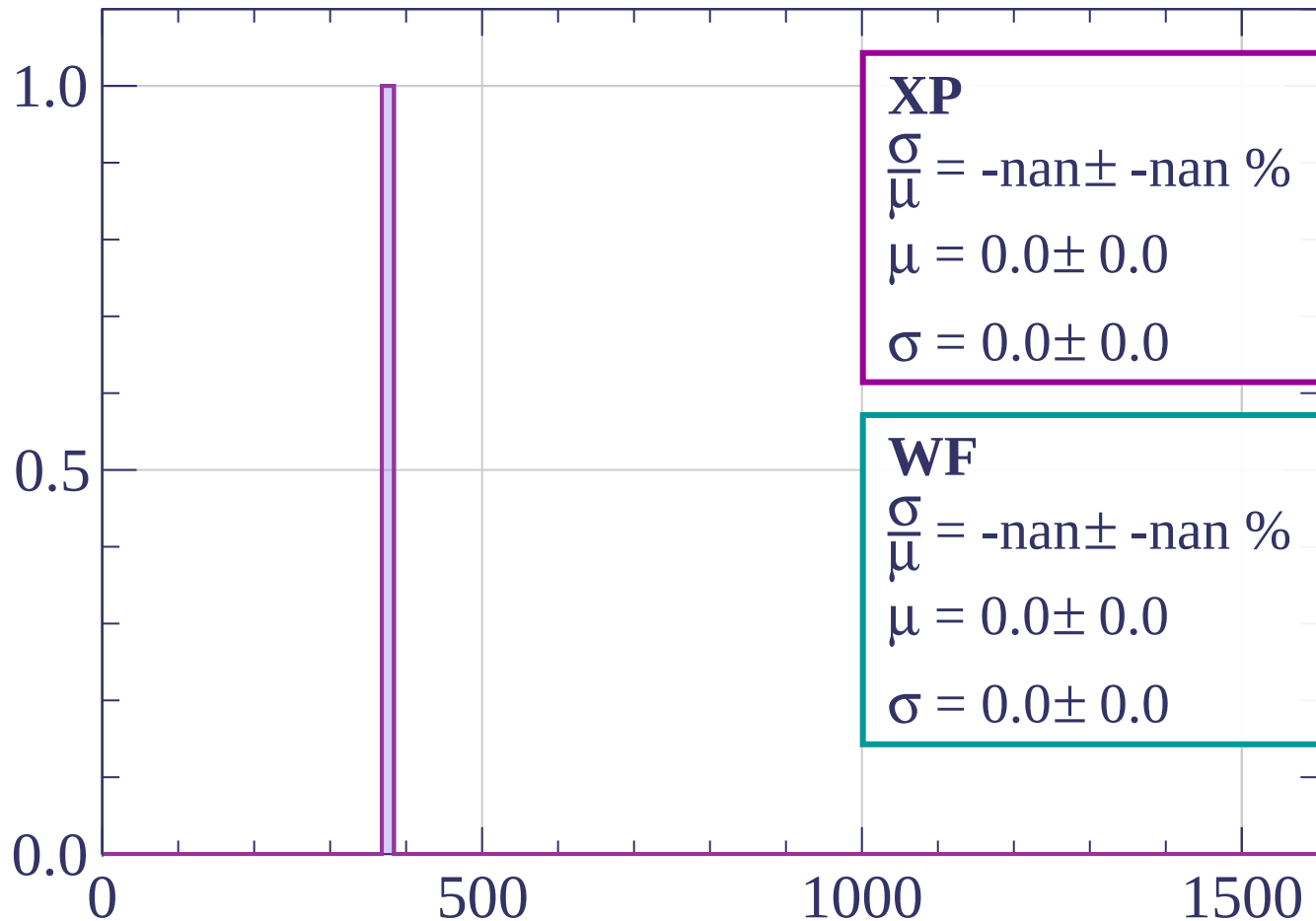
$$\mu = 376.0 \pm 5.2$$

$$\sigma = 10.8 \pm 4.5$$

dE/dx [ADC counts/cm]

Energy loss | $66 < \theta < 69$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

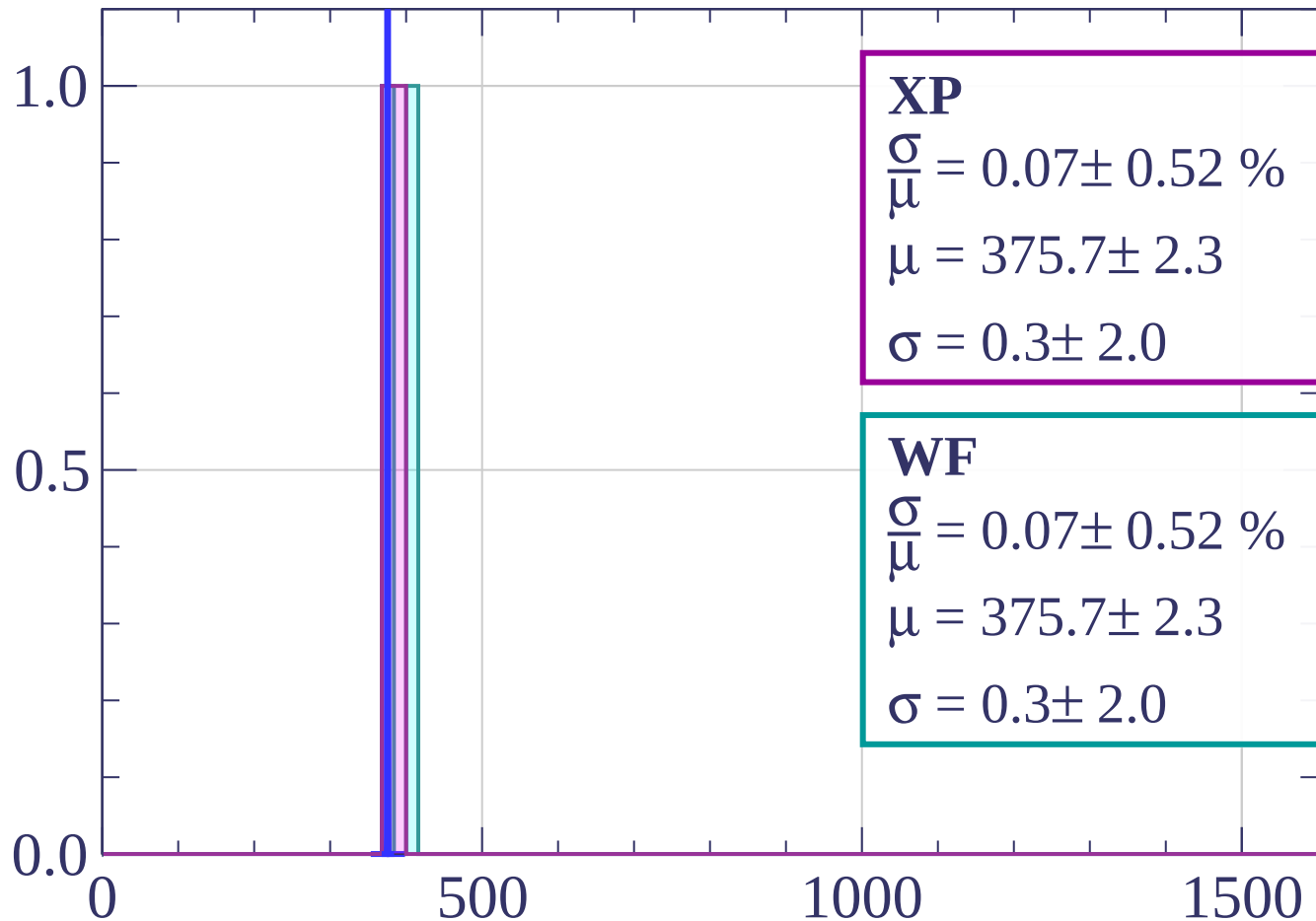
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

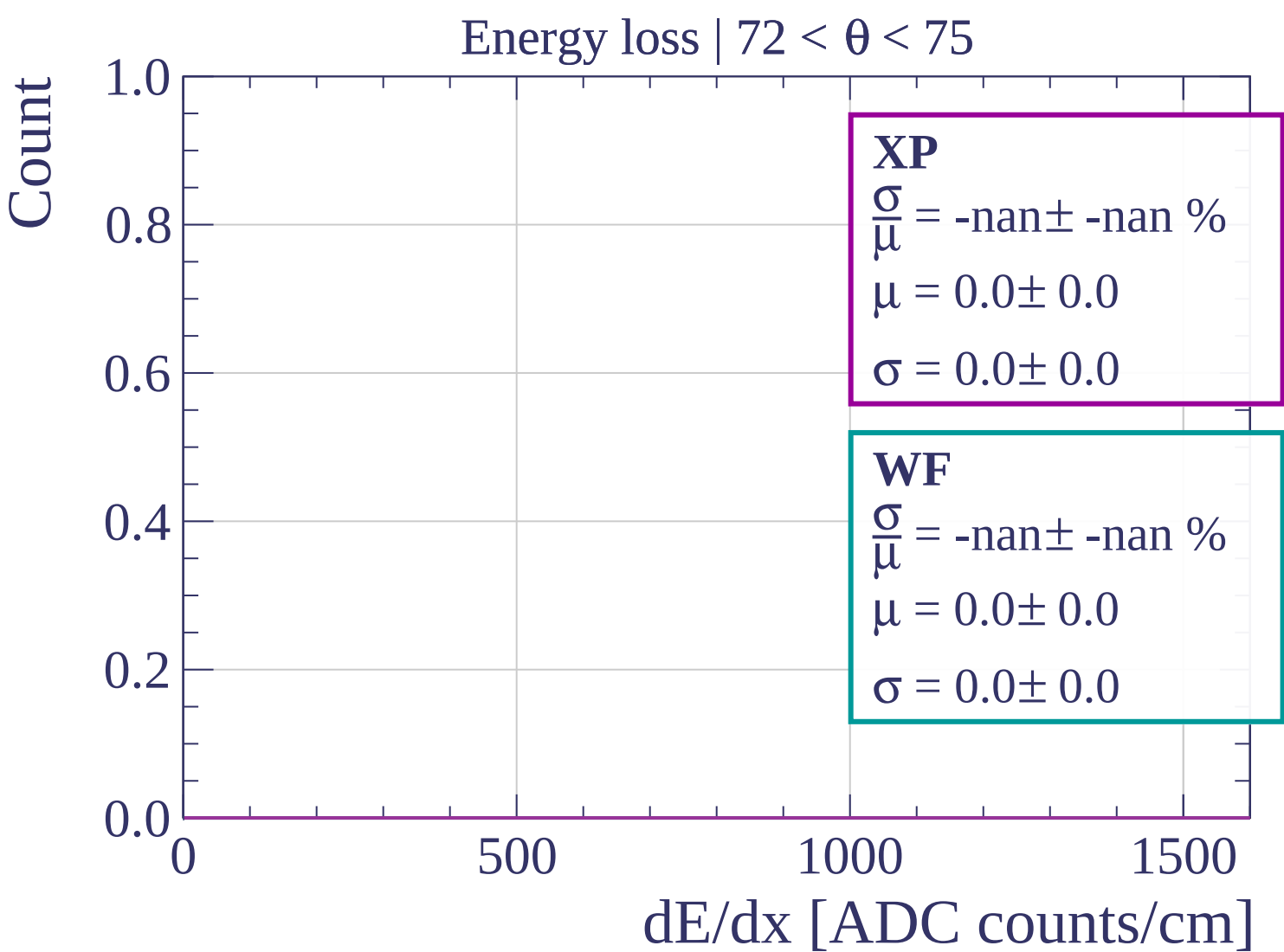
dE/dx [ADC counts/cm]

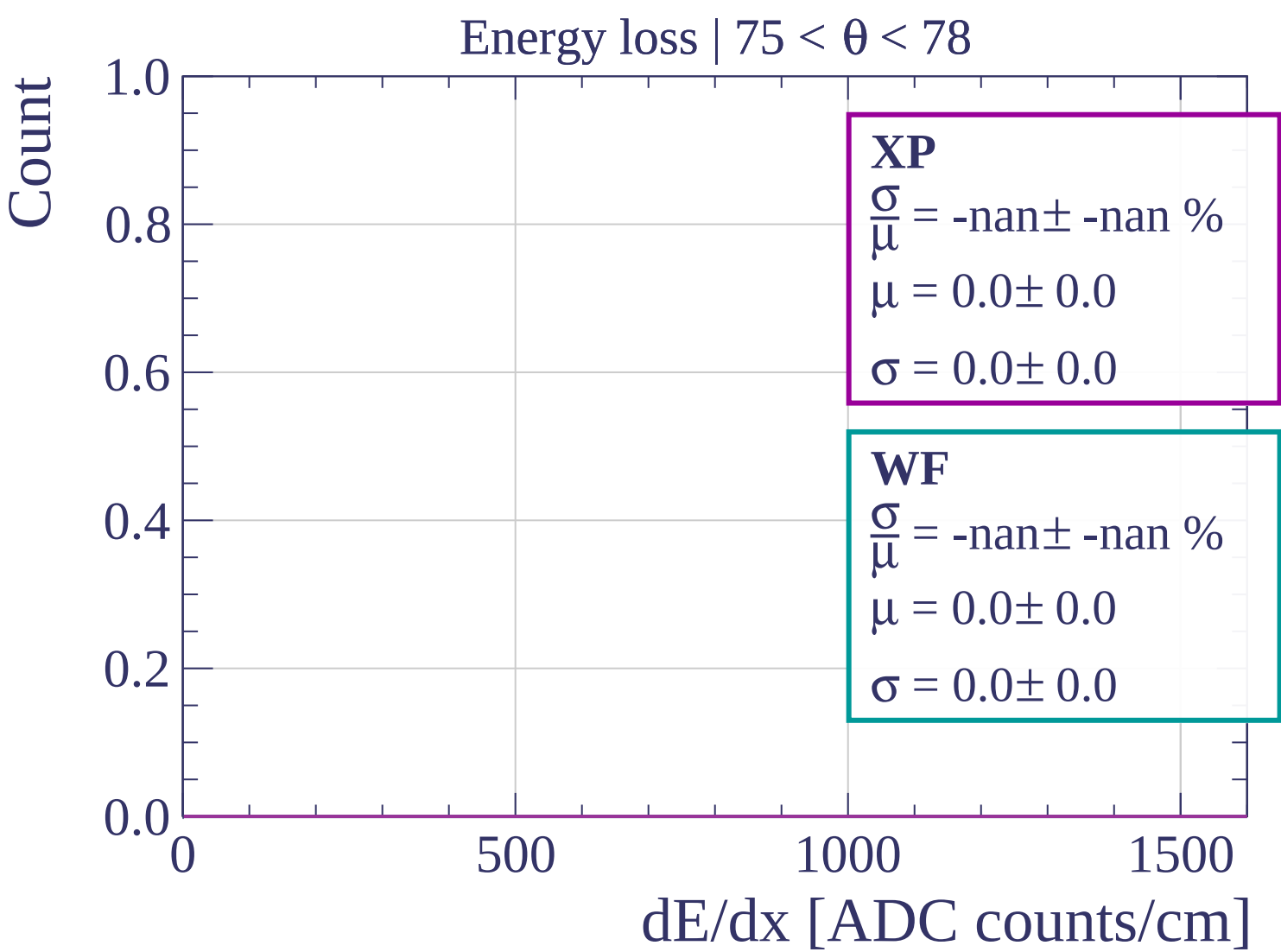
Energy loss | $69 < \theta < 72$

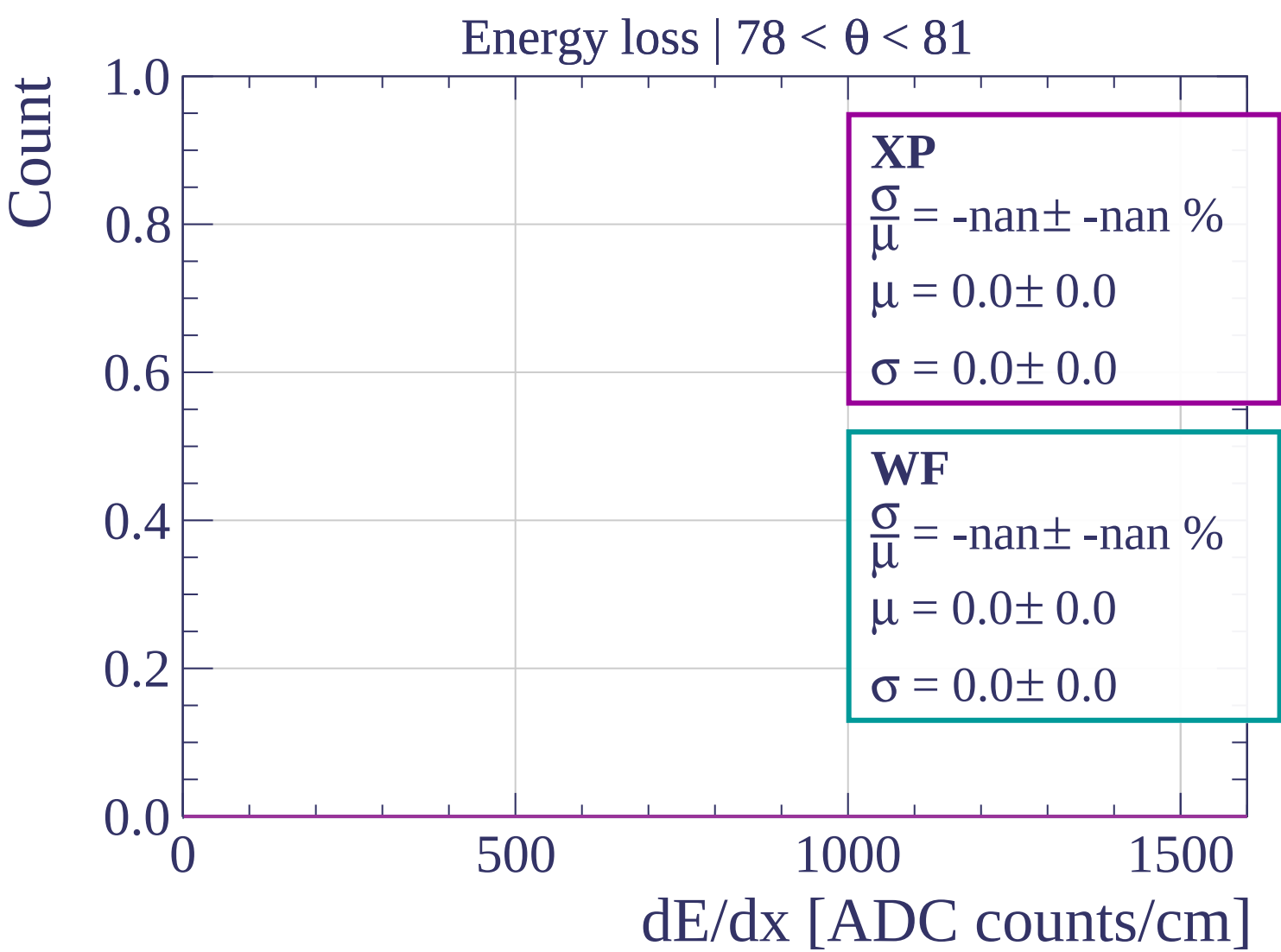
Count

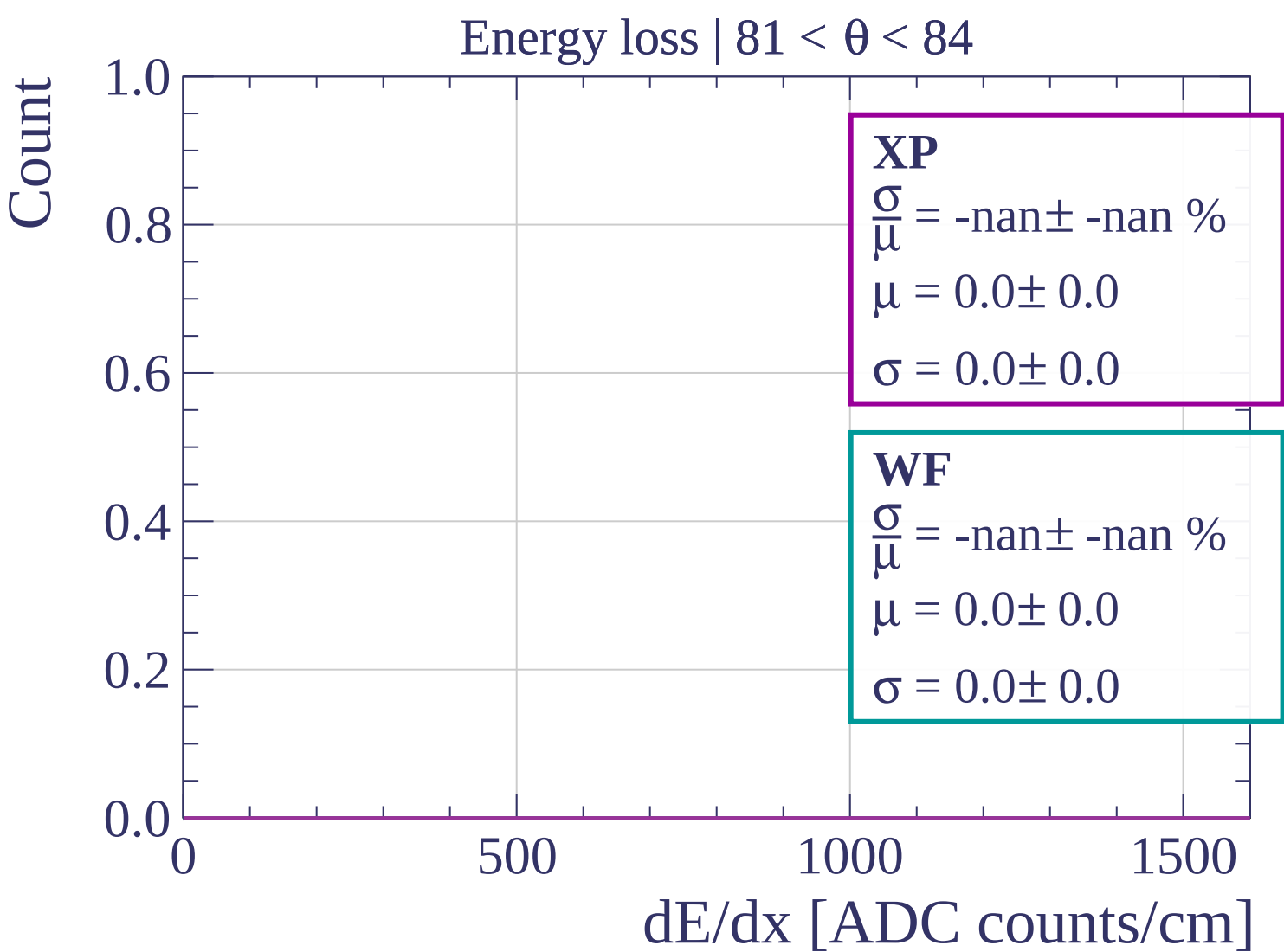


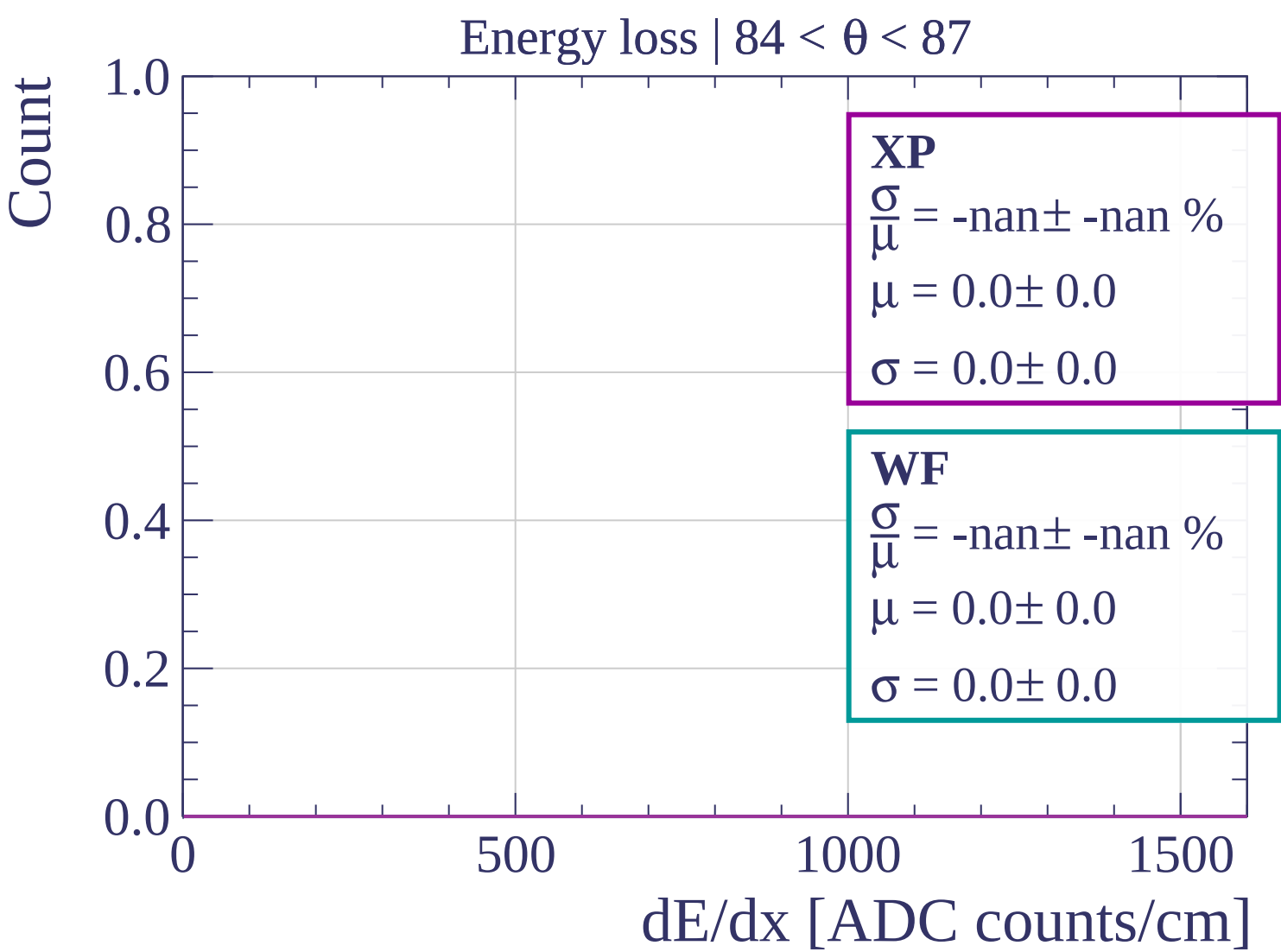
dE/dx [ADC counts/cm]

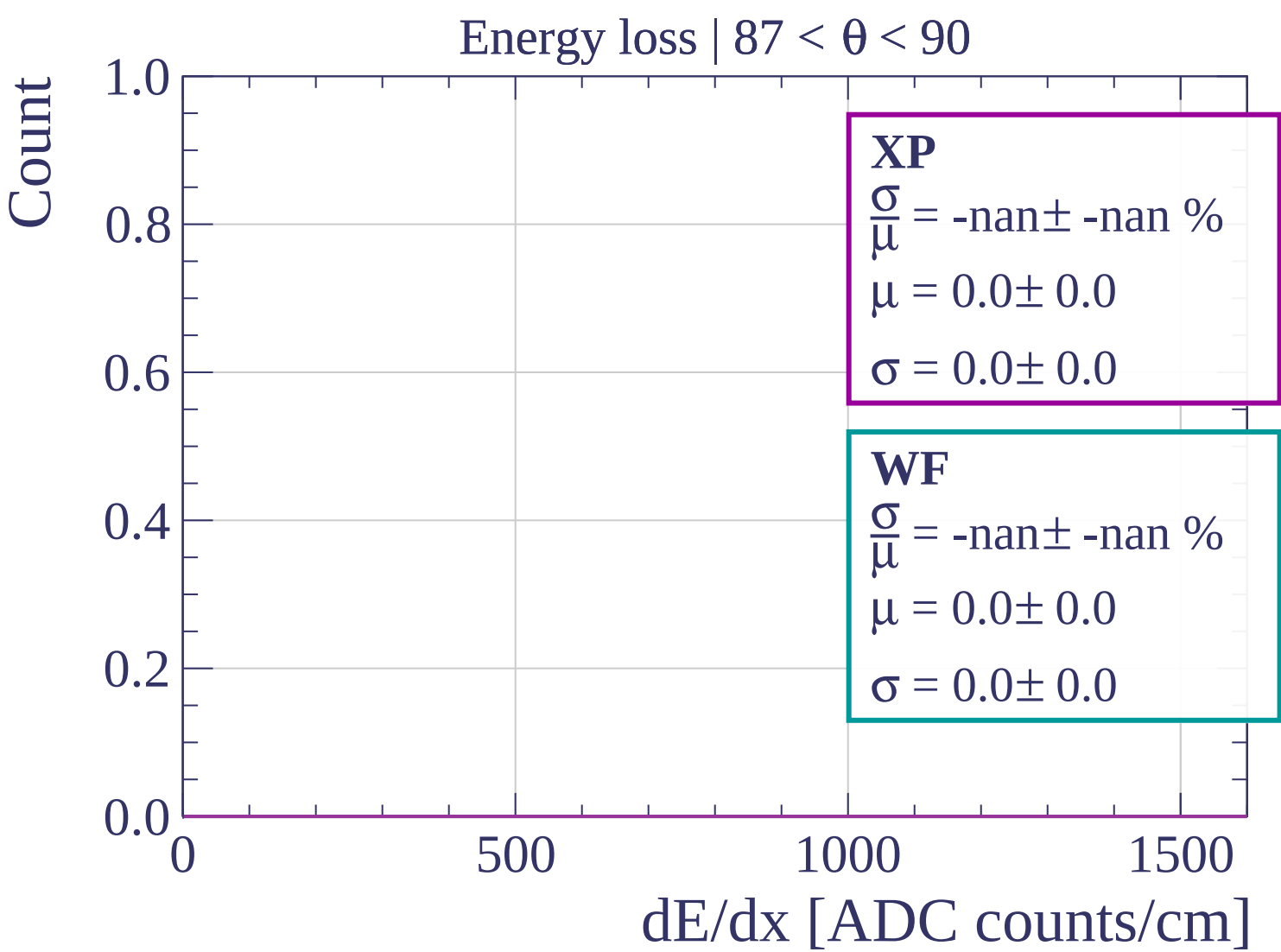




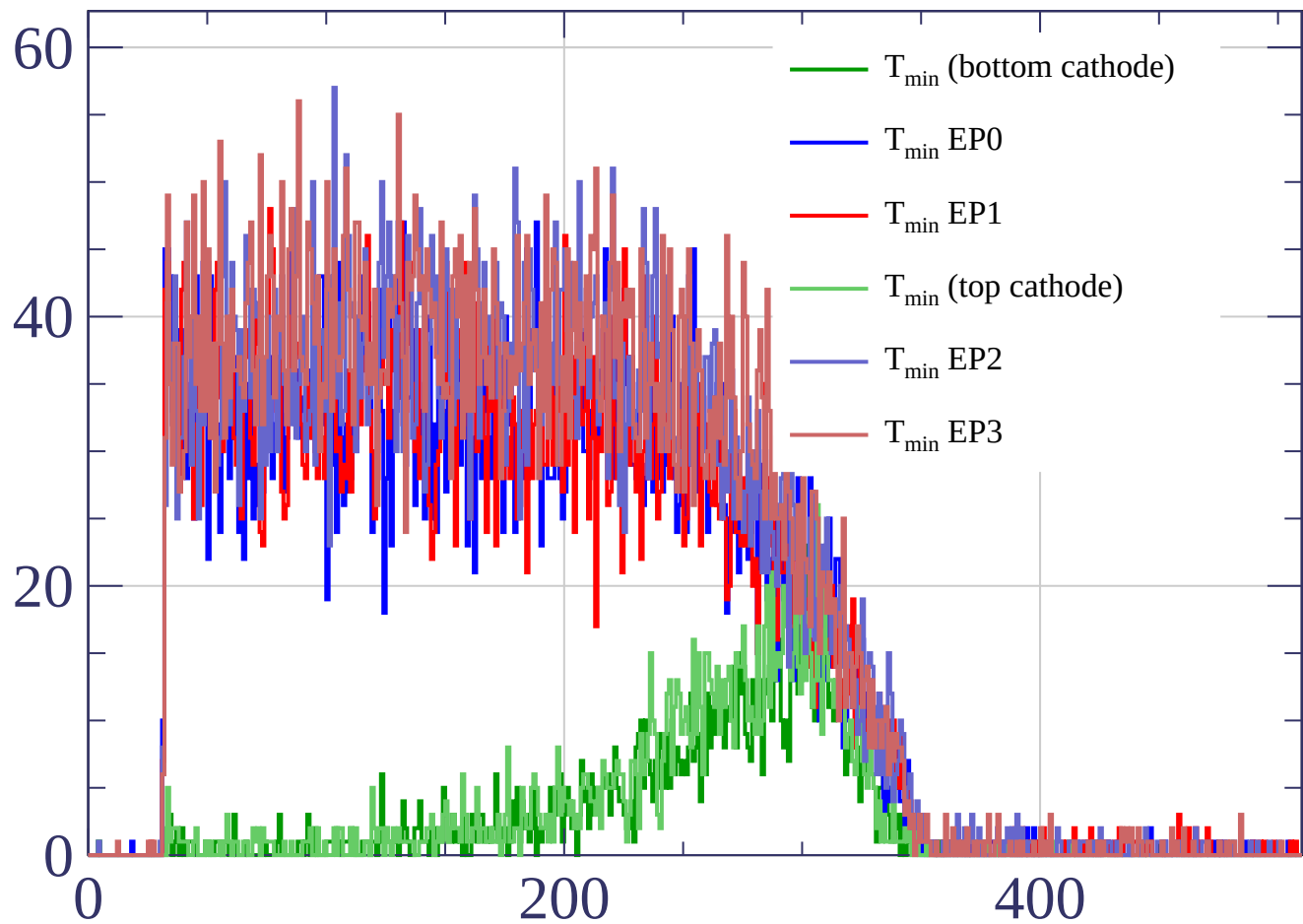








Count



Count

